MODELLING OF CRACKED REINFORCED CONCRETE CORROSION IN SERVICE ENVIRONMENT

A Thesis Submitted to the College of

Graduate and Postdoctoral Studies

In Partial Fulfillment of the Requirements

For the Degree of Doctor of Philosophy

In the Department of Civil, Geological and Environmental Engineering

University of Saskatchewan

Saskatoon

By

GANG LI

© Copyright Gang Li, August, 2023. All Rights Reserved.

Unless otherwise noted, copyright of the material in this thesis belongs to the author.

PERMISSION TO USE

In presenting this thesis in partial fulfilment of the requirements for a Postgraduate degree from the University of Saskatchewan, I agree that the Libraries of this University may make it freely available for inspection. I further agree that permission for copying of this thesis/dissertation in any manner, in whole or in part, for scholarly purposes may be granted by the professor or professors who supervised my thesis/dissertation work or, in their absence, by the Head of the Department or the Dean of the College in which my thesis/dissertation work was done. It is understood that any copying or publication or use of this thesis/dissertation or parts thereof for financial gain shall not be allowed without my written permission. It is also understood that due recognition shall be given to me and to the University of Saskatchewan in any scholarly use which may be made of any material in my thesis/dissertation.

Requests for permission to copy or to make other use of material in this thesis/dissertation in whole or part should be addressed to:

Head of the Department of Civil, Geological and Environmental Engineering
University of Saskatchewan
Engineering Building, 57 Campus Drive
Saskatoon, Saskatchewan S7N 5A9 Canada

OR

Dean
College of Graduate and Postdoctoral Studies
University of Saskatchewan
116 Thorvaldson Building, 110 Science Place
Saskatoon, Saskatchewan S7N 5C9 Canada

ABSTRACT

Corrosion of reinforced concrete (RC) structures is a complex problem, and corrosion rate prediction is a challenging task. The impact of cracking on corrosion rates during the propagation phase is influenced by concrete properties and further complicated by moisture content fluctuation, chloride ion contamination and carbonation in the cracked reinforced concrete structures.

Numerical simulation is a versatile tool that allows an in-depth assessment of corrosion of RC structures. However, current numerical models face challenges in describing the corrosion cell of rebars in concrete in the presence of spatial variations of microenvironment due to cracking and service exposure conditions.

The objective of this research was to develop a comprehensive mechanistic model for predicting the corrosion rate of cracked concrete in realistic service environments. In this model, the homogenized corrosion kinetics of rebar were coupled with the evolution of the non-uniform microenvironment in concrete caused by cracking of concrete and varying exposure conditions. This objective was achieved by conducting a series of experimental programs and numerical simulations.

The first-phase experimental program quantified the corrosion parameters of rebar such as corrosion potential, corrosion current density and Tafel slopes in simulated pore solutions and mortar with various chloride content, carbonation, and moisture conditions. A novel two-step potentiodynamic test routine was developed, and great care was taken to ensure a realistic rebar condition history such as preserving the mill scale, passivation of rebar prior to exposure to the corrosive environments, and extended exposure periods.

The experimental results revealed the synergistic effect of chloride, moisture, and carbonation on the corrosion parameters of rebar. It was found that the role of water saturation on corrosion includes 1) an area effect, which affects the rebar-pore solution area, 2) an ohmic control effect by changing the bulk resistivity, and 3) an anode-to-cathode ratio effect by altering the anodic and cathodic site area ratio. These effects complicate the role of chloride and carbonation on rebar corrosion in mortar, rather than being solely dependent on chloride concentration and pH in simulated pore solutions.

The first-phase study provided a database, from which the corrosion kinetics model was developed, where the homogenized corrosion parameters were empirical functions of chloride content, water saturation, and carbonation. In addition, it provided the basis for modelling the corrosion kinetics in concrete from corrosion in solution.

In the modelling phase, a circuit model was developed to simulate the polarization behaviour of a rebar experiencing non-uniform surface conditions. It started from localized corrosion kinetics for active and passive rebar-concrete units (mesoscopic–macroscopic level) and arrived at a simulated scan curve representing the overall performance of the rebar, from which the homogenized corrosion parameters were derived. The simulation results confirmed the hypothesis that the homogenized corrosion parameters are functions of concrete resistivity and the active-to-passive rebar-concrete interface area ratio.

Building upon the circuit model, a simulation approach was developed to derive the homogenized corrosion parameters of the rebar-concrete interface (mesoscopic–macroscopic level) from the corrosion kinetics of the rebar-solution interface (microscopic pore level). It featured in describing the allocation of active and passive rebar-solution corrosion sites in the representative elementary volume based on the non-uniform water and chloride distribution in concrete pores. The simulated homogenized corrosion kinetics were functions of the total chloride content and water saturation of concrete, and they captured the patterns of the empirical functions from the first-phase experimental study.

To simulate the microenvironment evolution at the rebar depth of a concrete structure, a comprehensive 3-D transport model (environment model) was developed. In the case study of an uncracked RC structure — the University Bridge subjected to service conditions with the source of chloride, carbonation, varying relative humidity, and exposure to unevenly sheltered run-off water, this model predicted the corrosion rates of century-old rebar in concrete by accurately modelling the microenvironment at rebar depth with a realistic depiction of the microclimate as well as various speculated climate scenarios, including elevated CO_2 levels and changes in precipitation.

Finally, a comprehensive corrosion model was formulated by incorporating crack configuration into the environment model and coupling it with the corrosion kinetics model. This model

simulated the transport of species such as water, chloride ion, oxygen, and carbon dioxide in the concrete domain and the electrochemical reaction on the rebar-concrete interface and solved for the potential and current fields. A second-phase experimental program was designed to validate this model, using segmented rebars in cracked concrete specimens subjected to wetting and drying cycles with chloride salt solutions.

The validated model enabled a numerical investigation of the mechanisms behind varying field observations of rebar corrosion in cracked concrete. It explained the complex effects of cracks on corrosion, including corrosion reduction over time, crack size and healing effects.

Under submerged conditions, cracks affect corrosion by affecting the diffusion rate of chloride ions through the crack. Thinner cracks lead to less chloride influx, resulting in a reduced peak corrosion current in the cracking zone. The corrosion rate decreases over time and eventually reaches a similar level limited by oxygen diffusion that is independent of crack size.

Under wetting and drying conditions, the corrosion kinetics of rebar in the cracking zone are affected by chloride and moisture, with the possibility of rebar acting as a cathode during drying cycles. The average corrosion rate in the cracking zone peaks during wetting cycles and decreases during subsequent drying cycles, with the peak values diminishing with repeated wetting and drying cycles. Micro-cell corrosion eventually becomes the primary contributor to rebar corrosion in the cracking zone and the rate depends on the chloride content on the rebar and oxygen concentration.

Crack healing results in reduced transport properties, dilution of high chloride content near the crack, and limited oxygen diffusion. It accelerates corrosion rate reduction during wetting and drying cycles. Less permeable concrete, such as high performance concrete, benefits more from crack healing with greater chloride dilution. The critical crack width below which it is considered harmless should be determined based on the effects of the crack on corrosion with specific material properties and exposure conditions.

CO-AUTHORSHIP

All the experimental and numerical modelling work that is presented in this thesis was conducted by Gang Li. Moh Boulfiza and Richard Evitts provided valuable input and contributions to the analytical work. All the manuscripts included in this thesis are the result of collaborative efforts between the authors.

The authors would like to express their sincere appreciation to the Natural Sciences and Engineering Research Council of Canada (NSERC), the City of Saskatoon, Mitacs, and the Department of Civil, Geological, and Environmental Engineering at the University of Saskatchewan for their financial support of this project.

The manuscript corresponding to Chapter 5 (The Impact of Non-uniformity and Resistivity on the Homogenized Corrosion Parameters of Rebars in Concrete - a Circuit Model Analysis) was published in Corrosion Engineering, Science and Technology. The manuscript was reformatted and included in the thesis with written permission. It is cited as

"Li, G., Evitts, R., & Boulfiza, M. (2023). The impact of non-uniformity and resistivity on the homogenised corrosion parameters of rebars in concrete – a circuit model analysis. *Corrosion Engineering, Science and Technology*, *58*(4), 399–409. https://doi.org/10.1080/1478422X.2023.2190442"

The manuscript corresponding to Chapter 3 (On the Corrosion Parameters Acquired through Potentiodynamic Scans of Carbon Steel Rebar in Simulated Pore Solution and Mortar) and the manuscript corresponding to Chapter 4 (Carbon Steel Rebar Corrosion in Mortar: the Interactive Effect of Moisture, Chloride, and Carbonation on Rebar Corrosion) were submitted to Construction and Building Materials in August 2023.

The manuscript corresponding to Chapter 7 (Effect of Accurate Microclimate Representation on the Corrosion Performance of a Bridge Subjected to Carbonation and Chloride Attack) was submitted to Case Studies in Construction Materials in September 2023.

Note: The authors declare no competing financial interest and no conflict of interest.

ACKNOWLEDGEMENTS

I would like to express my heartfelt gratitude to my co-supervisors, Dr. Moh Boulfiza and Dr. Richard Evitts, for their consistent encouragement and invaluable guidance in the fields of materials and structural engineering, corrosion engineering, and engineering ethics and practices throughout my Ph.D. journey.

I am deeply thankful to the members of my Committee, Dr. Leon Wegner, Dr. WonJae Chang, Dr. Jerzy Szpunar, and Dr. Chris Hawkes, for their inspiring advice and support.

A special thanks goes to Dr. Jim Zacaruk for his mentorship and collaborative research on industrial projects, which involved the development of innovative hardware and software tools for testing ageing reinforced concrete infrastructures.

I would like to express my appreciation to Dr. Carey Simonson for his patience during our lengthy discussions while taking his course on heat and mass transfer in porous media. Additionally, I would like to thank Dr. Andrew Ireson for his exceptional teaching in the programming of numerical modelling.

I am grateful to the Department of Civil, Geological, and Environmental Engineering, the Department of Chemical and Biological Engineering, and the Department of Computer Science for providing teaching opportunities that have enriched my academic journey.

I would also like to extend my sincere appreciation to the following individuals: Dr. Glyn Kennell and Regan Gerspacher from the corrosion lab for their guidance on electrochemical tests; Serge Nazarenko and Brennan Pokoyoway from the Structural Engineering Laboratory for their assistance in sourcing materials and constructing the carbonation chamber; Helen Yin from the environmental engineering lab for her support with chloride content testing; Rob Peace from the mechanical engineering machine shop for granting me the access to machining facilities for rebars; Daniel Vessey from the Engineering Shop for his prompt assistance in building sensors and customized apparatus. I must emphasize that this project would not have been practically feasible without the invaluable assistance of all those mentioned above.

Lastly, I wish to extend my appreciation to the Natural Sciences and Engineering Research Council of Canada (NSERC), the City of Saskatoon, Mitacs, and the Department of Civil, Geological, and Environmental Engineering at the University of Saskatchewan for their generous financial support, without which this project would not have been possible.

DEDICATION

To my beloved wife Yimeng,

To my son William,

To my parents,

I am eternally grateful for your love, understanding, and patience throughout this academic pursuit. Your presence and respect in my life have made all the difference and are always my source of strength.

TABLE OF CONTENTS

PERMISSION TO USE ..... **i**

ABSTRACT ..... **ii**

CO-AUTHORSHIP ..... **v**

ACKNOWLEDGEMENTS ..... **vi**

DEDICATION ..... **viii**

TABLE OF CONTENTS ..... **ix**

LIST OF TABLES ..... **xv**

LIST OF FIGURES ..... **xvii**

LIST OF ABBREVIATIONS ..... **xxx**

LIST OF SYMBOLS ..... **xxxi**

1 Introduction **1**

1.1 Background **1**

1.2 Limitations in current corrosion models **2**

1.3 Statement of the research question and significance **5**

1.4 Objectives **5**

1.5 Methodology and scope **6**

1.6 Organization of chapters **8**

2 Literature Review **11**

2.1 Non-uniform corrosion in cracked concrete **11**

2.1.1 Effect of cracks on corrosion - experimental observation 12

2.1.2 Numerical modelling of corrosion 14

2.1.3 Corrosion parameters used in models 22

2.2 Microenvironment in concrete and natural exposure conditions **26**

2.2.1 Moisture transport 26

2.2.2 Transport of gas 29

2.2.3 Transport of dissolved solutes 30

2.2.4 Natural exposure boundary conditions 31

3 On the Corrosion Parameters Acquired through Potentiodynamic Scans of Carbon Steel Rebar in Simulated Pore Solution and Mortar 33

3.1 Preface 33

3.2 Introduction 34

3.3 Material and methods 42

3.3.1 Sample preparation 42

3.3.2 Test procedure 44

3.4 Data processing 46

3.5 Results and discussion 47

3.5.1 IR compensation noise 47

3.5.2 Open Circuit Potential (OCP) drift with mortar surface wetting 48

3.5.3 Two-step scan vs. one-step scan 49

3.5.4 Depassivation time of rebar in the simulated solutions 51

3.5.5 Corrosion parameters for SPS and CSPS: the effect of chloride concentration and pH 53

3.5.6 Corrosion parameters in mortar: the effect of chloride content and conditioning RH 58

3.6 Conclusions 64

3.7 Epilogue 66

4 Carbon Steel Rebar Corrosion in Mortar: the Interactive Effect of Moisture, Chloride, and Carbonation on Rebar Corrosion 68

4.1 Preface 68

4.2 Introduction 69

4.2.1 Dependence of water saturation on RH in concrete 70

4.2.2 Dependence of rebar corrosion on water saturation in concrete 72

4.2.3 Objective 74

4.3 Resistivity test 74

4.4 Results and discussion 77

4.4.1 The effect of chloride content and carbonation on water retention 77

4.4.2 The effect of saturation, carbonation and chloride on Resistivity 79

4.4.3 The effect of chloride and carbonation: comparing rebar corrosion between mortar and simulated pore solutions 80

4.4.4 The effect of water saturation on rebar corrosion in mortar (ohmic control effect vs area effect) 82

4.5 Conclusions 95

4.6 Epilogue 96

4.7 Appendix for chapter 4 97

4.7.1 Statistical significance of the fitted lines of $icorr, norm$ vs Se at various chloride and saturation levels. 97

4.7.2 Non-linear regression of empirical models for corrosion parameters 99

5 The Impact of Non-uniformity and Resistivity on the Homogenized Corrosion Parameters of Rebars in Concrete - a Circuit Model Analysis 101

5.1 Preface 101

5.2 Introduction 102

5.3 Model development.....104

5.3.1 Theory.....104

5.3.2 Model setup 106

5.3.3 Homogenized potential 111

5.3.4 Simulated potentiodynamic scan 112

5.4 Results and discussion 113

5.4.1 Significance of IR compensation 114

5.4.2 Deviation of localized potential affected by resistance.....115

5.4.3 Polarization curve affected by resistance 117

5.4.4 Effect of the area ratio of activated and passivated rebar area and resistance on the fitted corrosion parameters 120

5.5 Conclusions 124

5.6 Limitations and suggestions for future works.....125

5.7 Epilogue 126

6 Chapter 6 A Pore-scale Corrosion Model for Rebar in Concrete: the Effect of Water and Chloride Distribution at the Pore Level on Rebar Corrosion in Concrete 127

6.1 Preface 127

6.2 Introduction 128

6.3 Methodology.....130

6.3.1 Homogenized corrosion kinetics....130
6.3.2 Corrosion modelling from microscopic (pore-scale) to mesoscopic scale (REV)....131
6.4 Pore model development....133
6.4.1 Pore size distribution and water distribution....133
6.4.2 Chloride ion in the pore solution....147
6.4.3 Allocation of the active and passive rebar-pore solution interface....154
6.5 Implementation of the corrosion model at the REV level....155
6.5.1 Corrosion circuit model....155
6.5.2 On the limitation of the corrosion circuit model....156
6.6 Corrosion kinetics in solution....157
6.6.1 Mortar REV resistivity....157
6.6.2 Equivalent resistors in the circuit model....159
6.7 Simulation results....160
6.7.1 The effects of water saturation and chloride content on active rebar-solution interface area ratio in concrete 160
6.7.2 Simulated homogenized corrosion parameters for rebar-concrete interface....161
6.8 Conclusion....163
6.9 Epilogue....164
7 Effect of Accurate Microclimate Representation on the Corrosion Performance of a Bridge Subjected to Carbonation and Chloride Attack....165
7.1 Preface....165
7.2 Introduction....166
7.3 Research significance....167
7.4 Field and experimental investigation....168
7.5 Numerical modelling....172
7.5.1 Transport model....172
7.5.2 Corrosion model....183
7.6 Simulation results and discussion....186
7.6.1 Climate change: the effect of elevated CO_2 concentration and change in precipitation on rebar corrosion....190
7.7 Conclusions....195

7.8 Epilogue 196

7.9 Appendix for Chapter 7 196

7.9.1 Pore solution extraction with centrifuge technique 196

7.9.2 Pore solution composition results 198

7.9.3 Water retention test 198

8 Comprehensive Corrosion Model: Presentation and Validation 200

8.1 Preface 200

8.2 Introduction 200

8.3 Experimental program 203

8.3.1 Material and sample preparation 203

8.4 Numerical corrosion model 209

8.4.1 Geometry setup 209

8.4.2 Water flow and transport of species 209

8.4.3 Electrochemical process 216

8.5 Validation of the corrosion model 220

8.5.1 Disconnected mode: open-circuit corrosion potential 222

8.5.2 Disconnected mode: corrosion current density 223

8.5.3 Connected mode: rebar segment potential 223

8.5.4 Connected mode: net current of each rebar segment 224

8.6 Conclusion 225

8.7 Epilogue 226

9 Numerical Investigation of Effects of Crack on Rebar Corrosion in Concrete 227

9.1 Preface 227

9.2 Summary of typical and peculiar observed behaviours (Lab and/or field) 227

9.3 Numerical investigation of effects of crack on corrosion 227

9.3.1 Submerged condition 229

9.3.2 Wetting and drying condition 236

9.3.3 Crack healing and critical crack width 248

9.3.4 Critical crack width 254

9.4 Conclusion 254

9.4.1 Submerged condition 255

9.4.2 Wetting and drying condition 255

9.5 Epilogue 256

10 Conclusions and Recommendations 258

10.1 Summary of work 258

10.2 Conclusions 259

10.2.1 The synergistic effect of chloride, moisture, and carbonation on the corrosion parameters of rebar in concrete. 260

10.2.2 Homogenized rebar corrosion parameters in concrete as a function of the variable microenvironment 263

10.2.3 Simulation of the evolution microenvironment at the rebar depth of a concrete structure in service conditions. 265

10.2.4 Validation of comprehensive corrosion model 265

10.2.5 Numerical investigation of effects of crack on rebar corrosion in concrete 266

10.3 Recommendations for future research 268

10.3.1 Effect of temperature on corrosion parameters 268

10.3.2 Enhancing interpretation of half-cell potential mapping results 269

10.3.3 Dynamic link between corrosion and concrete fracture models 269

REFERENCES 270

LIST OF TABLES

Table 2.1 Corrosion parameters for active and passive rebar (Sohail, 2013) 23

Table 2.2 corrosion parameters used by various macro-cell models 24

Table 3.1 Corrosion parameters of rebar in concrete 37

Table 3.2 Corrosion parameters of rebar in simulated pore solutions 38

Table 3.3 Corrosion parameters used in corrosion models 39

Table 3.4 NaCl contamination levels in mortar 43

Table 3.5 Dry chemicals used for the simulated pore solution, SPS (pH=13.5) 44

Table 3.6 Dry chemicals used for the simulated carbonated pore solution, CSPS (pH=8.0) 44

Table 3.7 Potentiodynamic scan test control settings 46

Table 4.1 Chloride content levels in mixing water and in mortar samples 75

Table 4.2 Compared with baseline type [T.7N0] 98

Table 4.3 Compared with baseline type [T.7N1] 98

Table 4.4 Compared with baseline type [T.7N2] 98

Table 4.5 Compared with baseline type [T.7C0] 99

Table 4.6 Compared with baseline type [T.7C1] 99

Table 4.7 Parameters for the fitted functions 100

Table 5.1 Symbol descriptions of the circuit model 109

Table 5.2 Corrosion parameters of an active/passive unit, with data from (Sohail, 2013) 113

Table 5.3 Base parameters for rebar area, area ratio, and resistance 113

Table 6.1 Length scales in a mortar 145

Table 6.2 Coefficient of variation of the 30 repeatedly sampled total porosities for selected REV sizes 147

Table 6.3 Classification of chloride, summarized from (Glass et al., 1996; Pargar et al., 2017) 148

Table 6.4 Rebar-solution interface types 154

Table 6.5 Corrosion parameters of an active/passive unit 157

Table 7.1 Pore solution composition of the arch concrete 184

Table 7.2 Climate scenarios regarding $\mathbf{CO2}$ emission and precipitation over the next 50 years 191

Table 7.3 Comparison of Rebar Section Loss in Three Climate Scenarios over 50 Years 193

Table 7.4 Pore solution composition 198

Table 8.1 Moisture transport parameters 215

Table 8.2 Chloride transport parameters 216

LIST OF FIGURES

Figure 2.1 Evans diagrams for a single-electrode (a) and a mixed-electrode (b) reaction. 18

Figure 2.2 Schematic example of Evans Diagram and Stern-Geary Diagram of the coupled corrosion reactions (adapted from Perez, 2004). 18

Figure 3.1 Stern-Geary diagrams plotted with the reversible potentials and exchange current densities of the half-cell reactions utilized by different modellers for reinforced concrete corrosion simulation, as reported in Table 3.3. 40

Figure 3.2 Schematic for rebar and mortar sample dimensions. Rebar sample is a NO. 10 deformed rebar and mortar sample has a water-cement ratio of 0.7. 42

Figure 3.3 The setup of three-electrode potentiodynamic scan test for (a) rebar in solution, and (b) rebar in mortar cylinder. Both setups are placed in a faraday shield during testing. 46

Figure 3.4 The effect of Faraday shield on noise reduction caused by IR compensation, (a) an example of rebar sample in 3% (wt.%) NaCl solution and (b) an example of rebar cast in a mortar with 3% (wt.%) NaCl mixing water at ambient dry environment. The control datasets were acquired without the use of a Faraday shield. 48

Figure 3.5 An example of open circuit potential (OCP) drift of the rebar embedded in a carbonated mortar sample when the surface is wrapped with a wet sponge. 49

Figure 3.6 Potentiodynamic scan curves of rebar in CSPS with 3% (wt%) NaCl solution: (a) One-step scan and (b) two-step scan. Note: the small difference in $\boldsymbol{Ecorr}$ (less than 10 mV) of the two-step scan branches was offset in the plot for better comparison. 50

Figure 3.7 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: corrosion potential of pre-passivated rebar in (a) SPS for 72 hours and 20 days and (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 52

Figure 3.8 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: corrosion current density of pre-passivated rebar in (a) SPS for 72 hours and 20 days and (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 53

Figure 3.9 Comparison of corrosion potential (a) and corrosion current density (b) of pre-passivated rebar in SPS and CSPS with chloride after 20 days of conditioning. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results..... 54

Figure 3.10 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: anodic Tafel slopes of pre-passivated rebar in (a) SPS for 72 hours and 20 days (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 56

Figure 3.11 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: cathodic Tafel slopes of pre-passivated rebar in (a) SPS for 72 hours and 20 days (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 58

Figure 3.12 The effect of the chloride content and conditioning RH on the corrosion potential of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 60

Figure 3.13 The effect of the chloride content and conditioning RH on the corrosion current density of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 61

Figure 3.14 The effect of the chloride content and conditioning RH on the anodic Tafel slope of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 62

Figure 3.15 The effect of the chloride content and conditioning RH on the cathodic Tafel slope of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of

conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results. 63

Figure 4.1 Schematic representation of the meniscus formation of water in a capillary pore..... 71

Figure 4.2 Resistivity test of mortar cylinder samples using the direct-current method. 76

Figure 4.3 Porosity of non-carbonated and carbonated mortar samples (w/c=0.7) with varying chloride levels 0-5 as listed in Table 4.1. Chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement. The mean value point is shown, and the error bars represent the range of the three repeated test results. 77

Figure 4.4 Water saturation level of carbonated and non-carbonated mortar conditioned at various RH 65% 85% 95% and 100% (submerged). Legend: N (Non-carbonated), C (Carbonated), 0-5 (chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results. 78

Figure 4.5 Resistivity of non-carbonated and carbonated concrete conditioned at various saturation levels. Legend: N(Non-carbonated), C(Carbonated), 0-5 (chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results. 80

Figure 4.6 The path diagram of quaulitative influence of chloride ion and carbonation on rebar corrosion in pore solution and concrete. 82

Figure 4.7 Corrosion potential vs. saturation of mortar samples with chloride contents (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results. 83

Figure 4.8 icorr vs. saturation for mortar samples with various chloride content (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean

value point is shown, and the error bars represent the range of the three repeated test results. 88

Figure 4.9 Linear dependence of normalized corrosion current density, $\boldsymbol{icorr,norm}$ with water saturation level for rebar in mortars. (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement. The dashed lines represent linear regression lines... 89

Figure 4.10 First derivative of normalized corrosion current density with respect to water saturation for rebar corrosion in mortars. (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The error bars represent a 95% confidence bound of the fitted parameter. 90

Figure 4.11 Hypothetical corrosion current density, $\boldsymbol{icorr,pore}$ in saturated pores in non-carbonated mortar and corrosion current density in the simulated pore solutions for 20 days. Legend: Cl level 0-5 represent chloride content levels in mortar 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement, and in simulated pore solution 0: 0%, 1: 0.5%, 2: 1.0% 3: 2.0%, 4: 3.0%, 5: 5.0%, NaCl wt.% by weight of solution. The mean value point is shown, and the error bars represent the range of the three repeated test results. 93

Figure 4.12 Hypothetical corrosion current density, $\boldsymbol{icorr,pore}$ in saturated pores in carbonated mortar and corrosion current density in the carbonated simulated pore solutions for 20 days. Legend: Cl level 0-5 represent chloride content levels in mortar 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement, and in carbonated simulated pore solution 0: 0%, 1: 0.5%, 2: 1.0% 3: 2.0%, 4: 3.0%, 5: 5.0%, NaCl wt.% by weight of solution. The mean value point is shown, and the error bars represent the range of the three repeated test results. 94

Figure 5.1 Evans diagrams for a single-electrode (a) and a mixed-electrode (b) reaction. 105

Figure 5.2 Schematic of the multi-level treatment of rebar corrosion in concrete (n, l, m is the number of “pore-level” rebar segments). 107

Figure 5.3 Circuit diagram of the proposed model. 108

Figure 5.4 IR drop increases with an increase in resistance through concrete cover and a decrease in resistance between the dissimilar corrosion units. 115

Figure 5.5 The influence of resistance on localized potentials and homogenized potential..... 116

Figure 5.6 The influence of resistance on simulated polarization scan curves at "unit level" and homogenized "high-level". 118

Figure 5.7 Tafel extrapolation on a distorted polarization curve: Case 1- significantly distorted and Case 9- insignificantly distorted (same line legend as Figure 5.5 and Figure 5.6, omitted for simplicity)..... 119

Figure 5.8 Corrosion current density,$\boldsymbol{icorr}$, from the open circuit mode and scan curve fitting with varying active unit area ratio. a) high resistance: $\boldsymbol{R0}$, $\boldsymbol{Re}$ =1000 $\boldsymbol{\Omega}$, b) low resistance: $\boldsymbol{R0}$, $\boldsymbol{Re}$ = 10 $\boldsymbol{\Omega}$. 120

Figure 5.9 Curve fitting with the Butler-Volmer equation that showed the largest deviation in $\boldsymbol{icorr}$. ($\boldsymbol{R0}$, $\boldsymbol{Re}$ = 1000 $\boldsymbol{\Omega}$, active unit area/total area is 0.5, data range: $\boldsymbol{Ecorr} \pm 0.25$ V). 121

Figure 5.10 Corrosion potential, $\boldsymbol{Ecorr}$, from the open circuit mode and scan curve fitting with varying active unit area ratio. a) high resistance: $\boldsymbol{R0}$, $\boldsymbol{Re}$ =1000 $\boldsymbol{\Omega}$, b) low resistance: $\boldsymbol{R0}$, $\boldsymbol{Re}$ = 10 $\boldsymbol{\Omega}$. 122

Figure 5.11 Effect of the active unit area ratio and resistance on the fitted anodic Tafel slope. 122

Figure 5.12 Effect of the active unit area ratio and resistance on the fitted cathodic Tafel slope. 123

Figure 6.1 Schematic of rebar condition and chloride concentration distribution: localized pore level non-uniformity vs. REV-level homogenization. 130

Figure 6.2 A schematic depicting the corrosion of the rebar surface, from pore level to REV. The rebar condition within the REV encompasses a range of conditions, whereas at the pore level, it is binary..... 132

Figure 6.3 Proposed model for characterizing corrosion kinetics in reinforced concrete: schematic representation of the process flow from pore size distribution to corrosion kinetics. 133

Figure 6.4 Adsorption-condensation model of a capillary pore: Illustration of the conceptualization of pores as a single pore with varying diameter. The change in adsorbed water ($\boldsymbol{\Delta\theta a}$) and condensed water content ($\boldsymbol{\Delta\theta c}$) is due to an increase in relative humidity ($\boldsymbol{\Delta h}$). 134

Figure 6.5 Schematic depicting the transition from adsorption to adsorption + condensation regime as relative humidity increases in pores. 135

Figure 6.6 Flowchart of the iterative solving procedure to determine the pore distribution from the water retention curve. 140

Figure 6.7 Determination of the pore size distribution with the adsorption-condensation model and water retention data. The condensation regime begins at a relative humidity of 0.4... 141

Figure 6.8 Solved pore size distribution (a) surface area per unit volume of mortar for pore size greater than r. (b) pore volume per unit volume of mortar for pore size less than r. 142

Figure 6.9 The probability density function of the pore size distribution derived from the water retention curve of mortar. 143

Figure 6.10 Effect of the selected REV size on the scatteredness of volume-averaged porosity at the rebar-mortar interface. Each violin plot shows the probability density of the porosity of the sampled volume. 146

Figure 6.11 Chloride binding isotherm representing the relationship between chloride concentration in solution and chloride content in hardened cement paste. The point marked with a circle is excluded in the curve. 149

Figure 6.12 Schematic of the EDL effect on the 1-D infinite concentration profile in the diffusion layer from OHP. 150

Figure 6.13 Flowchart of the iterative solving procedure to determine the chloride concentration distribution in pore solution. 152

Figure 6.14 Chloride concentration distribution in saturated pores for concrete with a total chloride content of 0.6% by weight of cement. 153

Figure 6.15 The illustration of the updated corrosion circuit model that homogenize the corrosion kinetics in REV from pore level. 156

Figure 6.16 Equivalent circuit of the corrosion model at the REV level. 160

Figure 6.17 Active rebar-solution surface area ratio as a function of water saturation and total chloride content. 161

Figure 6.18 Corrosion parameters of rebar-concrete interface affected by the total chloride content and water saturation. (a) simulated corrosion potential $\boldsymbol{Ecorr}$, (b) experimental corrosion potential $\boldsymbol{Ecorr}$, (c) simulated corrosion current density $\boldsymbol{icorr}$, (d) experimental corrosion current density $\boldsymbol{icorr}$. Marks represent experimental data points. 162

Figure 6.19 The relative error of corrosion current density generated from the model and the corrosion current density fitted from the simulated potentiodynamic scan curve using the Butler-Volmer equation. (a) relative error – $\boldsymbol{icorr}$ relationship (b) relative error- water saturation and chloride content relationship. 163

Figure 7.1 Field program (a) location of Arch C and D of the University Bridge (b) location of water content and oxygen content probes and pore solution extraction cores on Arches C and D. 169

Figure 7.2 The field program shows the distribution of (a) oxygen content, (b) moisture content, and (c) half-cell potential over the arches. 171

Figure 7.3 Water saturation as a function of pressure head. The experimental data were fitted with van Genuchten-Mualem model. 175

Figure 7.4 Sheltering categories of the arch: Sheltered, Unsheltered, and Partially sheltered areas based on field investigation findings. 178

Figure 7.5 Moisture boundary conditions for the sheltered portion of an arch: (a) Raw data of pressure head obtained from weather stations; (b) averaged data fitted from the raw data. 179

Figure 7.6 Moisture boundary conditions for the unsheltered portion of the arch (eg, Year 2013). 180

Figure 7.7 Workflow to get the equivalent boundary conditions for the arch top surface exposed to ambient humidity and rain. 181

Figure 7.8 Comparison of the saturation results simulated from the equivalent boundary conditions vs. the spiky boundary conditions of year 2009-2014 for the unsheltered portion of the arch. 182

Figure 7.9 Corrosion current density as a function of pH and chloride concentration in the simulated pore solutions. 185

Figure 7.10 Reinforced concrete corrosion rate reduction factor relative to saturated condition, adapted from Tuutti's work (Dangla & Dridi, 2009; Tuutti, 1982). 185

Figure 7.11 pH distribution in Arch C (Bay 5, Bay 6, Bay 7) after 100 years of carbonation. . 186

Figure 7.12 Chloride concentration distribution in Arch C (Bay 5, Bay 6, Bay 7) after 100 years of exposure. 187

Figure 7.13 View of the rebar in Arch C (Bay 5, 6, and 7) showing corrosion loss in thickness after 100 years of exposure, plotted in log scale: (a) 3D isometric view, (b) plan view of arch. 188

Figure 7.14 View of the rebar in Arch C (Bay 5, 6, and 7) showing additional corrosion loss during 101 years –150 years of exposure, plotted in a log scale: (a) 3D isometric view, (b) plan view of arch. 190

Figure 7.15 Field observation showing concrete patches for corrosion-induced damage on Arch C in the unsheltered area where high corrosion section loss/rust accumulation was predicted. 191

Figure 7.16 Time series plot of monthly mean $\mathbf{CO2}$ concentration from 1958 to 2021 (US Department of Commerce). The dotted line extrapolates with a linear upward trend in $\mathbf{CO2}$ concentration. 192

Figure 7.17 Rebar section loss across the arch over 150 years in service. The last 50 years has different climate scenarios. Scenario 1: Baseline, Scenario 2: increased $\mathbf{CO2}$and wet (20% more precipitation), Scenario 3: increased $\mathbf{CO2}$ and drought (20% less precipitation). Each violin represents the distribution density of the section loss over all rebars at a given time. Mean values of section loss are connected by dashed lines. 194

Figure 7.18 Vacuum-saturation of the extracted concrete cores. 197

Figure 7.19 Centrifuge technique (a) Two concrete cores placed in the swing buckets in a centrifuge machine (BECKHAM J6-HC) (b) Pore solution collected from two concrete cores. 197

Figure 7.20 Water retention test setup (a) suspended sliced concrete cores to be placed in a humidity chamber (b) humidity chambers with varying relative humidity. 199

Figure 8.1 Cracked reinforced concrete beam sample and rebar configuration (a) reinforced concrete beam dimensions (b) segmented rebar connection design (c) Photo of the segmented rebar. Dimension unit: mm. 204

Figure 8.2 Three-point loading to create vertical cracks. 204

Figure 8.3 Epoxy-coated beam samples and cylinder samples. 205

Figure 8.4 Water loss of beam samples during the wetting and drying. Legend: w/c is the water/cement ratio, and crack width is the surface crack opening size. 206

Figure 8.5 Open circuit potential test in the disconnected mode. 207

Figure 8.6 Linear polarization test in the disconnected mode. 207

Figure 8.7 Current test between first and second rebar segment. 208

Figure 8.8 Definition of the positive current direction for the connected mode. 208

Figure 8.9 3-D model geometry implementing half of the beam sample with the segmented rebar. 209

Figure 8.10 Water and chloride transport boundary conditions for exposed and sealed concrete surface in wetting and drying cycles. 214

Figure 8.11 Calibration of water transport parameters using a 1-D model implementation with the water content change in cylinder samples. 215

Figure 8.12 Calibration of chloride penetration model using a 1-D model implementation with the experimental chloride profile in mortar cylinder samples. (a) w/c = 0.7, (b) w/c = 0.5. 216

Figure 8.13 Corrosion potential as a function of chloride content and water content. 218

Figure 8.14 Corrosion current density as a function of chloride content and water content. 219

Figure 8.15 Example of solved models in (a) disconnected mode for rebar #4 and (b) connected mode for rebar #1 – 4 at Day 8 (one day after the first wetting cycle). Surface plot shows the electrode potential and streamline plot shows the electrolyte current density vector… 221

Figure 8.16 The locations of potential data extraction in the 3-D model during the post-processing procedure. 221

Figure 8.17 Validation of open circuit potential in the disconnected mode. 222

Figure 8.18 Validation of the current density of the rebar segments in the disconnected mode. 223

Figure 8.19 Validation of the potential of each rebar segment in the connected mode. 224

Figure 8.20 Validation of the net current of each rebar segment in the connected mode. 225

Figure 9.1 Geometric configuration of the 2-D corrosion model of the cracked beam. 228

Figure 9.2 Model boundary condition settings for the investigated exposure conditions: submerged condition and wetting and drying cycles. 229

Figure 9.3 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the submerged condition at the early stage (day 30) and developed stage (day 180). 230

Figure 9.4 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. 231

Figure 9.5 Corrosion current density distribution along the rebar at different times. 232

Figure 9.6 Averaged corrosion current in the cracking zone at different times for the submerged condition with 1 mm crack. 232

Figure 9.7 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the submerged condition at the early stage (Day 30) and developed stage (Day 180). Case: 0.1 mm crack. 233

Figure 9.8 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack. 234

Figure 9.9 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack. 235

Figure 9.10 Averaged corrosion current in the cracking zone decreases with time due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack. 235

Figure 9.11 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the wetting cycles at the early stage (day 28.5) and developed stage (day 175.5). 237

Figure 9.12 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 1 mm crack, wetting cycles. 238

Figure 9.13 Corrosion current density distribution along the rebar at different times. Case: 1 mm crack, wetting cycles. 238

Figure 9.14 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the drying cycles at the early stage (day 32) and developed stage (day 182). 239

Figure 9.15 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 1 mm crack, drying cycles. 240

Figure 9.16 Corrosion current density distribution along the rebar at different times. Case: 1 mm crack, drying cycles. 241

Figure 9.17 Averaged corrosion current in the cracking zone during wetting and drying cycles. Case: 1 mm crack, wetting and drying cycles. 242

Figure 9.18 Maximum corrosion current during wetting and drying cycles. Case: 1 mm crack, wetting and drying cycles. 242

Figure 9.19 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the wetting cycles at the early stage (Day 28.5) and developed stage (Day 175.5). 243

Figure 9.20 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack, wetting cycles. 244

Figure 9.21 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack, wetting cycles. 244

Figure 9.22 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the drying cycles at the early stage (day 32) and developed stage (day 182). 245

Figure 9.23 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack, drying cycles. 246

Figure 9.24 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack, drying cycles. 246

Figure 9.25 Averaged corrosion current in the cracking zone decreases in the drying cycle and decreases overtime due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack, wetting and drying cycles. 247

Figure 9.26 Maximum corrosion current on the rebar fluctuates in the wetting and drying cycles and decreases with time due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack, wetting and drying cycles. 248

Figure 9.27 Linear reduction of transport properties such as hydraulic conductivity and chloride and oxygen diffusivity as a function of time when crack width with a healable crack width of 0.08 mm. 250

Figure 9.28 The hydraulic conductivity along the crack line changes due to crack healing. 251

Figure 9.29 The diffusivity of chloride ions and oxygen gas along the crack line changes due to crack healing. 251

Figure 9.30 The evolution of chloride content profile (wt.% Cl by weight of cement) around the crack (0.1 mm surface opening) after complete crack healing (Day 60) with a width up to

0.08 mm and crack width greater than 0.08 mm remains unhealed. Left: high performance concrete (HPC), Right: control concrete. 252

Figure 9.31 The effects of crack healing on the reduction of corrosion rate in the cracking zone. The surface crack width is 0.1 mm; the salt solution in the wetting condition is 3.0 wt.% NaCl. HPC (high performance concrete) has lower hydraulic conductivity and chloride diffusivity by a reduction factor of 0.2 to the control concrete. Low oxygen diffusivity $\boldsymbol{Deff O2}$ refers to the diffusivity that could have caused oxygen to be depleted eventually on the rebar if there is no crack healing effect. 253

LIST OF ABBREVIATIONS

ACI	American Concrete Institute
ANCOVA	Analysis of Covariance
ASTM	American Society for Testing and Materials
BC	Boundary Condition
C-S-H	Calcium Silicate Hydrates
C_3A	Tricalcium aluminate
C_4AF	Tetracalcium aluminoferrite
CSE	Copper–Copper Sulfate Electrode
CSPS	Carbonated Simulated Pore Solution
CV	Coefficient of Variation
DC	Direct Current
EDL	Electrical Double Layer
FEM	Finite Element Method
GFRP	Glass Fibre Reinforced Polymer
HPC	High Performance Concrete
LPR	Linear Polarization Resistance
MIT	Mercury Intrusion Test
NSERC	Natural Sciences and Engineering Research Council of Canada
OCP	Open Circuit Potential
OHP	Outer Helmholtz Plane
RC	Reinforced Concrete
REV	Representative Elementary Volume
RH	Relative Humidity
SCE	Saturated Calomel Electrode
SCETI	Saskatchewan Centre of Excellence for Transportation and Infrastructure
SPS	Simulated Pore Solution
TDR	Time Domain Reflectometry

LIST OF SYMBOLS

Symbol	Description	Unit
$\langle \dot{m}_{CO_2} \rangle^g$	rate of the mass density change of the CO_2 in the gas phase	$kg/m^3/s$
$\langle \rho_{CO_2} \rangle^g$	CO_2 density in the gas phase	kg/m^3
$\bar{c}_{c0}$	mean concentration	mol/m^3
$\langle E_{rebar} \rangle$	homogenized area-averaged potential	V
M_{CO_2}	molar mass of CO_2	kg/mol
$\langle \epsilon_p \rangle$	volume-averaged porosity	$-$
$\langle \sigma_{sol} \rangle$	local volume-averaged electrical conductivity of the pore solution	S/m
a_{O_2}	activity of oxygen gas	$-$
a_{OH^-}	activity of hydroxide ion	$-$
A_{react}	The total rebar-pore solution area in saturated and unsaturated pores per unit volume of the porous medium	$-$
A_{sat}	The rebar-pore solution area in saturated pores per unit volume of the porous medium	$-$
A_{unsat}	The rebar-pore solution(adsorbed) area fraction in unsaturated pores per unit volume of the porous medium	$-$
A_{X-REV}	cross-sectional area of the REV perpendicular to the pore direction in the bundled capillary pore model	m^2
c_0	O_2 concentration in bulk solution	mol/m^3
c_b	concentration in bulk solution	mol/m^3
c_{Cl^-}	chloride concentration in solution	mol/m^3
c_i	concentration of i_{th} ion	mol/L
c_{OH^-}	hydroxide ion concentration	mol/m^3
c_s	O_2 concentration adjacent to the rebar surface	mol/m^3
d_+	dimensionless length of the pore diameter	$-$
D^d	dispersion coefficient	m^2/s
D_{eff}^g	effective diffusivity of CO_2 in the gas phase	m^2/s
D_{eff}	effective diffusivity	m^2/s
D_l	liquid water diffusivity	m^2/s
D_s	effective diffusivity under fully saturated conditions	m^2/s

D_{toal}	total diffusivity of liquid and vapour	m^2/s
D_v	vapour diffusivity	m^2/s
$E^o_{O_2/OH^-}$	reversible potential of O_2/OH^- half cell reaction under standard condition	V
E_{O_2/OH^-}	reversible potential of O_2/OH^- half cell reaction	V
$e_a(r)$	water adsorption thickness as a function of pore radius	m
E^{act}	localized electrode potential on activated rebar unit	V
E_{app}	applied electrode potential on the concrete cover surface	V
$E_{corr,measured}$	corrosion potential reported by a voltmeter	V
E_{corr}	corrosion potential, equilibrium potential	V
$E_{eq}^{O_2}$	equilibrium potential of OH^-/O_2 reversible reaction	V
E_{eq}^{Fe}	equilibrium potential of Fe^{2+}/Fe reversible reaction	V
E_{eq}	equilibrium potential	V
E^{pas}	localized electrode potential on passivated rebar unit	V
E_{rebar}^{IRcomp}	corrected rebar potential measured from concrete cover surface	V
G_i	conductivity coefficient	$(mol/L)^{-1/2}$
h_c	critical RH, at which the condensation occurs in the smallest pores	$-$
H_p	pressure head	m
$I_{(n-1),n}$	current flowing between the $(n-1)^{th}$ and the nth rebar segments	A
$i_\perp$	$\boldsymbol{i}$ component perpendicular to the electrode-electrolyte interface	A/m^3
$i_0^{O_2}$	exchange current density of OH^-/O_2 reversible reaction	A/m^2
i_0^{Fe}	exchange current density of Fe^{2+}/Fe reversible reaction	A/m^2
i_0	exchange current density	A/m^2
I_0	"high-level" internal current near the rebar surface between rebar units	A
I_1, I_2	currents in the circuit	A/m^2
$i_a^{O_2}$	anodic current density of OH^-/O_2 reversible reaction	A/m^2
i_a^{Fe}	anodic current density of Fe^{2+}/Fe reversible reaction	A/m^2

I^{act}	"high-level" total ("unit-level" external) current on activated rebar unit	A
$i_c^{O_2}$	cathodic current density of OH^-/O_2 reversible reaction	A/m^2
i_c^{Fe}	cathodic current density of Fe^{2+}/Fe reversible reaction	A/m^2
i_c	cathodic current (negative)	A/m^2
$i_{corr,mortar}$	volume averaged corrosion current density at the rebar-mortar interface	A/m^2
$i_{corr,pore}$	hypothetical corrosion current density in saturated pores (Chapter 4)	A/m^2
$i_{corr,pore}$	local volume averaged corrosion current density at rebar-pore solution interface (Chapter 6)	A/m^2
$i_{corr,sat}$	corrosion current density of rebar in fully saturated mortar	A/m^2
$i_{corr,norm}$	normalized corrosion current density	$-$
i_{corr}	corrosion current density	A/m^2
I_e^{act}	"high-level" external current from activated rebar unit	A
I_e^{pas}	"high-level" external current from passivated rebar unit	A
i_{F_net}	current density corresponding to rust film growth (negative) or dissolution (positive)	A/m^2
I_m	ionic strength	mol/L
I_n	current flowing out from the nth rebar segment	A
$i_{net}^{O_2}$	net current density of OH^-/O_2 reversible reaction	A/m^2
i_{net}^{Fe}	net current density of Fe^{2+}/Fe reversible reaction	A/m^2
i_{net}	net current density	A/m^2
I^{pas}	"high-level" total ("unit-level" external) current on passivated rebar unit	A
k_{CO_2}	Carbonation reaction coefficient	$1/s$
K_f	hydraulic conductivity of the crack	m/s
K_l	hydraulic conductivity or coefficient of permeability	m/s
k_l	permeability	m^2
K_n	Knudsen number	$-$
k_r	relative permeability	$-$
K_s	hydraulic conductivity of concrete at full saturation	m/s
K_{total}	total hydraulic conductivity of liquid and vapour	m/s

m_l	total volume-averaged liquid water mass content	kg/m^3
$m_{l\to v}$	liquid-to-vapour phase change mass content	kg/m^3
m_{l+v}	total mass content of liquid water and water vapour	kg/m^3
m_{od}	mass of the sample under oven-dry condition	kg
m_{sample}	mass of the sample	kg
m_{ssd}	mass of the sample under saturated-surface-dry condition	kg
m_v	total volume-averaged water vapour mass content	kg/m^3
$m_{v\to l}$	vapour-to-liquid phase change mass content	kg/m^3
$\mathrm{N}_{(\mathrm{C_3A})_{\mathrm{eq}}}$	residual quantity of equivalent aluminates per unit weight of cement	mol/kg
$\mathrm{N}_{\mathrm{C_3A}}$	quantity of tricalcium aluminate per unite weight of cement	mol/kg
$\mathrm{N}_{\mathrm{C_4AF}}$	quantity of tetracalcium aluminoferrite per unit weight of cement	mol/kg
$\mathrm{N}_{\mathrm{C-S-H}}$	calcium silicate hydrates quantity per unit weight of cement	mol/kg
p_c	capillary pressure	Pa
p_{sat}	saturated vapour pressure above a flat surface	Pa
p_v	vapour pressure	Pa
q_{fit}	fitted parameter for the linear relationship between the volumetric water content θ by pure adsorption and the logarithm of the metric potential of capillary water	$-$
r'	the inner radius of pipe-like adsorbed water layer	m
R_0	equivalent resistance between rebar units in the horizontal direction	Ω
R_1	equivalent resistance in the concrete cover above activated rebar unit in the vertical direction	Ω
R_2	equivalent resistance in the concrete cover above passivated rebar unit in the vertical direction	Ω
r_a	area ratio of the activated rebar surface to the whole rebar surface	$-$
r_c	pore radius where condensation starts to occur at a given relative humidity	m
$R_{concrete}$	resistance of concrete in the ionic path of the measurement circuit	Ω
r_{curve}	radius of the meniscus curvature (negative value for concave meniscus)	m
R_e	equivalent resistance in concrete cover between rebar surface and concrete cover surface	Ω

r_i	the i_{th} of the total n sampled pore radius from the pore size distribution	m
R_L	electrical resistance	Ω
r_{max}	maximum pore radius	m
r_{min}	smallest pore radius	m
r_p	pore radius	m
r_p	area ratio of the passivated rebar surface to the whole rebar surface (Chapter 5)	$-$
r_{sat}	largest saturated pore radius	m
$R_{voltmeter}$	internal resistance of the voltmeter	Ω
$S'(r)$	the area of the interior surface of adsorbed water in pores with radii greater than r per unit volume of the porous medium	m^2/m^3
S_0	total pore surface area per unit volume of porous medium	$1/m$
s_{Cl}^{c}	chemically bound chloride by Friedel's salt formation per unit weight of cement	mol/kg
s_{Cl}^{P}	physically bound chloride by C-S-H adsorption per unit weight of cement	mol/kg
s_{Cl}	total bound chloride per unit weight of cement	mol/kg
S_e	effective water saturation	$-$
s_{fit}	fitted parameter for the linear relationship between the volumetric water content θ by pure adsorption and the logarithm of the metric potential of capillary water	$-$
T_f	transmissivity	m^2/s
V_1, V_2	applied voltages	V
V_a	empirical parameter account for rust film growth	V
V_c	empirical parameter (negative) account for rust film dissolution	V
V_m	molar volume	m^3/mol
$V_{m.Ca(OH)_2}$	molar volume of $Ca(OH)_2$	m^3/mol
$V_{m.CaCO_3}$	molar volume of $CaCO_3$	m^3/mol
V_{sample}	volume of sample	m^3
V_{sol}	total volume of pore solution	m^3
z_i	charge number of i_{th} ion	$-$

$\beta_a^{O_2}$	anodic Tafel slope of OH^-/O_2 reversible reaction	V/dec
β_a^{Fe}	anodic Tafel slope of Fe^{2+}/Fe reversible reaction	V/dec
β_a	anodic Tafel slope	V/dec
$\beta_c^{O_2}$	cathodic Tafel slope of OH^-/O_2 reversible reaction	V/dec
β_c^{Fe}	cathodic Tafel slope of Fe^{2+}/Fe reversible reaction	V/dec
β_c	cathodic Tafel slope	V/dec
ε_0	dielectric permittivity of vacuum, $8.8541878128 \times 10^{-12}$	F/m
ϵ_0	initial solid volume fraction	$-$
ϵ_g	gas volume fraction	$-$
ε_p	porosity	$-$
ϵ_s	solid volume fraction	$-$
ϵ_{smax}	maximum solid volume fraction	$-$
η_{conc}	concentration polarization overpotential	V
θ_l	liquid volume fraction	$-$
θ_r	residual volumetric water content	$-$
θ_s	volumetric water content under fully saturated condition or the total connected porosity	$-$
θ_{sat}	volumetric water content of water in saturated pores per unit volume of the porous medium	$-$
θ_{unsat}	volumetric water content of adsorbed water in unsaturated pores per unit volume of the porous medium	$-$
λ_i^0	equivalent conductivity of i_{th} ion at infinite dilution	cm^2S/mol
λ_i	equivalent conductivity of i_{th} ion	cm^2S/mol
ρ_l	liquid water density	kg/m^3
ρ_r	electrical resistivity	Ωm
ρ_v	vapour density	kg/m^3
σ_{rev}	electrical conductivity of the porous medium	S/m
σ_{sol}	electrical conductivity of pore solution	S/m
ϕ^{act}	electrolyte potential on activated rebar unit	V
ϕ_{app}	applied electrolyte potential on the concrete cover surface	V

ϕ_l	electrolyte potential	V
ϕ^{pas}	electrolyte potential on passivated rebar unit	V
Ω_{sol}	domain of pore solution	$-$
A	area of the cylinder top/base surface	m^2
a	fitted parameter accounts for the baseline pore connectivity and pore solution volume fraction	$-$
a	aperture of the crack (Chapter 8)	m
b	fitted parameter accounting for the change in interconnectivity due to the change in water saturation level	$-$
C	average pore size or interparticle clearance (Chapter 2)	m
C	change in storage due to change in pressure head	$1/m$
c	liquid phase-averaged solute concentration	kg/m^3
d	diameter of the cylinder sample	m
d	pore diameter (Chapter 6)	m
F	Faraday's constant, 96485.3321	C/mol
$\boldsymbol{g}$	gravitational acceleration	m^2/s
h	Relative humidity	$-$
i	measured or applied current density	A/m^2
$\boldsymbol{i}$	current density vector	A/m^3
$\boldsymbol{J}$	total volume-averaged mass flux of solute in concrete	kgm^2/s
L	height of the cylinder sample	m
l	empirical van Genuchten parameter	$-$
m	empirical van Genuchten parameter	$-$
n	number of electrons transferred in the reaction	$-$
n	number of sampled pore radius from the pore size distribution (Chapter 6)	$-$
n	empirical van Genuchten parameter (Chapter 7, 8)	$-$
Pe	Peclet number	$-$
q	surface charge density of electrical double layer (EDL)	C/m^2
R	universal gas constant, 8.3144598	$J/K/mol$

r	radius of a droplet (negative value for concave meniscus)	m
$R(h)$	The critical pore radius for condensation to occur at the relative humidity of h	m
$S(r)$	the surface area of pores with radii greater than r per unit volume of the porous medium	$1/m$
T	absolute temperature	K
t	time	s
$\boldsymbol{u}$	Darcy's velocity	m/s
Z	elevation	m
α	anodic charge transfer coefficient (Chapter 2)	$-$
α	contact angle	$°\,, rad$
A	calibration parameter in the analytical chloride bonding model (Chapter 6)	$-$
α	empirical van Genuchten parameter (Chapter 7, 8)	$1/m$
β	cathodic charge transfer coefficient	$-$
β	calibration parameter in the analytical chloride bonding model (Chapter 6)	$-$
γ	surface tension	N/m
Δe_a	incremental change of adsorbed water in thickness	m
$\Delta\theta_a$	incremental change of adsorbed water in volume fraction	$-$
$\Delta\theta_c$	incremental change of condensed water in volume fraction	$-$
$\Delta L(r)$	pore length per unit volume with diameter ranges from $[r, r+\Delta r\,]$	m/m^3
Δp	pressure decrease caused by the addition of solute	Pa
$\Delta\eta$	Incremental change of saturated pore volume per unit volume of the porous medium	$-$
$\Delta\theta$	incremental change of water volume fraction	$-$
ε	dielectric permittivity of solution	F/m
η	overpotential	V
$\eta(r)$	the pore volume of the pores with radii smaller than r per unit volume of the porous medium	$-$
θ	volumetric water content	$-$
κ	Debye constant, $1/\kappa$ represents the extension length of the diffuse layer	$1/m$

λ	mean free path of gas molecules	m
μ	dynamic viscosity of water	$Pa \cdot s$
σ	electrical conductivity of the electrolyte	S/m
Φ	metric potential of capillary	Pa
$\Phi(\epsilon_s)$	factor considering the effect of carbonation completeness on carbonation reaction rate	$-$
$\Phi'(S_e)$	factor considering the effect of moisture content on carbonation reaction rate	$-$
$\Phi(S_e)$	reduction function for effective diffusivity that accounts for discontinuity in the liquid phase	$-$
ψ	pressure head	m

1 Introduction

1.1 Background

Reinforced concrete (RC) corrosion damage is a multibillion-dollar problem in Canada and many other countries. In 2001, the Federal Highway Administration (2002) reported that the total direct cost of corrosion was U.S.$239 billion per year which was 3.2 percent of the U.S. gross domestic product (GDP). The corrosion of reinforced concrete structures has become a more urgent problem in the last few years with an increasing number of aging RC structures. According to a recent report published by Transportation for America (Davis et al., 2013), one in nine bridges in the U.S. have sustained a major defect requiring closure, significant maintenance, rehabilitation, or replacement of one of the major components. Prediction and early diagnosis of corrosion have become very important tasks for construction engineers, repair practitioners, and owners of buildings and bridges. It allows preventive action to take place and avoids unnecessary major expensive repairs, and hence, a reduced life cycle cost. It would also help policymakers to regulate the evaluation criteria and choose optimum rehabilitation strategies for RC structures exceeding their design service life that do not necessarily need to be replaced.

Corrosion of RC structures is a complex problem and accurate prediction of the corrosion rate is a challenging task. The alkalinity (pH>12.5) of concrete provides a suitable microenvironment to generate and maintain a passive layer on steel, which protects the rebar from corrosion. Therefore, in the absence of cracks, steel reinforcement in concrete structures usually exhibits a prolonged corrosion initiation period. However, the existence of cracks complicates the micro-environment in concrete near the rebar which makes the assessment of RC corrosion much more difficult. The cracks in aged RC structures can occur as a result of many phenomena such as restrained shrinkage, thermal shocks, freeze-thaw cycles, and alkali-aggregate reactions, and can also be induced by external loadings. In most cases, cracks do not immediately affect the load-carrying capacity of the concrete structure, but may adversely affect its durability by providing easier access to oxygen, carbon dioxide, water and other aggressive agents such as chloride ions in marine environments or from the use of de-icing salts, which allows the corrosion process to initiate earlier and possibly propagate much faster than in the case of uncracked concrete.

Many studies (Arya & Ofori-Darko, 1996; Lu et al., 2017; Pettersson et al., 1996) have found that the effects of cracking on corrosion rate in the propagation period are affected by the properties of cracks, such as crack width, crack frequency, and the properties of concrete, and cover depths. Moreover, these effects may be further complicated by the service environment. For example, under various exposure conditions, there exists a different critical crack width below which the effect of the crack is negligible. Therefore, the empirical corrosion model drawn from experiments is only limited to the application of similar scenarios: similar concrete structure layout, similar concrete quality, and similar exposure conditions. Given the interactive multi-factor nature of the development of the problem, a numerical model that can simulate the transport phenomena and electrochemical processes of the cracked RC structures seems to be a promising way to capture the necessary complication. Such a versatile mechanistic model can then be used to assess the corrosion of cracked concrete in various service environments. However, the currently available numerical models still suffer from serious limitations to capture the spatial variations of corrosion in cracked concrete in their service environment.

1.2 Limitations in current corrosion models

Not being limited by the practical constraints of the field assessment, numerical simulation is a versatile tool that allows a more in-depth assessment of corrosion in RC structures. Over the past couple of decades, numerous studies have been carried out to develop models for assessing corrosion rates in RC structures.

Empirical models (Alonso et al., 1988; DURACRETE, 1999; Liu & Weyers, 1998; Vu et al., 2005) or closed-form regression equations from numerical models (Pour-Ghaz et al., 2009a; Yu et al., 2014) correlate the corrosion rate with corrosion affecting factors such as water/cement ratio, concrete resistivity, moisture content, carbonation degree, cover thickness, temperature etc. The empirical models reveal how the material properties and environmental factors affect the corrosion rate. However, these models are only close to their experimental setup conditions. The limitations could be found when applied to a different corrosive environment or different corrosion patterns such as non-uniform corrosion in cracked concrete under wetting and drying conditions: the fitted model parameters lack physical meaning, and thus cannot be properly selected.

Other mechanistic numerical models (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kim & Kim, 2008; Kranc et al., 1992; Muthulingam & Rao, 2015; Sohail, 2013; Yu, Liu, & Li, 2017; Zacaruk, 2013)(Ghods et al., 2007; Isgor & Razaqpur, 2006; Kim & Kim, 2008; Kranc et al., 1992; Sohail, 2013; Yu, Liu, & Li, 2017; Zacaruk, 2013) are based on solving the electrochemical electrode process on a rebar-concrete interface and the current distribution in a concrete domain. The electrochemical process simulation may be coupled with the solution of transport of oxygen, moisture, and chloride ions in concrete. The model parameters include the electrochemical properties of steel and the transport properties of concrete. Although these electrochemistry-based numerical models have rationally defined parameters, the guidance of parameter selection is lacking; usually, a large scatter of values is used among different researchers and both parameters from solution and concrete have been used.

However, there are various drawbacks and limitations of the current models that prevent their application in a more realistic situation: non-uniform corrosion of cracked/uncracked RC structures under different environmental conditions.

Firstly, the corrosion kinetics are usually taken as constant although it was observed that in simulated pore solution, it is a function of pH value, chloride content and oxygen content. There are some attempts to empirically correlate the corrosion parameters with chloride concentration anodic Tafel slope and anodic equilibrium potential (Maekawa et al., 2008) due to the anodic control assumption (Glass et al., 1991) or by making critical chloride content independent of anodic Tafel slope (Sagüés et al., 2008) or by relating equilibrium potential with the concentration of reacting ions such as OH^-, Fe^{2+} , oxygen content by Nernst's equation (Isgor & Razaqpur, 2006). However, the dependence of these critical corrosion parameters (i.e., equilibrium potential, exchange current density, Tafel slope) on the chemical environment is usually not explicitly represented.

Secondly, more often than not, the role of saturation is not properly modelled. The experimental observation suggests that the corrosion rate increases with increasing water saturation until the cathodic current reaches a maximum value due to the limiting transport of dissolved oxygen. In most corrosion models, saturation does not contribute to the rebar-concrete corrosion kinetics but is usually only reflected as the change in bulk resistivity, which then translates to the resistance

between anodes and cathodes. However, in macro-cell corrosion models, an increase in resistivity will only affect the "macro-cell" current by increasing the resistance between anodes and cathodes. However, the reduction of corrosion rate due to water saturation could be explained in two ways: (1) resistance in the corrosion cell circuit (micro- and macro-) controls the corrosion rate; (2) the overall corrosion rate on the rebar-concrete interface is affected by the actual rebar-pore solution interface area, which is the effective corroding surface. There is little research on the modelling of the above-mentioned mechanism regarding the corrosion-saturation relationship.

Thirdly, the corrosion kinetics are typically assigned to questionable pre-defined anodes and cathodes (macro-cell corrosion model) as input information, which limits the application of these models to problems where anodic site and cathodes are separated, such as chloride-induced pitting corrosion in cracked concrete with the assumed sized anode in the cracking areas and the rest areas as the cathode to couple with. Although in some cases, the allocation of anode and cathode is determined by certain threshold criteria such as pH, and chloride ion concentration, this treatment of the corrosion cell is not suitable in more uniformly chloride-contaminated concrete or carbonation-induced concrete where rebar is not strongly polarized to discrete anodic and cathodic sites. Such scenarios may exist after the carbonation or contamination of chloride propagates around the cracks for an extended period in aged structures. More importantly, there is no clear cut-off between these conditions.

Finally, the effect of realistic service exposure conditions as boundary conditions is rarely included. The micro-environment of concrete around the crack will be greatly affected by weather conditions such as ambient relative humidity, temperature, precipitation, and other factors such as the seasonal application of de-icing salts. Although the chloride diffusion, carbonation and moisture transport in concrete were extensively simulated in numerical models, few corrosion models have incorporated the evolving microenvironment around the rebar from these models. In current models, the microenvironment around the rebar is typically described by a chloride diffusion problem with a constant chloride concentration on the surface of RC (Cao, 2014). The resultant chloride content is merely used to trigger the assignment of discrete anodes and cathodes.

1.3 Statement of the research question and significance

A comprehensive mechanistic prediction model is developed to predict the corrosion rate of cracked concrete under different service environments. In this model, rational and measurable electrochemical properties of rebar in solution are to be used to develop homogenized corrosion kinetics (both anodic and cathodic) at every point on the rebar surface with surrounding concrete in different non-uniform micro-environment caused by cracking of concrete and varying exposure conditions. The electrochemical processes are coupled with the evolution of the micro-environment.

This research improves our understanding of the corrosion kinetics of rebar in cracked concrete and the comprehensive model provides an objective and quantitative assessment of the real-world corrosion problem in RC structures with/without cracking, under different weather and climate conditions. An accurate assessment of the corrosion performance of a given RC structure would help the engineers and property owners to make informed decisions on the frequency of inspection of existing structures, rehabilitation timelines and strategies. The inclusion of region-dependent weather and climate data will help policymakers to improve their local structural design/construction code, for example, maximum crack length/width allowed, minimum cover depth, concrete types etc. and municipal policies such as the amount of the application of de-icing salts.

1.4 Objectives

The main objective of this research is to develop a comprehensive mechanistic corrosion model for rebar corrosion in non-uniform corrosive microenvironments caused by cracking and exposure conditions. These sub-objectives include

1) to investigate the synergistic effect of chloride, moisture, and carbonation on the corrosion parameters of rebar in concrete.

2) to develop a model that describes the homogenized rebar corrosion parameters in concrete as a function of the variable microenvironment (moisture level, chloride concentration, carbonation state)

3) to develop a model that simulates the microenvironment evolution at the rebar depth of concrete structure in its service conditions that involves sources of chloride, carbonation and moisture fluctuation.

4) to integrate the homogenized rebar corrosion kinetics model and the microenvironment evolution model into a comprehensive corrosion model.

5) to validate the comprehensive corrosion model with the experimental corrosion test of cracked beam samples exposed to wetting and drying cycles with chloride salt solution.

6) Utilize the validated comprehensive model to explore the intricate relationship between cracks, microenvironmental factors, and rebar corrosion, with particular emphasis on addressing some controversial aspects in the literature. This includes investigating the relative importance of microcell versus macrocell corrosion, examining corrosion reduction over time, and analyzing the effect of crack size (opening) on corrosion.

1.5 Methodology and scope

The main goal of this project is to model the rebar corrosion in cracked/uncracked RC structures under realistic service exposure conditions. The core of the research question is to understand the realistic corrosion kinetics of the rebar-concrete interface under varying conditions of moisture, chloride, and carbonation and make it applicable in large-scale FEM models. To break it down, one needs to know the effect of the pore solution composition on the corrosion kinetics of the rebar-solution interface, and how moisture level—one distinctive characteristic brought by concrete's pore system, affects this behaviour at a representative elementary surface area of the rebar-concrete interface.

These objectives of this study were achieved by a combination of laboratory experiments and numerical modelling.

The laboratory tests were used to study the synergistic effects of moisture, chloride, and carbonation on the rebar corrosion in both simulated pore solution and mortar.

The laboratory experiments were divided into two phases. Phase one consists of the concrete property tests and electrochemical tests of rebar corrosion. The concrete properties include porosity, chloride diffusivity, carbonation resistance and water retention property, and resistivity. Knowledge of these properties was used to impose a controlled corrosive environment in the reinforced concrete samples that involve varying moisture levels, chloride concentration, and carbonation state. The electrochemical tests are then carried out to test the corrosion parameters of the embedded rebar in those formulated conditions. With the cause—controlled corrosion

microenvironment condition in concrete and the response—tested rebar corrosion kinetics, we may analyze the role of moisture, chloride, carbonation, and resistivity in contributing to the corrosion of the rebar-concrete interface. Phase two is a parallel validation test. It tests the corrosion condition of the cracked beams under controlled conditions and validation the performance of the developed comprehensive corrosion model.

The numerical modelling part aims at developing the comprehensive mechanistic corrosion model that simulates the evolution of moisture, chloride, and carbonation in the cracked concrete under the natural-like exposure condition and the electrochemical processes taking place on the rebar exposed to the non-uniform microenvironment. This comprehensive model can be broken down into three parts:

“Environment model”; a mass transfer model that simulates the evolution of micro-environment in uncracked concrete under service exposure conditions, which then can be extended to incorporate the effects of cracking. In this part, data analysis of historical meteorological information, field monitoring and laboratory testing of moisture, chloride, and carbonation evolution in concrete structures exposed to natural and simulated environments were used to come up with realistic boundary conditions.

“Corrosion kinetics model”; a model that describes the homogenized corrosion kinetics for rebar in concrete. This homogenized model is essentially dependent on the microenvironment near the rebar while treating the rebar-concrete interface as a continuum, and hence can be applied to the whole rebar surface without having to specify discrete anodic and cathodic areas. The homogenized corrosion kinetics provides a link between the rebar corrosion at the non-uniform rebar-pore solution interface and the overall corrosion behaviour at the rebar-concrete interface. The corrosion kinetics of the rebar-concrete is not just a quantity like the corrosion rate, it also contains the polarization behaviour or in other words the dependence of the corrosion rate on the overpotential distribution in the representative elementary rebar-concrete interface. Therefore, such kinetics, classically-described by the Butler-Volmer equation, cannot be acquired by simply volume-averaging or homogenization. In this paper, this homogenization was done through numerical modelling of the corrosion kinetics of rebar embedded in concrete at a microscopic level starting from the corrosion kinetics in pore solution. This will provide a numerical method of

acquiring electrochemical parameters of rebar in the concrete of different saturation and different pore solution composition. At the same time, the empirical homogenized corrosion kinetics can also be derived from the laboratory corrosion test of the reinforced concrete samples under controlled conditions.

In the comprehensive corrosion model, the homogenized corrosion kinetics will be used to accommodate both micro- and macro-corrosion, so that both micro-cell and macro-cell corrosion kinetics are dependent on the varying concrete conditions. This will enable the corrosion modelling rebar exposed to microenvironments with spatial variations caused by the existence of cracks, which could not be properly done by using separated pure anodes and cathodes.

The RC structures studied in this research do not include RC structures with surface treatments (such as coating, and fibre-reinforced polymer wrapping) and the rebar is not subject to cathodic protection (galvanic protection and/or impressed current protection) or stray current. The steel reinforcement is regular carbon steel; stainless steel or galvanized rebar is not considered.

1.6 Organization of chapters

Chapter 1 is the introduction to the project. It presents the background of the corrosion of reinforced concrete structures and points out the limitations of current corrosion models. The research objectives and the methodologies are then proposed to target these limitations.

Chapter 2 reviews the existing literature related to the non-uniform corrosion phenomenon in reinforced concrete structures in detail. It covers the experimental observation of the behaviour of rebar corrosion in cracked concrete and the corrosion modelling methods. The electrochemical basics of the rebar corrosion cell were revisited. Different treatments of the corrosion cell in the numerical models were critically discussed. It also reviews the modelling of moisture transport, chloride penetration, and carbonation in concrete that affect the corrosive microenvironment on the rebar.

Chapter 3 and Chapter 4 present the experimental studies on the corrosion parameters of the carbon steel rebar in mortar samples and simulated pore solutions. The synergistic effects of chloride, carbonation and water on rebar corrosion were investigated.

Chapter 3 presents an improved three-electrode potentiodynamic test performed on the as-received carbon steel rebar in four media: simulated pore solution, carbonated simulated pore solution, non-carbonated mortar, and carbonated mortar. There are six levels of chloride contamination and two levels of carbonation state for both rebar in simulated pore solution and mortar. The carbon steel rebar corrosion parameters, including corrosion potential, corrosion current density, and Tafel slopes were measured. The effects of chloride and carbonation on corrosion were compared between mortar samples and simulated pore solution.

Chapter 4 covers an in-depth analysis of the results of the corrosion parameters from mortars with various chloride content, carbonation state and water content alongside their water retention curves and bulk resistivity. It discusses the roles of water saturation on corrosion concluded as 1) an area effect by the influenced rebar-pore solution area, 2) an ohmic control effect by the influenced bulk resistivity, and 3) an anode-to-cathode ratio effect. The corrosion kinetics parameters are described as empirical functions of the microenvironmental conditions.

Chapter 5 and Chapter 6 use the modelling method to explore how the overall corrosion parameters are related to the non-uniform corrosion parameters at mesoscopic–macroscopic level and microscopic level.

Chapter 5 presents the concept of the homogenized corrosion parameters of rebar in concrete with non-uniform surface conditions with a circuit model. This model starts from localized corrosion kinetics for active and passive rebar-concrete units (mesoscopic–macroscopic level) described by the Butler-Volmer equation and arrives at the simulated scan curve representing the overall performance of the rebar, from which, the homogenized corrosion parameters can be derived. It explored the validity of describing the overall scan curve with the Butler-Volmer equation, and how much deviation can be caused by the non-uniformity of rebar condition and concrete resistivity.

Chapter 6 presents a modelling approach to simulate the activation-controlled corrosion kinetics of the rebar-concrete interface (mesoscopic–macroscopic level) from the corrosion kinetics of the rebar-solution interface (microscopic pore level). The proposed model describes the area of active and passive rebar-solution corrosion sites at the representative elementary volume based on the

non-uniform water and chloride distribution in the concrete. The simulated homogenized corrosion kinetics of rebar in concrete as derived from this model are functions of the total chloride content and water saturation of concrete, similar to the empirical functions outlined in Chapter 4. These corrosion kinetics can be applied to every point on the rebar surface with surrounding concrete of non-uniform microenvironment caused by cracking of concrete and varying exposure conditions.

Chapter 7 presents a comprehensive 3-D transport model that simulates the microenvironment evolution at the rebar depth for a reinforced concrete structure. This model was applied to an old concrete arch bridge subjected to service conditions with the source of chloride, carbonation, varying air relative humidity, and exposure to unevenly sheltered run-off water. The model uses the bridge inspection data, weather data, moisture and oxygen content in concrete monitored by in-situ installed sensors to predict the microenvironment in the concrete arch. Results for the past 100 years of service showed good agreement with the bridge's current condition. Simulations for the next 50 years under different climate scenarios, such as elevated CO_2 level and change in precipitation, reveal implications of climate change for rebar section loss in the arch.

Chapter 8 presents a comprehensive corrosion model that simulates the evolution of moisture and chloride content on embedded rebars in cracked concrete. The model utilizes corrosion kinetics as functions of the microenvironmental conditions on the embedded rebar to solve the potential and current fields in the concrete domain and rebar surface. An experimental program was developed to quantify material properties, and a corrosion test of segmented rebars in cracked concrete under wetting and drying cycles was employed to validate the model.

Chapter 9 presents a numerical exploration of the potential mechanisms causing the marked differences in observations of rebar corrosion in cracked concrete. It explains the complex effect of cracks on observed corrosion, corrosion reduction over time, and the effect of crack size and crack healing.

Finally, Chapter 10 concludes the main findings of this work and recommendations for future work.

2 Literature Review

In this chapter, a critical review of the published literature on the corrosion of cracked reinforced concrete is provided. Firstly, the experimental and field observation of non-uniform corrosion in cracked concrete will be summarized. Secondly, various numerical treatments of the corrosion cell will be critically reviewed. Thirdly, special attention will be paid to the modelling of the corrosive micro-environment affected by the natural exposure conditions.

2.1 Non-uniform corrosion in cracked concrete

Steel reinforcement embedded in concrete is protected from corrosion by both physical and chemical barriers. Concrete contains a highly alkaline pore solution (pH 12–13) within its hardened cement paste. The high alkalinity of the pore solution is due to the presence of sodium, potassium and calcium hydroxides produced by the hydration of cement particles with mixing water. In this alkaline environment, a protective iron oxide film forms and maintains the steel bars in a passive condition with a very low corrosion rate. This protected state of steel in concrete against corrosion, known as passivity, requires a minimum pH value of approximately 11.5 to be maintained. The quality and the thickness of concrete cover have been shown to affect the stability of this passive layer around the reinforcement. The loss of passivity is often a result of the loss of alkalinity of the surrounding concrete due to carbonation or the presence of chloride ions. Cracks in concrete serve as fast pathways for the ingress of chloride onto the rebar. When the chloride content on the rebar reaches a critical value, and/or the pH value drops to a certain value, the protective film will be destroyed, which will cause the onset of active corrosion.

Non-uniform corrosion will take place wherever there is a non-uniform microenvironment. Non-uniform corrosion can be observed both around the circumference of the rebar and along the rebar. In aged RC structures, where adequate chloride penetration and/or carbonation front has just reached the rebar through an uncracked concrete cover, more corrosion products are found on the rebar surface closer to the cover surface (Ji et al., 2013; Muthulingam & Rao, 2014, 2015). In cracked RC structures, higher corrosion rates should be expected in the cracking areas affected by chloride penetration and/or carbonation. However, experimental observations have shown that this is not necessarily the case (i.e., the cracks may not cause a high corrosion rate), and hence

reinforcement corrosion in cracked concrete is more complex than corrosion in the absence of cracks. Although a certain number of previous studies (Alekseev et al., 1993; Lu et al., 2017; Pettersson et al., 1996; Schießl & Raupach, 1997; Subramaniam & Bi, 2010; Wilkins & Lawrence, 1980) have focused on the effect of cracking on steel reinforcement corrosion in concrete, this area still remains relatively unexplored, despite its practical importance since it is virtually impossible to eliminate cracking in real structures. Currently, there is a lack of clear understanding of the corrosion process in such situations and how the properties of the surrounding concrete affect the rate of local metal loss at the crack location. A review of the current state of knowledge in this area is provided in the following sections.

2.1.1 Effect of cracks on corrosion - experimental observation

The most significant effect of concrete cracking on rebar corrosion is the formation of a macro-cell that may significantly increase the corrosion rate. It has been observed that passive reinforcement in the non-cracked part of an RC structure and bare steel can act as effective cathodes if there is galvanic contact to the reinforcement at the cracks and when these cathodic areas have sufficient oxygen supply. This is typically the case, for cathodic areas located in the splash zone or exposed to air where the corrosion around cracks can be accelerated due to macro-cell corrosion. However, in submerged conditions, the cathodic oxygen reduction reaction is limited by the transport of dissolved oxygen to the rebar surface (Hussain, 2011). It was found that the main effect of macro cracks in submerged concrete is a reduction of the initiation time. Once the active rebar corrosion was initiated, the corrosion rate started high, but rapidly decreased and approached the active corrosion rate typically found in uncracked concrete (Alekseev et al., 1993). According to Wilkins & Lawrence (1980), the cross-sectional static cracks, even with up to 1 mm crack width, would not lead to corrosion at a significant level in the case of submerged concrete.

The effect of cracks on the corrosion rate in cracked concrete could also depend on the crack width and concrete type. For high performance concrete in partially submerged conditions (splash zone), it has been observed that although the chloride threshold level and the initiation time may be close to zero for crack widths up to 0.8 mm and a concrete cover of 30 mm, the active corrosion rate will become so small that it may be considered as practically harmless (Pettersson et al., 1996). In a recent study (Lu et al., 2017), the effects of transverse cracks on chloride-induced rebar corrosion

under wetting and drying cycles were found to depend on both the cover depth and crack parameters such as width and frequency/spacing. The corrosion rate increases more rapidly with increasing crack width and frequency and decreasing cover depth. Under various cover depth and crack frequency scenarios, it has been observed that there exists a certain critical crack width below which the effect of the crack is negligible. However, the reasons for the existence of that critical width and its dependence on exposure conditions and the crack parameters are not clear.

Two different mechanisms of corrosion, the micro-cell and the macro-cell have been identified in cracked concrete specimens (Subramaniam & Bi, 2010). The two mechanisms primarily differ in the length scales and the physical distance between the underlying electrochemical reactions. In the microcell mechanism, the anodic and cathodic reaction sites are microscopic and are located adjacent to each other. In macro-cell corrosion, the anode and cathode are spatially separated by a finite macroscopic distance resulting in a flow of electric charges over a physical distance. In the case of chloride-induced corrosion, the anodic reaction is localized, confined to the area of steel where the chloride threshold level for depassivation is exceeded. The cathodic reaction may significantly occur in all areas that are not permanently water-saturated and cathodic current density decreases as the distance from the crack increases (Raupach, 1996). That explains the experimental observation that crack frequency could be identified as a governing factor in the rate of corrosion (Arya & Ofori-Darko, 1996), as it enables more passive areas near those cracks to be polarized to its cathodic direction and hence larger total cathodic current. Poursaee et al. (2010) studied corrosion of multi-rebar mortar samples subject to various types of de-icing salt solutions, where an active rebar near the surface was subject to the solution and the other three were at a deeper location. It was found the micro and macro corrosion rates were similar. However, another similar test (Hansson et al., 2006) suggested that macro-cell corrosion is the main mechanism of rebar corrosion in ordinary corrosion. Although these tests are not in cracked concrete, these findings could suggest that the contributions from micro-cell and macro-cell depend on the area of active rebar and passive rebar and their arrangement. The contribution of micro-cell corrosion could be the dominant mechanism when the active rebar area is comparable to the passive rebar and the distance between these two are large.

In conclusion, most of the previous research on corrosion in cracked concrete has been limited to measuring either potentials or corrosion rates in either field structures (submerged or splashed

zone) or preconditioned samples (often subjected to wetting/drying cycles) in the lab but often without monitoring of the distribution around the crack of any of the environmental parameters that can affect the corrosion process, such as water saturation, oxygen, and chloride. Under different exposure conditions, the properties of cracks such as crack width, crack frequency, the properties of concrete, and cover depths would have very different effects on the corrosion rate. A clear understanding of the relative magnitudes of the microcell and macro-cell mechanisms in a corroding system and what creates the conditions for any of the two to become the dominant mechanism is still currently not available. The role of each mechanism, its contribution to the total corrosion rate and its dependence on the local microenvironments (moisture, oxygen, and chloride) created by the very existence of cracks are not clear yet.

2.1.2 Numerical modelling of corrosion

In the last few decades, corrosion of rebar in concrete has been extensively studied. Numerous studies have been focusing on the factors that affect the rate of rebar in concrete, such as water-cement ratio, cover thickness, moisture content, chloride content, pH value, oxygen availability etc. A variety of numerical models were proposed to predict the corrosion rate in the propagation phase and mechanical behaviour of reinforced concrete structures under different exposure conditions (Otieno et al., 2012). Despite different levels of complexity, direct modelling of the electrochemical process of rebar embedded in concrete remains the most versatile way to calculate the corrosion current, and hence the rate of loss of metal.

2.1.2.1 Basics of corrosion kinetics

When modelling corrosion in reinforced concrete, concrete is usually treated as a continuous homogeneous material, where the corrosion kinetics are generally modelled by the Butler-Volmer equation or by the Tafel equation for larger over-potentials, and their variations incorporating concentration polarization.

Although corrosion is not a reversible process, it is worth mentioning that in some literature, occasionally, terms such as equilibrium potentials (reversible potential of half-cell reactions: iron dissolution and oxygen reduction) and mixed potential (corrosion potential of coupled full reaction) are used interchangeably, and accordingly exchange current density and corrosion current density.

The confusing term usage could be due to the fact the Butler-Volmer equation when applied to the single electrode and mixed electrode, the mathematical expression takes the same form (Tan, 2012):

Single electrode with Half-cell reaction:

$$i = i_0 \left[exp\left(\frac{\alpha nF}{RT}\eta\right) - exp\left(\frac{-\beta nF}{RT}\eta\right)\right] \tag{2.1}$$

$$\eta = E - E_{eq} \tag{2.2}$$

where η $[V]$ is the overpotential, the deviation from equilibrium potential E_{eq} $[V]$; i $[A/m^2]$ is the measured or applied current density; i_0 $[A/m^2]$ is the exchange current density, F is Faraday's constant, R $[J/K/mol]$ is the universal gas constant, T $[K]$ is the absolute temperature, n is the number of electrons transferred in the reaction. α and β are the anodic and cathodic charge transfer coefficients, respectively. They are related to the potential drop through the electrochemical double layer.

when η is positive and large enough in the anodic direction:

$$\exp\left(\frac{\alpha nF}{RT}\eta\right) \gg \exp\left(\frac{-\beta nF}{RT}\eta\right) \tag{2.3}$$

$$i_a = i_0 \left[\exp\left(\frac{\alpha nF}{RT}\eta\right)\right] \tag{2.4}$$

or in the form of the Tafel equation:

$$\eta = \beta_a \log\left(\frac{i_a}{i_0}\right) \tag{2.5}$$

$$\beta_a = \frac{2.303RT}{\alpha nF} \tag{2.6}$$

where $\beta_a[V/dec]$ is anodic Tafel slope. When η is negative and $|\eta|$ is large enough in the cathodic direction:

$$\exp\left(\frac{\alpha nF}{RT}\eta\right) \ll \exp\left(\frac{-\beta nF}{RT}\eta\right) \tag{2.7}$$

$$i_c = i_0\left[-\exp\left(\frac{-\beta nF}{RT}\eta\right)\right] \tag{2.8}$$

where the cathodic current $i_c[A/m^2]$ has a negative sign, or in the form of the Tafel equation:

$$\eta = \beta_c \log\left(-\frac{i_c}{i_0}\right) \tag{2.9}$$

$$\beta_c = -\frac{2.303RT}{\beta nF} \tag{2.10}$$

where $\beta_c[V/dec]$ is cathodic Tafel slope. A similar mathematical form also applies to corrosion, for a mixed electrode system:

$$i = i_{corr}\left[\exp\left(\frac{\alpha nF}{RT}\eta\right) - \exp\left(\frac{-\beta n'F}{RT}\eta\right)\right] \tag{2.11}$$

$$\eta = E - E_{corr} \tag{2.12}$$

where η $[V]$ is the overpotential, the deviation from equilibrium potential E_{corr} $[V]$; i $[A/m^2]$ is the measured or applied current density; $i_{corr}[A/m^2]$ is the corrosion current density, F is Faraday's constant, $R[J/K/mol]$ is the universal gas constant, $T[K]$ is the absolute temperature, n and n' the number of electrons transferred in the anodic and cathodic reactions, respectively. α and β are anodic and cathodic charge transfer coefficients, respectively, that are related to the potential drop through the electrochemical double layer.

when η is positive and large enough in the anodic direction:

$$\exp\left(\frac{\alpha nF}{RT}\eta\right) \gg \exp\left(\frac{-\beta nF}{RT}\eta\right) \tag{2.13}$$

$$i_a = i_{corr}\left[\exp\left(\frac{\alpha nF}{RT}\eta\right)\right] \tag{2.14}$$

or in the form of the Tafel equation:

$$\eta = \beta_a \log\left(\frac{i_a}{i_{corr}}\right) \tag{2.15}$$

$$\beta_a = \frac{2.303RT}{\alpha nF} \tag{2.16}$$

when η is negative and $|\eta|$ is large enough in the cathodic direction:

$$\exp\left(\frac{\alpha nF}{RT}\eta\right) \ll \exp\left(\frac{-\beta nF}{RT}\eta\right) \tag{2.17}$$

$$i_c = i_{corr}\left[-\exp\left(\frac{-\beta nF}{RT}\eta\right)\right] \tag{2.18}$$

where the cathodic current i_c has a negative sign, or in the form of the Tafel equation:

$$\eta = \beta_c \log\left(-\frac{i_c}{i_{corr}}\right) \tag{2.19}$$

$$\beta_c = -\frac{2.303RT}{\beta nF} \tag{2.20}$$

Despite the similarity of the Butler-Volmer equation when applied to a single electrode with a reversible half-cell reaction and when applied to an electrode with a coupled reaction, there is a distinct difference in the meaning of the net current density calculated. Taking the corrosion of steel as an example: in a reversible half-cell reaction (Figure 2.1a), at the reversible potential, $E_{0,Fe}$, the exchange current, $i_{0,Fe}$ has no contribution to the change of iron (dissolution and deposition at the same rate). When the potential is polarized to E, the net current, i_{net} represents the total change of iron as dissolution as shown in Figure 2.1a. However, in a coupled full reaction (Figure 2.1b), at the mixed potential, E_{corr}, the exchange current, i_{corr} represents iron dissolution at the same rate as the oxygen reduction reaction. When polarized from its mixed potential, the net current, i_{net} (assume positive) does not include all the change of iron: the "internal current, i_{O_2}" also contributes to a portion of the total change of iron – coexisting macro-cell (net current, i_{net}) and micro-cell corrosion (internal current, i_{O_2}).

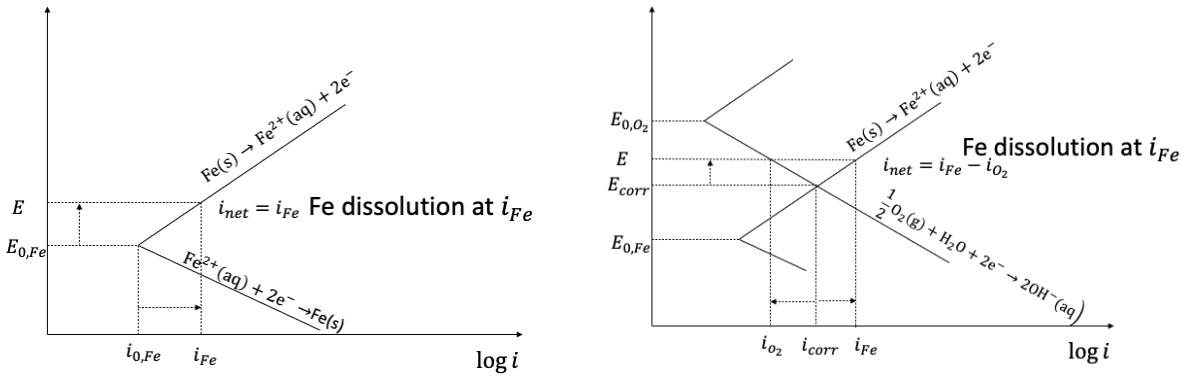


Figure 2.1 Evans diagrams for a single-electrode (a) and a mixed-electrode (b) reaction.

In Butler-Volmer/Tafel equation-based numerical models, corrosion parameters depicted by Evan's diagram (Figure 2.2) i.e. anodic branch of steel dissolution (E_{0a}, i_{0a}, β_a) and cathodic branch of oxygen reduction (E_{0c}, i_{0c}, β_c) are often used. It is mathematically equivalent to use both anodic and cathodic branches (E_{corr}, i_{corr}, β_a, β_c) depicted in Stern-Geary diagram (Figure 2.2), which, by linear extrapolation, gives a portion of the Evans diagram.

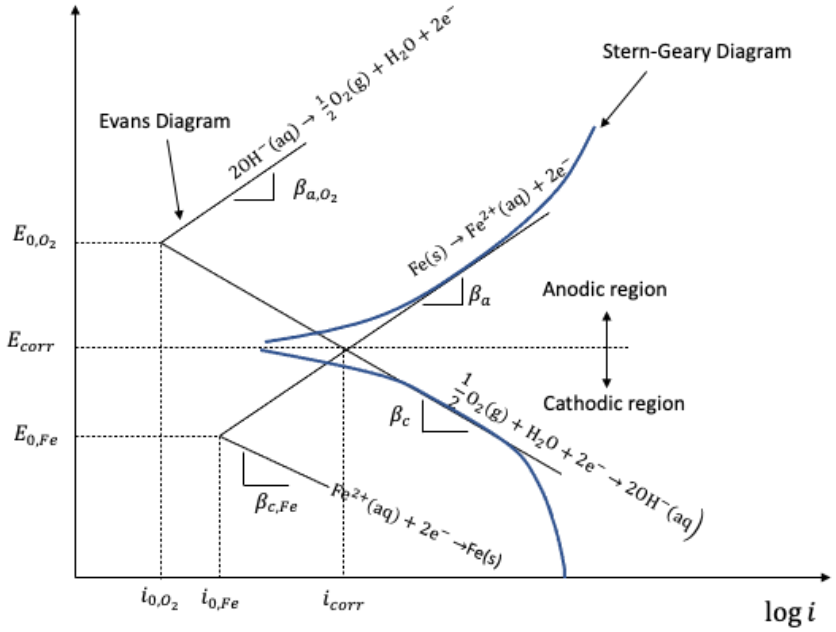


Figure 2.2 Schematic example of Evans Diagram and Stern-Geary Diagram of the coupled corrosion reactions (adapted from Perez, 2004)**.**

However, the latter set of parameters is more accessible in regular corrosion tests such as a three-electrode potentiodynamic scan of a corroding rebar. Evan's diagram is a simplified illustration of

the internal polarization over a corroding metal surface, and cannot be experimentally measured directly (Tan, 2012). Therefore, it is practical to estimate E_{0a}, i_{0a}, β_a, E_{0c}, i_{0c}, β_c from measured E_{corr}, i_{corr}, β_a, β_c, or write the equivalent polarization equations in terms of E_{corr}, i_{corr}, β_a, β_c, as they render the same straight lines in a log-scale diagram.

2.1.2.2 Treatment of corrosion cells

Fundamentally, there are three types of corrosion cells: pure micro-cell, pure macro-cell, and a combination of the two, macro-and-micro-cell. Non-Uniform corrosion in RC has been modelled both in the longitudinal direction (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc et al., 1992; Sohail, 2013) and around its circumference (Kim & Kim, 2008; Muthulingam & Rao, 2015; Zacaruk, 2013) of rebar using one of these three types of cells. The following sections will discuss the pros and cons of each approach together with their implementation in practice.

- Micro-cell model: no separation of anode and cathode

The corrosion process is divided algebraically into separate anodic and cathodic reactions with no net accumulation of electrical charge. Uniform kinetics of the anodic reaction and cathodic reaction are both prescribed on the surface of a rebar that may undergo corrosion. In this case, as the physical circuit is not explicitly modelled, the resistivity of the concrete will not play a role in determining the corrosion current although a conceptualized resistance between anodes and cathodes can be used to account for the ohmic polarization.

The micro-cell model has two major limitations: Firstly, the effect of resistivity on corrosion rate cannot be captured. In aqueous solutions, the resistance between micro-anodes and micro-cathodes can be neglected because the resistivity is very low, and the distance between the anodic and cathodic sites is very short. In porous media, such as concrete, however, this short distance does not necessarily lead to negligible resistance, and the anode-cathode separation might be larger than in the case of an aqueous solution. This non-negligible resistance as a function of the degree of saturation plays a significant role in the case of uniform corrosion, which cannot be captured with a simple mixed-potential uniform corrosion model. The effect of resistance between anodes and cathodes can only be included from the original experiment test where the corrosion kinetics are determined.

Secondly, the intermediate active to passive transition kinetics is indeterminate and usually not available. Rebar condition falls solely into either passive or active zones. For a given finite length of rebar (e.g. 5-10 cm), a binary characterization with either "active" or "passive" is not accurate. Appreciable variation has been found with repeated testing of strictly conditioned reinforced concrete samples (Sohail, 2013). It is believed that in aged concrete structures, due to varying exposure environments, different compositions, material non-uniformity (e.g. micro-cracks), etc. the rebar condition of a given finite segment may fall in between being active and passive.

- Macro-cell model: separation of anode and cathode

Macro-cell model is the most widely used approach with localized corrosion (often crack-induced) (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc et al., 1992; Yu et al., 2014). In these models, the active rebar is considered a pure anode and the passive rebar is considered a pure cathode. These pure anodic and cathodic sites are prescribed by the user or determined based on other corrosion initiation criteria (i.e. critical chloride content, pH value). By separating the anode and cathode on a rebar, we assume the active rebar portion assigned as anode only generates electrons ($Fe \rightarrow Fe^{2+} + 2e^{-}$) and the passive rebar portion assigned as cathode only receives electrons ($2H_2O + O_2 \rightarrow 4OH^{-}$). Occasionally, the reverse process such as iron deposition is also considered for anode (Yu, Liu, & Li, 2017). Physically separating anodes and cathodes allows the computation of current distribution in concrete so that concrete resistivity becomes an influential factor in determining the corrosion current. The macro-cell model is particularly useful for localized corrosion where the area of active rebar is small, and the corrosion rate is mainly contributed by macro-cell corrosion.

Having said that, the macro-cell corrosion model has difficulties when dealing with non-uniform corrosion where the micro-cell current is not negligible. Although neglecting the oxidation on passive rebar may be acceptable, the oxygen reduction in active rebar cannot be neglected when it is not largely polarized to its anodic direction, given the fact that, in concrete when the contribution of the macro-cell is limited due to high resistivity (Kranc & Sagüés, 1994). Particularly, when a long segment of rebar is treated as an anode, the micro-cell corrosion at the location out of the reach of the macro-cell effect is rather large and cannot be neglected.

The macro-cell corrosion model also has greater limitations in uniform corrosion modelling. In some studies (Ghods et al., 2007; Isgor & Razaqpur, 2006; Pour-Ghaz et al., 2009a; Yu, Liu, & Li, 2017), uniformly distributed anodes and cathodes with prescribed A/C ratios are used to model uniform corrosion. It was not clear whether the reported corrosion current density is only for the anodic rebar area, or it had been lumped onto the total rebar surface area. Nonetheless, this treatment would raise three questions: First, the cathodic reaction in this type of macro-cell model is presumably oxygen reduction in the passive zone of the rebar. Therefore, it is not proper to model the cathodic reaction on an active rebar and a different cathodic kinetics representation should be used. Second, the choice of the proper size of the anodes and cathodes is rather guessed, and hence, is questionable. Ideally, when uniform corrosion is treated in a macro-cell fashion, the assigned separated anodic and cathodic sites should reflect the physical size of these micro-anodes and micro-cathodes, which could be extremely computationally costly due to the small grid size needed for this representation. Finally, although this macro-cell treatment of uniform corrosion has allowed saturation to affect the corrosion rate, this effect only reflects on the resistivity change, without consideration of the reduction of corroding area due to drainage of the pores.

- Macro-and-micro-cell model

The macro-and-micro-cell model is a combination of the micro-cell model and macro-cell model and is by far the most comprehensive treatment of corrosion cells. In this model, the previously separated pure anode and cathode in a macro-cell model are assigned as segments of active rebar and passive rebar, where both anodic and cathodic current are enabled (i.e., anodic current on active rebar, cathodic current on active rebar, anodic current on passive rebar, and cathodic current on passive rebar) (Cao et al., 2013; Sohail, 2013). Sometimes the anodic current on a passive rebar is neglected due to its low value (Kim & Kim, 2008; The et al., 2010).

This model can capture both the macro- and the micro-corrosion currents, and hence the contribution of each component to the total non-uniform corrosion. However, its micro-cell feature and macro-cell feature still suffer from the previously mentioned limitations.

2.1.3 Corrosion parameters used in models

The parameters used in corrosion modelling are typically measured from a corrosion test involving rebar in a simulated pore solution or directly from a rebar embedded in concrete. Due to the obvious advantages of repeatability and simplicity, the effects of pH and chloride content on rebar corrosion parameters were mostly studied in simulated concrete pore solutions. The effects of chloride and carbonates on the polarization cures have been studied mostly qualitatively (Moreno et al., 2004) as there are only a very small number of studies that reported dependence of these parameters as a function of the chemical composition of the solution, such as chloride content and pH value, in a quantitative way.

It is worth noticing that although the main objective is to study the corrosion of rebars in RC structures, only a few of the reported corrosion parameters were obtained by the direct measurement of rebar corrosion in concrete. A review of the various corrosion parameters used in corrosion models from the literature shows that values with a large range were used.

Locke & Siman (1980) analyzed the polarization curves of steel rebar in chloride-contaminated concrete using a galvanostatic polarization technique. They found that with different levels of chloride content, the observed Tafel slopes were in the range of 320-1570 mV/dec and the cathodic Tafel slopes were in the range of 380-1080 mV/dec. and the calculated B value would be in the range of 82-278 mV.

Alonso et al. (1998a) tested the Tafel slopes of rebar corrosion in aged concrete using the potentiodynamic polarization technique. The chloride salt was added to the concrete mix for the rebar segment under testing and the test was performed on the concrete samples under wet and dry conditions. The forward scan from -400 mV to 400 mV (relative to E_{corr}) was used to curve fit the Tafel slopes. They set the anodic Tafel slope to a fixed large value (1×10^{20} mV) to account for the sharp slope observed in the curve and fitted cathodic Tafel slopes as 289 and 441 mV/dec for wet concrete and concrete with ambient humidity.

Chang et al. (2008a) measured the corrosion parameters in RC made with concrete containing flash and slag submerged in seawater. They used a “two-test procedure” in which the potentiodynamic scan was done separately in the anodic direction and cathodic direction to eliminate the initial

current surge and its influence on the distortion of scan curves. They noticed that the experimentally acquired polarization curve deviates from the Butler-Volmer equation form. In their companion paper (Chang et al., 2008b), they proposed a model based on the assumption of two parallel kinetic processes occurring at the steel/rust-film/concrete interfaces: an active corrosion process that follows the Buter-Volmer equation, and the possibility of growth or dissolution of a rust film on the steel surface. The latter process was used to account for that deviation. Some typical anodic Tafel slopes reported for fly ash concrete were 450, 600, 800 mV and the corresponding cathodic Tafel slopes were 240, 192, 263 mV/dec; for slag concrete, the anodic Tafel slopes were 205, 230, 295 mV/dec and the correspond cathodic Tafel slopes were 480, 250, 235 mV/dec. For direct exposure to seawater, a new bar had Tafel slopes of 89 mV/dec and 1215 mV/dec, and a corroded rebar had Tafel slopes of 115 mV/dec and 900 mV/dec.

Sohail (2013) directly measured the corrosion parameters for rebar in partially submerged carbonated and non-carbonated concrete by the potentiodynamic scan with explicit IR compensation between the reference electrode and the working electrode (corroding rebar). A similar "Two-test procedure" was also employed to test 24 replicate samples for each case: carbonated and non-carbonated to capture the uncertainty. The results are listed in Table 2.1.

Table 2.1 Corrosion parameters for active and passive rebar (Sohail, 2013)

parameter	Active(carbonated)	Passive(uncarbonated)
E_{corr}vs. SCE	-543 – -650 mV average value -640 mV	-89 – -319 mV average value -214 mV
i_{corr}	0.3–0.26 $\mu A/cm^2$ 5–0.9 $\mu A/cm^2$ (partial polarization)	0.01–0.15 $\mu A/cm^2$ 0.008-0.01 $\mu A/cm^2$ (partial polarization)
β_a	180 to 280 mV/dec average value 244 mV/dec	374 to 700 mV/dec average value 505 mV/dec
β_c	190 to 350 mV/dec average value 260 mV/dec	70 to 160 mV/dec average value N/A

The anodic and cathodic Tafel slopes for active rebar were 244 mV/dec (range 180-280 mV/dec) and 260 mV/dec (range 190-350 mV/dec). For passive rebar, the anodic Tafel slopes were 505 mV (range 374–700 mV/dec) and the cathodic Tafel slope ranges from 70 mV/dec to 160 mV/dec.

Despite the complexity of the dependence of corrosion parameters on concrete-rebar conditions, many researchers have selected a single set of values for the Tafel slopes to model the corrosion kinetics. For some Tafel slopes used by different modellers (Table 2.2), the calculated B values are 19 mV (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc et al., 1992; Kranc & Sagüés, 1994), 26 mV (Gulikers, 2005) with 19mV being closer to the case of an active rebar (26 mV). However, an 89 mV B value calculated using the averaged value of both Tafel slopes has also been used to model the active corroding rebar (Martın-Pérez, 1999).

Table 2.2 corrosion parameters used by various macro-cell models

Reference / Parameter	(Kranc et al., 1992)	(Kranc & Sagüés, 1994)	(Martın-Pérez, 1999)	(Isgor & Razaqpur, 2006)	(Ghods et al., 2007)	(Gulikers, 2005)
E_{oa} [mV] vs. SCE	-600	-780	-682	-780	-780	-887 (-906 vs CSE)
E_{oc} [mV] vs. SCE	0	160	160	160	160	480 (407 vs CSE)
$i_{0a}[\mu A/cm^2]$	N/A	1.875×10^{-2}	3.75×10^{-2}	2.75×10^{-2}	3×10^{-2}	1×10^{-4}
i_{0c} $[\mu A/cm^2]$	2×10^{-2}	6.25×10^{-4}	1.25×10^{-3}	6×10^{-4}	1×10^{-3}	1×10^{-5}
β_a [mV/dec]	60	60	(60–524) 409	60	60	91
$\lvert\beta_c\rvert$ [mV/dec]	160	160	(160–394) 350	160	160	176

There are a lot of cross-references of the corrosion parameters used, among which, some sources of highly referred Tafel slope values are presented below.

- Source of 60 and 160 mV/dec

Kranc et al. (1992) pointed out that the cathodic Tafel slopes of corrosion in concrete should be in 100 to 200 mV range when measured by very slow potential scan rates, and this range agrees with the Tafel slopes of about 160 mV commonly observed for oxygen reduction of steel in alkaline solutions (Fabjan et al., 1980). The iron dissolution reaction in concrete was assumed to be similar to the case in aqueous solution which is activation controlled with a relatively small Tafel slope: 60 mV/dec, typically, although the clear information of the aqueous solution was not stated (original article not accessible, cited as "Kaesche, H, Metallic Corrosion, NACE, Houston. Tex. 1985" in the work of Kranc et al., 1992).

- Source of 91 and 176 mV/dec

Some studies found that the cathodic Tafel slope was 176 mV/dec from the testing of reinforced concrete in dry conditions (Gulikers, 2005; Warkus et al., 2006). The original test data source is not accessible. It was not stated whether the anodic Tafel slope of 91mV/dec was obtained from the same test.

It has been well known that the experimentally acquired Tafel slopes from reinforced concrete tests remain remarkably uncertain, and their values are very different from the values used in some of the current models (Ghods et al., 2007; Gulikers, 2005; Isgor & Razaqpur, 2006; Kranc et al., 1992; Kranc & Sagüés, 1994) where the corrosion parameters employed were similar to the parameters found for steel corrosion in aqueous solutions (see the above-mentioned source of 60/160 mV/dec). However, there are also some electrochemical corrosion models (Martın-Pérez, 1999; Sohail, 2013) that used the corrosion parameters directly from reinforced concrete measurements. To conclude, to improve the prediction of the corrosion rate in cracked concrete, the proper corrosion parameters to use should be carefully and rationally determined. Generally speaking, corrosion parameters directly measured in concrete with a uniform-corrosion-assumption-based technique such as the potentiodynamic scan, should not be used in macro-cell models when microcell corrosion is not negligible; such parameters should be used in micro-cell or the micro-portion of micro-and-macro-cell models. A rational and systematic procedure for the determination of the corrosion parameters is needed to comply with the complicated issues associated with corrosion in cracked RC structures under various exposure conditions.

2.2 Microenvironment in concrete and natural exposure conditions

Moisture penetration, carbonation and chloride ingress are the major processes that affect the microenvironment near the rebar such as pH value and chloride content, which determine the onset of the propagation stage and affect the ongoing corrosion rate. These processes have been studied by a lot of researchers, and a good collection of numerical models were proposed to simulate the process of moisture transport, carbonation, and ingress of chlorides in concrete under different boundary conditions. In this chapter, the state of the art in this area will be summarized with a focus on the treatment of cracks in RC structures and the natural service conditions.

2.2.1 Moisture transport

The modelling of moisture evolution in cracked concrete is the most important task as moisture content and the transport of moisture affect both the ingress of chloride ions and carbonation. Water transport in partially saturated concrete has been long investigated and a large number of models have been proposed (Bažant & Najjar, 1972; DaÏan & Saliba, 1993; Isgor & Razaqpur, 2004; Künzel, 1995; Pel et al., 1996; Ramadani, 2013; Sun et al., 2018). Although it has been shown to be usually in good agreement with experimental observations, the treatment of the moisture transport process could be confusingly different from one study to another. In order to avoid further confusion, the following review will start with the most generic treatment in general porous media before considering some concrete-specific studies.

In porous media, such as concrete, water may transport in both vapour and liquid forms. Liquid and vapour flow are usually treated separately with respect to the different driving forces (Scheffler & Plagge, 2010): The vapour diffusion flow is proportional to the gradient of vapour pressure/phase-averaged density with the water vapour permeability/diffusivity as transport coefficient; the liquid water flow, on the other hand, is proportional to the pressure gradient with hydraulic conductivity/ permeability as transport coefficient. The general equations are typically written as:

Water vapour diffusion models:

$$\frac{\partial m_v}{\partial t} = \nabla \cdot \left[\frac{D_v(\theta_l)}{R_v T}\nabla p_v\right] + \frac{\partial m_{l\to v}}{\partial t} = \nabla \cdot [D_v(\theta_l) \cdot \nabla \rho_v] + \frac{\partial m_{l\to v}}{\partial t} \qquad (2.21)$$

Liquid water flow conduction models:

$$\frac{\partial m_l}{\partial t} = \nabla \cdot [\rho_l K_l(\theta_l)(\nabla\psi + 1)] + \frac{\partial m_{v\to l}}{\partial t} = \nabla \cdot \left[\rho_l \frac{k_l(\theta_l)}{\mu}(\nabla p_c + \rho_l \boldsymbol{g})\right] + \frac{\partial m_{v\to l}}{\partial t} \quad (2.22)$$

where $m_v[kg/m^3]$ is total volume-averaged water vapour mass content, $D_v[m^2/s]$ is vapour diffusivity, $R_v[J/K/kg]$ is specific gas constant of vapour, $\theta_l[-]$ is liquid volume fraction, $p_v[Pa]$ is vapour pressure, $\rho_v[kg/m^3]$ is vapour density, $m_{l\to v}[kg/m^3]$ is liquid-to-vapour phase change mass content, $m_{v\to l}[kg/m^3]$ is vapour-to-liquid phase change mass content, $t[s]$ is time, $m_l[kg/m^3]$ is total volume-averaged liquid water mass content, $\rho_l[kg/m^3]$ is liquid water density, $K_l[m/s\ ]$ is hydraulic conductivity or coefficient of permeability as a function of water content, $k_l[m^2]$ is permeability as a function of water content, $\psi[m]$ is pressure head, $p_c[Pa]$ is capillary pressure, $\mu[Pa \cdot s]$ is dynamic viscosity of water, $\boldsymbol{g}[m^2/s]$ is gravitational acceleration.

Concrete generally has very high capillary pressure due to its fine pore structure, the gravity effect becomes negligible. If we neglect gravity:

$$\frac{\partial m_l}{\partial t} = \nabla \cdot [\rho_l K_l(\theta_l)\nabla\psi] + \frac{\partial m_{v\to l}}{\partial t} = \nabla \cdot \left[\rho_l \frac{k_l(\theta_l)}{\mu}\nabla p_c\right] + \frac{\partial m_{v\to l}}{\partial t} \quad (2.23)$$

In analogy with the diffusion treatment of water vapour transport, the transport of liquid water can also be modelled by using a diffusion model. Neglecting the gravity effect, the driving force is the gradient of water concentration and the diffusivity which is a function of water concentration is the transport coefficient.

Liquid water diffusion:

$$\frac{\partial m_l}{\partial t} = \nabla \cdot [\rho_l D_l(\theta_l)\nabla\theta_l] + \frac{\partial m_{v\to l}}{\partial t} \quad (2.24)$$

where the liquid water diffusivity

$$D_l(\theta_l) = \frac{k_l(\theta_l)}{\mu}\frac{dp_c}{d\theta_l} \quad (2.25)$$

The derivation of moisture transport models in both the vapour and the liquid phases of water has also been examined using homogenization techniques where it was shown that transport of moisture in the vapour and liquid phases can be unified using the Richards equation through the concept of "total diffusivity" as shown in Equation(2.27) (Samson et al., 2005). Considering $\rho_v = f(\theta_l)$ (Daian, 1988), and the mutual phase change vanishes:

$$m_{l\to v} + m_{v\to l} = 0 \tag{2.26}$$

Equation (2.21) and (2.22) can be combined as

$$\frac{\partial m_{l+v}}{\partial t} = \nabla \cdot [\rho_l D_{toal}(\theta_l) \nabla \theta_l] \tag{2.27}$$

where, the total diffusivity of liquid and vapour

$$D_{toal}(\theta_l) = \frac{\rho_l D_l + \frac{d\rho_v}{d\theta_l} D_v}{\rho_l} \tag{2.28}$$

This unified equation fully describes the movement of both vapour and liquid water with one single variable. A similar unification of Richards' equation could also be derived in the form of a liquid water flow conduction model, i.e. Equation (2.21) and (2.23) with a total hydraulic conductivity, K_{total} (DaÏan & Saliba, 1993).

$$\frac{\partial m_{l+v}}{\partial t} = \frac{\partial m_{l+v}}{\partial \psi} \cdot \frac{\partial \psi}{\partial t} = \nabla \cdot [\rho_l K_{total}(\theta_l)(\nabla \psi)] \tag{2.29}$$

However, a separate solution of the liquid and the vapour transport is often needed in the case where the liquid water flow needs to be coupled with the transport of ions in the pore solution through advection.

When the liquid water flow is modelled, it is worth noticing that the diffusion model in the form of Equation (2.21) and the conduction model in the form of Equation (2.22) can be considered equivalent, provided that the proper choice of transport coefficient and the driving force are made. The distinguished difference between those two approaches is related to the choice of the driving force: gradient of water content vs. capillary pressure. The physical driving force of liquid water

transport, however, is the gradient of pressure. The diffusion approach, by introducing an artificial equivalent driving force (gradient of water content), is limited in its extendibility to hysteresis and solute transport modelling (Scheffler & Plagge, 2010).

In various moisture transport models, the moisture flow (liquid, vapour, or combined) is proportional to the gradient of various variables such as mass water content, volumetric water content, hydraulic head, pressure head (neglecting gravity), relative humidity (Bažant & Najjar, 1971), vapour density, and accordingly various transport coefficient were developed to be compatible with the selected form of the driving force.

Cracks, depending on their morphology and density, can be treated as discrete cracks (e.g., load-induced) or smeared cracked material (e.g., freeze-thaw induced). When treated discretely, cracks can be treated as free surface or equivalent porous media (Boulfiza et al., 2003; Li, 2014). When the cracks are treated as equivalent porous media, the concept of the relationship between the equilibrated relative humidity (RH) and the radius of the smallest empty pore given by the Kelvin equation is extended to the cracks. Given a distribution of the openings between the lips of a crack, the permeability of the equivalent porous medium can be estimated under different exposure RH conditions.

$$\ln(RH) = \ln\left(\frac{p_v}{p_{sat}}\right) = \frac{2\gamma V_m}{rRT} \tag{2.30}$$

where $p_v[Pa]$ is vapour pressure, $p_{sat}[Pa]$ is saturated vapour pressure above a flat surface at temperature $T[K]$, $\gamma[N/m]$ is surface tension, $V_m[m^3/mol]$ is molar volume, $r[m]$ is radius of a droplet (negative value for concave meniscus), which is related to the largest capillary pores size or a crack opening size of $|r_p|/cos\alpha$. The parameter α is the contact angle determined by the type of liquid and the solid surface.

2.2.2 Transport of gas

The gases of most interest in concrete are carbon dioxide and oxygen. Similar to water vapour, the diffusion of oxygen and carbon dioxide is driven by the gas density/partial pressure in the gas phase. Because the bulk movement of gas in concrete is rare in practical scenarios, the advection and dispersion effects are often neglected. Knudsen diffusion is only significant when the mean

free path of gas is of the same order or greater than the pores or interparticle space (Kaviany, 2012). The Knudsen number is defined as

$$K_n = \frac{\lambda(\text{mean free path of gas molecules})}{C(\text{average pore size or interparticle clearance})} \tag{2.31}$$

When $K_n \gg 1$, the flow is Knudsen flow and the diffusion is Knudsen diffusion. The mean free path of air at room temperature and at ambient pressure is 68 nm (S. G. Jennings, 1988) which could be comparable to the size of smaller capillary pores (10 nm – 5 μm). Therefore, at very dry conditions where concrete small pores were emptied, the Knudsen diffusion needs to be considered and the diffusivity becomes a combination of molecular and Knudsen diffusion.

For carbon dioxide and oxygen, the phase change term $\frac{\partial m_{l\to v}}{\partial t}$ in Equation (2.21) becomes a sink term of the consumption rate. The consumption rate of carbon dioxide is determined by the saturation level and pH in the pore solution and the consumption rate of oxygen is determined by the oxygen-reduction cathodic reaction of the rebar corrosion.

2.2.3 Transport of dissolved solutes

- Diffusion-advection-dispersion

The solutes in concrete pore solution such as chloride ions and dissolved gases are transported by diffusion, advection, and dispersion. The driving force for the diffusion of dissolved ions is the ion concentration gradient in the pore solution. Advection is the ion movement carried by the pore solution through the liquid phase flow during wetting and drying process. Dispersion is a mechanical mixing phenomenon as the ions move through the porous network. The dispersion effect can be neglected when Peclet number Pe < 1 – 10 (Kaviany, 2012). In concrete, the dispersion effect is often neglected due to the slow water movement under water content gradient (Samson et al., 2005). The transport of salts in concrete can be described by the diffusion-advection equation, whereby:

$$\frac{\partial(\theta c)}{\partial t} = \nabla \cdot \left(D_{eff}\theta \frac{\partial c}{\partial z} + \boldsymbol{u}c \right) + \theta \left(\frac{\partial c}{\partial t} \right)_{Sink/Source} \tag{2.32}$$

where c $[kg/m^3]$ is liquid phase-averaged solute concentration, $\boldsymbol{u}$ $[m/\mathrm{s}]$ is Darcy's velocity, D_{eff} $[m^2/\mathrm{s}]$ is effective diffusivity, term $\left(\frac{\partial c}{\partial t}\right)_{Sink/Source}$ may be introduced as chloride binding (negative) and releasing (positive) or the sulphate consumption (negative).

When dispersion is not neglected, such as the fast sorption of solution into dry concrete with a large Peclet number (Puyate et al., 1998), The dispersion coefficient $D^d[m^2/s]$ needs to be added to the effective diffusivity.

2.2.4 Natural exposure boundary conditions

Compared to a laboratory-controlled environment, natural exposure conditions feature daily fluctuation of RH, temperature, radiation, seasonal rainfall, snow melt, and scheduled application of de-icing salts. The moisture content and the movement of water in RC structures may also fluctuate dramatically which may have a significant impact on the chloride ingress, carbonation rate, and corrosion rate.

In many studies on concrete moisture, the internal RH is used to describe the saturation status of concrete given the measured RH-moisture content retention/isotherm curves. However, the continuous non-steady state moisture transport is caused by the ever-changing micro-climate immediately surrounding a given RC structure. Therefore, a transient model must be employed to capture the evolution of water content in naturally exposed RC structures. Measurements of the internal RH of concrete samples under natural and simulated environments showed that for sheltered concrete, temperature was the main influential factor and for unsheltered concrete the dominant factor was rain periods (Andrade et al., 1999). For known hysteresis and the effect of temperature on the retention curve, direct measurement of water content is more preferred than RH because a given RH may correspond to different water contents depending on the temperature, pore structure and wetting-and-drying history. Ryu et al. (2011) measured the relative moisture content of transversely cracked/uncracked concrete under simulated environments, it was found that the affected depth by 24-hour rain was within 30 cm for concrete (w/c=0.6) and within 10 cm for concrete (w/c=0.3) and that the rain intensity had little effect. For cracked concrete, it was found that larger cracks would lead to slightly faster drying of the surrounding concrete, but the effect became less significant for less dense concrete (w/c=0.6).

Given the availability of a good understanding of the mechanisms governing the transport of liquid water and water vapour in porous media, numerical simulation is considered a robust tool to accurately assess the multi-factor affected moisture evolution in naturally exposed RC structures. Having said that, there remain a few challenging tasks: Firstly, proper boundary conditions need to be developed for wetting (liquid/moist air) and drying. Secondly, hysteresis needs to be considered during wetting and drying. Thirdly, the effects of multiple environmental factors such as wind speed, radiation, and condensation need to be properly addressed.

3 On the Corrosion Parameters Acquired through Potentiodynamic Scans of Carbon Steel Rebar in Simulated Pore Solution and Mortar

3.1 Preface

It was introduced in Chapter 1 that one of the drawbacks in current numerical corrosion models on non-uniform corrosion of cracked/uncracked RC structures is that the corrosion kinetics described in these models are often constant, and the dependence of these critical corrosion parameters (i.e., equilibrium potential, exchange current density, Tafel slopes) on the chemical environment is usually not explicitly represented. In Chapter 2 (Section 2.1.3), A review of the corrosion parameters used in corrosion models from the literature shows that values with a large range were used. These parameters were either measured from a corrosion test involving rebar in a simulated pore solution or directly from a rebar embedded in concrete. Only a few of the reported corrosion parameters were obtained by the direct measurement of rebar corrosion in concrete. However, these tested reinforced concrete samples had diverse and separate conditions. Typically, these samples had limited variations in chloride content, moisture content, or carbonation state, with only one or two treatment levels. Although it is feasible to make qualitative assumptions about the impact of these parameters on the corrosion rate, quantifying it accurately is extremely difficult. Determining the specific causes of variations in the data due to the particular conditions of the concrete samples is a challenging task.

The work presented in this chapter aims at an accurate measurement of the corrosion parameters in simulated pore solutions and mortar and a quantitative analysis of the impact of the major environmental parameters on the corrosion parameters. This chapter starts with examining the challenges of the accurate measurement and interpretation of corrosion parameters due to measuring techniques and unrealistic sample conditions, which will be addressed in this chapter.

Realistic rebar conditions were used. The as-received rebar retained the mill scale and was prepassivated before exposure to chloride and carbonation. The long-term effect of depassivation in simulated pore solutions was investigated.

Then a modified three-electrode potentiodynamic test procedure was carried out using a Faraday shield and floating ground to reduce the large electrical potential oscillations caused by real-time IR compensation, testing the partially saturated rebar embedded in mortar in a non-submerged

setting to avoid the open circuit potential drifting due to extensive surface wetting of samples, using two-step scans with adequate rest time to minimize the distortion of scan curves.

The corrosion potential (E_{corr}), corrosion current density (i_{corr}), and Tafel slopes (β_a, β_c) were measured in four media: simulated pore solution (SPS); carbonated simulated pore solution (CSPS); non-carbonated mortar; and carbonated mortar with various chloride concentrations.

This study examines the impact of depassivation, rust build-up, and chloride concentration on corrosion parameters. The investigation includes the evaluation of rebar in SPS and CSPS over different periods, as well as tests conducted on non-carbonated and carbonated mortars. The analysis explores the effects of chloride content, carbonation, conditioning duration, and relative humidity on corrosion parameters in these media.

The manuscript of this chapter was submitted to "Construction and Building Materials". Gang Li is the primary author who designed and performed the experimental study and data analysis. Dr. Moh Boulfiza and Dr. Richard Evitts contributed to the review of the manuscript and provided feedback.

3.2 Introduction

Rebar corrosion is one of the most critical durability issues associated with reinforced concrete structures. Corrosion mitigation measures include selecting proper materials, on-site/laboratory corrosion testing and protection, and predictive corrosion modelling. Corrosion parameters such as corrosion potential, corrosion current density, and Tafel data are needed to evaluate corrosion rate, design cathodic protection, and develop predictive electrochemical corrosion models.

It is a well-known fact that the measured corrosion rate and corrosion potential of a given reinforced concrete (RC) structure are strongly dependent on the severity (harshness) of the exposure service conditions (temperature, RH, moisture content, presence of aggressive chemical species such as chloride ions, and carbonation, cracks, etc.) as well as on the accuracy of the measurement procedure used. Hence, it is expected that estimation of how the measured quantities, at a given time, could fluctuate if some of those parameters change over time (seasonal variations, use of de-icing salts in winter, etc.) would be very helpful. Similarly, the accuracy of a corrosion measurement depends on the kind of precautions taken to minimize interference of "parasitic

phenomena" such as the use of IR compensation or the use of Faraday's shield. Here also, measurements are often carried out without all these precautions, and it is often unclear how much improvement one should expect if they were used. Hence, the existence of a practical and reliable method remains rather elusive due to the large number of parameters that can affect the measured rate.

In this paper, the protective dense oxide film that forms on the rebar surface in high alkalinity is termed as "passive film", although it is not a passive film in the traditional sense. This passive film serves as a diffusion barrier that limits the corrosion rate, when the rebar is in its "passive" state. The destruction of this oxide film due to low pH and/or the presence of chloride ions is called "de-passivation", and the rebar undergoes "active" corrosion. The minimum chloride content in concrete or the chloride ion concentration in a solution that leads to significant active corrosion is called critical chloride content or critical chloride ion concentration.

Rebar corrosion parameters are typically measured using electrochemical testing of rebar in simulated pore solution or directly from rebar embedded in concrete. Due to repeatability and simplicity of experiment, the effects of pH and chloride content on rebar corrosion have been mostly studied in simulated concrete pore solutions (Figueira et al., 2017; Ghods et al., 2010; Moreno et al., 2004; Shi et al., 2016). In these research works there were various types of steel rebar samples used. Some potentiodynamic tests were performed on polished cross-sections of rebar (Figueira et al., 2017; Shi et al., 2016). It was found that the roughness of the polished surface and the pre-oxidization in aerated NaCl solution affect the critical chloride concentration threshold (Figueira et al., 2017). To mimic the passivated state of steel rebars in concrete, the polished rebar samples were pre-passivated in a simulated pore solution before the tests (Shi et al., 2016). In some tests, the mill scale (pre-existing oxide layer) was preserved, and the tests were performed on a cylindrical surface of the as-received rebar with no pre-passivation (Moreno et al., 2004) or with adequate pre-passivation (Ghods et al., 2010). The synergistic effects of chloride and carbonates on polarization characteristics have been studied qualitatively in solutions (Moreno et al., 2004). It is also worth mentioning that the chloride-induced depassivation state could be unstable in the first 10 days (Ghods et al., 2010). To include the above-mentioned considerations, the ideal potentiodynamic tests of rebar in simulated pore solutions need to be performed on rebar samples with mill scale, passivated in simulated pore solution before being subject to chloride and/or

carbonation attack. Also, the exposure conditioning period should be longer than a few days before the test to allow for the possible depassivation to reach a stable condition.

The corrosion parameters in aqueous solutions could be helpful with the qualitative comparison of the corrosion resistance of rebar in different environments. However, these parameters might not be representative of rebar corrosion in concrete, which leads to inaccurate prediction of corrosion rate in corrosion models. A variety of numerical models were proposed to predict the corrosion rate in the propagation phase and mechanical behaviour of reinforced concrete structures under different exposure conditions(Otieno et al., 2012). Despite the complexity, modelling the electrochemical process of rebar embedded in concrete is the core component required to calculate the corrosion current and the corresponding rate of loss of metal. A parametric study with a separated anode and cathode and a fixed concrete bulk resistivity shows that the corrosion rate is most sensitive to the anodic and cathodic Tafel slopes (Ge & Isgor, 2007). Therefore, the accurate prediction of corrosion rate relies on the correct choice of these two corrosion parameters.

There have been only a few corrosion parameters reported by the direct measurement of rebar corrosion in concrete (Alonso et al., 1998b; Chang et al., 2008b; Elsener, 2005; Garcés et al., 2005; Gulikers, 2005; Locke & Siman, 1980; Sohail, 2013). These corrosion parameters had a significant range of variation as they seem to be sensitive to different conditions in concrete, such as moisture content, pH, and chloride content. The reported corrosion parameters for rebar cast in concrete/mortar and their conditions are summarized in Table 3.1.

Table 3.1 Corrosion parameters of rebar in concrete

E_{corr}[mV] vs.SCE	i_{corr}[μA/cm^2]	β_a[mV/ dec]	$\lvert\beta_c\rvert$[mV/ dec]	Remarks	Reference
–	1.81* – 5.52*	400 – 830	180 – 320	NaCl wt.% concrete (0-1%) Ambient dry Machined Rebar	(Locke & Siman, 1980)
–	5.18*–488.33*	320 – 1570	380 – 1080	NaCl wt.% concrete (0-1%) Ambient dry Rebar with scale	(Locke & Siman, 1980)
-91	0.56	1×10^{20}	441	3% $CaCl_2$ mixing water Ambient dry	(Alonso et al., 1998b)
-238	0.61	1×10^{20}	289	Damp (not specified) 3% $CaCl_2$ mixing water	(Alonso et al., 1998b)
-568	0.32*	450	240	Concrete with fly ash Submerged in seawater for 9 months	(Chang et al., 2008b)
-260	0.076*	205	480	Concrete with slag Submerged in seawater for 9 months	(Chang et al., 2008b)
-702, -664	0.66*, 0.52*	600, 800	192, 263	Concrete with fly ash Submerged in seawater for 4.7 years	(Chang et al., 2008b)
-283, -296	0.032*, 0.034*	230, 295	250, 235	Concrete with slag Submerged in seawater for 4.7 years	(Chang et al., 2008b)
-543 – -650	0.3 – 2.6	180 – 280	190 – 350	Carbonated concrete Saturated Polished rebar	(Sohail, 2013)
-89 – -319	0.01 – 0.15	374 – 700	70 – 160	Non-carbonated concrete Saturated Polished rebar	(Sohail, 2013)
–	–	180	102	Steel fiber added depassivated by Wetting and drying with 3.5% NaCl for 2.5 years	(Nguyen et al., 2018)
–	–	562	19.8	Steel fiber added Passivated rebar	(Nguyen et al., 2018)
–	–	–	110	Periodic exposure to NaCl solution for 30 months	(Glass & Chadwick, 1994)

* converted from corrosion rate [mm/year]

There is very little Tafel data reported for rebar in simulated concrete pore solutions. Some of the reported data on corrosion potential and corrosion current density are listed in Table 3.2.

Table 3.2 Corrosion parameters of rebar in simulated pore solutions

E_{corr}[mV] vs.SCE	i_{corr}[µA /cm^2]	β_a[mV/ dec]	β_c[mV/ dec]	Remarks	Reference
-576	10.18*	89	1215	Rebar in seawater for 1 hour	(Chang et al., 2008b)
-640	34.68*	115	900	Corroded rebar in seawater for 3.9 years	(Chang et al., 2008b)
-467**	0.15**	–	–	$[Cl^-]/[OH^-]$ = 6.11 pH = 13.5 Rebar cross section polished	(Figueira et al., 2017)
-583**	2.0**	–	–	$[Cl^-]/[OH^-]$= 2.62 pH = 11.6 Rebar cross section polished	(Figueira et al., 2017)
-494	2.0**	–	–	pH 13.9 1.0 % Cl ($[Cl^-]$=0.28 mol/L) 100-150 hours	(Moreno et al., 2004)
-474	8.0	–	–	pH 9 50 hours 1.0 % Cl ($[Cl^-]$=0.28 mol/L)	(Moreno et al., 2004)
-500**	–	–	–	Rebar pre-passivated 10 days $[Cl^-]$=1.0 mol/L	(Shi et al., 2016)
-380**	–	–	–	Rebar not pre-passivated $[Cl^-]$=1.0 mol/L	(Shi et al., 2016)
-450**, -550**	–	–	–	Rebar pre-passivated original mill scale pH = 13.5 $[Cl^-]$=0.3 mol/L, 1.0 mol/L	(Ghods et al., 2010)

* converted from corrosion rate [mm/year]

** estimated from the graphs from the sources by the authors

The reversible potentials and exchange current densities of the half-cell reactions used by various modellers for reinforced concrete corrosion simulation are listed in Table 3.3 and visualized in a Stern-Geary diagram shown in Figure 3.1.

Table 3.3 Corrosion parameters used in corrosion models

E_{oa} [mV] vs. SCE	E_{oc}[mV] vs. SCE	i_{0a}[μA/ cm^2]	i_{0c} [μA/ cm^2]	E_{corr}* [mV] vs. SCE	i_{corr}* [μA /cm^2]	β_a [mV/ dec]	$\lvert\beta_c\rvert$ [mV/dec]	Reference
-600	0	–	2.00×10^{-2}	–	–	60	160	(Kranc et al., 1992)
-780	160	1.88×10^{-2}	6.25×10^{-4}	-675	1.04	60	160	(Kranc & Sagüés, 1994)
-780	160	2.75×10^{-2}	6.00×10^{-4}	-683	1.12	60	160	(Isgor & Razaqpur, 2006)
-780	160	3.00×10^{-2}	1.00×10^{-3}	-675	1.66	60	160	(Ghods et al., 2007)
-834**	480**	1.00×10^{-4}	1.00×10^{-5}	-772	4.60×10^{-4}	91	176	(Gulikers, 2005)
-682	160	3.75×10^{-2}	1.25×10^{-3}	-679	0.038	409	350	(Martın-Pérez, 1999)
–	–	–	–	-640	0.7	244	260	(Sohail, 2013)

*Calculated based on Evan's diagram with the corresponding E_{oa} E_{oc} i_{0a} i_{0c}

** Converted from Standard Hydrogen reference electrode

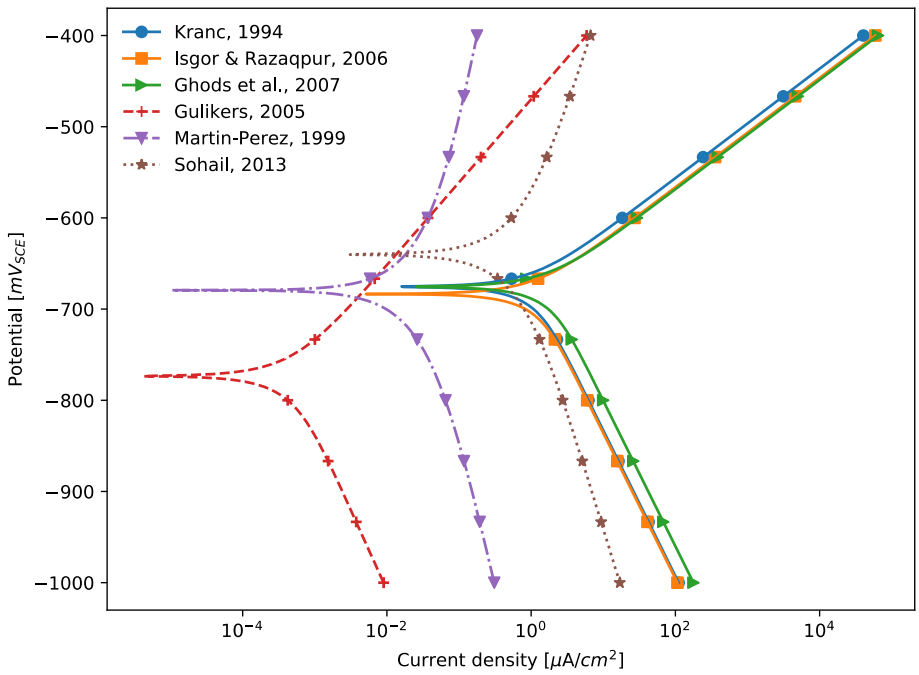


Figure 3.1 Stern-Geary diagrams plotted with the reversible potentials and exchange current densities of the half-cell reactions utilized by different modellers for reinforced concrete corrosion simulation, as reported in Table 3.3.

In this table, the corrosion potentials and corrosion current densities were derived from the half-cell reversible potentials (E_{oa}, E_{oc}) and exchange current densities (i_{0a}, i_{0c}) by the mixed-potential theory treating corrosion as uniform corrosion with the same total area for anodic and cathodic reactions. Some corrosion current densities are comparable to the active rebar corrosion in solutions having the same order of magnitude (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc & Sagüés, 1994), and some are significantly smaller (Gulikers, 2005; Martın-Pérez, 1999), where corrosion parameters are based on measured parameters in concrete. Moreover, the Tafel slopes, especially the anodic Tafel slopes measured from rebars in simulated pore solutions are significantly smaller than that observed in concrete. When macro-cell corrosion is modelled with smaller Tafel slopes, the macro-cell corrosion rate becomes overly sensitive to the overpotential on the rebar surface, leading to inaccurate large prediction of corrosion rate. Such overestimation would be most likely corrected by prescribing proper or actual discrete anodic and cathodic areas on the rebar, which are not feasibly accessible in real concrete structures.

The corrosion parameters in concrete are the critical component in non-destructive testing and numerical simulations as the calculated corrosion rate depends on the corrosion parameters used. Although some studies have reported corrosion parameters, direct measurement of rebar corrosion in concrete is still challenging. In tests on rebar in concrete (Alonso et al., 1998b; Chang et al., 2008b; Elsener, 2005; Garcés et al., 2005; Gulikers, 2005; Locke & Siman, 1980; Sohail, 2013), regardless of various concrete proportioning and different concentrations of chloride contamination, there are limitations in the utility of the data. The limitations are originated from measuring technique, and unrealistic sample preparation. For the measuring technique, the IR drop between the rebar and the reference electrode is either neglected or post-compensated by measuring the resistance through the concrete cover separately. Real-time IR compensation has been attempted (Chang et al., 2008a; Sohail, 2013) and reportedly causes scan curves to have large oscillations and noise, which makes the results very difficult to interpret. For sample preparation, when sodium chloride is added during concrete mixing, the embedded rebar does not have a chance to be pre-passivated before it comes into contact with the chloride ions. In addition, the reported data were acquired from reinforced concrete samples with scattered and isolated conditions, with only one or two treatment levels of chloride content, moisture content, or carbonation state. It would be possible to speculate qualitatively on how the increase or decrease of any of those parameters may affect the corrosion rate, a quantitative estimation is very challenging. It is difficult to interpret how the variation in the data was caused by these concrete sample conditions.

In this research work, a quantitative analysis of the impact of the major environmental parameters and a measurement procedure is carried out. This present research explores a suitable setup of the three-electrode potentiodynamic test procedure to quantify the corrosion parameters of rebar in simulated pore solutions and mortar. The improvements in the setup are aimed at overcoming the above-mentioned experimental challenges by eliminating the large potential oscillation during the real-time IR compensation, the corrosion potential drift and the distortion of the scans. The comprehensive experimental program tests the as-received carbon steel rebar samples in four media: simulated pore solution (SPS); carbonated simulated pore solution (CSPS); non-carbonated mortar; and carbonated mortar. The corrosion potential, corrosion current density, and Tafel data were measured. With this new corrosion data, the effect of chloride and carbonation on rebar depassivation and corrosion rate in simulated pore solutions and mortar are analyzed.

3.3 Material and methods

3.3.1 Sample preparation

No.10 deformed carbon steel rebar (nominal diameter: 11.3 mm) was used in this research. The rebar was cold cut into the desired lengths (3 cm and 14 cm) with a metal cutting band saw. The original mill scale of the rebar was preserved to represent the form in its practical application. A 2.5 mm diameter hole was drilled into one end of the rebar to allow a lead wire to be attached. Any loose red rust visible on the rebar surface was gently removed by a wire brush. The rebar samples were rinsed with distilled water and then degreased with acetone to remove any contamination due to the cutting and drilling process. An electrical connection was secured by wedging a wire lead inside the drilled hole and sealing the hole with epoxy. All rebars used in this study for both rebar samples and mortar samples were preconditioned in the simulated pore solution (pH=13.5) for one week to develop a stable passive film on the surface, which would achieve adequate pre-passivation according to a study (Poursaee & Hansson, 2007).

Two types of test samples were prepared: bare rebar samples and samples with rebar cast in mortar. Figure 3.2 presents a schematic of these samples.

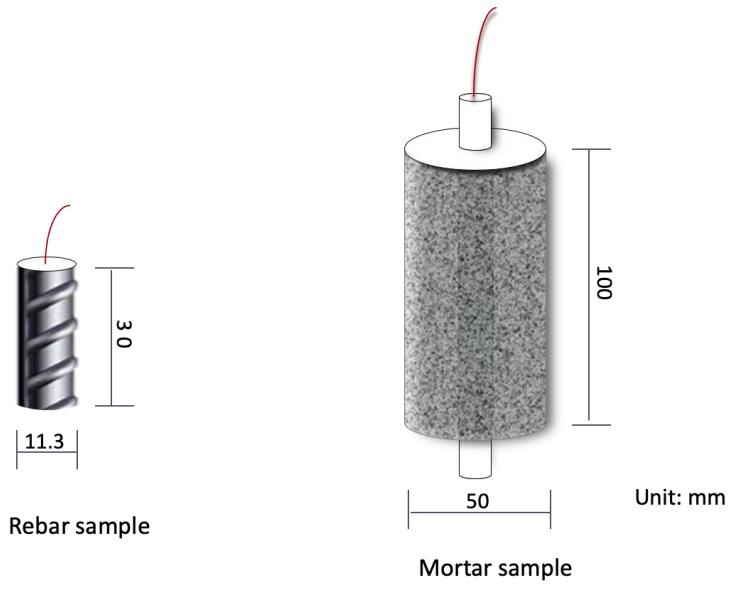


Figure 3.2 Schematic for rebar and mortar sample dimensions. Rebar sample is a NO. 10 deformed rebar and mortar sample has a water-cement ratio of 0.7.

Both ends of the rebar sample and mortar sample were coated with epoxy resin, with only the lateral or cylindrical surface being exposed to the electrolyte.

Type GU cement was used for casting 36 mortar samples. The mix proportioning (cement: water: sand) was 1:0.7:1.65 by weight. Dry NaCl crystals (Manufacturer: Fisher Chemical, purity≥99.0%) were added to the mixing water as a source of chloride contamination. Six levels of contamination were considered ranging from 0% to 5% (wt%, by weight of mixing solution); the equivalent chloride contents with respect to cement mass are listed in Table 3.4. The mortar samples were demolded after 48 hours and cured in sealed plastic wrap for 14 days. Then two coats of epoxy were applied to the end surfaces of the cylinders and extended rebar surfaces; the lateral surfaces of the mortar cylinders remained wrapped in plastic film. The epoxy coatings were fully cured after another 14 days. The wrapped cylinders were cured in the ambient environment until they reached a total age of 60 days.

Table 3.4 NaCl contamination levels in mortar

Chloride level	0	1	2	3	4	5
wt.% NaCl mixing solution	0%	0.5%	1.0%	2.0%	3.0%	5.0%
wt.% Cl by weight of cement	0%	0.21%	0.43%	0.87%	1.31%	2.24%

In order to create carbonated mortar samples, half of the coated samples were then put into a carbonation chamber of 50% (vol%) CO_2 and 65% relative humidity (RH); the other half were stored in a humidity chamber of 65% RH. The samples were exposed for one month before the corrosion tests were performed. The carbonation progress was checked by spraying 1% (vol%) phenolphthalein solution on a freshly broken surface of the mortar samples. The mortar samples were fully carbonated after one month. The mortar samples (both non-carbonated and carbonated) were conditioned at various moisture levels, which includes the relative humidity of 65%, 85% and 95% for two months and finally submerged in fresh water for seven days. A uniform moisture distribution is preferred so that the moisture content at the rebar depth can be calculated by the overall moisture content of the mortar samples and a gradual wetting procedure was chosen to achieve a uniform moisture distribution; drying of wet mortar is a slow process, and it produces a moisture profile along the depth of each sample. However, the wetting process proceeds much

faster than drying, with a consistent moisture front sweeping through the sample, creating a consistent moisture condition throughout the samples. In addition, with a shortened conditioning period, the reduced rust accumulation on the rebar will be less likely to crack the samples.

Two types of simulated pore solutions were formulated to provide corrosive electrolytes for the rebar samples: non-carbonated simulated pore solution (SPS) and carbonated simulated pore solution (CSPS). The SPS and CSPS have a pH of 13.5 and 8.0, respectively. The composition of SPS follows previous studies (Figueira et al., 2017; Ghods et al., 2010) (Table 3.5), and the composition of CSPS (Table 3.6) is based on the measured ionic concentration of calcium and potassium in carbonated concrete (Pu et al., 2012). Sodium chloride was added to the pore solutions with the same molar concentration as the mixing solution used for mortar casting.

Table 3.5 Dry chemicals used for the simulated pore solution, SPS (pH=13.5)

Dry chemicals added	$Ca(OH)_2$	NaOH	KOH	$CaSO_4 \cdot 2H_2O$
Concentration [mol/L]	0.012	0.1	0.2	0.002

Table 3.6 Dry chemicals used for the simulated carbonated pore solution, CSPS (pH=8.0)

Dry chemicals added	$KHCO_3$	$CaSO_4 \cdot 2H_2O$
Concentration [mol/L]	0.1	0.002

In summary, there were six levels of chloride contamination and two levels of pH/carbonation state for both rebars in simulated pore solution and mortar. For the mortar samples, there are also four levels of moisture conditions: relative humidity of 65%, 85%, 95% and submerged. Three replicate samples were prepared for each condition level to capture the uncertainty of the test data.

3.3.2 Test procedure

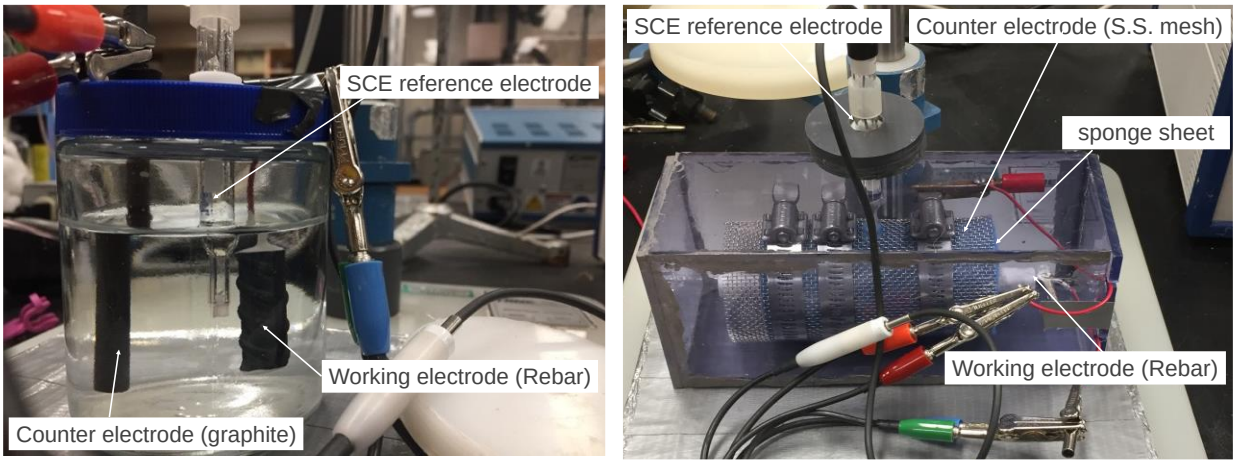
The three-electrode potentiodynamic test was used to measure the corrosion parameters. The scan range was $\pm$150 mV around the open circuit potential (OCP). A graphite rod or a stainless steel mesh was used as the counter electrode. A saturated calomel electrode (SCE) was used as the reference electrode; all potentials reported in this paper are with respect to the saturated calomel

electrode (SCE) with real-time IR compensation activated on the Gamry Reference 600 potentiostat that was used to perform the tests.

The three-electrode system for rebar in simulated pore solution and mortar are shown in Figure 3.3. The first series of tests showed that the counter electrode material type did not affect the potentiodynamic curve beyond normal experimental uncertainty. For the test of bare rebar samples in solution, a graphite rod was used as the counter electrode. For the test of mortar cylinders, the electrode was not submerged, and a stainless steel mesh was used as a counter electrode. A sponge sheet slightly moistened with simulated pore solution served as electrolytic contact between the stainless steel mesh and the mortar cylinder surface. The stainless steel mesh was wrapped around the lateral surface of the mortar cylinder and secured with stainless steel hose clamps to achieve a uniform current distribution over the embedded rebar surface. The reference electrode had a contact point with the sponge through a small cut-away area in the stainless steel mesh. The whole mortar sample testing setup was installed in a sealed PVC box to prevent the sponge from completely drying out during the scans. Both setups were enclosed in a Faraday shield connected to floating ground. The scan settings are listed in Table 3.7.

It was noted that, although often used, epoxy-coated rebar ends were reported to be susceptible to crevice corrosion in a solution. A setup of the rebar cut end held tightly with Teflon washers was reported to eliminate visible signs of crevice corrosion (Ghods et al., 2010). This setup was attempted. However, the rebar cut end is very sharp, where the rebar rib and the cut surface form a small angle. These sharp, thin edges create hot spots for corrosion, and any chamfering to remove the sharp edges would expose bare steel surfaces without mill scale and create gaps under the Teflon washers. The authors acknowledge that possible crevice corrosion near the epoxy coating may contribute to the source of error in the rebar test in a solution. Additionally, the potential gradient around the rebar by using a rod counter electrode in the solution could also be a source of error.

The pre-passivated rebar samples were conditioned for 20 days in the chloride-contaminated test solutions (SPS and CSPS) to allow adequate time for the breakdown of the passive film.



(a) (b)

Figure 3.3 The setup of three-electrode potentiodynamic scan test for (a) rebar in solution, and (b) rebar in mortar cylinder. Both setups are placed in a faraday shield during testing.

Table 3.7 Potentiodynamic scan test control settings

Sample type	Scan rate	Scan range VS. OCP	Counter electrode	Ref. electrode	Use of Faraday shield	Real-time IR compensation
Rebar sample	0.166 mV/s	±150 mV	Graphite rod	SCE	Yes	Yes
Mortar sample	0.1 mV/s	±150 mV	Stainless steel mesh	SCE	Yes	Yes

3.4 Data processing

The experimental scan data were fitted to the Butler-Volmer Equation

$$i_{net} = i_{corr} \cdot \left[10^{(E-E_{corr})/\beta_a} - 10^{(E-E_{corr})/\beta_c}\right] \quad (3.1)$$

where, i_{net} is net current density, E_{corr} is corrosion potential, i_{corr} is corrosion current density, β_a is anodic Tafel slope, and β_c is cathodic Tafel slope.

The data did not provide an exact fit to the equation and the deviation is thought to be related to the development of the oxide film that was formed during the conditioning of rebar in SPS, and similarly for rebar cast in mortar. Thus, the model was changed to incorporate this. Chang et al. (2008b) developed an empirical kinetic equation that considers the growth and dissolution of the

rust film. The net current density due to the growth/dissolution of rust film is described by Equations (3.2) and (3.3):

Anodic polarization:

$$i_{F_net} = i_{corr} \cdot \sqrt{\frac{E - E_{corr}}{V_a} \cdot 10^{-(E-E_{corr})/\beta_a}} \tag{3.2}$$

Cathodic polarization:

$$i_{F_net} = -i_{corr} \cdot \sqrt{\frac{E - E_{corr}}{V_c} \cdot 10^{-(E-E_{corr})/\beta_c}} \tag{3.3}$$

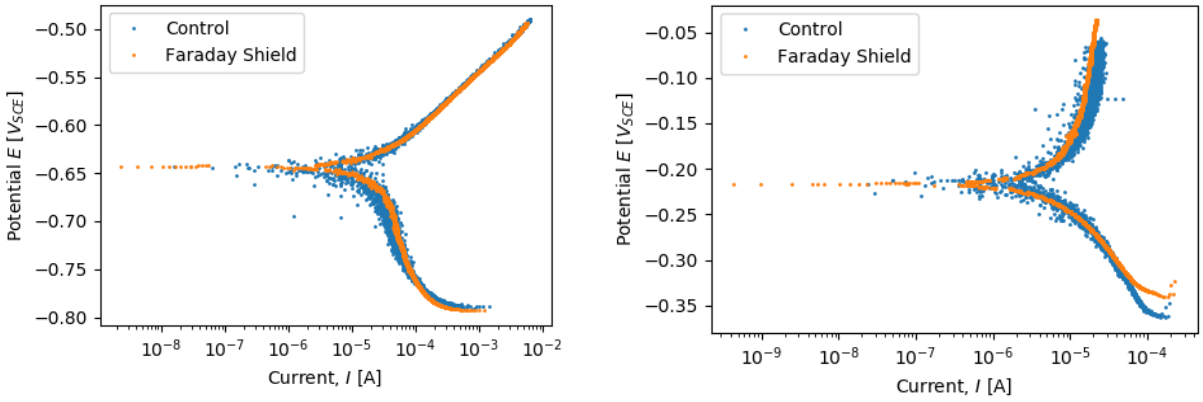
where, E_{corr}, i_{corr}, β_a, β_c(negative) are corrosion parameters from the Bulter-Volmer equation. V_a, V_c (negative) are parameters indicating the rust film growth at a rate of i_{F_net} (negative) and dissolution at a rate of i_{F_net} (positive).

The total net current is the summation of i_{net} (Butler-Volmer equation) and i_{F_net} (film growth/dissolution equation), and is termed, in this paper, the Butler-Volmer-Film equation. All the scan data in this study was modelled with this equation using open-source software for interactive curve fitting on potentiodynamic scan data that had been previously developed by the authors (Li, Evitts, Boulfiza, et al., 2018).

3.5 Results and discussion

3.5.1 IR compensation noise

Some preliminary tests were performed to develop the test settings for data acquisition. It was found that the use of a Faraday shield and floating ground will significantly reduce the large potential oscillations caused by real-time IR compensation. Two examples of the use of these for rebar in 3% (wt%) NaCl solution are shown in Figure 3.4.



(a) (b)

Figure 3.4 The effect of Faraday shield on noise reduction caused by IR compensation, (a) an example of rebar sample in 3% (wt.%) NaCl solution and (b) an example of rebar cast in a mortar with 3% (wt.%) NaCl mixing water at ambient dry environment. The control datasets were acquired without the use of a Faraday shield.

3.5.2 Open Circuit Potential (OCP) drift with mortar surface wetting

Submersion or extensive surface wetting of the air-dried mortar samples was found to significantly alter the original open circuit potential. An example of this is presented in Figure 3.5, where the carbonated mortar sample was wrapped with a soaking-wet sponge while taking the OCP readings. The OCP drifted more than 100 mV within one hour and continue to drop afterwards. In fact, the OCP was measured at around -100 mV and drifted rapidly to -225 mV before the continuous reading started. The time to reach steady-state readings was denoted as the stabilization time. The stabilization time was found to vary from 30 minutes to more than 2 hours. Generally, it took longer to stabilize if the sample was drier and had higher chloride content. A stabilization time is normal in a standard polarization test where a freshly polished metal is submerged in the test solution; it normally takes some time for the metal to develop a stable or steady-state corrosion state. However, in the case of a mortar sample, the mortar is the electrolyte, and the corrosion state should have been stabilized prior to the external wetting of the exterior of the mortar. Within the stabilization period, the moisture front would likely have insufficient time to reach the rebar surface from the wetted external surface, and there is sufficient oxygen in the mortar to sustain the oxygen reduction half-reaction during the initial OCP process. In the OCP logging process, the consumption rate of oxygen corresponds to the corrosion current density (i_{corr}), which is low. The partially wet concrete cover does not directly affect the microenvironment on the rebar surface, but it may affect the current distribution within that 2 cm thick mortar cover by altering the ionic

path and mass transfer (and corresponding electrical) resistance between the coupled anodic and cathodic sites. This could shift the localized potential of these sites and cause the potential at the reference electrode contact point to drift. The use of a slightly moistened sponge thus minimized this OCP drift.

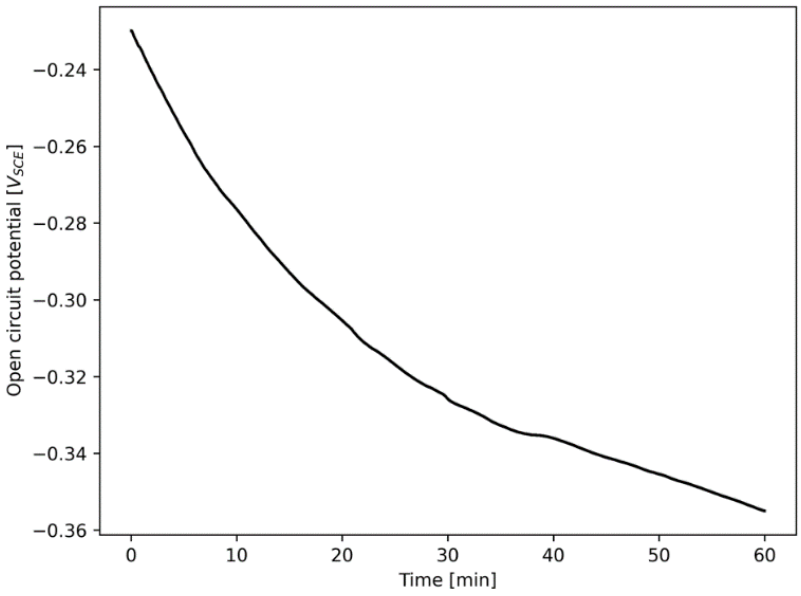


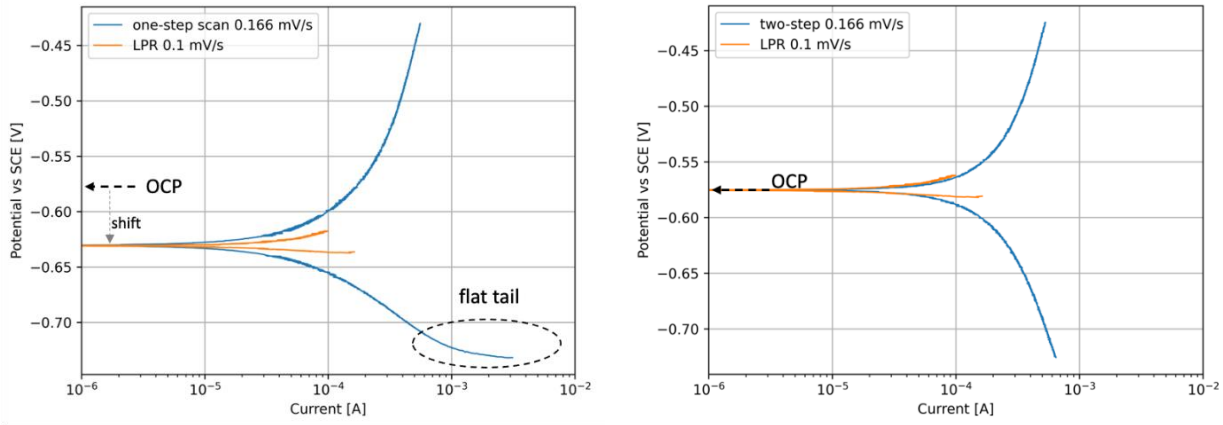
Figure 3.5 An example of open circuit potential (OCP) drift of the rebar embedded in a carbonated mortar sample when the surface is wrapped with a wet sponge.

3.5.3 Two-step scan vs. one-step scan

A one-step scan procedure is the normal potentiodynamic scan procedure of ASTM G5-14, where the one-direction potential scan starts at the low end of the scan range from the cathodic branch and with a positive potential increment, scans towards the high end of the anodic branch. A two-step scan procedure consists of two separate anodic and cathodic scans. Both scans start from the OCP and scan in the positive direction for the anodic branch and in the negative direction for the cathodic branch. Figure 3.6 presents examples of scan curves from both scan procedures for the rebar samples in CSPS with 3% (wt%) NaCl. The potentiodynamic scan range is OCP$\pm$150 mV. The scan curves were compared to the linear polarization resistance (LPR) scan curves, which is a one-step scan over a small-range potential (OCP$\pm$10mV) and considered to be “non-destructive”.

In this research the LPR scans were performed first and then followed by the potentiodynamic scans. The same sample was used to perform these scans. The sample was left to rest up to 24 hours between the two potentiodynamic scans to allow the OCP to recover to the previous values.

As shown in Figure 3.6, the one-step scan curve exhibits a "flat tail" where the cathodic branch scan starts. This flat tail occurs because a much greater current is required to sustain the -150 mV overpotential in the one-step scan procedure compared to the two-step scan procedure. This current surge, also observed by Chang et al. (2008a), seems to significantly change the rebar surface condition. It caused E_{corr} to significantly shift from OCP in the negative direction, and distorted the cathodic scan curve to such an extent that the anodic branch was affected. The distortion effect on the anodic is confirmed by comparing it to the small range of "non-destructive" LPR scan curves. For the one-step scan (Figure 3.6a), the anodic branch deviates from the anodic portion of the LPR scan curve, which suggests that the rebar surface condition has significantly changed during the cathodic scan history in the one-step scan. For the two-step scan (Figure 3.6b) because there was no immediate cathodic scan history, the anodic branch overlaps well with the LPR scan curve.



(a) (b)

Figure 3.6 Potentiodynamic scan curves of rebar in CSPS with 3% (wt%) NaCl solution: (a) One-step scan and (b) two-step scan. Note: the small difference in E_{corr} (less than 10 mV) of the two-step scan branches was offset in the plot for better comparison.

When performing a one-step test with a scan range of from OCP-150 mV to OCP+150 mV, the reported E_{corr} shifts by 20–50 mV towards the negative direction from the OCP for rebar samples in both solutions and mortars, and sometimes even larger (about 100 mV). This shift is more

pronounced for the tests in solutions. In addition, the distorted scan curve would also provide an inaccurate corrosion current density, i_{corr}.

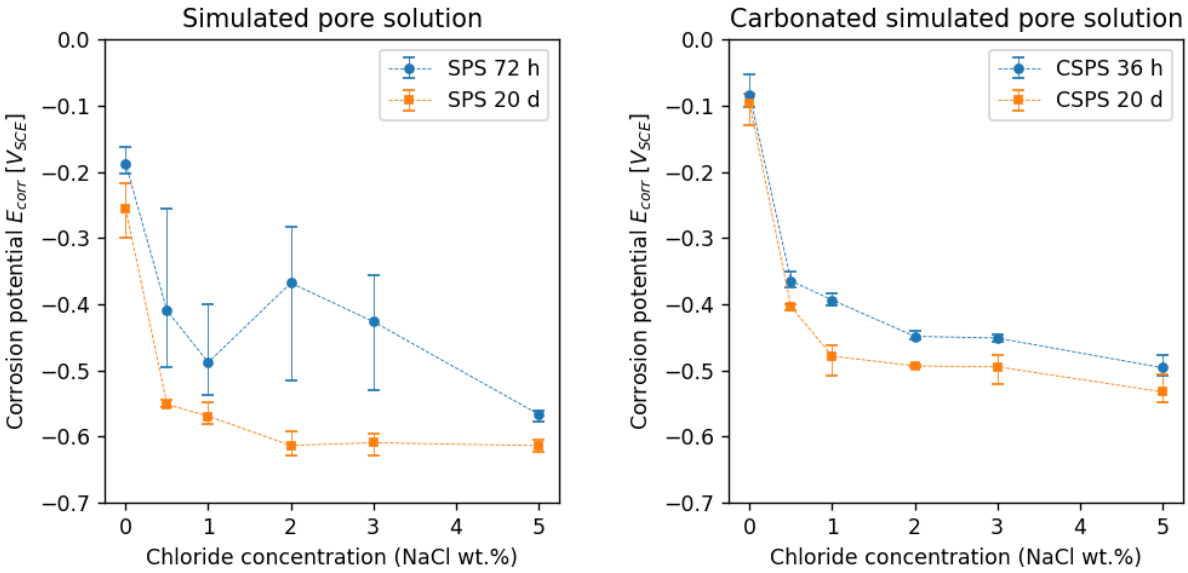
In order to avoid these drawbacks, the two-step scan procedure was used in this research. The cathodic branch was scanned followed by the anodic branch. Between each scan, ample time was given for the samples to restore to their original OCP naturally with a difference of less than 10 mV. The ample conditioning period was found to be four hours for the tests in solutions and 24 hours for the tests in mortar with all air relative humidity conditions. The negligibly small drift in potential indicates the rebar surface change brought by the cathodic scan branch is reversible, but not within a short scan period. This reversible recovery of the rebar surface is further confirmed because there is no discrepancy between the LPR anodic branch and the anodic branch from a two-step test as shown in Figure 3.6b.

3.5.4 Depassivation time of rebar in the simulated solutions

All the rebar samples were pre-oxidized in the simulated pore solution to form the passive film prior to exposure to chloride contamination and carbonation. It would take some time for the chloride ions to destroy the passive film. According to previous experiments by Ghods et al. (2010) where chloride was gradually added to the simulated pore solution, E_{corr} started to decrease when a critical chloride concentration of 0.1mol/L was reached and continued to decrease with an increase of chloride concentration of up to 1 mol/L. At each concentration increment, the corrosion potential fluctuated over a range of $\pm 0.1\,V$ within 100-160 hours. The stabilization of the depassivation of the passive film likely depends on both chloride ion concentration and conditioning time. In the present study, the tests were performed for two depassivation periods: 72 hours and 20 days for SPS, and 36 hours and 20 days for CSPS.

For the SPS tests, as shown in Figure 3.7a, E_{corr} had higher variability for the repeated samples at 72 hours, but at 20 days, this variability is dramatically reduced. After 20 days, E_{corr} became less noble E_{corr} and was not sensitive to chloride ion concentration greater than or equal to 2% (NaCl wt%). Above the 2% threshold, in SPS, a longer period of time is needed to achieve a stable depassivation of rebar in SPS with lower chloride ion concentrations compared to higher concentrations. However, eventually, both lower and higher concentrations will lead to the same depassivation result. This was not captured in the previous experiments (Ghods et al., 2010) where

the chloride was added in steps, and there was insufficient time between steps to enable changes in E_{corr} to be measured. Corresponding to the stabilized less noble E_{corr} after 20 days, the variation of i_{corr} for replicate samples is also decreased after 20 days. However, the i_{corr} value showed a slight decrease instead of a significant increase (Figure 3.8a). This suggests that in SPS, for chloride equal or greater than 2%, despite the ongoing depassivation indicated by E_{corr}, there is no corresponding increase in i_{corr}.



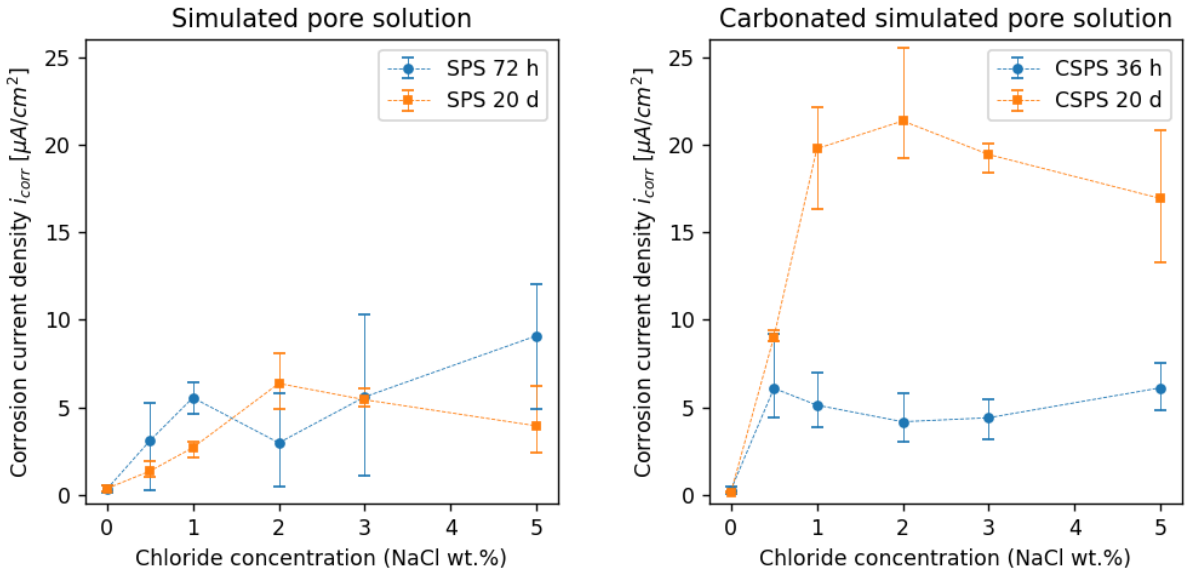
(a) (b)

Figure 3.7 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: corrosion potential of pre-passivated rebar in (a) SPS for 72 hours and 20 days and (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

For the CSPS tests, as shown in Figure 3.7b, E_{corr} appeared to be stable with small variation for repeated tests both 36 hours and 20 days for various chloride concentrations. From 36 hours to 20 days, there is an approximately 50 mV drop in E_{corr} which corresponds to a significant increase in i_{corr} (Figure 3.8b): it is about a three times increase in i_{corr} for chloride concentration larger than 1% (wt% NaCl).

Both i_{corr} in SPS and CSPS at 20 days levels off and slightly decrease from chloride concentration at 2% (wt.% NaCl solution). However, there was a trend that shows an increase in i_{corr} for the shorter exposure period (72 h for SPS and 36 hours for CSPS, Figure 3.8). This suggests that the chloride concentration equal to or greater than 2% (wt.% NaCl solution) has reached the maximum

effect of depassivation in both SPS and CSPS. It is important to note that continuous depassivation occurs in CSPS, as indicated by a large increase in i_{corr} at 20 days. In conclusion, after the short term depassivation, the long-term depassivation additionally slightly decreases the E_{corr} for rebar in both SPS and CSPS but only significantly increases i_{corr} for rebar in CSPS.



(a) (b)

Figure 3.8 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: corrosion current density of pre-passivated rebar in (a) SPS for 72 hours and 20 days and (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

3.5.5 Corrosion parameters for SPS and CSPS: the effect of chloride concentration and pH

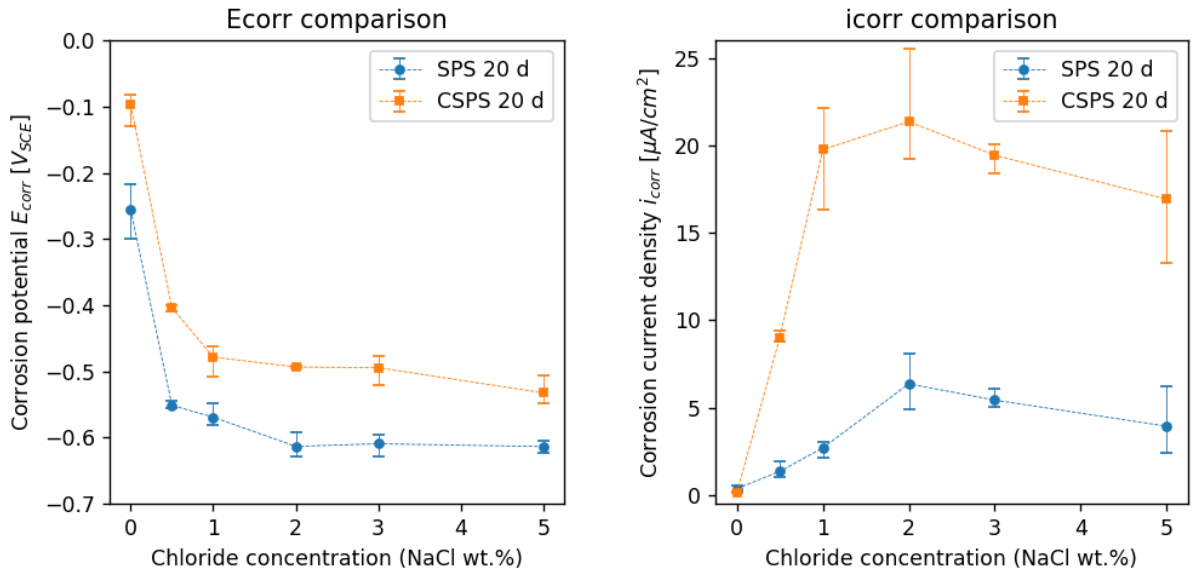
3.5.5.1 Corrosion potential and corrosion current density in SPS and CSPS

The measured corrosion potentials and corrosion current densities for the rebars conditioned in solutions for 20 days are shown in Figure 3.9. The corrosion potential is lower at higher chloride ion concentrations (greater than 1% wt% NaCl) than that at lower chloride ion concentrations for both SPS and CSPS types. At approximately 2% (wt%) NaCl concentration and above, the corrosion potential is less dependent on the chloride ion concentration. It is interesting to notice

that the concentration invariant corrosion potentials for rebar in SPS are 100-150 mV more negative than that for rebar in CSPS.

The corrosion current densities of the pre-passivated rebar in both SPS and CSPS remain small (0.15 $\mu A/cm^2$ for CSPS and 0.36 $\mu A/cm^2$ for SPS) in the absence of chloride ions after 20 days of exposure comparing to their values during early short-term exposure (72 h for SPS and 36 hours for CSPS). After 20 days, for both SPS and CSPS, the rebar corrosion current densities increase with the chloride ion concentration and peak at 2% (wt%) and then slightly decrease at higher chloride ion concentration.

In general, the rebar corrosion current densities in CSPS with chloride contamination was found to be approximately 3–10 times that of the SPS, even though the corrosion potentials are slightly more noble.



(a) (b)

Figure 3.9 Comparison of corrosion potential (a) and corrosion current density (b) of pre-passivated rebar in SPS and CSPS with chloride after 20 days of conditioning. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

3.5.5.2 Anodic Tafel slopes in SPS and CSPS

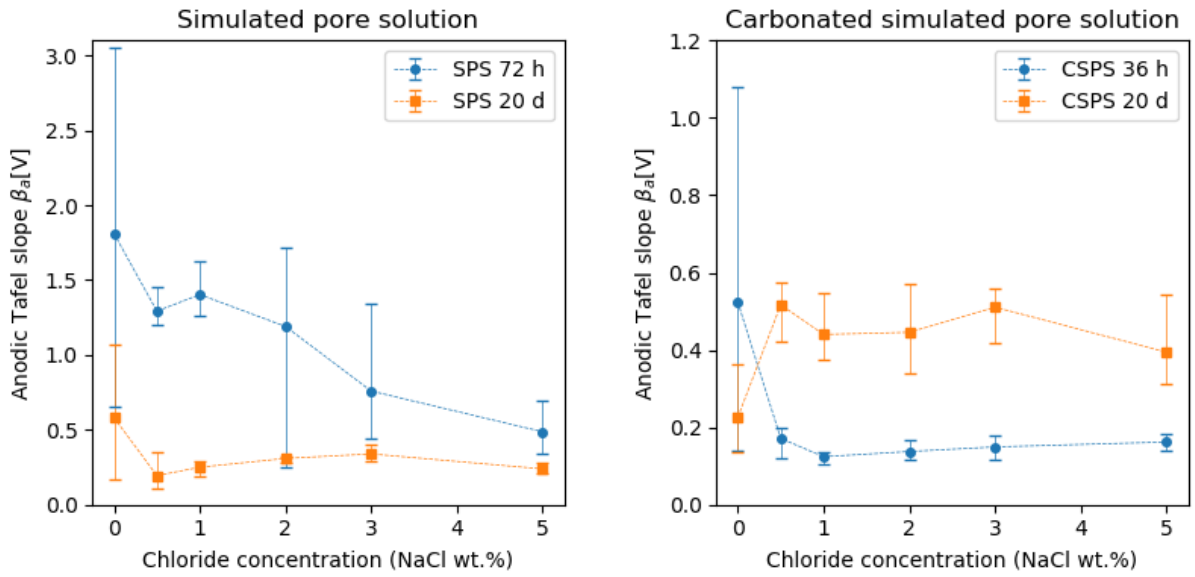
The anodic Tafel slopes of rebar corrosion in SPS and CSPS are shown in Figure 3.10. The fitted Tafel slopes for the rebar exclude the secondary reaction, i.e., film growth/dissolution Va/Vc. In SPS, for the first 72 hours, the anodic Tafel slope decreases with chloride ion concentration (Figure 3.10a) but eventually converges to around 0.2–0.4 V/dec after 20 days of exposure.

The anodic Tafel slope is the change of current corresponding to the change of overpotential, contributed by both the dissolution of steel and the Fe^{2+}/ Fe^{3+} conversion in the passive rust film. In SPS, at 20 days, when there is no chloride, the anodic slopes measured from the repeated tests varies from 0.2 V/dec to more than 1.0 V/dec. This large variation in the anodic Tafel slope could be caused by the competition between the iron dissolution reaction and the change of passivated film and the latter could be more significant when there was no chloride ion in the solution and the rebar sample remained passivated. When chloride ion is present in SPS, the anodic Tafel slopes measured from the repeated tests become more uniform and exhibit a range of around 0.2–0.3 V/dec, and it slightly increases with chloride concentration up to 3.0% (wt.%) NaCl solution and then slightly decreases at higher chloride ion concentrations. This value is significantly larger than 89 mV for non-prepassivated rebar immersed in seawater for one hour (Chang et al., 2008b). This suggests the protective passivated film on the rebar significantly increases the anodic Tafel slope in chloride-containing solution with high pH.

In CSPS, the anodic Tafel slope is shallow when there is no chloride ion, but it does not indicate the depassivation of the rebar as E_{corr} is about -0.1 V (Figure 3.9). Instead, it shows the current density on the rebar, excluding the contribution of the oxide film change, is sensitive to the overpotential before the net current density reached the passivated current density. When chloride ion is present, the anodic Tafel slope tends to stay near 0.4–0.5 V/dec, which is higher than that in SPS.

It is noteworthy that the presence of chloride ions resulted in a higher anodic slope for the rebar in CSPS, which has a lower pH of 8.0 compared to SPS with a pH of 13.5. It is hypothesized that the corrosion accumulation on the rebar in CSPS with chloride ions led to an increase in the anodic Tafel slope. As shown in Figure 3.10b, at 36 hours, the anodic Tafel slope is around 0.1–0.2 V/dec

and it increased to 0.4–0.5 V/dec at 20 days with the rust buildup observed on the rebar surface in CSPS with chloride ions. However, it is noticed in another study (Chang et al., 2008b) that the four-year rust build-up on the rebar in neutral sea water only slightly caused an increase in the anodic Tafel slope from 0.089 V/dec to 0.115 V/dec. Therefore, the larger anodic Tafel slope observed in CSPS could be attributed to both the rust buildup and the pH-buffering carbonate-bicarbonate couple that is absent in the sea water.



(a) (b)

Figure 3.10 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: anodic Tafel slopes of pre-passivated rebar in (a) SPS for 72 hours and 20 days (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

This anodic Tafel slope increase accompanied by rust build-up was not significant for rebar in SPS at 20 days, but the depassivation process was. After 20 days of conditioning, the anodic slope did not increase but instead decreased to 0.2–0.3 V/dec, which is significantly lower than that at 72 hours(Figure 3.10a). This suggests that chloride-induced depassivation is an ongoing process between 72 days and 20 days. The difference in the degree of depassivation at these two times (72 days and 20 days) is greater when the concentration of chloride ions is low. This also shows that it needs a longer time for the rebar's continuing depassivation to develop for a small chloride ion concentration.

To summarize, the anodic Tafel slope behaves differently in SPS and CSPS with chloride ions due to a competing effect of depassivation and rust build-up: In SPS, chloride slowly depassivate the rebar and finally reaches down to a similar value, but in CSPS, chloride quickly depassivate the rebar, and the anodic Tafel slope drifts to a higher value with the rust build-up.

3.5.5.3 Cathodic Tafel slope in SPS and CSPS

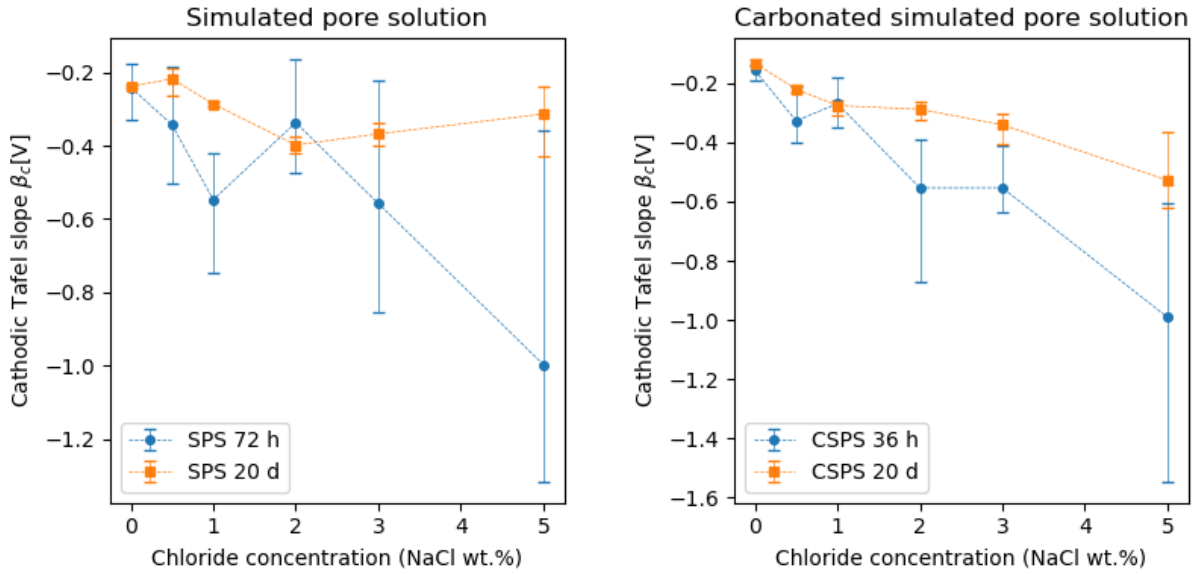
The cathodic Tafel slope is related to the oxygen reduction reaction. The cathodic Tafel slope value is affected by both the chloride ion concentration and the conditioning time. As shown in Figure 3.11, for the short-term conditioning time (72 hours for SPS and 36 hours for CSPS), the cathodic Tafel slope $|\beta_c|$ tends to increase with the chloride content from around 0.2 V/dec to 1.0 V/dec. The large slope at high chloride ion concentration indicated the concentration polarization, where the oxygen concentration was significantly reduced on the rebar surface due to the limited mass transfer rate of the dissolved oxygen.

For the rebar in the chloride-containing solutions for both SPS and CSPS, the cathodic Tafel slope $|\beta_c|$ decreased after a long-term condition of 20 days, when the disruption of passive oxide film stabilized over time. At 20 days, $|\beta_c|$ was 0.23 V/dec in SPS with no chloride ions, and increased to 0.40 V/dec with chloride concentration increasing up to 2% (wt%) NaCl and then levelled off with a slight decrease to 0.38 V at 3% and further to 0.33 V/dec at 5% with a larger variance. In CSPS with no chloride ions, $|\beta_c|$ is 0.13 V/dec, and it gradually increased to about 0.55 V/dec with chloride concentration increasing from 0.5% to 5% (wt%) NaCl.

The change of $|\beta_c|$ can be explained by the change of mill scale and protective film. In the short term, when the chloride ions are present, oxygen in the mill scale is consumed faster due to a higher corrosion rate, which caused the concentration polarization, hence a large $|\beta_c|$ in the chloride-rich solutions. With continuous depassivation and corrosion, the protective film and mill scale were destroyed, which facilitated the transport of dissolved oxygen to the steel surface, hence a decreased $|\beta_c|$. This oxygen-obstructing effect is more significant with a protective film formed at high pH. In neutral solution such as seawater, where no protective film is formed on the rebar surface, the large $|\beta_c|$ only decreased slightly from 1.215 V/dec to 0.9 V/dec for rebar with 1-hour and 3.9-year exposure (Chang et al., 2008b). In CSPS, The decrease of $|\beta_c|$ is less significant for

the rebar sample with high chloride concentration (5% wt.%) due to a significant build rust on the surface after the disruption of mill scale and protective film.

To summarize, the cathodic Tafel slope $|\beta_c|$ increases with the chloride concentration in short-term exposure in SPS and CSPS. This correlation becomes less dramatic and $|\beta_c|$ overall decreases with the gradual disruption of the mill scale and protective film after long-term exposure.



(a) (b)

Figure 3.11 The effect of the chloride concentration on the depassivation of rebar in simulated pore solutions: cathodic Tafel slopes of pre-passivated rebar in (a) SPS for 72 hours and 20 days (b) CSPS for 36 hours and 20 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

3.5.6 Corrosion parameters in mortar: the effect of chloride content and conditioning RH

3.5.6.1 Corrosion potential

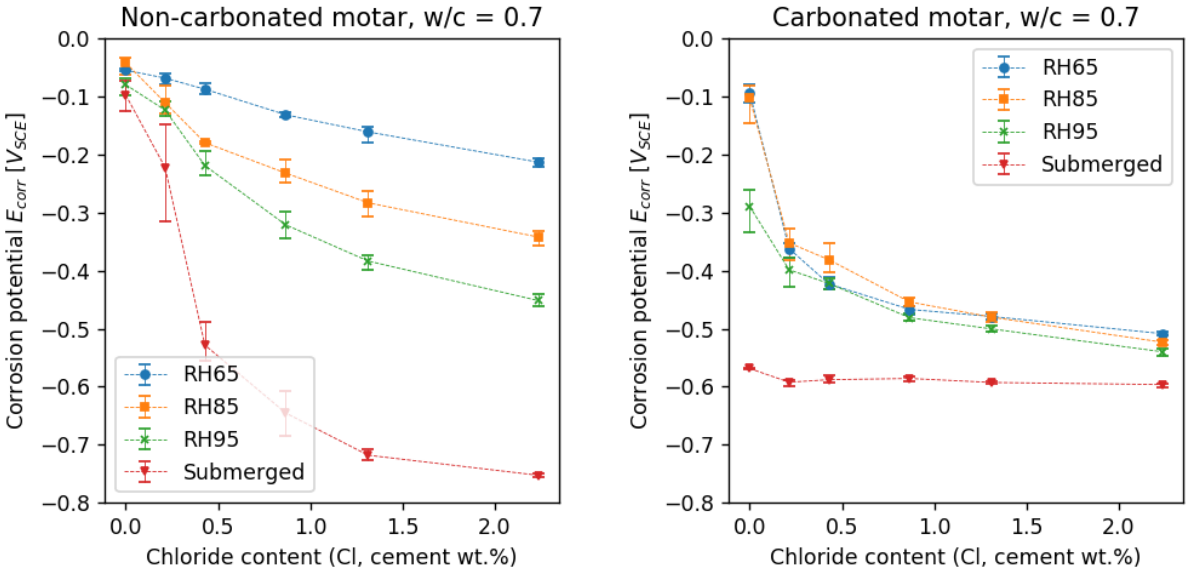
For both non-carbonated and carbonated mortars, the corrosion potential drops to more negative with the increase of chloride content, which indicates more active corrosion. The relative humidity conditioning levels significantly influence the effect of chloride content on the corrosion potential. This partially explains the large scatter of the critical chloride values reported in the literature (U. Angst et al., 2009).

As shown in Figure 3.12a, in the non-carbonated mortar samples, the corrosion potential decreases more rapidly with the increase of chloride content at a larger RH. For a given chloride content, the corrosion potential is a function of the conditioning RH of the mortar: the higher the RH, the less noble the rebar is, and the RH effects are more pronounced with higher chloride content. In the drier mortars (e.g. conditioned at 65% RH) with a chloride content of 2.24% (level 5) by weight of cement will only drop the E_{corr} to -200 mV, which can only be characterized as "uncertain" by ASTM criteria (ASTM International, 2015), whereas, at a relative humidity of 95%, the same chloride content brings corrosion potential to "corrosion probability greater than 90%" category, and furthermore, in the saturated condition, only chloride content level 2 (0.43% by weight of cement) will cause the E_{corr} to fall in "corrosion probability greater than 90%", which means the E_{corr} is greatly affected by the moisture content. This raises the precautions to the selection of the field half-cell survey timing to capture the worst-case scenario, where seasonal fluctuation of moisture content is suspected.

For carbonated mortar (Figure 3.12b), the corrosion potential E_{corr} changes with the chloride content in two stages: it drops more rapidly with a small amount of chloride added (0.21%–0.43% by weight of cement) than that in a non-carbonated mortar. When the chloride content is higher (0.43%–2.24% by weight of cement), the decrease of E_{corr} with chloride content in the carbonated mortar become less significant. That is because the carbonated mortar has a pH of around 8, and at this pH the passive film becomes unstable. The additional chloride has less contribution in "activating" the corrosion on the rebar since the rebar has already been largely activated.

The effect of chloride content on E_{corr} are also influenced by the conditioning RH for the mortar samples. Generally, the increase in the conditioning moisture reduced E_{corr}, but with a different pattern between carbonated and non-carbonated mortar samples. For the carbonated mortars (Figure 3.12b), the E_{corr} is similar for the samples with the same chloride content conditioned at RH of 65% and 85%. When the conditioning RH increased to 95%, E_{corr} slightly decreased for the chloride-containing carbonated samples, and significantly decreased in the carbonated mortars with no chloride contamination. Under the submerged condition, the E_{corr} is reduced to around -550 mV – -600 mV for carbonated samples with all chloride levels, indicating active corroding.

For non-carbonated mortar samples, as shown in Figure 3.12a, the E_{corr} declines with the increase in conditioning RH at all chloride levels. The greater the chloride content, the greater reduction in E_{corr} caused by the increase in RH. The most dramatic E_{corr} reduction was found in the mortar under the submerged condition when the chloride content increased from 0.21% to 0.43% (by weight of cement), appearing as a threshold chloride content to cause depassivation.



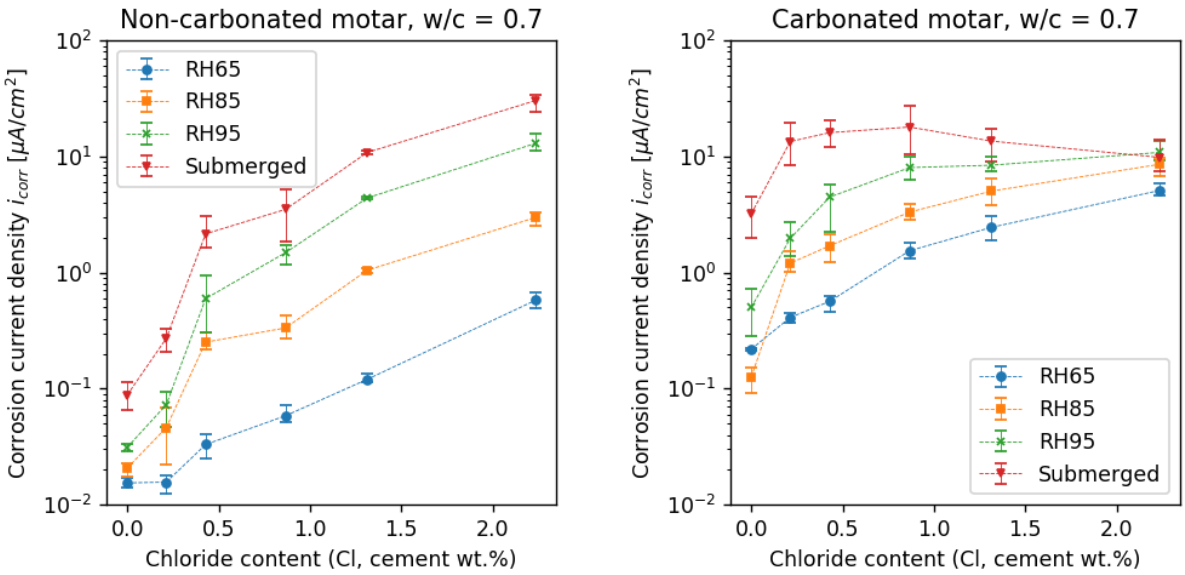
(a) (b)

Figure 3.12 The effect of the chloride content and conditioning RH on the corrosion potential of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

In summary, for non-carbonated mortar under partially saturated conditions, the effect of chloride does not show a "threshold" effect on depassivation, but under saturated conditions, it shows a "threshold" effect by a dramatic decrease in E_{corr} with chloride content. Whereas it is the opposite in carbonated mortars, the threshold level is evident under partially saturated conditions, but there was not much change in the E_{corr} under the saturated condition. It is worth mentioning that the carbonated mortar alone effectively depassivated the rebar in the saturated condition while the CSPS did not depassivate the rebar when there is no chloride.

3.5.6.2 Corrosion current density

The corrosion current density of the rebar embedded in mortar increased with the chloride content. However, the conditioning humidity affects the sensitivity of the current density's dependence on the chloride contents in different ways for the carbonated mortar samples and the non-carbonated samples. For the non-carbonated mortar samples, as shown in Figure 3.13a, the corrosion current density increases with the chloride content in all humidity condition levels, and the trends are similar to the inverse of the E_{corr} in Figure 3.12a.



(a) (b)

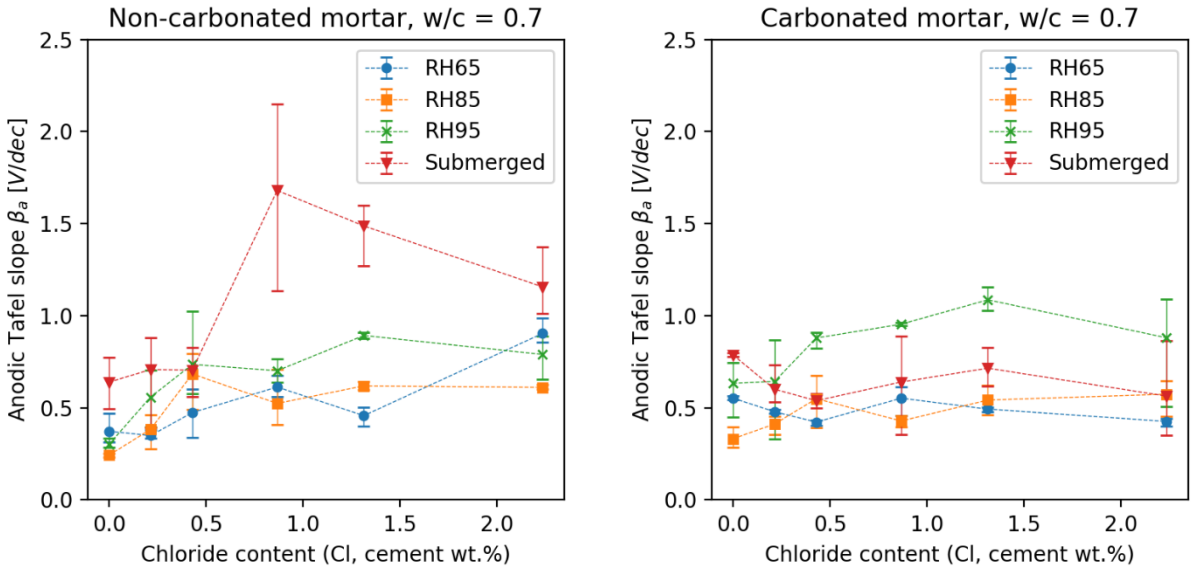
Figure 3.13 The effect of the chloride content and conditioning RH on the corrosion current density of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

For the carbonated mortar samples, as shown in Figure 3.13b, the "two-stage" trend found in the E_{corr} does not appear for i_{corr} under drier conditions (RH of 65–85%). The corrosion current density continues to increase for large chloride content (0.43%–2.24% by weight of cement), where the E_{corr} tends to level off. This phenomenon suggests that besides the depassivation effect by chloride, there are other factors that contribute to the corrosion current density and at the same time have less effect on E_{corr} . These factors are thought to be the moisture content and the resistivity. The moisture content and resistivity are concurring phenomena, but they may have

different mechanistic effects on the corrosion rate. These effects will be discussed in the next chapter (Chapter 4).

3.5.6.3 Anodic Tafel slopes

For the non-carbonated mortar samples (Figure 3.14a), under the partially-saturated conditions (RH of 65%–95%), the anodic Tafel slope increases with the chloride content until level 2 (0.43% by weight of cement), and after that (0.43% – 2.24% by weight of cement), the Tafel slopes fluctuates with no clear trend. For the submerged condition, the Tafel slopes significantly increase to a large value (1.0–2.2 V/dec) with a large variance at chloride level 3 (0.87% by weight of cement) where the E_{corr} drops to -650 mV, and then decreases with the increase of chloride content.



(a) (b)

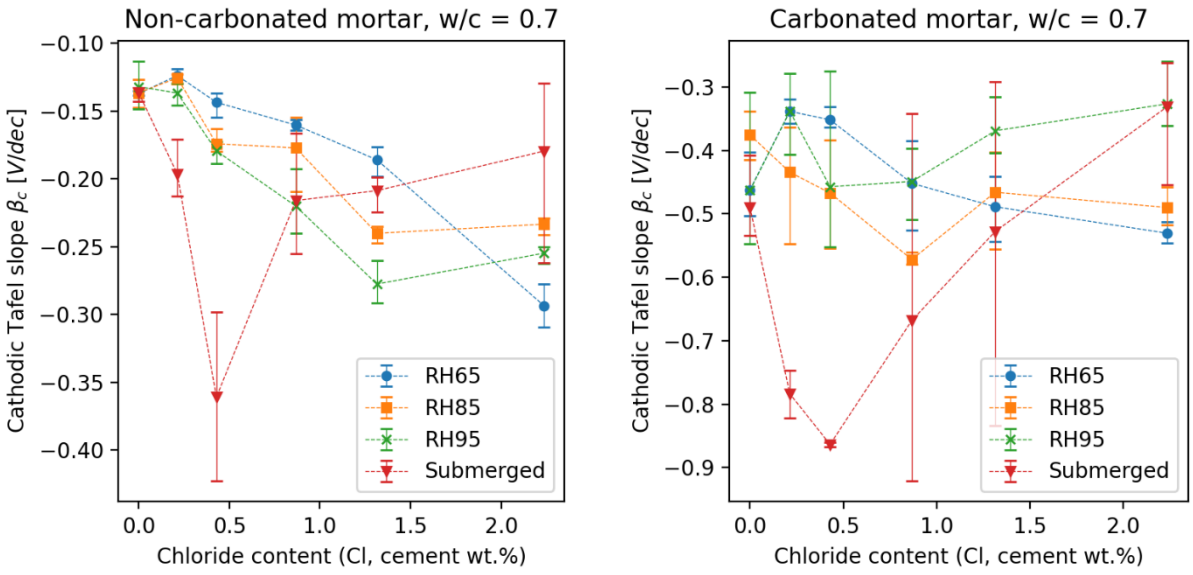
Figure 3.14 The effect of the chloride content and conditioning RH on the anodic Tafel slope of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

For the carbonated mortar samples (Figure 3.14b), the anodic Tafel slope does not appear a significant change with chloride content for the submerged condition. Samples conditioned at RH of 65% and 85% have similar Tafel slopes ranging from 0.3–0.7 V/dec. The samples stored at RH

of 95% have an overall higher anodic Tafel slope (0.8–1.0 V/dec) than the rest when the chloride content is larger than level 2 (0.43% by weight of cement). Under the submerged condition, the Tafel slope does not increase significantly when the chloride content is greater than level 3 (0.87% by weight of cement).

3.5.6.4 Cathodic Tafel slopes

For the non-carbonated mortar samples (Figure 3.15a) under partially saturated conditions, the absolute value of cathodic Tafel slope, $|\beta_c|$, increases with chloride content, but for RH of 85% and 95%, it tends to level off around 0.28 V/dec after chloride content reaches level 4 (1.13% by weight of cement). Under the saturated condition, the $|\beta_c|$ increases to its peak value (0.30-0.42 V/dec) at the chloride content level 2 (0.43% by weight of cement) and then decreases to around 0.22 V/dec for higher chloride content.



(a) (b)

Figure 3.15 The effect of the chloride content and conditioning RH on the cathodic Tafel slope of rebar in (a) non-carbonated mortar and (b) carbonated mortar, after two months of conditioning under RH of 65%, 85%, 95%, and submerged for 7 days. The mean values are indicated by markers, and the error bars represent the range of the three repeated test results.

For the carbonated mortar samples (Figure 3.15b) under partially saturated conditions, there is no clear trend of $|\beta_c|$ with respect to chloride content. $|\beta_c|$ ranges from 0.30–0.55 V/dec, higher than

the non-carbonated counterpart. Under the submerged conditions, it shows a similar pattern of the non-carbonated samples. It peaked at the chloride content level 2 (0.43% by weight of cement) and then declines with a large variation to around 0.32 V/dec.

3.6 Conclusions

An assessment of the impact of the major environmental parameters such as chloride concentration, carbonation state and exposure moisture and duration, on the measured corrosion parameters (corrosion potential, corrosion rate, and Tafel slopes) of rebars in a simulated concrete pore solution and in mortar has been carried out. An improved three-electrode potentiodynamic test procedure was developed to measure the corrosion parameters of rebar in simulated pore solutions and mortar with various chloride concentrations and pH. The impact of using a Faraday shield, real-time IR compensation, and a two-step scan routine was also evaluated. The findings of this study are as follows:

It was found that the use of a Faraday shield and floating ground significantly reduced the large electrical potential oscillations caused by real-time IR compensation. The air-dried rebar-embedded mortar samples were tested in a non-submerged setting to avoid the open circuit potential drifting due to extensive surface wetting of samples. It was also found that it was necessary to perform separate anodic and cathodic scans with adequate rest time in order to accurately determine the corrosion parameters.

In chloride-contaminated SPS and CSPS

1) The depassivation and rust build-up were simultaneous processes, and they both affected the corrosion parameters over the observed periods (36, 72 hours and 20 days). Relative to the short term depassivation (36 hours for CSPS, 72 hours for SPS), the long-term (20 days) depassivation slightly decreased the E_{corr} for both SPS and CSPS but only significantly increased i_{corr} for rebar in CSPS. There existed a chloride concentration (2% wt.% for SPS, and 1% wt.% for CSPS), above which higher chloride concentration had an insignificant long-term (20 days) effect on E_{corr} and i_{corr}. At 20 days the corrosion current densities in CSPS were approximately 3-10 times those of the SPS, even though the corrosion potentials were slightly more noble.

2) The competing effect of depassivation and rust build-up affects the anodic Tafel slope. The anodic Tafel slope in SPS decreased with the chloride concentration at 72 hours and reached a similar value (0.2–0.3 V/dec) for all chloride concentrations at 20 days. In CSPS, the anodic Tafel slope reached similar values for all chloride concentrations and drifted to a higher value (0.4–0.5 V/dec) at 20 days with the rust build-up.

3) The gradual disruption of the mill scale and protective film affects the cathodic reduction reaction. The cathodic Tafel slope $|\beta_c|$ increased with chloride concentration in both SPS and CSPS but with a large variance for short-term exposure. More stable and less scattered values were found at 20 days. The $|\beta_c|$ increased with the chloride concentration in CSPS but only increased with chloride concentration up to 2% (wt.%) from 0.23 V/dec to 0.38 V/dec in SPS followed by levelling off at higher chloride concentrations.

In the non-carbonated mortar and carbonated mortar contaminated with chlorides

1) For both non-carbonated and carbonated mortars, the corrosion potential E_{corr} decreases to more negative with the increase of chloride content. In non-carbonated mortars under partially saturated conditions with chloride content up to 2.24% by weight of cement, there was not a threshold chloride content as seen in SPS. However, in saturated conditions, it showed a "threshold" chloride content (0.21% – 0.43% by weight of cement), similar to the behaviour in SPS. For the carbonated mortar samples, the chloride content threshold showed under the partially saturated conditions, but the E_{corr} were similar in all chloride levels under the saturated condition and were more noble than the non-carbonated mortars just like the case in SPS and CSPS.

2) An increase in the conditioning RH results in a decrease in E_{corr}, but its extent depends on the carbonation state and chloride content. The most significant decrease in E_{corr} was found in the non-carbonated mortars with large chloride content and the carbonated mortars with no chloride.

3) Compared to the non-carbonated samples, the corrosion current densities (i_{corr}) were significantly higher in the carbonated ones in drier conditions with smaller chloride content. Increasing chloride content and RH minimized the difference.

4) In non-carbonated mortar, i_{corr} increased with higher chloride content across all humidity conditions, similar to the inverse trend of E_{corr}. However, in carbonated mortar under drier conditions (RH 65-85%), the two-stage level off trend observed in E_{corr} did not appear in i_{corr}. i_{corr} continued to rise with larger chloride content (0.43%-2.24% by weight of cement), while E_{corr} levelled off. This indicates that factors other than chloride depassivation, such as moisture content and resistivity, contribute to corrosion current density with a lesser impact on E_{corr}. Further discussion on these factors will be presented in the upcoming chapter (Chapter 4).

5) In both non-carbonated and carbonated mortar, the anodic Tafel slopes β_a did not show a clear dependence on chloride content and conditioning RH and had a range from 0.3-1.0 V/dec. The exceptional high β_a values were found in non-carbonated mortar under the submerged condition with higher chloride content.

6) Under partially saturated conditions, for non-carbonated mortar, the cathodic Tafel slope $|\beta_c|$ increased with the chloride content and levelled off at higher chloride content. The $|\beta_c|$ is significantly larger in carbonated mortars (0.30–0.55 V/dec) than that in non-carbonated mortars (0.12–0.30 V/dec), but there is no clear concluded trend in carbonated mortar.

7) Under saturated conditions in both types of mortars, $|\beta_c|$ had a peak value when chloride content reaches 0.43% by weight of cement and then decreased with higher chloride content.

3.7 Epilogue

This chapter investigates the effects of chloride content, carbonation, and moisture conditions on the corrosion parameters of rebar in simulated pore solutions and mortar. It sheds light on the underlying mechanisms of how these factors affect corrosion kinetics, which are essential

considerations for the numerical simulation phase. Additionally, it uncovers plausible synergistic effects of chloride, water saturation, and carbonation on the corrosion parameters of rebar in mortar, which will be further discussed in the next chapter.

4 Carbon Steel Rebar Corrosion in Mortar: the Interactive Effect of Moisture, Chloride, and Carbonation on Rebar Corrosion

4.1 Preface

In Chapter 3, we explored the influence of environmental parameters, including chloride content, carbonation, and relative humidity, on the corrosion rate and potential of rebars in mortar. The depassivation effects of chloride content and carbonation were found to be influenced by the relative humidity to which the samples were exposed. Building upon these findings, Chapter 4 investigates the combined effects of chloride content, carbonation, and moisture conditions on rebar corrosion parameters in mortar.

The porous nature of mortar and the variable water saturation levels are the distinctive features that may cause the different corrosion behaviour of rebar in mortar than in bulk simulated pore solutions. The roles of water saturation in mortar are the change of volume fraction of the pore solution, the change of bulk resistivity of mortar, and the change of the chemical concentration. All of them can affect the apparent corrosion behaviour of rebar in mortar. However, to develop a mechanistic corrosion model, the root cause should be unravelled and considered in the model.

This chapter will have a detailed analysis of the previous corrosion parameters measured in mortar by examining the water retention curves and bulk resistivity of these mortar samples. In this analysis, the effect of the water saturation level on corrosion was isolated from the combined effect brought by chloride contamination, carbonation and their effects on bulk resistivity and water retention capacity.

It was determined that water saturation level affects corrosion rates due to three causes: 1) an area effect by the influenced rebar-pore solution area, 2) an ohmic control effect by the influenced bulk resistivity, and 3) an anode-to-cathode ratio effect.

In non-carbonated mortar, when there is no chloride, the main contributor to rebar corrosion rate is the area effect. When there is chloride present, in addition to the area effect, there also exists an ohmic control effect and anode-to-cathode ratio effect. The chloride-induced rebar depassivation leads to non-uniform corrosion with a varying anode-to-cathode ratio dependent on water

saturation. This non-uniform corrosion cell is at a scale that is under ohmic controlled with a varying bulk resistivity which is also dependent on water saturation.

In carbonated mortar, the rebar corrosion behaviour is affected by both the anode-to-cathode ratio and the surface area. The rust scale on the rebar affects the anode-to-cathode ratio and is influenced by the moisture level. When the rust is dry, it is in a passivation state, and when it is wet, it is in an active state, which results in the dependence of the corrosion potential on the water saturation level in carbonated mortar. This dependence is more pronounced when the chloride content is low, as the conductivity of the rust is highly dependent on water content, and less significant when the chloride content is high. The non-uniform corrosion cell occurs on a scale that is not sensitive to bulk resistivity, hence less ohmic controlled.

The manuscript of this chapter was submitted to "Construction and Building Materials" as a companion paper to the manuscript of Chapter 3. Gang Li is the primary author who designed and performed the experimental study and data analysis. Dr. Moh Boulfiza and Dr. Richard Evitts contributed to the review of the manuscript and provided feedback.

4.2 Introduction

In Chapter 3, the impact of environmental parameters on the corrosion rate and potential of carbon steel rebar cast in mortar was examined. A modified three-electrode potentiodynamic test was performed on the pre-passivated carbon steel rebars cast in mortar samples with six levels of chloride content that each contained a different concentration of chloride ion. The mortar samples were also carbonated or non-carbonated at each chloride ion concentration. The samples were conditioned at three relative humidity (RH 60%, 80%, 95%) and under fully submerged conditions. The corrosion potential and corrosion current density were tested by a potentiodynamic scanning technique that separately measured the cathodic and anodic current-voltage relationship around open circuit potentials.

It was found that the depassivation effects of chloride content and carbonation on rebar corrosion in mortar were influenced by the relative humidity that the test samples were exposed to. An increase in the conditioning RH resulted in a decrease in corrosion potential E_{corr} and in increase in corrosion current density i_{corr}, but its extent depended on the carbonation state and chloride

content. For non-carbonated mortar samples, the decrease of E_{corr} and the increase of i_{corr} with the increase of RH was more pronounced in mortar with large chloride content. For carbonated mortar samples, on the contrary, the decrease of E_{corr} and the increase of i_{corr} with the increase of RH was more pronounced in mortar with no chloride.

4.2.1 Dependence of water saturation on RH in concrete

In this paper, instead of the conditioning RH, the water saturation and volumetric water content are used to investigate the effect of moisture on corrosion potential, E_{corr} and corrosion current density, i_{corr} under the varying chloride contents and carbonation states. The purpose is to steer away from correlating the relationship of E_{corr} and i_{corr} with RH, which does not represent the actual moisture level in mortar because the equilibrium moisture contents of the mortars under the conditioning environment is influenced by the ion concentrations in pore solution and porosity.

For mortar samples, the ion concentrations in pore solution increases when sodium chloride is added to the mortar mix. When carbonation occurs, the carbonation reaction significantly reduces the ionic concentration of OH^-, and the resultant hydrous silica gel from decalcification of C-S-H absorbs ions such as Na^+, K^+ (Pu et al., 2012). Accompanying the reduction in ion concentrations, carbonation also leads to possible porosity change, most likely a decreased porosity because the molar volume of the carbonation product, calcite is about 10% greater than that of calcium hydroxide, but it could also be an increased porosity with coarsened pore structure, especially in concrete made with pozzolanic supplementary cementitious materials where the calcium hydroxide content is relatively low and the excessive consumption of calcium hydroxide leads to significant decalcification of C-S-H (Borges et al., 2010; Ngala & Page, 1997; Shah et al., 2018).

There exist interactions between the retained moisture content, RH, ion concentration, pore size is discussed as follows. The relationship between the equilibrated relative humidity (RH) and the radius of the smallest empty pore given by the Kelvin equation (Kontogeorgis & Kiil, 2016).

$$\ln(RH) = \ln\left(\frac{p_v}{p_{sat}}\right) = \frac{2\gamma V_m}{rRT} \tag{4.1}$$

where $p_v[Pa]$ is vapour pressure, $p_{sat}[Pa]$ is saturated vapour pressure above a flat surface at temperature $T[K]$, $\gamma[N/m]$ is surface tension, $V_m[m^3/mol]$ is molar volume, $r[m]$ is radius of a

droplet (negative value for concave meniscus). As illustrated in Figure 4.1, r is related to the critical pore radius r_p i.e. the radius of the largest saturated capillary pores by

$$r_p = |r|/cos\alpha \tag{4.2}$$

where, the parameter α is the contact angle determined by the type of liquid and the solid surface.

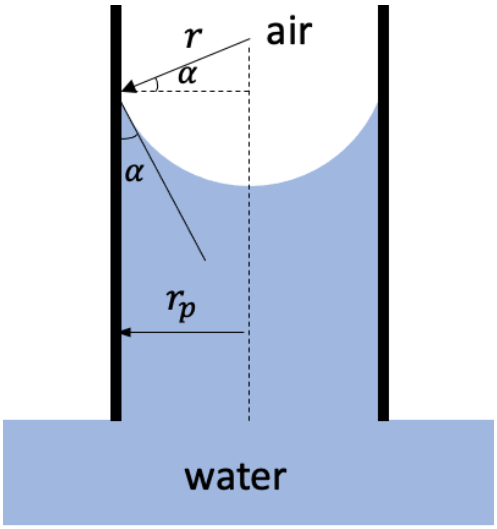


Figure 4.1 Schematic representation of the meniscus formation of water in a capillary pore.

For a given relative humidity (RH $= p_v/p_{sat}$), the capillary pores with a pore size smaller than r_p will be saturated. Therefore, concrete with a refined pore structure would retain more water at a certain RH. Therefore, the carbonation process of mortar will increase the water retention if it results in a reduced porosity and a refined pore structure.

The concrete pore solution is not pure water. It contains various salt solutes of different concentrations. According to Raoult's Law:

$$\Delta p = p_v x_B \tag{4.3}$$

where, $p_v[Pa]$ is vapour pressure of water; $\Delta p[Pa]$ is the decrease in the vapour pressure of water by the addition of solute B with a molar fraction x_B. A solution with a higher ion concentration

will have a lower saturated vapour pressure at which the equilibrium between vapour and liquid water is established on the solution surface (Schmitz, 2017). The pore solution has a lower equilibrium vapour pressure, and the measured RH with respect to the saturated vapour pressure above a flat surface p_{sat} is lower than 100% above the solution-air surface. In Equation (4.1), this is reflected as a lower p_{sat}, or an increased $\frac{p_v}{p_{sat}}$ in the pores, which leads to an increased r and an increased r_p in Equation (4.2). This means that for a given RH, additional larger pores will be saturated due to the existence of salt. Therefore, for the same pore structure, concrete will retain more solution with a lower RH as defined with respect to a flat surface of pure water.

In summary, for a given RH, an increase in chloride content will result in mortar retaining more water. Carbonation may reduce the salt concentration in mortar pore solution leading to reduced water retention, but it is possible that its refinement of the pore structure cause increased water retention.

4.2.2 Dependence of rebar corrosion on water saturation in concrete

In the context of rebar corrosion, the two main effects brought by water content variation in concrete is the change of the bulk resistivity and the rebar-pore solution interface area. In drier concrete, the reduced number of saturated pores causes the reduction in the rebar-pore solution interface, and the reduced interconnection between the saturated pores is the main reason for the dramatic increase in bulk resistivity, although the ion concentration in pore solution may be increased.

The simultaneously occurring change in rebar-pore solution interface area and the change in mortar bulk resistivity have distinct mechanistic effects on the corrosion rate. Firstly, area effect: corrosion mainly occurs on rebar surfaces exposed to water-saturated pores, and the corrosion current density is expressed over the entire surface area of the rebar at a macroscopic level. Therefore, for drier concrete, a lower water saturation level means less rebar surface area exposed to the pore solution, which leads to a lower corrosion current density over the entire rebar surface. Secondly, ohmic control effect: the increase in bulk resistance of concrete between anodic and cathodic sites on the corroding rebar, leads to a decrease in corrosion current through ohmic control. Ohmic control takes place whenever the ionic current of a corrosion cell flows through the porous,

saturated pore network. It can be reasoned that this type of ohmic control in porous medium diminishes when corrosion is nearly uniform, and the ionic current path of coupling anodic and cathodic reactions occur locally in the same pore without going through the network. However, non-uniform corrosion at a macroscopic scale often happens in chloride-contaminated, non-carbonated concrete with or without cracks (Angst et al., 2011; Raupach, 1996).

The area effect and ohmic control effect are concurring. For example, in a corrosion study of rebar embedded in mortar samples with and without 2% Cl (López & González, 1993), when the saturation level is below 0.75, the resistivity of mortar increased exponentially with decreasing water content which coincides with a linear decrease in corrosion rate. The authors correlated the corrosion current density change to resistivity citing ohmic control, but the reduced rebar-pore solution interface area should also contribute to the corrosion current density change.

In empirical corrosion models, corrosion is often related to the resistivity of concrete (Alonso et al., 1988; Liu & Weyers, 1998; Yu, Liu, & Chen, 2017). The area effect can be considered mathematically lumped into resistivity since rebar-pore solution area and resistivity are correlated. In numerical models that model electrochemical reactions, the area reduction is usually not considered distinctly (Cao et al., 2013; Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc et al., 1992; Yu et al., 2014). This effect is mathematically covered through the selected size and/or ratio of the anodic and cathodic sites and corrosion parameters. Only a few models have considered this effect by factoring the anodic/cathodic exchange current density by the liquid volume fraction (Dangla & Dridi, 2009).

Admittedly, these two controlling mechanisms are ubiquitous in the rebar corrosion context, but the significance of each varies. Generally, it is expected that ohmic control effect is less significant in carbonated reinforced concrete compared to the area effect. In a setup of a thin carbon steel wire (Ø0.5 mm) embedded in a thin carbonated mortar sample (6 mm), the area reduction mechanism was found to control the corrosion of carbon steel in a porous medium with near-neutral pore solutions (Stefanoni et al., 2019). However, even for sound carbonated concrete where the pore solution is considered more uniform, as-received carbon steel rebars usually come with non-uniform surface conditions (Stefanoni et al., 2015), and also the rebar-concrete interface exhibits non-uniform conditions (Angst et al., 2019). Nonuniformity would most likely create coupling of

the corrosion cells across different pores at a distance where, depending on the saturation level, the resistance change between anodic and cathodic sites might not be negligible, hence ohmic control. The unanswered question is what the extent of the influence of saturation's area effect and ohmic control effect is on controlling the corrosion rate in concrete with varying chloride content and carbonation states.

4.2.3 Objective

This research aims for investigating the role of moisture on rebar corrosion will be highlighted in terms of ohmic control and the rebar-pore solution surface area. The relationship will be explored between corrosion parameters such as corrosion potential E_{corr}, and corrosion current density i_{corr}, and microenvironmental factors such as water saturation, chloride content, and carbonation. The microenvironmental factors will be reduced to rebar-pore solution area and bulk resistivity which govern the area effect and ohmic control effect. This reduction can be achieved by analyzing the dependence of equalized water saturation level (area effect-linked) and bulk resistivity (ohmic control effect-linked) on the exposure RH, and pore solution ionic species concentration, and porosity affected by chloride ion addition and carbonation. The exhaustive treatments with combinations of RH, chloride content, and carbonation allows emphasizing one of the concurring rebar-pore solution area effect and bulk resistivity effect on corrosion, therefore making the isolation of these two effects possible.

4.3 Resistivity test

In Chapter 3, the corrosion parameters such as corrosion potentials and corrosion current densities were measured from pre-passivated, as-received rebars embedded in mortar. In this chapter, the corrosion parameters will be analyzed along with the resistivity test results. An equal number of plain cylinder mortar samples were prepared at the same time from the same batch of raw materials and conditioned alongside the mortars with embedded rebar. The plain mortar samples also had six levels of chloride content that each contained a different concentration of chloride ion (as listed in Table 4.1). The mortar samples were also carbonated or non-carbonated at each chloride ion concentration. The samples were conditioned at three relative humidity (RH 60%, 80%, 95%) and under fully submerged conditions to achieve varying volumetric water content in the rebar-

embedded mortars. The detailed sample preparation and improved corrosion measurement methods used for the rebar-embedded mortar samples are described in Chapter 3.

Table 4.1 Chloride content levels in mixing water and in mortar samples

Chloride level	0	1	2	3	4	5
NaCl wt.% mixing solution	0%	0.5%	1.0%	2.0%	3.0%	5.0%
Cl wt.% by weight of cement	0%	0.21%	0.43%	0.87%	1.31%	2.24%

The water saturation level was determined by measuring the mass of a sample under a given condition, and then comparing the mass of the sample when it was fully saturated and when it was oven dried. The saturation level of the sample is expressed as

$$S_e = \frac{m_{sample} - m_{od}}{m_{ssd} - m_{od}} \tag{4.4}$$

where, $m_{sample}[kg]$ is the mass of sample under a given condition, $m_{ssd}[kg]$ is the mass of the sample under saturated-surface-dry condition, $m_{od}[kg]$ is the mass of the sample after being oven dried at 110 ± 5°C to a constant mass. The volumetric water content, $\theta[-]$ can be computed as

$$\theta = S_e \varepsilon_p \tag{4.5}$$

where $\varepsilon_p[-]$ is porosity which is equal to the volumetric water content under fully saturated condition, θ_s

$$\varepsilon_p = \theta_s = \frac{(m_{ssd} - m_{od})}{\rho_l V_{sample}} \tag{4.6}$$

where $\rho_l[kg/m^3]$ is density of liquid water, $V_{sample}[m^3]$ is the volume of a mortar sample.

The bulk resistivity was tested using a direct current (DC) method to eliminate the potential capacitive effect of the mortar matrix that can occur when using alternating current. Two applied DC voltages (5 V and 10 V) were selected to offset the electrode polarization effect (Monfore,

1968). The test setup is illustrated in Figure 4.2. The mortar cylinder sample was sandwiched between moist sponge pads, carbon felt, and copper plates. The selected voltages were applied across the copper plates. The applied voltage was regulated by the DC power supply and monitored by a voltmeter. The moist sponge pads were used to form low ionic resistance contact between the ends of the cylindrical mortar sample and the carbon felt electrodes, which was connect to the power circuit through the copper plates. Carbon felt electrodes are inert electrodes and served as a transition from moist sponges to dry copper plates. They were used to eliminate the electrode changes such as corrosion if the copper plates were used directly in contact with electrolyte in moist sponge pads.

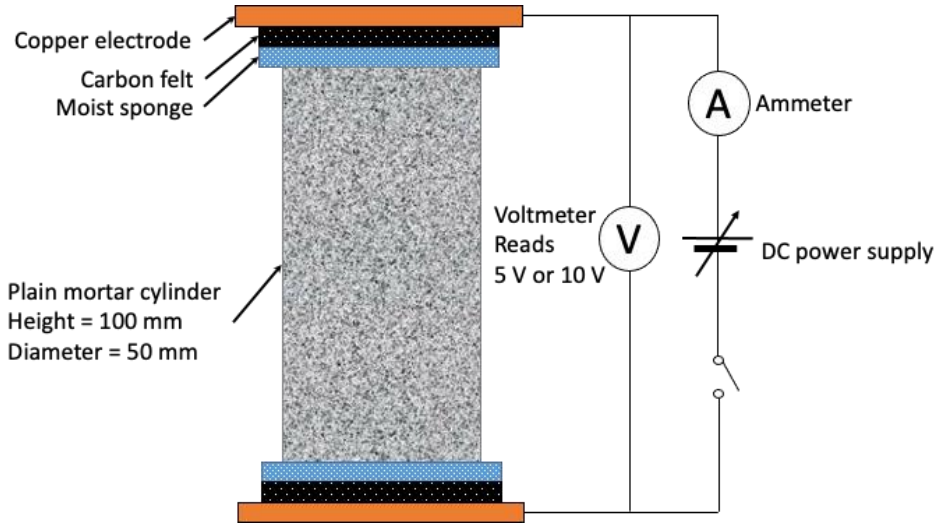


Figure 4.2 Resistivity test of mortar cylinder samples using the direct-current method.

The resistivity, ρ_r can be calculated as

$$\rho_r = \frac{R_L A}{L} = \frac{V_2 - V_1}{I_2 - I_1} \cdot \frac{\pi d^2}{4L} \tag{4.7}$$

where R_L is resistance across the two ends of the mortar cylinder, A is area of the cylinder top/base surface, V_1 and V_2 are applied voltages ($V_1 = 10$ V, $V_2 = 5$ V), I_1 and I_2 are currents in the circuit, d is diameter of cylinder and L is height of the cylinder.

4.4 Results and discussion

4.4.1 The effect of chloride content and carbonation on water retention

As summarized in introduction, for a given RH, higher salt concentration in mortar pore solution and finer pore structure could increase water retention. As shown in Figure 4.3, the carbonation process has decreased the porosity of the mortar significantly from approximately 0.30 to 0.22. A slight increase in porosity with the increase in chloride content was observed in the chloride-contaminated samples.

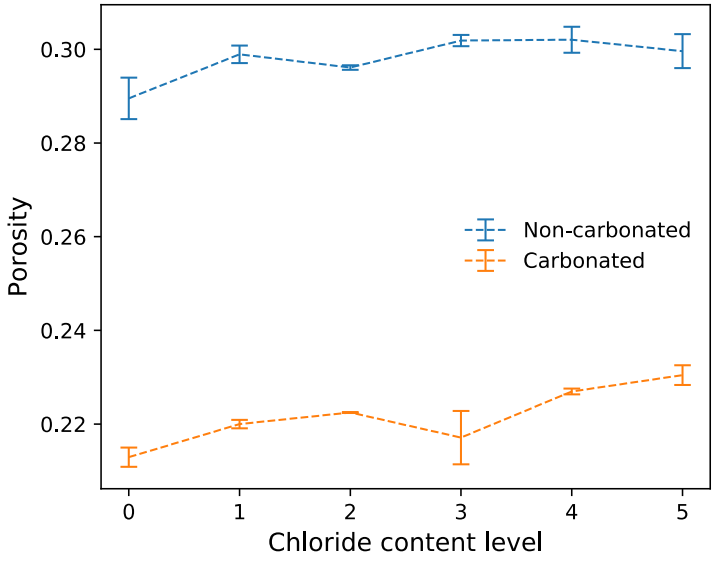


Figure 4.3 Porosity of non-carbonated and carbonated mortar samples (w/c=0.7) with varying chloride levels 0-5 as listed in Table 4.1. Chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement. The mean value point is shown, and the error bars represent the range of the three repeated test results.

The effect of chloride and reduced porosity on the equalized moisture level of the mortar samples conditioned at different RH is shown in Figure 4.4. Both non-carbonated and carbonated samples exhibit an increase in water saturation level with an increase in RH. When at the same RH, samples with higher chloride content have a higher saturation level. The effects of added chloride salt and porosity change are discussed in the following subsections.

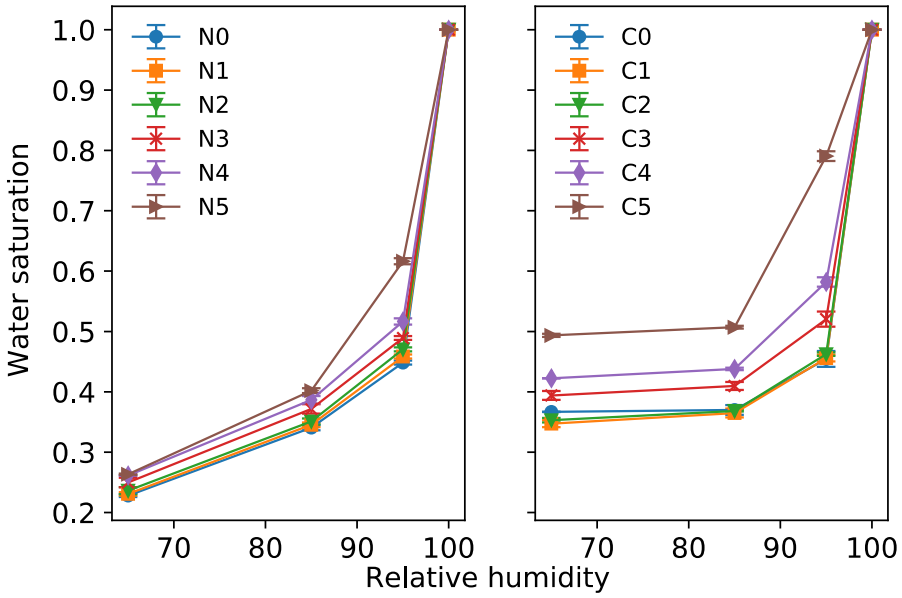


Figure 4.4 Water saturation level of carbonated and non-carbonated mortar conditioned at various RH 65% 85% 95% and 100% (submerged). Legend: N (Non-carbonated), C (Carbonated), 0-5 (chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results.

4.4.1.1 The effect of salts on water retention

The effect of chloride content on water saturation level is more pronounced in carbonated samples (as shown in Figure 4.4) compared to non-carbonated samples. In carbonated samples, the ion concentration is more sensitive to the addition of sodium chloride because the carbonation reaction transforms calcium hydroxide into calcium carbonate, which decreases the original ion concentration of the pore solution and simultaneously releases the bonded chloride ions (Mays, 2004).

4.4.1.2 The effect of capillary pore size on water retention

According to the Kelvin-Laplace Equation (4.1), the saturated vapour pressure is lower in narrower pores. As a result, the carbonated mortar samples with a refined pore structure would retain more moisture at a given RH. With the porosity reduction due to carbonation, in finer pores, according to the Kelvin-Laplace equation, the condensation of water vapour will occur at a lower RH, which explains the slightly higher saturation level at 85% RH and 95% RH for carbonated samples

compared to non-carbonated samples. This carbonation-induced saturation increase is more significant especially at higher chloride content levels, where the reduced saturated vapour pressure reduced the critical pore radius, r_p in Equation (4.2), so that more pores can be saturated at a given RH. At 65% RH, the carbonated mortar has a significantly higher saturation level compared to non-carbonated mortar samples. Besides the effect of the refined pore structure, the carbonated samples went through a different internal wetting history due to the water released during the carbonation process at 65% RH. Therefore, its re-wetting curve influenced by hysteresis (H. M. Jennings et al., 2015) would additionally contribute to a higher water content at 65% RH, and this influence is reduced at 85% and 95% RH.

4.4.2 The effect of saturation, carbonation and chloride on Resistivity

Although the water retained at various RH levels are not the same for mortar with different chloride content and carbonation states, we can observe the dependence of resistivity on the water saturation level by plotting the resistivity against saturation.

Resistivity as a function of saturation is affected by chloride and carbonation. As shown in Figure 4.5, the resistivity of carbonated mortar is very sensitive to chloride content, while the non-carbonated mortar is almost only affected by the saturation level. This observation confirms the reasoning that the pore solution of non-carbonated mortar contains abundent dissolved ions (Ca^{2+}, OH^-, Na^+ K^+,Cl^-), but in carbonated mortars, Cl^- becomes the major charge-bearing ion. This phenomenon made it possible to isolate the two effects brought by saturation level, namely, ohmic control effect and rebar-solution area effect. The isolated ohmic-control effect is expected to be observed from the chloride-contaminated carbonated mortars This is because for a given saturation level, the rebar-solution area becomes a controlled variable, and various resistivity is obtained with different chloride contents in mortar samples at the given saturation level.

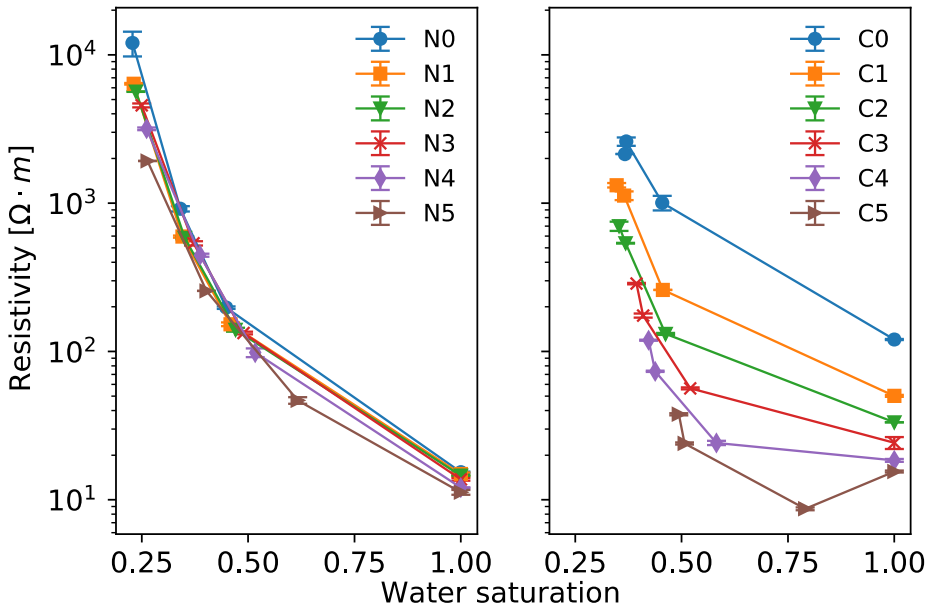


Figure 4.5 Resistivity of non-carbonated and carbonated concrete conditioned at various saturation levels. Legend: N(Non-carbonated), C(Carbonated), 0-5 (chloride levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results.

4.4.3 The effect of chloride and carbonation: comparing rebar corrosion between mortar and simulated pore solutions

As discussed in Chapter 3, the rebar corrosion potentials in carbonated mortar showed similar trends and values to the carbonated simulated pore solution (CSPS). However, unlike the case in CSPS, the i_{corr} in drier conditions (RH 65–85%) does not reach a plateau with the increase of chloride content. This may be because in CSPS, chloride is the main inducer to corrosion and this effect levels off when it reaches a specific concentration, so is the case in saturated carbonated mortar. However, in the partially saturated mortar, the water saturation level also significantly affects the corrosion rate, even when the rebar is at a similar corrosion tendency (indicated by similar corrosion potential as in CSPS). This i_{corr}'s dependence on chloride is probably affected by the increased saturation due to presence of chloride ions (as discussed in section 4.4.1), which results in increased exposed rebar surface to solution-filled pores (area effect) and the reduced resistance (as discussed in section 4.4.2) between the anodic and cathodic sites (ohmic control effect). The discussion of the contribution from the two mechanisms will be elaborated in Section 0.

The companion study (Chapter 3) has shown the significant effect of saturation on both E_{corr} and i_{corr}. Now we use simulated pore solution (SPS) as a reference to investigate the additional effect of chloride in mortar samples. When comparing the results between the non-carbonated mortar and the simulated pore solution (SPS), E_{corr} in the partially saturated non-carbonated mortars (RH 65-85%) gradually decreased with increasing chloride content. In both SPS and the saturated non-carbonated mortar, a threshold chloride content (critical chloride content) was observed, with a significant decrease in E_{corr} and a drastic increase in i_{corr}. However, a similar pattern of E_{corr} drop with chloride was not translated to a similar i_{corr} increases with chloride: i_{corr} reaches a plateau in SPS but increases with chloride continuously in both the saturated and the partially saturated mortars. To sum up, with the increase in chloride content,

a) The trends in E_{corr} between SPS and saturated mortar are similar, but the trends in i_{corr} are different,
b) The trends in i_{corr} between partially saturated mortars and saturated mortars are similar but the trends in E_{corr} are different,
c) The trends in E_{corr} and i_{corr} between SPS and partially saturated mortars are both different.

The role of chloride in the carbonated mortar samples is also present in the non-carbonated samples. In the non-carbonated mortar samples, the increase in chloride content also induces saturation increase (as discussed in section 4.4.1). To a lesser extent, that increased saturation level will reduce the mortar's resistivity, although the chloride ion itself does not significantly reduce the resistivity of the pore solution in non-carbonated mortar (as discussed in section 4.4.2) at a given saturation level. These effects of chloride on saturation level (referred as "indirect" effect) for a given conditioning relative humidity at least partially caused the gradual increase of i_{corr} with the increase in chloride content in partially saturated mortars, especially in the drier ones.

The gradual drop of E_{corr} with chloride is probably caused by the depassivating effect of chloride on rebars, referred as "direct" effect. Firstly, in mortar samples, there could be reduced chloride concentration in pores. Although the chloride concentrations are the same between the SPS and the mixing solution of the mortar, the free chloride concentration in the pores is reduced due to the bonding effect of chloride during cement hydration process. Secondly, when the non-carbonated mortar samples have lower water content (conditioned at 65% or 85% RH), the chloride salts in larger empty capillary pores may crystalize, and the local chloride concentration in the pore

solution has not reached to the "threshold" level given a pH of 12.5 – 13.5, hence contribute little to the active corrosion of rebar. Thirdly, with the variable depassivated rebar area, the ohmic-control mechanism also affects the E_{corr}. This depassivation mechanism complements the partial contribution to the i_{corr} change. The chloride's "indirect" and "direct" mechanisms overshadow each other, especially in terms of i_{corr}. The effects of these two mechanisms will be discussed in the following section.

4.4.4 The effect of water saturation on rebar corrosion in mortar (ohmic control effect vs area effect)

The previous section discussed how the effects of chloride and carbonation (pH) on rebar corrosion in mortar are influenced by water saturation level, which distinguishes it from the case in the simulated pore solutions and adds complexity to the corrosion in concrete. Figure 4.6 illustrates the path of influence of chloride ions and carbonation on the pore structure, pore solution, water retention in concrete, and rebar corrosion in concrete.

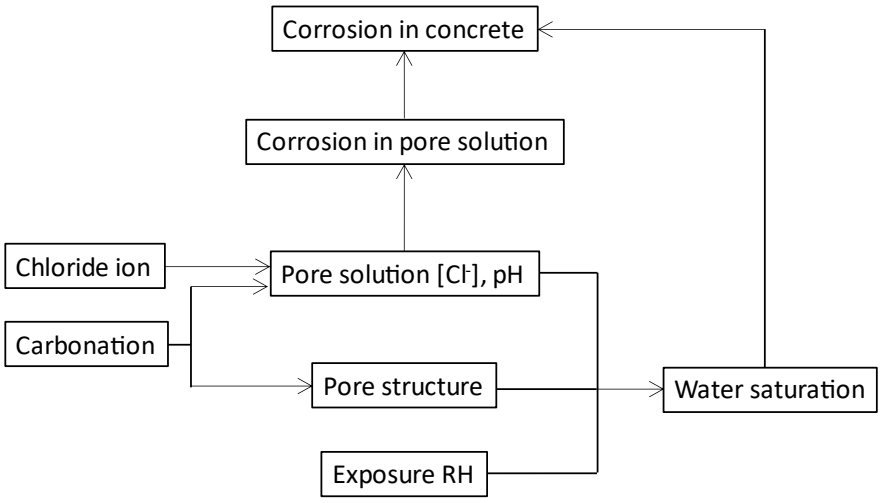


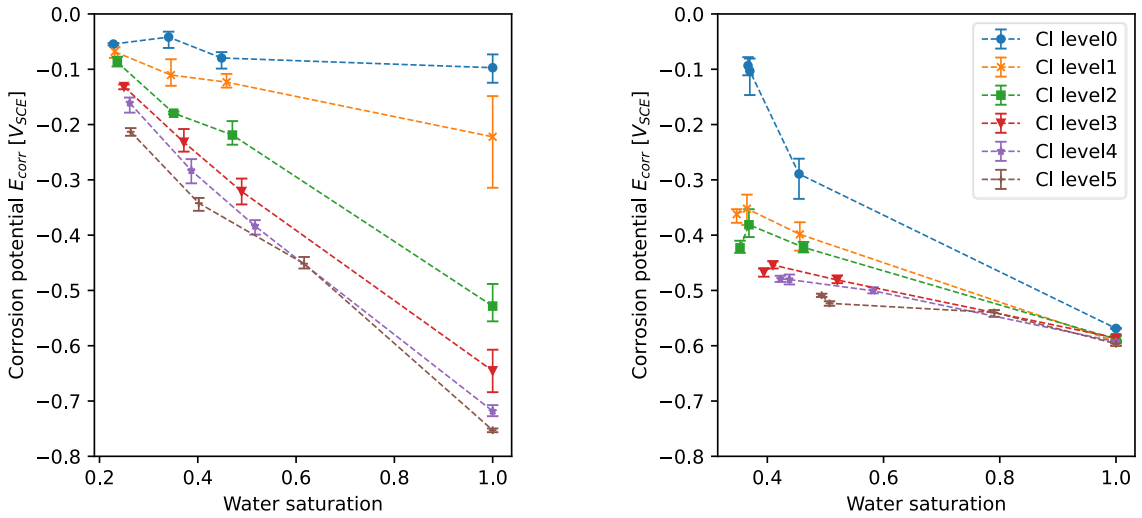
Figure 4.6 The path diagram of quaulitative influence of chloride ion and carbonation on rebar corrosion in pore solution and concrete.

The effects of saturation on the corrosion potential, E_{corr} and corrosion current density, i_{corr} are explored by plotting them against the saturation levels for various treatments of chloride and carbonation.

4.4.4.1 The relationship between corrosion potential and water saturation

4.4.4.1.1 Non-carbonated mortar

In the non-carbonated mortars (Figure 4.7a), when no chloride ions were present, the E_{corr} did not change significantly with the saturation level, indicating that the rebar remained in a passivated state. In the mortar with chloride ions, E_{corr} decreased with the increase in saturation level, which indicates a gradually increased extent of depassivation with more rebar surface exposed to depassivating pore solution (high localized chloride ion concentration).



(a) (b)

Figure 4.7 Corrosion potential vs. saturation of mortar samples with chloride contents (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results.

As shown in Figure 4.7a, the mortar samples with a lower chloride content (level 1) showed a slight linear decrease in E_{corr} from -80 mV to -150 – -300 mV. With higher chloride contents (level 2–5), E_{corr} decreased linearly with a similar larger derivative with the increase in saturation level, shown as parallel lines. This parallel trend represented a more negative corrosion potential for higher chloride contents at all observed saturation levels. The chloride content that differentiates the two observed patterns lies between level 1 (0.21 wt% cement) and level 2 (0.43 wt% cement), which is consistent with the critical chloride content range of 0.2% to 2.0% wt% cement (Schiessl et al., 2005). To conclude, there was a synergistic effect between saturation and

chloride content on E_{corr}. Higher water saturation facilitated chloride-induced rebar depassivation, leading to a larger depassivated area ratio of rebar and a more negative E_{corr}. The negative correlation between the area ratio of depassivated rebar and E_{corr} measured from concrete cover was simulated in a circuit model (Li et al., 2023).

4.4.4.1.2 Carbonated mortar

The effect of saturation on corrosion for chloride-contaminated carbonated mortar is very different from that for chloride-contaminated non-carbonated mortar samples. As shown in Figure 4.7b, in carbonated mortar samples, the corrosion potential dropped rapidly with the increase of the water saturation level when there was no chloride present. For samples with chloride (levels 1-5), the corrosion potential dropped steadily with the increase in saturation level but the steepness decreased with the increase in chloride content. The E_{corr} converged to a similar value in fully saturated conditions for all carbonated mortar samples regardless of the chloride content level. Compared to the synergistic effect of chloride content and saturation level in non-carbonated mortar samples, in carbonated samples, the saturation had a decreased effect on decreasing the E_{corr} when chloride ions were present and became less significant at higher chloride content. This observation suggested that in carbonated mortar, the depassivated rebar surface area ratio does not significantly increase with saturation when chloride content is high.

To summarize, the water saturation level has an effect on the E_{corr} in chloride-contaminated mortar. In non-carbonated mortars, the corrosion potential decreases with the saturation level when the chloride content exceeds a critical value, but is less affected by the saturation level if the chloride content is less than the critical value. In contrast, in carbonated mortars, the effect of the saturation level on the E_{corr} is most significant when there is no chloride present. As the chloride content increases, the effect of saturation becomes gradually less significant.

4.4.4.2 Explaining E_{corr}'s dependence on saturation

The influence of moisture content on corrosion potential (half-cell potential) has been noticed for a long time (Andrade et al., 2003). Various explanations for this phenomenon are critically discussed here.

4.4.4.2.1 Preferential water adsorption

As the water saturation increases, more rebar surface becomes exposed to solution-filled pores. If the active-to-passive rebar surface area ratio does not change, E_{corr} would remain the same while the corrosion current density, i_{corr} averaged over the whole rebar surface would be scaled up by that increased rebar-pore solution surface area. However, a decrease in E_{corr} following a saturation increase indicates an increase in the active-to-passive rebar surface area ratio, which means the increased rebar-pore solution interface is depassivated to a greater extent with a greater active-to-passive rebar surface area ratio. The dependence of E_{corr} on active or passive rebar ratio was demonstrated with a circuit model (Li et al., 2023).

The reasoned dependence of the active-to-passive rebar area ratio on mortar saturation can be explained by preferential water adsorption in concrete i.e. the increased saturation water is preferably adsorbed or condensed in the pores with higher local chloride content because pore solution with a higher ionic concentration has a lower saturated vapour pressure according to Raoult's Law, Equation (4.3). This effect on the equilibrated water content in mortar has been shown and discussed in Section 4.4.1. This chloride-induced preferential water adsorption synergistic effect on E_{corr} is more significant in the non-carbonated mortar, as chloride is a competing factor to the high pH, whereas, in the carbonated mortar with lower pH, the chloride concentration in water-saturated pores readily surpasses the critical content and the additional depassivated rebar surface ratio does not add up much to the existing active rebar surface ratio.

There is one case that can not be explained by the preferential water adsorption mechanism. In the carbonated mortar samples with no chloride ions, the E_{corr} was about -0.1 V_{SCE}, much more noble than the ones with chloride. This indicated the depassivation was barely happening at its starting saturation level (conditioned at 65% RH), more significant depassivation occurred with the increased saturation and finally achieved the similar E_{corr} with other chloride-contaminated samples at the full saturated condition. It seems that water saturation facilitates the depassivation in the carbonated mortar with no chlorides. This can be explained by the rust scale effect, elaborated in the next section.

4.4.4.2.2 Rust scale effect

The iron corrosion potential depends on the moisture state of the iron oxides. For atmospheric corrosion of iron, oxygen reduction predominantly occurs on the oxides and the Fe^{2+}/Fe^{3+} solid-state conversion of the oxide scale during the wetting and drying process affects the oxygen reduction rate (Stratmann, 1990). At the beginning of drying, with good oxygen availability, oxygen is reduced on the rust cathode with a high surface area, which results in fast corrosion, and the dense rust precipitation passivates the steel. Without a fresh source of Fe^{2+}, the rust is all oxidized to Fe^{3+} which leads to poor electrical conductivity and a compromised rust cathode with a reduced oxygen reduction rate. The passivation state of iron with the dried oxidized rust scale (Fe^{3+} state) leads to more noble E_{corr}. In the wetting process, the Fe^{3+} is reduced to Fe^{2+} state with an improved conductivity, which again allows the fast reduction of oxygen on the rust cathode doped with Fe^{2+} states. Besides the effect of the rust scale state on oxygen reduction rate, this Fe^{2+}/Fe^{3+} redox reaction was also regarded as a secondary cathodic reaction to interpret the E_{corr} being more noble than $-100\ mV_{SCE}$ in chloride-contaminated concrete (Alonso et al., 1998b).

It is likely that both the rust scale effect and the preferential water adsorption effect contribute to the dependence of E_{corr} on saturation. The passivated state of rebar in drier mortar would have a limited ionic path coupling with other anodic corroding surfaces due to the high bulk resistivity or high localized resistance. Therefore, the oxide scale effect explains corrosion in the individual pores or locally connected pores where the electrical resistance through the bulk mortar pore network can be neglected, whereas the preferential water adsorption effect explains the increase in rebar corroding surface of high active-to-passive ratio with ionic path coupling between non-uniform rebar-mortar interfaces.

4.4.4.2.3 Other factors to rule out

- **Higher oxygen concentration**

The small E_{corr} increase (100mV) with the drying of non-carbonated mortar was interpreted as forming differential aeration cells by (López & González, 1993). With higher oxygen concentration in some pores, according to the Nernst Equation, there should be a slight increase in the reversible potential of the oxygen reduction reaction, E_{O_2/OH^-}.

$$E_{O_2/OH^-} = E^o_{O_2/OH^-} - \frac{RT}{nF}\ln\frac{a^4_{OH^-}}{a_{O_2}} \quad (4.8)$$

where, $E^o_{O_2/OH^-}$ is reversible potential under standard condition, n is number electrons being transferred, T is temperature, F is Faraday constant, a_{OH^-} is activity of hydroxide ion, a_{O_2} is activity of oxygen gas.

For a given pH and temperature of 25 ℃, when the effective partial pressure of O_2 increase from 0.1 to 1, E_{O_2/OH^-} will increase by $\Delta E_{O_2/OH^-}$ =14.8 mV, the coresponding change in E_{corr} can be calculated by

$$\Delta E_{corr} = \frac{\Delta E_{O_2/OH^-}}{1 + \frac{|\beta_c|}{\beta_a}} \quad (4.9)$$

where, β_a is anodic Tafel parameters of the coupled iron dissolution reaction, $|\beta_c|$ is cathodic Tafel parameters of oxygen reduction reaction. The denominator $1 + \frac{|\beta_c|}{\beta_a} > 1$, $\Delta E_{corr} < \Delta E_{O_2/OH^-}$=14.8 mV. It can be concluded that he E_{corr} change due to the change in oxgyen concentration is insignificant in partially saturatred mortar samples.

- **Resistance in the measuring circuit**

The measured corrosion potential, is affected by the resistance of the voltmeter and the concrete cover. It can be expressed by

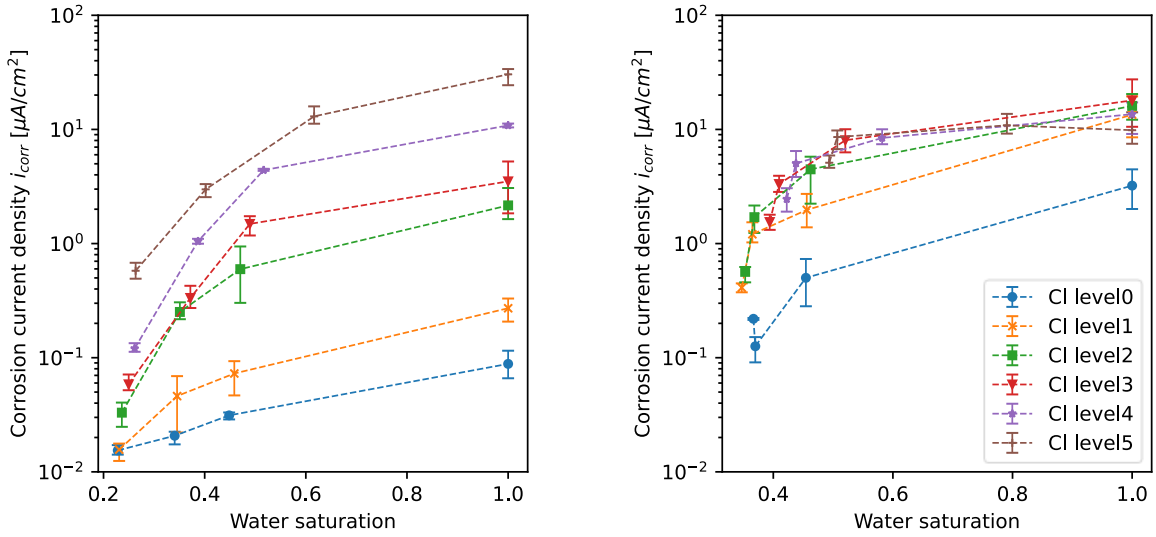
$$E_{corr,measured} = E_{corr} \cdot \frac{R_{voltmeter}}{R_{voltmeter} + R_{concrete}} \quad (4.10)$$

where $E_{corr,measured}$ is the corrosion potential reported by a voltmeter, E_{corr} is rebar corrosion potential, $R_{voltmeter}$ is internal resistance of the voltmeter, $R_{concrete}$ is the resistance of concrete in the ionic path of the measurement circuit. $E_{corr,measured}$ will be non-negligibly smaller than E_{corr} when the resistance in the ionlic path through the cover is not much less than the internal resistance of the voltage meter (Gu & Beaudoin, 1998). However, in our study, the $E_{corr,measured}$

is greater in drier mortar with larger resistivivty, which contradicts to the expected outcome from the influence of the resistance in the measuring circuit and therefore can not be explained by this theory. Nevertheless, this phenomenon has been avoided by using a moist sponge on the concrete cover and a large internal resistance of the voltmeter, e.g. the known resistance between the work sensor and reference of Gamry Reference 600 is 100 TΩ (Gamry Instruments, 2012).

4.4.4.3 The relationship between corrosion current density and water saturation

The saturation has been long thought to be the major influential factor contributing to rebar corrosion in concrete with chloride or carbonation. As shown in Figure 4.8a, the i_{corr} increased with the saturation level for all chloride and carbonation cases. The i_{corr} spread across two orders of magnitude for the observed saturation levels. It appeared that the chloride content and carbonated state determined a baseline corrosion rate, and that rate alters with the saturation level.



(a) (b)

Figure 4.8 icorr vs. saturation for mortar samples with various chloride content (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The mean value point is shown, and the error bars represent the range of the three repeated test results.

4.4.4.4 Effect of saturation: Normalized corrosion current density's dependence on saturation

In order to extract the effect of water saturation on i_{corr} at different baselines controlled by chloride content and carbonation state, the normalized corrosion current density, $i_{corr,norm}$, is introduced in Equation (4.11).

$$i_{corr,norm} = \frac{i_{corr}}{i_{corr,sat}} \tag{4.11}$$

It is the ratio of the corrosion current density, i_{corr} to the corrosion current density when the mortar is fully saturated, $i_{corr,sat}$. As illustrated in Figure 4.9a, the $i_{corr,norm}$ was linearly proportional to the water-saturated level for most cases including various chloride levels and carbonation states. The exceptional cases are the carbonated mortar samples with the highest chloride content (level 5), where the linear regression line is significantly different from others.

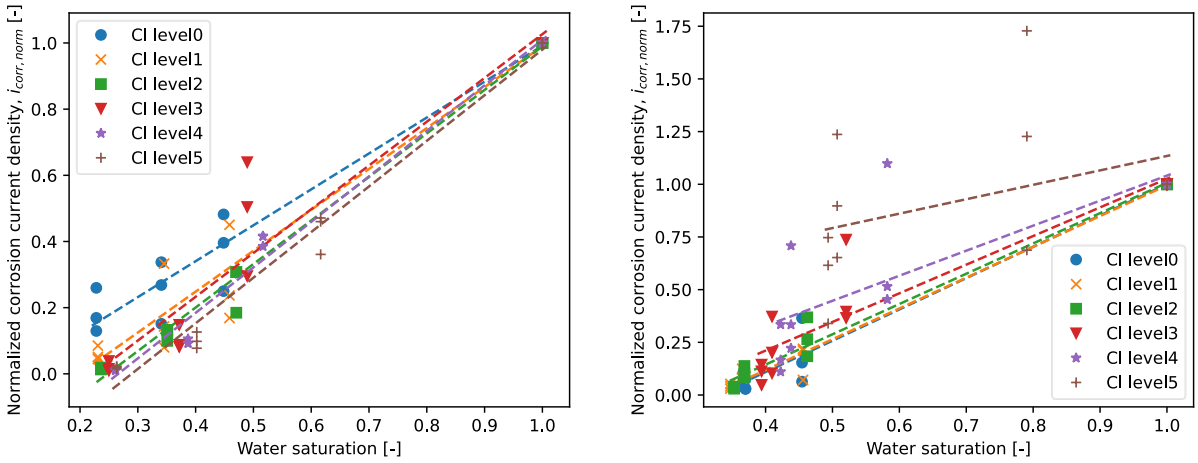
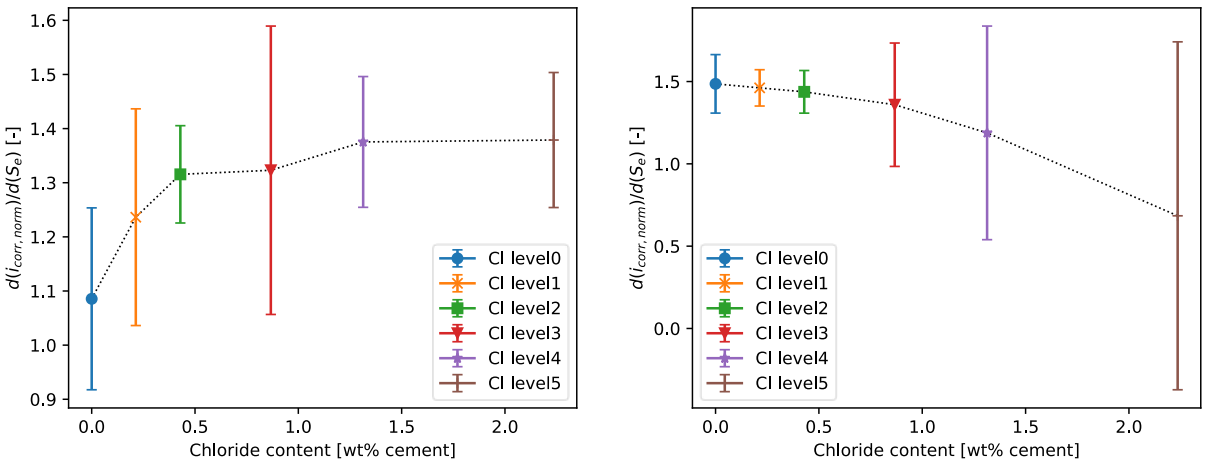


Figure 4.9 Linear dependence of normalized corrosion current density, $i_{corr,norm}$ with water saturation level for rebar in mortars. (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement. The dashed lines represent linear regression lines.



(a) (b)

Figure 4.10 First derivative of normalized corrosion current density with respect to water saturation for rebar corrosion in mortars. (a) non-carbonated, (b) carbonated. Legend: Cl level 0-5 represent chloride content levels 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement). The error bars represent a 95% confidence bound of the fitted parameter.

4.4.4.4.1 Non-carbonated mortar samples

It can be seen in Figure 4.9a, the effect of saturation on $i_{corr,norm}$ is linear for all non-carbonated mortar samples with chloride content ranging from level 0 to level 5. The statistical significance of the fitted regression lines shown in Figure 4.9a were analyzed in Appendix 4.7.1.1. The first derivatives of the regression lines are plotted in Figure 4.10a. The results show that for non-carbonated samples, the chloride content significantly affects the regressed linear relationship compared to the case with no chloride, but there is no significant difference between the regressed lines for various chloride levels. This suggests that saturation's effect on corrosion rate is related to the mechanisms independent of chloride content (level 1-5). One of the mechanisms is thought to be the area effect: the area of rebar surface exposed to solution-filled pores is determined by the saturation level.

The Ohmic control effect cannot be ruled out because the bulk resistivity at the same saturation is relatively less sensitive to the chloride content (as shown in Figure 4.5, and discussed in Section 4.4.2). This suggests that there could be a similar ohmic control applies to the chloride-containing

mortars. However, the regressed lines of chloride-containing mortars are significantly different from the chloride-free mortar, indicating a different ohmic control effect between the chloride-containing and chloride-free samples. The cause may be due to different allocation patterns of the anodic and cathodic sites. The chloride-free non-carbonated mortar retains an intact protective passivated film on rebar and provides a uniform environment with closely distributed micro-anodes and micro-cathodes. Therefore, the area effect is the dominant one for chloride-free samples, with a regression line of $d(i_{corr,norm})/d(S_e) = 1.08$, close to 1. For the mortar with chlorides, the $d(i_{corr,norm})/d(S_e)$ ranges from 1.24 – 1.37, which indicates besides "area effect", there are more effects of saturation on i_{corr}. Therefore, the controlling effects of saturation on i_{corr} in chloride-contaminated non-carbonaed mortar include the area effect, and possible ohmic control effect.

4.4.4.4.2 Carbonated mortar samples

For carbonated mortar samples, as shown in Figure 4.9b, Figure 4.10b, the linear dependence of the $i_{corr,norm}$ on saturation has no significant difference between the samples with no chloride present (level 0) and the ones with low to moderate chloride content (level 1–3). However, the carbonated samples with high chloride content (levels 4 and 5) showed significant difference, where level 4 is identified as "significantly different" with a higher p-value (less confidence) and level 5 appears not to follow the linear trend due to the reduced corrosion rate in the saturated condition (see appendix 4.7.1.2). It suggests that there are similar effects of saturation on $i_{corr,norm}$ in the carbonated mortar with chloride content (level 0–3) where the first derivative $d(i_{corr,norm})/d(S_e)$ varies between 1.36 and 1.48. It has been observed that the bulk resistivity of carbonated mortar sensitively depends on the chloride content (as shown in Figure 4.5, and discussed in Section 4.4.2). Since the area effect on rebar corrosion is independent of bulk resistivity, the remaining ohmic control effect, if any, should logically also be similar among these samples (level 0–3) and independent of bulk resistivity. This independence of the ohmic control effect on bulk resistivity is probably because the electrode potentials of rebar in the carbonated mortar do not change too much with chloride. With smaller potential differences, the different rebar surfaces will not polarize to near-discrete large anodes and cathodes and the micro anodes and cathodes will be closely located. Therefore, in carbonated samples, the ohmic control is

determined by the localized resistance between the neighbouring connecting pores, which will remain low and not change with chloride concentration as much as the bulk resistivity. In conclusion, for carbonated mortar samples, the area effect and ohmic control effect on corrosion current density brought by change in water saturation is similar among carbonated mortar with low to moderate chloride content (level 0–3).

4.4.4.5 Hypothetical corrosion current density in saturated pores

To exclude the area effect brought by water volume fraction, the concept of hypothetical corrosion current density in saturated pores, $i_{corr,pore}$, is introduced. The ratio of rebar-saturated pore area to total rebar area is expressed as the volumetric water content, then

$$i_{corr,pore} = \frac{i_{corr}}{\theta} \tag{4.12}$$

where, θ is the volumetric water content. It is noted that $i_{corr,pore}$ seems to share a similar concept of $i_{corr,norm}$. However, $i_{corr,norm}$ ignored the change in porosity caused by various chloride content and carbonation. It also set a baseline corrosion rate determined by the chloride content and carbonation of mortar. If the apparent corrosion current density is mainly scaled by the rebar-saturated pore surface area, then the hypothetical corrosion current density, $i_{corr,pore}$ should not be affected by the area effect. Any other dependence is a result of the ohmic control and the variation in the ratio of anodic and cathodic area.

4.4.4.5.1 Non-carbonated mortar samples

As shown in Figure 4.11, in the non-carbonated mortar with no chlorides, $i_{corr,pore}$ does not seem to depend on the water content and has a value similar to the value in the corresponding SPS. However, when the chloride is present, it becomes dependent on the water content, which indicates the corrosion is also “ohmic controlled” and has a changing anode/cathode ratio dependent on water content (as discussed in Section 4.4.4.1.1).

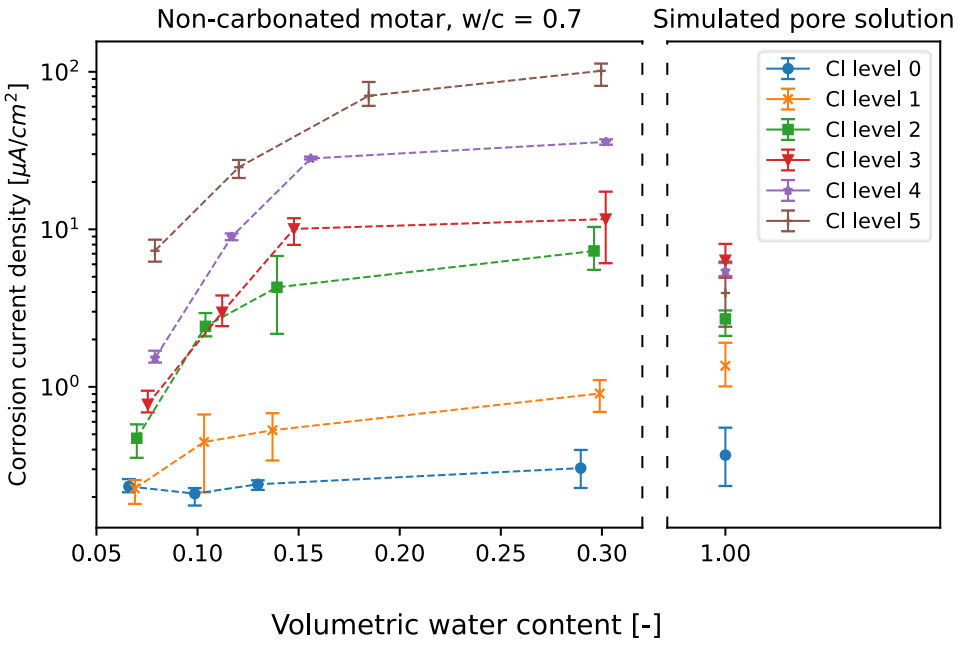


Figure 4.11 Hypothetical corrosion current density, $i_{corr,pore}$ in saturated pores in non-carbonated mortar and corrosion current density in the simulated pore solutions for 20 days. Legend: Cl level 0-5 represent chloride content levels in mortar 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement, and in simulated pore solution 0: 0%, 1: 0.5%, 2: 1.0% 3: 2.0%, 4: 3.0%, 5: 5.0%, NaCl wt.% by weight of solution. The mean value point is shown, and the error bars represent the range of the three repeated test results.

The $i_{corr,pore}$ deviates from the SPS when chloride is present: at low water content, $i_{corr,pore}$ was lower than i_{corr} in SPS for all chloride levels but increased significantly with water content. At full saturation, when chloride levels (level 3–5) exceeded the critical values, $i_{corr,pore}$ is significantly higher than the i_{corr} in SPS. Ohmic control with a fixed anode/cathode area ratio cannot explain this phenomenon. The explanation has to do with the changing in the anode/cathode area ratio. The ratio of the anodic area to total rebar area increases with the increase in the chloride concentration. At higher chloride levels (level 3–5) above the critical chloride content, the anodic area ratio in non-carbonated mortar is higher than that in SPS. This inference is possible, and the reasoning is as follows: for chloride-induced rebar corrosion, the anodic area is stabilized by the surrounding cathodic sites and the anodic area cathodically protect the surround area. In SPS, due to the low resistivity in solution, the anodic area can cathodically protect a larger area of rebar resulting in a limit of anodic area ratio. However, in mortar, the rebar surface is divided into smaller

areas confined in pores with greater resistance in the path to cathodically protect the rebar at a greater distance, which will lead to a higher anodic area ratio. The increased anodic area ratio and the resistance become the two factors controlling the $i_{corr,pore}$. The $i_{corr,pore}$ significantly increases with water content in the drier range where resistivity decreased dramatically.

4.4.4.5.2 Carbonated mortar samples

In the carbonated mortar samples, $i_{corr,pore}$ increased dramatically with water content in the drier range for all chloride levels. At higher water content, $i_{corr,pore}$ kept increasing with water content in no/low chloride samples but leveled off or decreased slightly with water content in high chloride content samples (Figure 4.12).

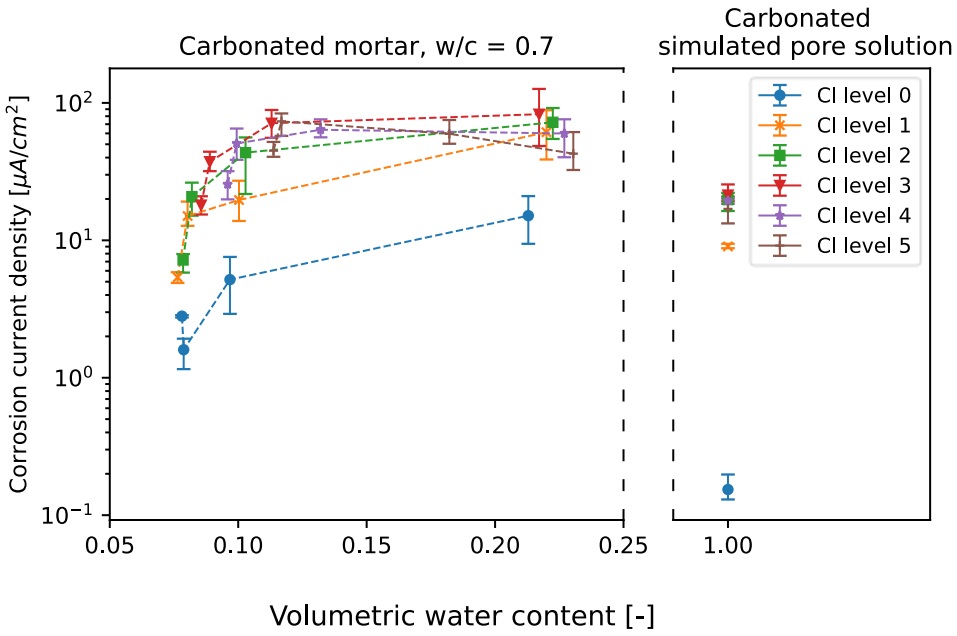


Figure 4.12 Hypothetical corrosion current density, $i_{corr,pore}$ in saturated pores in carbonated mortar and corrosion current density in the carbonated simulated pore solutions for 20 days. Legend: Cl level 0-5 represent chloride content levels in mortar 0: 0%, 1: 0.21%, 2: 0.43%, 3: 0.87%, 4: 1.31%, 5: 2.24%, Cl wt.% by weight of cement, and in carbonated simulated pore solution 0: 0%, 1: 0.5%, 2: 1.0% 3: 2.0%, 4: 3.0%, 5: 5.0%, NaCl wt.% by weight of solution. The mean value point is shown, and the error bars represent the range of the three repeated test results.

This trend is consistent with the resistivity trend found in carbonated mortar samples where resistivity decreased sensitively with water content in low-chloride samples, and decreased less dramatically or even increased with water content in high chloride samples at higher water content range.

This indicates the ohmic control effect is still at least one of the controlling factors. However, this ohmic control is not sensitive to the pore solution conductivity among samples with different chloride content (level 2-5) but is sensitive to water content, especially in drier conditions. This observation again suggests that the anode/cathode area ratio has changed with the increase in water content. Such ratio change is not due to the preferential adsorption, but due to the rust effect that passivates the steel in dry conditions and activates in wetting conditions (Section 4.4.4.2). It was also noted that the pre-passivated rebar remained passive in the carbonated simulated pore solution with no chlorides. One possibility could be the lack of disturbance in pH or external factors that would trigger the initiation of corrosion processes as it would occur in carbonated mortar.

4.5 Conclusions

Carbonation and chloride contamination affect the equalized water content and bulk resistivity of mortars. The effect of chloride content on water retention is more pronounced in carbonated mortars than non-carbonated ones. At a given RH, the carbonated mortars with higher chloride content significantly retain more water. At a given saturation level, the bulk resistivity of the carbonated mortars decreases significantly with increasing the chloride content, whereas for non-carbonated mortars, the dependence of bulk resistivity on chloride content is insignificant.

Water content affects E_{corr} in chloride-contaminated mortar. In non-carbonated mortar, the corrosion potential decreases with saturation when the chloride content exceeds a critical value, but is less affected if the chloride content is less than the critical value. In carbonated mortars, on the contrary, the effect of the saturation on the E_{corr} is most pronounced when there is no chloride and becomes less significant as chloride content increases.

The i_{corr} increases with the saturation for all samples with various chloride content and carbonation states. i_{corr} spreads across two orders of magnitude for the observed saturation levels.

It appears that the chloride content and carbonated state determine a baseline corrosion rate, and that rate alters with the saturation level.

The role of water saturation on corrosion includes 1) the area effect, which affects the rebar-pore solution area, 2) the ohmic control effect by changing the bulk resistivity, and 3) the anode-to-cathode ratio effect by the altering the anodic and cathodic site area ratio. These effects complicate the role of chloride and carbonation on rebar corrosion in mortar, rather than being solely dependent on chloride content and pH in simulated pore solutions.

In non-carbonated mortar with no chlorides, the area effect is dominant over the other two effects i.e., the ohmic control effect and the anode-to-cathode ratio effect. In a chloride-contaminated non-carbonated mortar and carbonated mortar with/without chlorides, the dependence of E_{corr} on the water content suggests the anode-to-cathode ratio effect. Different mechanisms are suggested for the change in the anode-cathode area ratio: preferential adsorption of moisture for the non-carbonated sample and rust-scale effect on the carbonated samples with the efficiency of the rust-cathode dependent on conductivity (moisture content).

With a change in the anode-to-cathode ratio, the ohmic control effect depends on the sensitivity of the resistance between the anodic and cathodic sites to bulk resistivity. In non-carbonated mortar with chloride, this ohmic control is significant. However, in carbonated mortar with/without chlorides, this ohmic control is less significant due to the closely located anodic and cathodic sites and weaker coupling between distant sites.

4.6 Epilogue

The findings in this chapter provide clarity on important questions regarding the interplay of water saturation, chloride contamination, and carbonation in relation to rebar corrosion in mortar. It highlights the effects of carbonation on water content and resistivity, the differential influence of chloride content on water retention, and the complex mechanisms that affect corrosion. Based on this analysis, the roles of water saturation on corrosion potential and corrosion current density of rebar in mortar with chloride contamination and carbonation are clarified.

With the analysis in this chapter, the measured corrosion parameters in the first phase of the experimental study (Chapter 3, 4) establish a database from which an empirical model of corrosion

parameters as a function of chloride content, water saturation, and carbonation can be developed. Furthermore, it provides a more implementable basis for the subsequent development of a mechanistic corrosion model in the numerical simulation phase.

4.7 Appendix for chapter 4

4.7.1 Statistical significance of the fitted lines of $i_{corr,norm}$ vs S_e at various chloride and saturation levels.

The fitted lines in Figure 4.9 are grouped with similar first derivative and intercepts. To determine if they are significantly different, an analysis of covariance (ANCOVA) was performed using categorical regression tests with the Statsmodels package (Seabold & Perktold, 2010). The data points from various treatments (chloride content level) were fitted with each categorical variables using treatment (dummy) coding. The significance of the coefficient for each treatment indicates whether this treatment significantly contributes to capturing the variance of the $i_{corr,norm}$. If not, the individually fitted line for that treatment is not significantly different from the baseline, which the fitted line was compared to.

4.7.1.1 Non-carbonated sample data

In the statistical results tables, the symbol T.7N(0-5) stands for the treatment of non-carbonated mortar samples with chloride level 0-5. Results show the treatments 1-5 are significantly different from 0, which suggests the added chloride at all studied levels (1-5) significantly affects the dependence of $i_{corr,norm}$ on saturation. However, among the chloride levels 1-5, there is no significant difference in the dependence of $i_{corr,norm}$ on saturation. In conclusion, the existence of chloride has a significant effect compared to no chloride, but the specific chloride content does not significantly affect the dependence of $i_{corr,norm}$ on saturation.

Table 4.2 Compared with baseline type [T.7N0]

	coef	std err	t	P>|t|	[0.025	0.975]
type[T.7N1]	-0.0765	0.033	-2.331	0.023	-0.142	-0.011
type[T.7N2]	-0.1157	0.033	-3.525	0.001	-0.181	-0.050
type[T.7N3]	-0.0819	0.033	-2.494	0.015	-0.147	-0.016
type[T.7N4]	-0.1225	0.033	-3.73	0.000	-0.188	-0.057
type[T.7N5]	-0.1507	0.033	-4.582	0.000	-0.216	-0.085

Table 4.3 Compared with baseline type [T.7N1]

	coef	std err	t	P>|t|	[0.025	0.975]
type[T.7N2]	-0.0394	0.031	-1.273	0.208	-0.102	0.023
type[T.7N3]	-0.0062	0.031	-0.199	0.843	-0.068	0.056
type[T.7N4]	-0.0474	0.031	-1.529	0.132	-0.109	0.015
type[T.7N5]	-0.0768	0.031	-2.475	0.016	-0.139	-0.015

Table 4.4 Compared with baseline type [T.7N2]

	coef	std err	t	P>|t|	[0.025	0.975]
type[T.7N3]	0.033	0.029	1.137	0.262	-0.025	0.091
type[T.7N4]	-0.009	0.029	-0.296	0.769	-0.067	0.050
type[T.7N5]	-0.039	0.029	-1.333	0.190	-0.097	0.020

4.7.1.2 Carbonated sample data

In the statistical results tables, the symbol T.7C(0-5) stands for the treatment of carbonated mortar samples with chloride level 0-5. The treatment 4-5 are significantly different from 0 where 4 has a p-value of 0.035 which is slightly smaller than 0.05 indicating a weaker significance. This suggests adding chloride does not significantly affect the dependence of $i_{corr,norm}$ on saturation unless there is a very high chloride content (4-5). When comparing among different chloride levels, only level 5 showed a significant difference and level 4 nearly reached the significance threshold, which means that higher chloride content affects $i_{corr,norm}$ dependence on saturation. In

conclusion, for carboned samples, only higher chloride content (4-5) significantly affects the dependence of $i_{corr,norm}$ on saturation.

Table 4.5 Compared with baseline type [T.7C0]

	coef	std err	t	P>|t|	[0.025	0.975]
type[T.7C1]	0.005	0.078	0.065	0.948	-0.150	0.160
type[T.7C2]	0.028	0.078	0.361	0.719	-0.127	0.183
type[T.7C3]	0.085	0.078	1.091	0.279	-0.070	0.240
type[T.7C4]	0.168	0.078	2.157	0.035	0.012	0.323
type[T.7C5]	0.402	0.079	5.112	0.000	0.245	0.559

Table 4.6 Compared with baseline type [T.7C1]

	coef	std err	t	P>|t|	[0.025	0.975]
type[T.7C2]	0.023	0.084	0.277	0.783	-0.144	0.191
type[T.7C3]	0.081	0.084	0.971	0.336	-0.086	0.249
type[T.7C4]	0.165	0.084	1.972	0.054	-0.003	0.333
type[T.7C5]	0.404	0.085	4.737	0.000	0.233	0.574

4.7.2 Non-linear regression of empirical models for corrosion parameters

This section presents the empirical fits of the surface functions for the corrosion potential (E_{corr}) data in Figure 4.7 and corrosion current density (i_{corr}) data in Figure 4.8, using saturation (Se) and chloride content (Cl) as variables. The expressions of the fits are given below, and the fit parameters are listed in Table 4.7.

4.7.2.1 Non-carbonated mortar with chloride

The $E_{corr} - Cl - Se$ data presented in Figure 4.7 is regressed by an empirical function named "Ecorr_func":

$$E_{corr} = p1 + p2 \cdot x + p3 \cdot y + p4 \cdot ln(x + p5) + p6 \cdot ln(x + p7) \cdot y \qquad (4.13)$$

The $i_{corr} - Cl - Se$ data presented in Figure 4.8 is regressed by an empirical function named

"icorr_func":

$$i_{corr} = p1 + p2 \cdot x + p3 \cdot y + p4 \cdot exp(p5 \cdot x \cdot y) \quad (4.14)$$

4.7.2.2 Carbonated mortar with chloride

The $E_{corr} - Cl - Se$ data presented in Figure 4.7 is regressed by an empirical function named "Ecorr_carb_func":

$$E_{corr} = p1 + p2 \cdot x + p3 \cdot y + p4 \cdot ln(x + p5) + p6 \cdot ln(x + p7) \cdot y \quad (4.15)$$

The $i_{corr} - Cl - Se$ data presented in Figure 4.8 is regressed by an empirical function named "icorr_carb_func":

$$i_{corr} = p1 + p2 \cdot x + p3 \cdot y + p4 \cdot x \cdot y + p5 \cdot ln(p6 \cdot x + p7) \cdot y^{p8} \quad (4.16)$$

where x is chloride content z V vs. SCE and i_{corr} has a unit of $\mu A/cm^2$.

Table 4.7 Parameters for the fitted functions

	Ecorr_func	icorr_func	Ecorr_carb_func	icorr_carb_func
p1	260.690864	-6.81350596	-0.8217863	-7.8192924
p2	3.81499597	-144.198819	-0.176338	563.282881
p3	-1.898141	0.068146096	0.22552484	31.7826539
p4	-69.43264	6.87586626	-0.0913669	-1216.4041
p5	42.730401	79.6234978	6.7105E-05	4.86663471
p6	-0.2838278	-	0.08867938	68.0514188
p7	0.00155937	-	6.1648E-05	0.01470596
p8	-	-	-	1.6312814
R^2	0.959	0.955	0.961	0.964

5 The Impact of Non-uniformity and Resistivity on the Homogenized Corrosion Parameters of Rebars in Concrete - a Circuit Model Analysis

5.1 Preface

In the previous chapters, we explored the intricate relationship between water saturation, chloride contamination, carbonation, and rebar corrosion in mortar. The findings form the basis for developing mechanistic corrosion models. In the numerical simulation phase starting from this chapter, we will use the modelling method to explore how the overall corrosion parameters are related to the non-uniform localized corrosion parameters at mesoscopic–macroscopic level and microscopic level.

Chapter 5 presents the concept of the homogenized corrosion parameters of rebar in concrete with non-uniform surface conditions with a circuit model. This model starts from localized corrosion kinetics for active and passive rebar-concrete units (mesoscopic–macroscopic level) described by the Butler-Volmer equation and arrives at the simulated scan curve representing the overall performance of the rebar, from which the homogenized corrosion parameters can be derived. It explored the validity of describing the overall scan curve with the Butler-Volmer equation, and how much deviation can be caused by the non-uniformity of rebar condition and concrete resistivity.

It suggests that the homogenized corrosion parameters are a function of concrete resistance, and the area ratio of the active or passive rebar-concrete surface area — the necessity to employ corrosion parameters as a function of the microenvironments. In addition, the non-uniform surface condition at mesoscopic–macroscopic scale and concrete electrical resistivity distort the polarization curve, which may significantly deviate from the Butler-Volmer equation — the necessity of measuring corrosion parameters in a uniform rebar-concrete interface condition at mesoscopic–macroscopic scale, as was done in the first phase experimental studies in Chapter 3 and Chapter 4.

With the nonuniformity at mesoscopic–macroscopic scale, the degree of deviation is more significant in concrete with higher electrical resistivity, particularly when the active and passive rebar surfaces have a similar area ratio. In such cases, corrosion parameters obtained from non-uniform reinforced concrete samples using potentiodynamic scans may not accurately represent

the actual corrosion kinetics. Hence, they cannot be considered valid homogenized corrosion parameters.

For nonuniform rebar at the meso-macroscopic scale, the reinforced concrete samples should represent field conditions, and the corrosion model should utilize potentiodynamic curves instead of the fitted corrosion parameters derived from it. Such application might be applicable for large-scale corrosion modelling but is not feasible for numerical simulation of rebar corrosion in cracked concrete with sharp spatial variation of microenvironment around the crack.

In the next chapter, we will zoom in scale and use this circuit model to simulate the homogenized corrosion kinetics of the rebar-concrete interface (mesoscopic–macroscopic level) from the corrosion kinetics of the rebar-solution interface (microscopic pore level).

Chapter 5 was adapted (with permission) from the manuscript " The Impact of Non-uniformity and Resistivity on the Homogenized Corrosion Parameters of Rebars in Concrete - a Circuit Model Analysis " published in Corrosion Engineering, Science and Technology[1]. Gang Li is the first author who designed and performed numerical simulation and data analysis. Dr. Richard Evitts and Dr. Moh Boulfiza contributed to the review of the manuscript and provided feedback. The manuscript was reformatted for inclusion in the thesis.

5.2 Introduction

The corrosion parameters of rebar, including corrosion potential, corrosion current density, and Tafel constants, are important parameters that are used to describe activation-controlled corrosion kinetics. These corrosion parameters are often used in models to predict the service life of reinforced concrete structures. The corrosion potential can be used to estimate the likelihood of active rebar corrosion (ASTM International, 2015). The widely used non-destructive linear polarization resistance method relies on the estimated values of the Tafel constants.

[1] Cited as "Li, G., Evitts, R., & Boulfiza, M. (2023). The impact of non-uniformity and resistivity on the homogenised corrosion parameters of rebars in concrete – a circuit model analysis. *Corrosion Engineering, Science and Technology*, *58*(4), 399–409. https://doi.org/10.1080/1478422X.2023.2190442"

Rebar corrosion parameters are often determined ex-situ using lab-created pore solutions due to ease of testing, but the results are expected to provide insights into the corrosion of rebar embedded in concrete. However, one immediate shortcoming is that in these solutions, the rebar surface is exposed to a uniform environment, but the microenvironmental conditions on the rebar in concrete tend to be non-uniform. For example, non-uniform chloride ion content can be found in uncracked concrete (Silva et al., 2012) and this non-uniformity is often exacerbated in the case of cracked concrete contaminated with chlorides. In addition, the as-received rebar itself often has rust spots that create non-uniform rebar-concrete interfaces with a different electrochemical behaviour than that of rust-free areas (Stefanoni et al., 2015). Regardless of the non-uniform rebar-concrete interface, the corrosion parameters and rate are frequently determined via a polarization curve acquired in a potentiodynamic scan by fitting the Butler-Volmer equation or its Tafel approximation to the test data. This method often involves testing a short rebar sample with the assumption that the sample will undergo uniform corrosion. However, this assumption is not frequently valid for rebar in-situ since most will experience distributed concentrations of chloride ions and varying levels of carbonation. Consequently, a potentiodynamic scan curve may not follow Butler-Volmer kinetics. This type of deviation has been noticed in a few studies: Elsener (2005) found the scan curves of rebar in concrete do not manifest Tafel behaviour (no linear portion found on the logarithmic current density scale); Chang et al. (2008b) reported that scan curve could only be fit by considering rust film growth/dissolution. For those who managed to apply Tafel extrapolation to determine the corrosion rate with a selected linear portion (Alonso et al., 1998b; Locke & Siman, 1980), or with modified Tafel equations (Chang et al., 2008b), the anodic Tafel slopes (205–1570 mV) can be significantly larger than the ones measured ex-situ in solutions (89–115 mV) (Chang et al., 2008b) and also ones used by a few modellers (60–91 mV) (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kranc & Sagüés, 1994). In this study, it is hypothesized that deviation of the polarization curve of the rebar in concrete from Butler-Volmer kinetics is caused by the non-uniformity of the environment surrounding the rebar.

For heat and mass transfer studies in concrete, non-uniformity in porous microstructure is often neglected. Concrete is treated as a continuum with the properties of its representative elementary volume (REV) using a homogenization/volume-averaging technique. When the REV size is properly selected, properties such as effective thermal conductivity, diffusivity, permeability and

other thermophysical properties can be used in their representative equations to solve heat and mass transfer problems at a larger scale. Similarly, if the non-uniform corrosion rate at a microscopic level is homogenized at the scale of the REV, then the homogenized corrosion properties can be used at the larger scale. However, Butler-Volmer kinetics may no longer hold for the homogenized potential-current density relation since they are dependent on the localized surface interaction between the rebar and its surroundings. This leads to the concerns raised by experimental observations (Chang et al., 2008b; Elsener, 2005).

In this research, a model is developed to relate the homogenized, or area-blended corrosion kinetics of rebar in non-uniform conditions in concrete to the corrosion kinetics of rebar in uniform conditions. The non-uniformity could be due to non-uniform chloride ion contamination and/or non-uniform carbonation along the rebar or around its circumference. This model can be used to identify systems that follow Butler-Volmer kinetics with negligible error, and those that do not. It simulates the homogenized kinetics in the form of a potentiodynamic scan curve. This curve can be used in numerical models that require the input of rebar corrosion kinetics to predict large-scale rebar degradation.

5.3 Model development

5.3.1 Theory

For rebar corrosion in concrete, the Butler-Volmer equation is used to describe the activation-controlled corrosion kinetics.

For a reversible half-cell electrode reaction (D. A. Jones, 1996):

$$i_{net} = i_0 \cdot \left[10^{(E-E_0)/\beta_a} - 10^{(E-E_0)/\beta_c}\right] \tag{5.1}$$

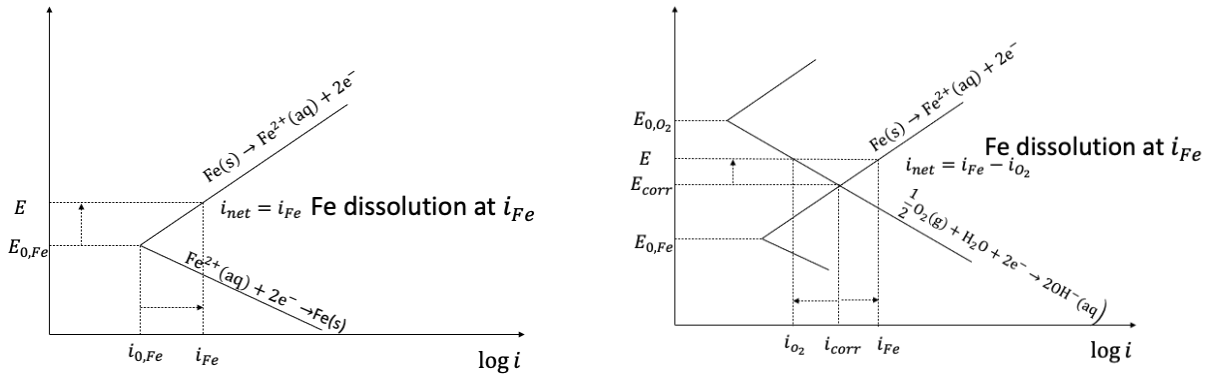
where, i_{net} is measured or applied current density, i_0 is exchange current density, E_0 is reversible potential, E is electrode potential, β_a is anodic Tafel constant, β_c is cathodic Tafel constant.

For a mixed electrode reaction for corrosion (D. A. Jones, 1996):

$$i_{net} = i_{corr} \cdot \left[10^{(E-E_{corr})/\beta_a} - 10^{(E-E_{corr})/\beta_c}\right] \quad (5.2)$$

where, i_{net} is measured or applied current density, i_{corr} is corrosion current density, E_{corr} is equilibrium potential or corrosion potential, E is electrode potential, β_a is anodic Tafel constant, β_c is cathodic Tafel constant.

Despite the mathematical similarity of the Butler-Volmer equation when applied to either a single electrode with a reversible half-cell reaction or a mixed electrode reaction, there is a distinctly different meaning of the net current density, i_{net}. Taking the corrosion of steel for example, with the reversible half-reaction ($Fe^{2+} + 2e^- \leftrightarrow Fe$), i_{net} in Equation (5.1) represents the total change of iron (either dissolution or deposition), dissolution case shown in Figure 5.1a. However, in a mixed electrode reaction (Figure 5.1b), when polarized anodically from its equilibrium potential, i_{net} in Equation (5.2) does not include all the change of iron: the "internal current", i_{O_2} also contributes to a portion of the total change of iron ($i_{Fe} = i_{net} + i_{O_2}$). This contribution could be significant when the anodic polarization overpotential is small, which implies a non-negligible coexisting macro-cell (net current) and micro-cell corrosion (internal current).



(a) (b)

Figure 5.1 Evans diagrams for a single-electrode (a) and a mixed-electrode (b) reaction.

Experimental observations (Hansson et al., 2006) have indicated the micro-cell contribution in a seemingly macro-cell-dominant corrosion could be significant. This is especially true when the

macro-cell current density is low due to small potential differences between the neighbouring rebar segments.

5.3.2 Model setup

The proposed model aims at the goal of deriving a homogenized corrosion parameter from the corrosion parameters at the pore level. There are two types of rebar surfaces being considered in this model: passive and active. A passive rebar surface has a dense protective oxide film due to high alkalinity in concrete, and this film is disrupted for the active rebar surface due to localized environmental factors such as acidity (carbonation) or chloride ions. The passive rebar surface has a more noble electrode potential than the active one. In this model, two single electrode half-cell reactions are considered: iron deposition/dissolution ($Fe^{2+} + 2e^{-} \rightleftarrows Fe$), oxygen reduction/evolution ($O_2 + H_2O + 4e^{-} \rightleftarrows 4OH^{-}$) on both the active and passive portions of the steel surface. In this model those two electrode reactions on both active and passive steel surfaces are simplified by neglecting iron deposition and oxygen evolution, leaving one anodic or dissolution reaction and one cathodic or oxygen reduction reaction, ($Fe \rightarrow Fe^{2+} + 2e^{-}$) and ($O_2 + H_2O + 4e^{-} \rightarrow 4OH^{-}$); these are the normal corrosion reactions. It is assumed that on the same surface condition (either activated or passivated) at the capillary pore size (10 nm – 5 μm (Mehta & Monteiro, 2006)), which is termed as "pore-level" (see Figure 5.2), the microscopic anodic and cathodic sites are physically close and uniform kinetics can be applied on the surface. Thus, there is no electric circuit modelled at this microscopic pore-level.

At the mesoscopic-macroscopic level (a few millimetres), a rebar piece embedded in concrete in a non-uniform environment is comprised of a number of active or passive pore-level rebar pieces and non-reactive pore-level rebar pieces. In this research, this is called "unit-level" (Figure 5.2). Within each type of unit, due to the uniform environmental condition, there is little microscopic level external current between pore-level rebar pieces; hence the effect of electrolytic resistance between these pore-level rebar pieces can be neglected and the corrosion is uniform, i.e., there is no physical separation of micro anodes and cathodes and thus no internal current is modelled between them.

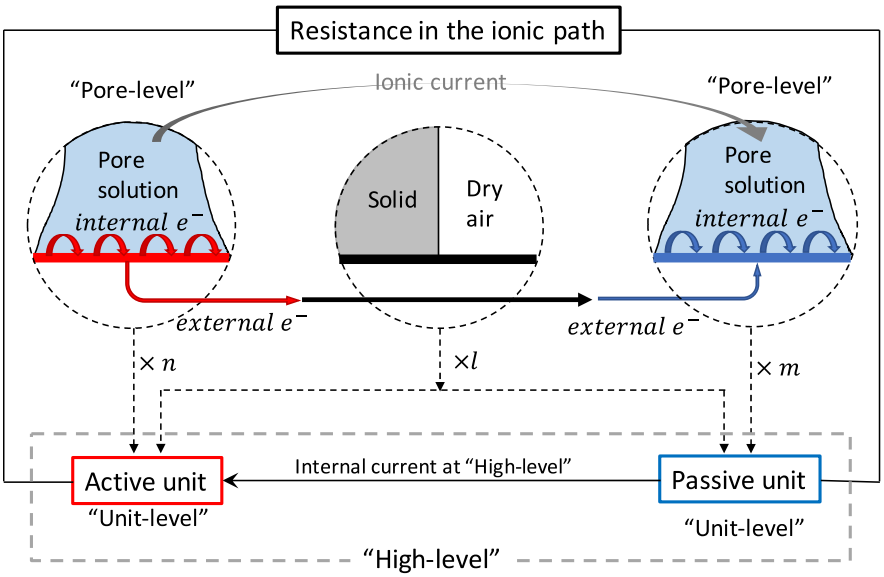


Figure 5.2 Schematic of the multi-level treatment of rebar corrosion in concrete (n, l, m is the number of "pore-level" rebar segments).

At the larger macroscopic level (a few centimetres), a selected rebar piece might be exposed to a concrete environment that has non-uniform environmental conditions. For example, flexural transverse cracks bring a non-uniform condition to the rebar in its longitudinal direction; or ongoing carbonation and chloride penetration through concrete cover may cause different exposure conditions on the rebar around its circumference. In these cases, a rebar piece is represented by a combination of active/passive corrosion units and there is current flow between the different units. Therefore, there is electrical resistance between these corrosion units, and this translates into possible ohmic control at the macroscopic level. In this research, rebar consisting of both active and passive units at the macroscopic level is termed "high-level" (Figure 5.2), the electrical current between units is termed as internal current at "high-level". At this level, the uniform corrosion assumption is invalid, and the rebar is treated as micro-and-macro cell corrosion.

The circuit model developed for this research work simulates a potentiodynamic scan test of reinforced concrete at its "high-level" from the known corrosion kinetics at the "unit-level". The schematic circuit that combines these cells to simulate corrosion in rebar is shown in Figure 5.3. Three battery symbols are used to represent electrochemical interfaces (red: active unit, blue: passive unit, and black: potentiostat-controlled concrete surface). When combined, the active unit and passive unit represent the rebar at "high level". Each interface follows a characteristic $E - I$

relationship defined by the Butler-Volmer equation, where E is the potential difference from metal to electrolyte and I is external current at the "unit level". The concrete cover electrolyte potential is independent of current and is maintained at ϕ_{app}. Three resistors (R_0, R_1, R_2) represent the resistances in the concrete cover. The high-level "internal current" (I_0) goes through resistance (R_0) between those rebar units, and the high-level "external current" (I_e^{act}, I_e^{pas}) goes through resistance (R_1, R_2) between the rebar units and the concrete cover.

The current is defined as positive when flowing outwards from the rebar surface, and negative when flowing inwards to the rebar surface. The positive direction of current and potential symbols and grounding are presented in Figure 5.3. A summary of the symbol description is shown inTable 5.1.

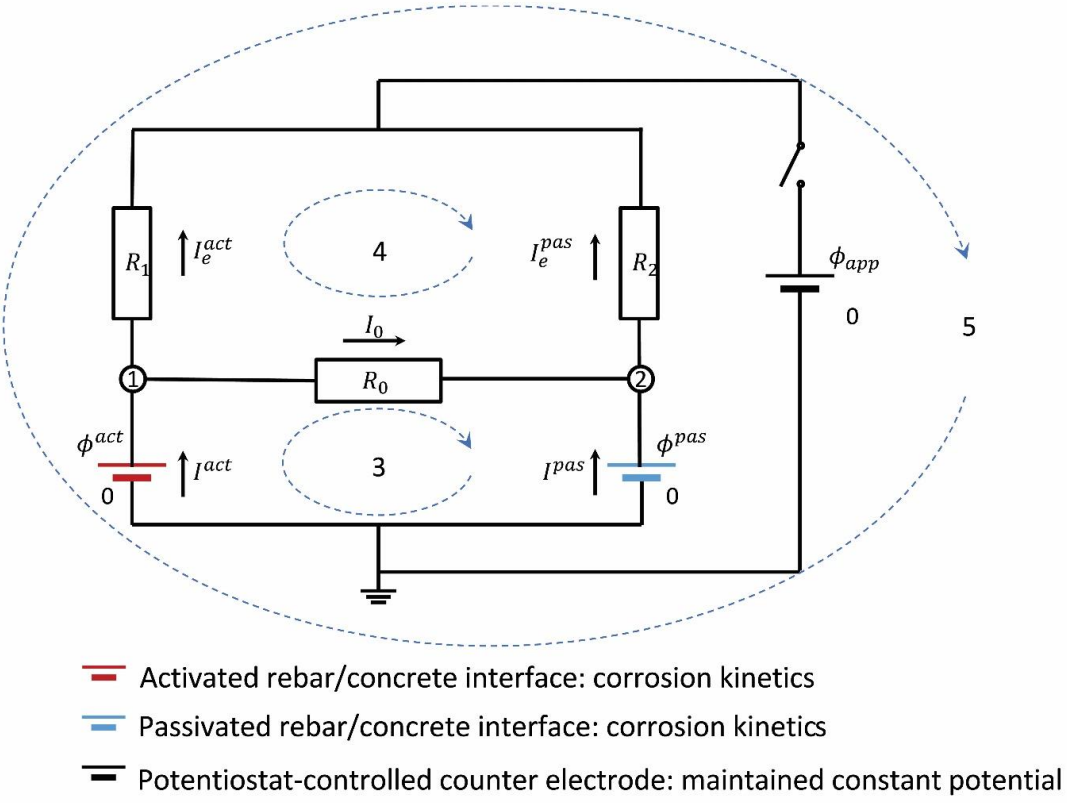


Figure 5.3 Circuit diagram of the proposed model.

Table 5.1 Symbol descriptions of the circuit model

Symbol	Description
ϕ^{act}	electrolyte potential on activated rebar unit
ϕ^{pas}	electrolyte potential on passivated rebar unit
I^{act}	"high-level" total ("unit-level" external) current on activated rebar unit
I^{pas}	"high-level" total ("unit-level" external) current on passivated rebar unit
I_0	"high-level" internal current near the rebar surface between rebar units
I_{e}^{act}	"high-level" external current from activated rebar unit
I_e^{pas}	"high-level" external current from passivated rebar unit
ϕ_{app}	applied electrolyte potential on the concrete cover surface
R_0	equivalent resistance between rebar units in the horizontal direction
R_1	equivalent resistance in the concrete cover above activated rebar unit in the vertical direction
R_2	equivalent resistance in the concrete cover above passivated rebar unit in the vertical direction
R_e	equivalent resistance in concrete cover between rebar surface and concrete cover surface, $\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2}$
pore-level	microscopic level (order of 10 nm – 5 μm) in the same concrete pore
unit-level	mesoscopic-macroscopic level (order of a few millimetres), a combination of a number of active/passive pore-level rebar pieces and non-reactive pore-level rebar pieces.
high-level	macroscopic level (order of a few centimetres), a combination of active/passive corrosion units

In this circuit, the following relationships are satisfied:

Kinetics on individual rebar units:

$$I^{act} = i_{corr}^{act} \cdot \left[10^{(0-\phi^{act}-E_{corr}^{act})/\beta_a^{act}} - 10^{(0-\phi^{act}-E_{corr}^{act})/\beta_c^{act}}\right] \cdot AREA \cdot r_a \quad (5.3)$$

$$I^{pas} = i_{corr}^{pas} \cdot \left[10^{(0-\phi^{pas}-E_{corr}^{pas})/\beta_a} - 10^{(0-\phi^{pas}-E_{corr}^{pas})/\beta_c^{pas}}\right] \cdot AREA \cdot r_p \quad (5.4)$$

The equation (5.3) and (5.4) is the Butler-Volmer equation with the corrosion parameters E_{corr}, i_{corr}, β_a, β_c of the active and passive units designated with superscripts "act" and "pas"; AREA is the total surface area of the rebar; r_a and r_p are the area ratios of the activated rebar surface and the passivated rebar surface to the whole rebar surface, respectively.

When the switch is in the "off" position, the circuit is in "open-circuit mode", where the current flows between the two corrosion units:

$$I^{act} + I^{pas} = 0 \quad (5.5)$$

The potential drop between the active and passive units is balanced with total resistance between these units being:

$$\phi^{act} - \phi^{pas} = I^{act} \cdot 1/\left(\frac{1}{R_0} + \frac{1}{R_1 + R_2}\right) \quad (5.6)$$

Four unknowns in open-circuit mode (i.e., I^{act}, I^{pas}, ϕ^{act}, ϕ^{pas}) can be solved with Equations (5.3) to (5.6). The area-averaged electrode potential or open circuit potential, OCP, is defined as $-(\phi_{act} r_a + \phi_{pas} r_p)$, and the corrosion current is I^{act} or $|I^{pas}|$.

When the switch is in "on" position, the circuit is at "scan mode", where a range of electrical potentials, ϕ_{app}, are applied to the concrete cover.

Applying Kirchhoff's Current Law at node 1, 2 yields:

$$I^{act} - I_0 - I_e^{act} = 0 \tag{5.7}$$

$$I^{pas} + I_0 - I_e^{pas} = 0 \tag{5.8}$$

Applying Kirchhoff's Voltage Law for closed-circuits 3, 4, and 5 yields:

$$(0 - \phi^{act}) + I_0 R_0 - (\phi^{pas} - 0) = 0 \tag{5.9}$$

$$I_e^{act} R_1 - I_e^{pas} R_2 - I_0 R_0 = 0 \tag{5.10}$$

$$\left(\phi_{app} - 0\right) + (0 - \phi^{act}) + I_e^{act} R_1 = 0 \tag{5.11}$$

The seven unknowns (i.e., ϕ^{act}, ϕ^{pas}, I^{act}, I^{pas}, I_e^{act}, I_e^{pas}, I_0) can be determined via the seven equations: Equation (5.3–5.4 and 5.7–5.11).

5.3.3 Homogenized potential

Due to the existence of the electrolytic resistance in the concrete or IR drop, for a "high-level" rebar piece, the localized electrode potentials of the active or passive units are different from the zero-resistance mixed potential of the two units. The potential difference between the two localized electrode potentials develops a potential and current field in the concrete. The mixed potential measured from the concrete cover lies between the two localized electrode potentials. During a simulated potential dynamic scan process, the following rebar potentials are present: the localized potentials on active and passive units (E^{act}, E^{pas}); homogenized area-averaged potential, $\langle E_{rebar} \rangle$; and corrected rebar potential measured from concrete cover surface, E_{rebar}^{IRcomp}. This corrected potential is equivalent to the IR compensated potential read from a potentiostat. The mathematical expression of these potentials can be found from Equations (5.12) to (5.15).

$$E^{act} = 0 - \phi^{act} \tag{5.12}$$

$$E^{pas} = 0 - \phi^{pas} \tag{5.13}$$

$$\langle E_{rebar} \rangle = 0 - \langle \phi_{rebar} \rangle = -\left(\phi_{act} r_a + \phi_{pas} r_p\right) \tag{5.14}$$

$$E_{rebar}^{IRcomp} = 0 - \phi_{rebar}^{IRcomp} = -\left[E_{app} - (I_e^{act} + I_e^{pas})\ R_e\right] \tag{5.15}$$

where the R_e is the total resistance in concrete cover with R_1, R_2 in parallel.

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \tag{5.16}$$

The total external current is $I_e^{act} + I_e^{pas}$, which is equal to total current $I^{act} + I^{pas}$ can be plotted against $\langle E_{rebar} \rangle$ to create the homogenized potentiodynamic scan curve.

5.3.4 Simulated potentiodynamic scan

The model is firstly solved in "open circuit mode" to find the open circuit potential (OCP). Then, a simulated scan starts from OCP ($-\phi_{app} = \text{OCP}$) and increases/decreases using small potential increments (e.g., 0.5 mV) to alter the homogenized area-averaged rebar electrode potential, $\langle E_{rebar} \rangle$. The scan stops when the prescribed scan range bounds are reached (e.g., OCP-200 mV or OCP+ 200 mV). The simulated potentiodynamic scan procedure was coded in Python and the program is documented in an open-source public repository (Li, Evitts, & Boulfiza, 2018). The group of equations in both "open circuit model" and "scan mode" were numerically solved with an iterative optimization algorithm from SciPy (Jones et al., 2001) in the Python ecosystem. To accelerate the convergence process, solutions from the previous scan step were used as initial guesses for the next scan step.

In this model, the base parameters used are listed in Table 5.2 and Table 5.3. The corrosion parameters listed in Table 5.2 are from the repeated tests of rebar embedded in both uncarbonated

and completely carbonated concrete, with both types of concrete subjected to water-saturated conditions (Sohail, 2013) respectively representing the corrosion kinetics of a passive rebar unit and an active rebar unit. At the "unit level", it was defined that the pore solution composition is similar within the active unit or within the passive unit and there is negligible internal current within each type of unit, and therefore the Tafel constant will be unaltered, and the corrosion current, i_{corr}, of each unit will be reduced by the ratio of the area exposed to the saturated pores to the total rebar-concrete interface area. In real structures, the moisture level is not uniform throughout the concrete cover, and the moisture content at the rebar-concrete interface tends to be more stable than that in the cover in wetting and drying scenarios. For isolating the effect of resistance on the electrical field in this equivalent circuit approach, a constant corrosion current at the "unit-level" assumption will be used, and the values of the equivalent resistors reflect the resistivity distribution and geometry of concrete cover.

Table 5.2 Corrosion parameters of an active/passive unit, with data from (Sohail, 2013)

Corrosion unit	E_{corr} [V_{SCE}]	i_{corr}[A/m^2]	β_a[V]	β_c[V]
Active	-0.650	8.6×10^{-2}	0.224	-0.260
Passive	-0.200	5×10^{-4}	0.505	-0.130

Table 5.3 Base parameters for rebar area, area ratio, and resistance

AREA [m^2]	r_{act}[-]	r_{pas}[-]	R_0[Ω]	R_e[Ω]
10^{-4}	0.5	0.5	10^3	10^3

5.4 Results and discussion

Parametric scan results from the model were conducted with various values of equivalent resistors across the concrete cover (R_e) and between corrosion units (R_0). Both resistors were prescribed discrete values of 100, 1000, and 10000 Ω to represent variable saturation levels and geometry of

the concrete cover. The resistance was approximated as a spreading resistance (Angst et al., 2011; Hornbostel et al., 2015) of a circular surface area with a radius of 1 cm. The "high-level" rebar was polarized to $\pm$200 mV from its open circuit potential (OCP).

5.4.1 Significance of IR compensation

IR compensation is needed when there is a significant difference between the homogenized potential on the rebar surface and the potential measured by a reference electrode. Due to field practicalities, the potential measured from concrete cover in a potentiodynamic scan is commonly used as rebar potential, although some solid-state embedded reference electrodes could be used to reduce the IR drop (Pech-Canul & Castro, 2002).

It can also be seen from Figure 5.4 that the IR compensation is crucial for cases1-3, where the resistance across the concrete cover (R_e) is large. The IR compensation effect is insignificant when R_e is small (case 7–9). For cases 4–5, with reduced R_e the IR compensation effect becomes less significant when the potential gets closer to the OCP because the total external current going through the concrete cover becomes smaller when the over-potential is small. This suggests that a smaller scanning range such as the ones done in linear polarization resistance tests (range: OCP$\pm$10 mV) can be performed with less IR compensation. When comparing with group 1–3 or group 4–6, with prescribed R_e, a smaller equivalent resistance between the corrosion units (R_0) increases the IR drop in the concrete cover. This is because a reduced R_0 would increase the corrosion current at OCP, and hence a higher current through the concrete cover would occur when polarized.

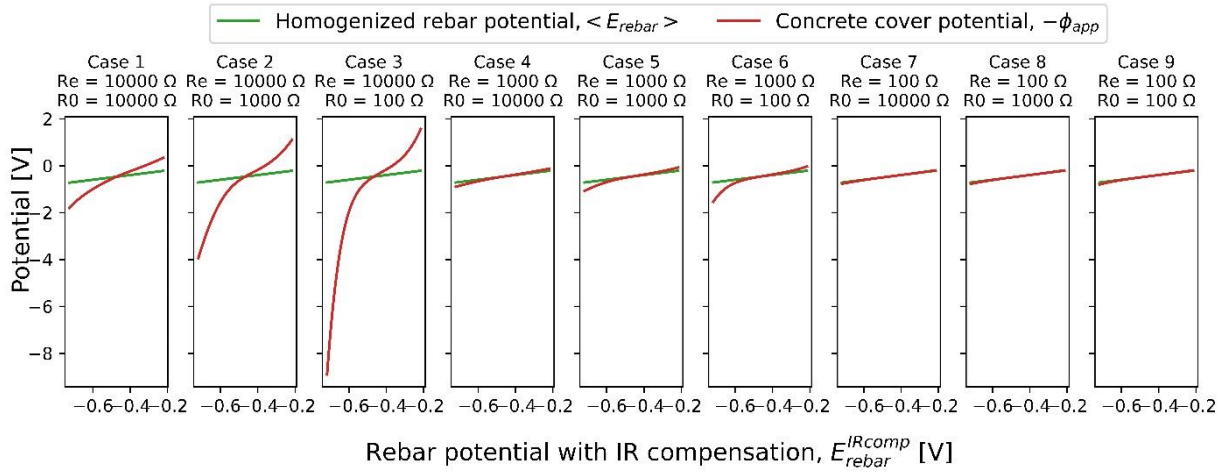


Figure 5.4 IR drop increases with an increase in resistance through concrete cover and a decrease in resistance between the dissimilar corrosion units.

5.4.2 Deviation of localized potential affected by resistance

As shown in Figure 5.5, the homogenized potential of the rebar, $\langle E_{rebar} \rangle$ equals the reference-electrode potential measured from the concrete cover with IR compensation (E_{rebar}^{IRcomp}) regardless of resistance. That means at the "high-level" scale, when the rebar is treated as being under uniform corrosion, the experimentally acquired potential of the rebar from the concrete cover corrected by IR compensation can be used to represent the homogenized potential of the rebar without the need of knowing the details of the localized potential distribution.

Nonetheless, the localized potential of the active and passive units (E^{act}, E^{pas}) deviate from the homogenized potential $\langle E_{rebar} \rangle$. As shown in Figure 5.5, this deviation significantly decreases with decreasing R_0, because a smaller resistance between the active and passive units means the rebar condition is more like the situation in the solution where there is uniform corrosion. In addition, R_e has a similar effect: this deviation would be minimized if the resistance through the concrete cover is reduced. The simulations also show that this deviation is more pronounced when the samples are more polarized.

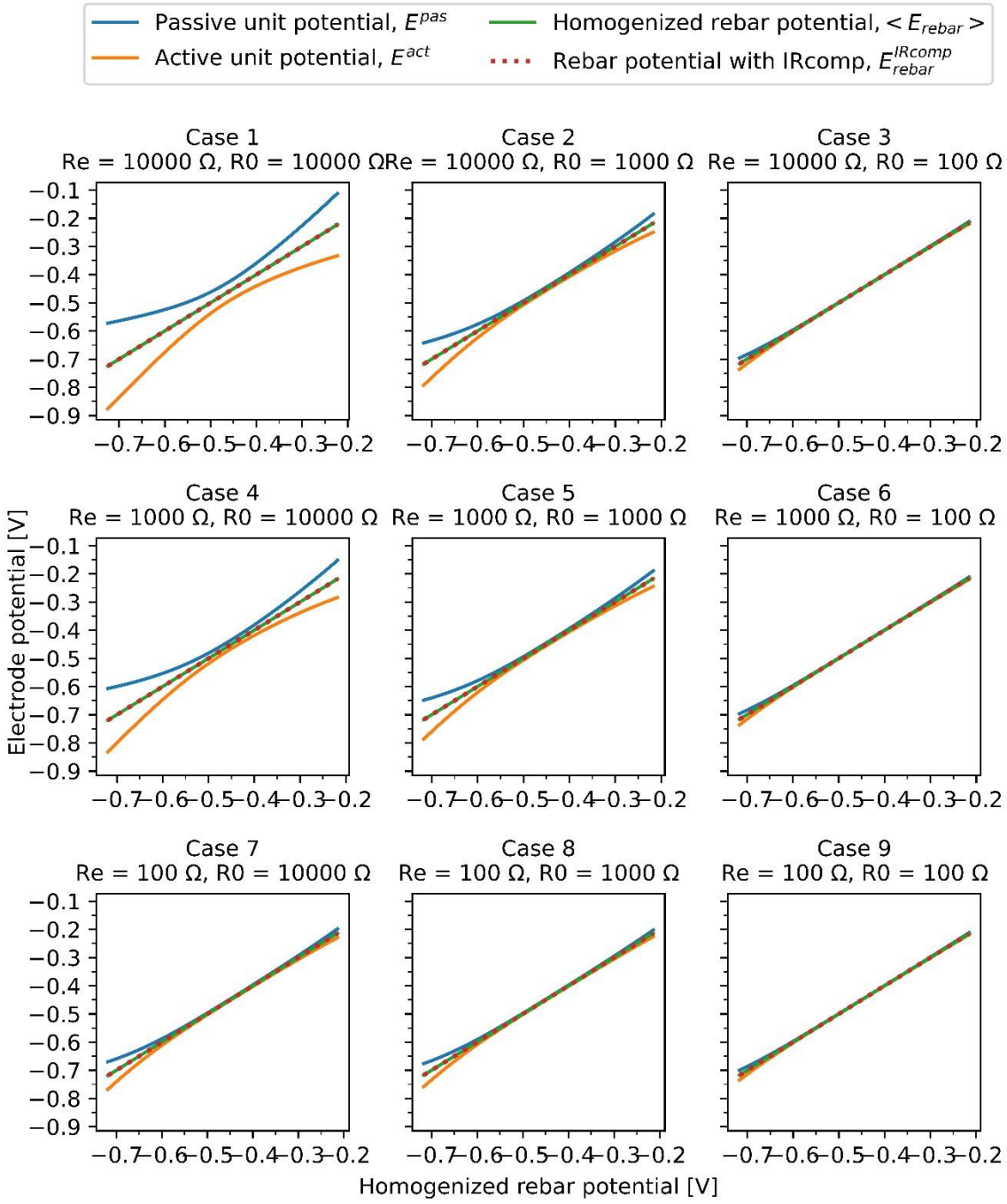


Figure 5.5 The influence of resistance on localized potentials and homogenized potential.

5.4.3 Polarization curve affected by resistance

Since there is a deviation between homogenized and localized potential, the homogenized potential-current relationship also deviates from that described by the Butler-Volmer equation. The degree of deviation is similar to the deviation between the localized potentials and the homogenized potential (Figure 5.5).

Such deviation distorts the polarization curve. For the case with a large concrete cover resistance, shown in Figure 5.6 (case 1), the homogenized potential-current plot (green curve) appears to have a non-linear slope change, and this distortion becomes more significant at larger polarization levels. When the resistance is small (Figure 5.6, case 9), this distortion becomes negligible, and the polarization curve becomes closer to a Butler-Volmer equation type, with the homogenized scan curve having a linear Tafel portion.

Consequently, if one directly applies the Butler-Volmer equation or Tafel extrapolation technique to the polarization curve, the homogenized corrosion current density of the rebar at the "high-level" will have large errors and is overestimated. As illustrated in Figure 5.7, for Case 1 (large concrete cover resistance), the estimated corrosion current density is significantly greater than the corrosion current density at the corrosion potential; this difference diminishes for cases where the resistance is lower.

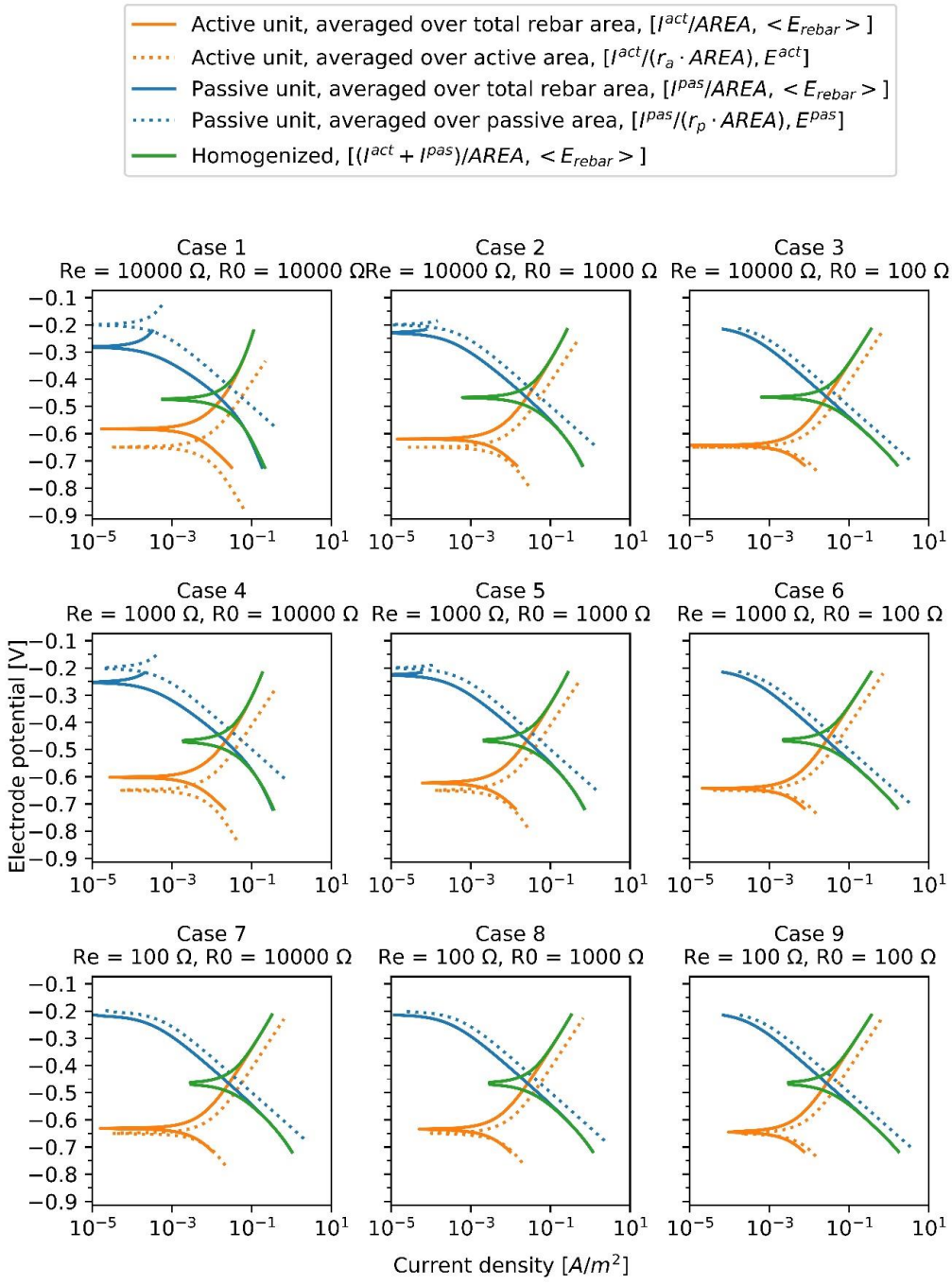


Figure 5.6 The influence of resistance on simulated polarization scan curves at "unit level" and homogenized "high-level".

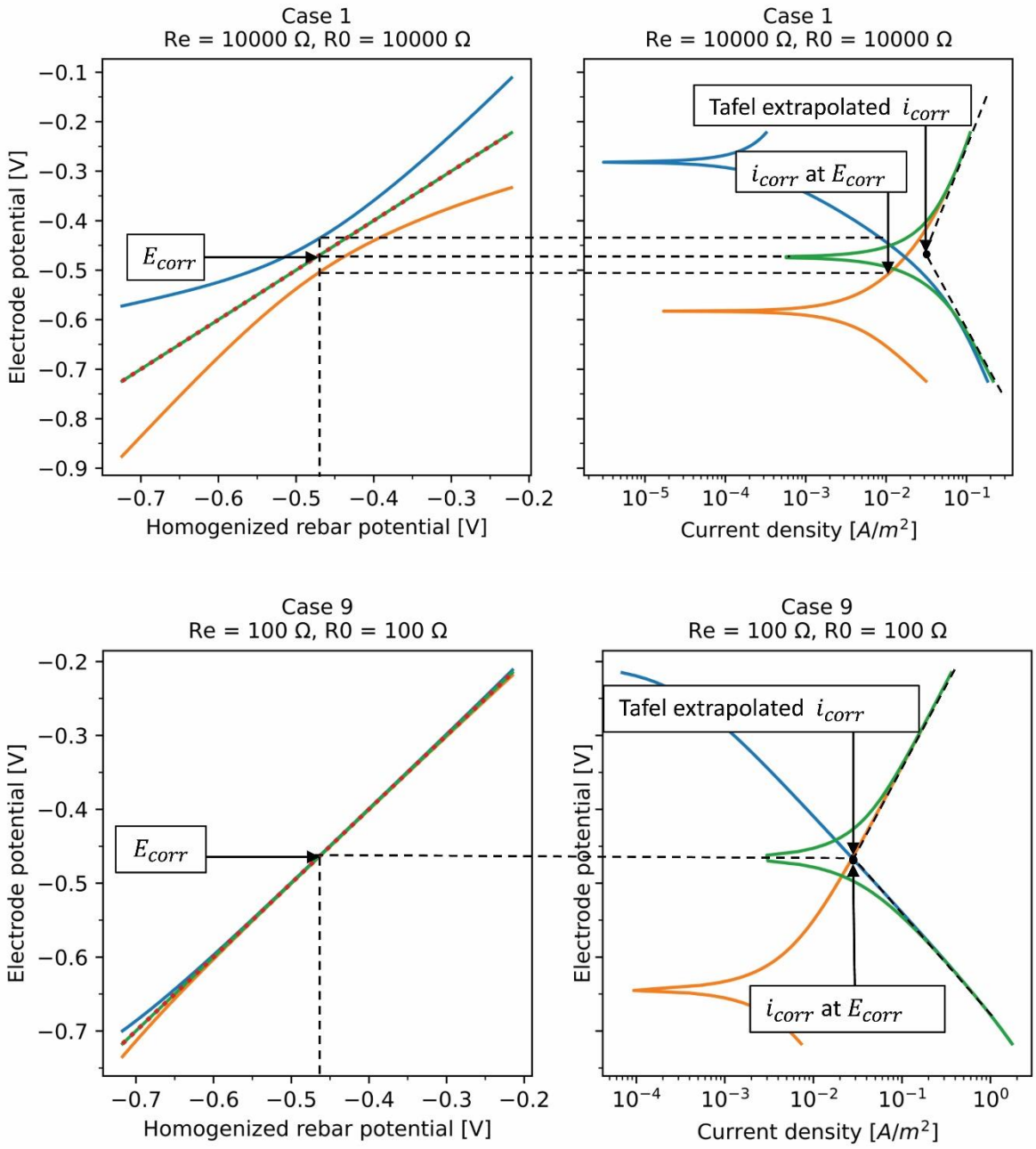
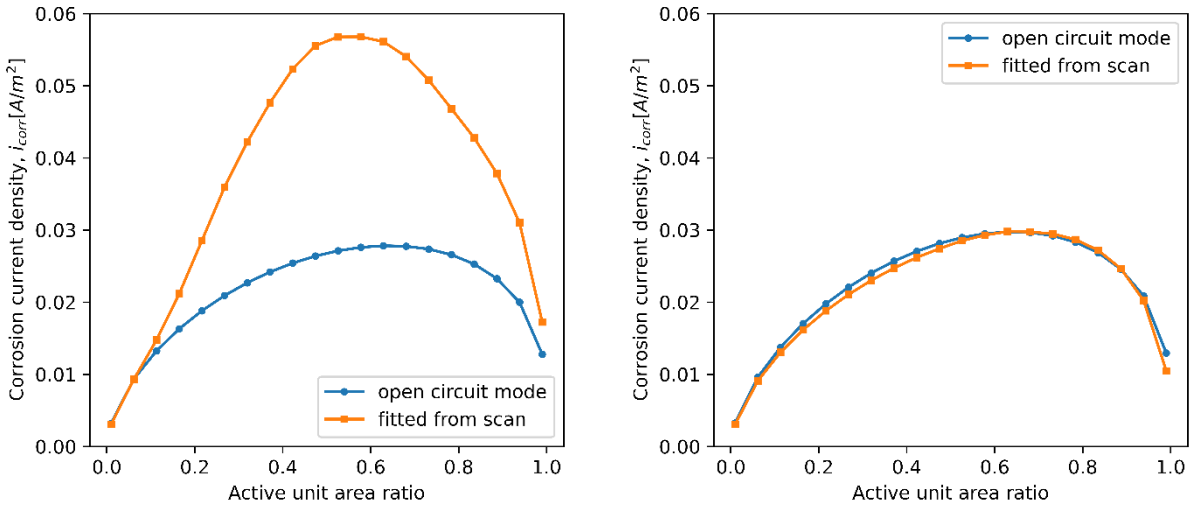


Figure 5.7 Tafel extrapolation on a distorted polarization curve: Case 1- significantly distorted and Case 9- insignificantly distorted (same line legend as Figure 5.5 and Figure 5.6, omitted for simplicity).

5.4.4 Effect of the area ratio of activated and passivated rebar area and resistance on the fitted corrosion parameters

To fully investigate how the distortion of the scan curve affects the fitted corrosion parameters (E_{corr}, i_{corr}, β_a, β_c), the scanned potential-current density curves were fitted with the Butler-Volmer equation using a non-linear regression method (Li, Evitts, Boulfiza, et al., 2018). The active area was set to have a ratio ranging from 0.01-0.99, and the remaining area was passive. Two values of resistance were used: R_0, R_e = 1000 Ω to represent concrete with higher resistivity and R_0, R_e = 10 Ω to represent concrete with lower resistivity.

The fitted i_{corr} from the scan curve is compared with i_{corr} from the open-circuit mode. As shown in Figure 5.8a, i_{corr} has a significant overestimation when the resistance is high especially when the active and passive area is comparable. When the resistance is low, the fitted i_{corr} does not significantly deviate from the actual i_{corr}value. (Figure 5.8b).



(a) (b)

Figure 5.8 Corrosion current density,i_{corr}, from the open circuit mode and scan curve fitting with varying active unit area ratio. a) high resistance: R_0, R_e =1000 Ω, b) low resistance: R_0, R_e = 10 Ω.

Figure 5.9 shows a plot of the fitted curve that gives the i_{corr} with the largest difference from the open-circuit model for the case when the active area/ total area is 0.5. The distorted curve causes greater Tafel slopes which further leads to a significant overestimation of i_{corr}.

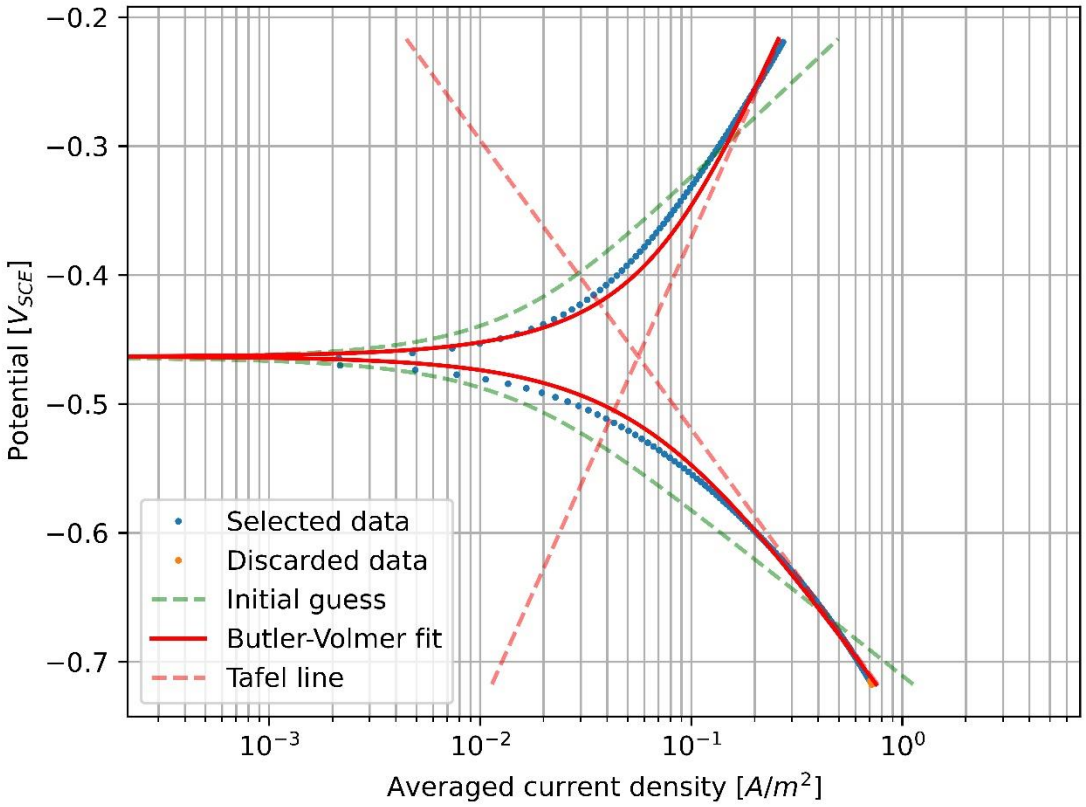
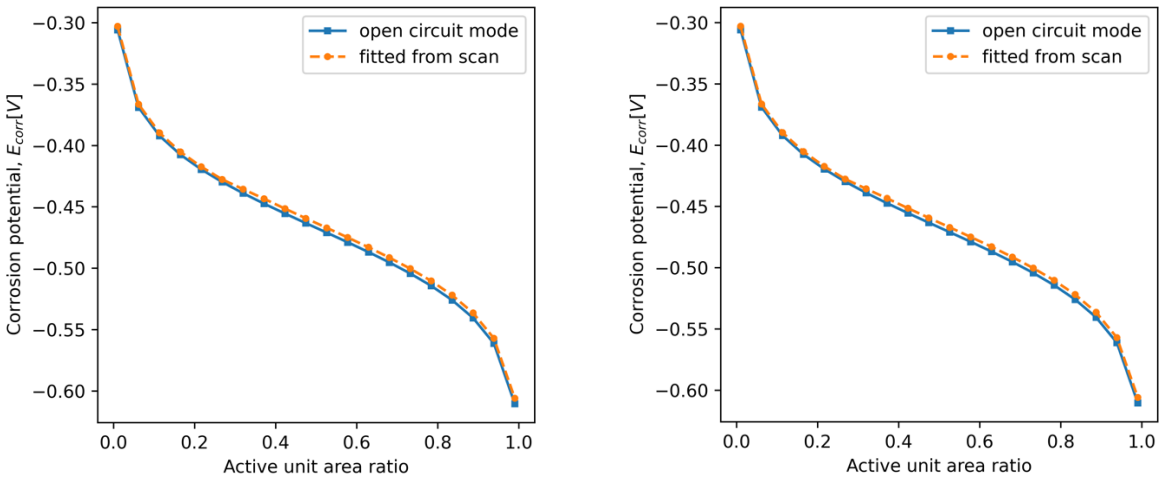


Figure 5.9 Curve fitting with the Butler-Volmer equation that showed the largest deviation in i_{corr}. (R_0, R_e = 1000 Ω, active unit area/total area is 0.5, data range: $E_{corr} \pm 0.25$ V).

However, the resistances between the rebar units and through the concrete cover do not significantly affect the corrosion potential, E_{corr}. The corrosion potential is mainly affected by the area ratio of the active or passive unit to the total area, as is shown in Figure 5.10.



(a) (b)

Figure 5.10 Corrosion potential, $E_{corr,}$ from the open circuit mode and scan curve fitting with varying active unit area ratio. a) high resistance: R_0, R_e =1000 Ω, b) low resistance: R_0, R_e = 10 Ω.

The modelled Tafel constants of the rebar at "high level" depend on both the resistance and active or passive area ratios. As shown in Figure 5.11, for a large concrete resistance, the apparent anodic Tafel constant of the high-level rebar obtained from the scan curve is greater than the anodic Tafel constant of the active unit (β_a^{act}).

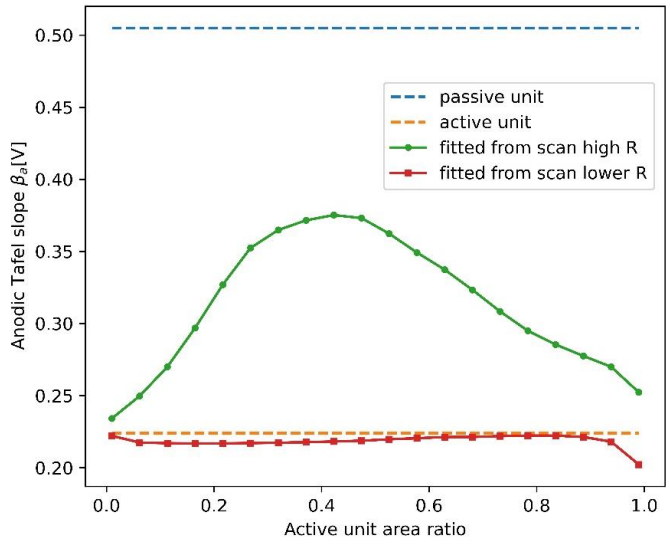


Figure 5.11 Effect of the active unit area ratio and resistance on the fitted anodic Tafel slope.

The difference reaches its maximum when the areas of the active and passive units are comparable. However, when the resistance between the active and passive units is low (10 **Ω**), the two Tafel constants are similar, regardless of the area ratios. This phenomenon explains why the anodic Tafel constant measured in concrete could be greater than the value in solutions.

The apparent cathodic Tafel constant (Figure 5.12) is also significantly affected by the area ratio of active unit to total rebar surface, when R_o, R_e are large (1000 **Ω**). At greater area ratios of the active unit to the total area, the contribution to the reduction of oxygen on the active unit becomes more important, which causes the cathodic Tafel constant to attain values that are close to that of the active units, β_c^{act}. However, when the active unit area is greater than 90% of the total rebar area, the cathodic Tafel constant changes in a direction towards the value of the cathodic Tafel slope of the passive unit, β_c^{pas}.. For the lower resistance case. The influence from the cathodic Tafel slope of the active unit is not significant unless of all the rebar surface is an active unit.

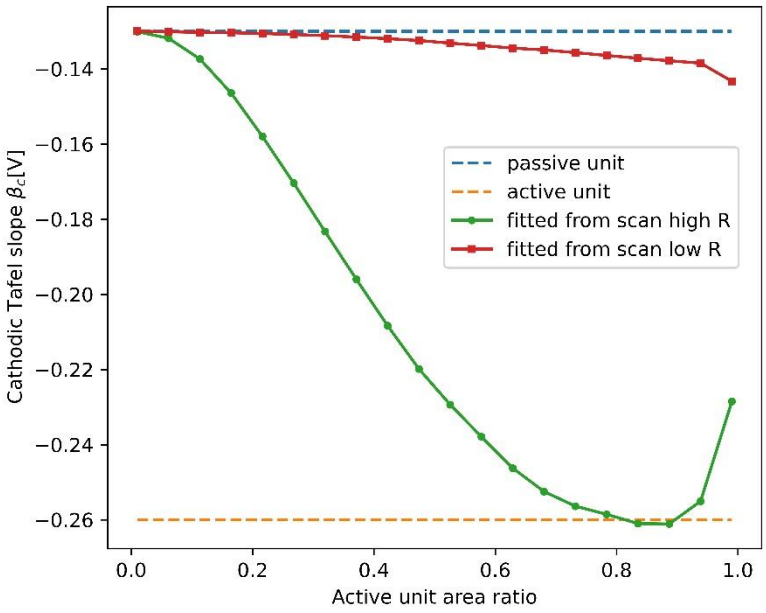


Figure 5.12 Effect of the active unit area ratio and resistance on the fitted cathodic Tafel slope.

In summary, when the concrete resistivity is low (low R_0 and R_e), the anodic Tafel constant is similar to that of the active unit, and the cathodic Tafel constant is similar to that of the passive

unit except when there are nearly no passive units (passive unit ratio = 0.01). For this case, a single set of Tafel slopes ($\beta_a^{act}, \beta_c^{pas}$) can be used for the rebar at the "high-level". However, when the concrete resistivity is high, the value of Tafel constant greatly depends on the area ratio of the active or passive area to that of the whole rebar. The combined effect from the resistance and non-uniformity (active/passive area ratio) explains why measured anodic Tafel constants are significantly larger and have a large scatter for concrete under partially saturated conditions (400-1570 mV or even infinite)(Alonso et al., 1998b; Locke & Siman, 1980). It is also worth mentioning that compared to the anodic Tafel constant changes, the increase in the cathodic Tafel constant is significantly smaller (180-480 mV) (Alonso et al., 1998b; Chang et al., 2008b; Locke & Siman, 1980). This is because the fitted cathodic Tafel constant falls between the cathodic Tafel constant of the basic active/passive units, which are not as different as the anodic ones.

5.5 Conclusions

An electrical circuit model was developed to simulate the potentiodynamic scan of a non-uniform reinforced concrete to capture the homogenized polarization behaviour at a "high-level" scale. A parametric study was carried out to capture the effect of the electrical resistance of the concrete cover on the homogenized polarization curve.

It is concluded that the area-averaged localized potential (homogenized potential) of the rebar is equivalent to the IR-compensated potential measured from the concrete cover. The difference between the localized potentials of the active and passive rebar portions may cause the apparent polarization curve to significantly deviate from the polarization curve of a rebar coupon in synthetic pore water that is described by the Butler-Volmer equation.

When fitted to the Butler-Volmer equation, the distorted scan curve exhibits larger Tafel constants, which leads to an overestimation of corrosion current density and corresponding corrosion rate. Such changes are more pronounced when the electrical resistance of the concrete is higher, especially when the active and passivated rebar units have comparable surface areas. The corrosion potential is not significantly affected by this distortion.

The homogenized corrosion parameters are thus a function of concrete resistance, and the area ratio of the active or passive rebar surface area. These factors cause the deviation of the

homogenized corrosion kinetics from those described by the Butler-Volmer equation for coupons immersed in a solution. When the deviation is small, the kinetics represented by homogenized corrosion parameters can be used in practical large-scale modelling when the non-uniform rebar is treated at the "high-level" scale. However, when the deviation is large, the potentiodynamic curves representing the kinetics for the anodic and cathodic reactions should be obtained directly from scans of reinforced concrete samples representative of field conditions or calculated using an averaging technique as per this model.

Similar to the polarization curves described by Butler-Volmer kinetics, the total anodic and total cathodic curve predictions approach the overall polarization curve when the overpotential from of the homogenized corrosion potential is large. Therefore, an experimental potentiodynamic scan curve can be used to represent the corrosion kinetics of the homogenized rebar at large overpotentials and often the directly accessible data from the experiment is the overall polarization curve (green curve, Figure 5.6). However, because the total polarization curve shows the net current density, it does not quantify the internal current density within the modelled rebar segment, which could be significant when the overpotential is small. For example, the current is near zero when the potential is near E_{corr} on the overall polarization curve. Therefore, for a corrosion model that has the capability of handling a full range of rebar potential, the complete anodic (orange curve, Figure 5.6) and cathodic (blue curve, Figure 5.6) curves need to be determined from the overall potential dynamic scan curve.

5.6 Limitations and suggestions for future works

In this model, the counter electrode on the concrete cover is used as the means of excitement providing or extracting electrons to get the overall polarization curve, which resembles the typical setup of potentiodynamic scan test and some cathodic protection configurations. However, the equivalent resistors used in this circuit model are affected by the concrete cover geometry and the setup location of the counter electrode. Therefore, when the homogenized corrosion kinetics are required for rebar with a different geometry, for example, a coupled neighbouring rebar, the value of the equivalent resistance may change which affects the homogenized corrosion kinetics. More investigation needs to be done to quantify this effect.

5.7 Epilogue

Chapter 5 introduced the concept of homogenized corrosion parameters for rebar in concrete with non-uniform surface conditions, utilizing a circuit model to simulate the polarization behaviour. The chapter emphasized the necessity of considering corrosion parameters as a function of microenvironments and the potential deviations caused by non-uniform surface conditions and concrete resistivity. This deviation is highly dependent on the area ratio of active passive rebar-concrete interface units and the resistance between them at the meso-macroscopic scale, which calls for a zoomed-in homogenization from the pore scale. In the upcoming chapters, we will delve further into the numerical simulation phase, utilizing the insights from experimental studies in Chapter 3 and Chapter 4 and the corrosion circuit model in Chapter 5 to explore the link between homogenized corrosion parameters with the localized corrosion parameters at the pore scale.

6 A Pore-scale Corrosion Model for Rebar in Concrete: the Effect of Water and Chloride Distribution at the Pore Level on Rebar Corrosion in Concrete

6.1 Preface

In the preceding chapters, we explored the concept of homogenized corrosion parameters for rebar in concrete, emphasizing the importance of considering microenvironments and the deviations caused by non-uniform surface conditions and concrete resistivity. Chapter 5 introduced a circuit model that simulated the polarization behaviour and highlighted the need for zoomed-in homogenization from the pore scale.

Building upon the insights from experimental studies conducted in Chapter 3 and Chapter 4 and the homogenization method with the corrosion circuit model from Chapter 5, Chapter 6 explores the mechanistic link between homogenized corrosion parameters and localized corrosion parameters at the pore scale.

The introduction of this chapter discusses the differences between rebar corrosion in bulk pore solution and concrete, particularly in terms of non-uniform chloride distribution and varying chemical environments at rebar-pore solution interfaces.

By selecting the representative elementary volume (REV) based on pore structure, a model is developed to quantitatively describe the allocation of active and passive rebar-solution corrosion sites at the representative elementary volume based on the non-uniform water and chloride distribution in the concrete. The model describes how water and chloride ions are distributed given the total water content and total chloride content taking into account the water adsorption-condensation phenomena, chloride binding effect, and chloride concentrate effects due to the electrical double layer. The corrosion circuit model from Chapter 5 is then employed to simulate the homogenized corrosion parameters of the rebar-concrete interface in REV from the corrosion parameters of the rebar-solution interface at the pore level. The simulated corrosion parameters capture the observed pattern from the empirical functions from the experimental data.

The homogenized corrosion parameters as a function of the environmental conditions can be applied to every point on the rebar surface with surrounding concrete of non-uniform microenvironment caused by cracking of concrete and varying exposure conditions.

The manuscript of this chapter will be submitted to “Construction and Building Materials”. Gang Li is the primary author who designed and performed the experimental study and data analysis. Dr. Moh Boulfiza and Dr. Richard Evitts contributed to the review of the manuscript and provided feedback.

6.2 Introduction

Rebar corrosion in concrete is not uniform. It is related to but quite different from rebar corrosion in bulk pore solution. Chloride contents in the pore solution at the steel-concrete interface are not uniform due to more restricted diffusivity. This non-uniformity leads to different non-uniform corrosion behaviour in concrete than bulk solutions.

In bulk solutions, actively corroding rebars have closely allocated microscopic anodic and cathodic sites with negligible resistance between them. These micro-sites are treated as uniform corrosion, and both the anodic and cathodic reactions happen on the same surface, with the corrosion current averaged over the whole steel-solution interface. In concrete, however, although the chloride test may suggest uniform chloride content with a specific sampling size, the steel surface may be exposed to pores with varying chloride concentrations. Therefore, the rebar-pore solution interfaces may have very different chemical environments and hence different corrosion states. If electrically isolated, the rebars would corrode at their own corrosion rates. However, they are all electrically connected with non-negligible resistance in concrete.

Suppose we still treat them as uniform corrosion by mathematically describing the anodic and cathodic reactions happening all over the steel-concrete interface, which is the basis for the commonly used electrochemical tests with rebar in concrete. In this case, several consequences must be considered: Firstly, the averaged corrosion kinetics would be sensitive to the rebar-pore solution interface area and the localized chemistry in pores. This limitation would make the averaged corrosion kinetics less generalizable when applied to concrete with different spatial moisture content and pore solution compositions commonly found in service environments.

Secondly, the possibly non-negligible resistance of the concrete cover also affects the overall corrosion behaviour making the test results unreliable. The previous circuit model (Chapter 5) shows that when non-uniform rebar is treated as a homogenized rebar piece, the homogenized

activation-controlled corrosion kinetics may deviate from the typical Butler-Volmer kinetics of a mixed electrode reaction where the resistance between the anodic and cathodic sites are not negligible. This deviation can be seen by the distorted polarization curve with a greater icorr and apparent Tafel slopes when the polarization curve is forced to be fitted using Butler-Volmer, or its Tafel extrapolation. The distortion is more pronounced when the concrete resistivity is high and the rebar severely non-uniformly corrodes. It suggests that the rebar's homogenized corrosion kinetics depends on the non-uniformity of the rebar and the resistance between the non-uniform rebar segments.

In the previous lab studies, we conditioned the rebar-embedded concrete samples with the known surface areas to a specific moisture content, chloride content, and carbonation condition in a lab (Chapter 4). For each given condition, the homogenized corrosion rate was found quite different from what is measured in the bulk solution, even after the correction of the rebar-pore solution area. These results suggest that the concrete pore structures alter the local electrochemistry process by potentially changing the active area ratio, the rebar-pore solution area, and the resistance between the anodic and cathodic sites.

The corrosion kinetics for rebar in concrete at a macroscopic level is of great interest from the service life prediction and corrosion mitigation perspectives. Without exercising experimental tests as was done in Chapter 3 and Chapter 4 with exhausting many major test factors, it is very beneficial to figure out the mechanistic link of the rebar corrosion kinetics (including its polarization behaviour) between bulk solution and concrete. The results from the corrosion test of rebar in a bulk solution can be translated into the rebar corrosion behaviour in concrete.

In this paper, the representative elementary volume (REV) will be selected based on the pore structure. A comprehensive model will be developed to link the rebar corrosion behaviour in the pore solution to the overall corrosion behaviour in the REV. The model will describe how water and chloride distribution in the concrete affects the distribution of the active and passive rebar corrosion sites at the REV scale.

6.3 Methodology

6.3.1 Homogenized corrosion kinetics

The concept is to seek for the averaged corrosion behaviour of rebar in the porous medium. For instance, in chloride-contaminated concrete, the localized pore solution may have a non-uniform chloride concentration, resulting in non-uniform corrosion states of rebar (Figure 6.1). The macroscopic representation of these characteristics involves treating the porous medium as a continuum with a selected representative elementary volume (REV). The goal is to smooth out the localized "spikiness" and obtain REV-level "smoothness".

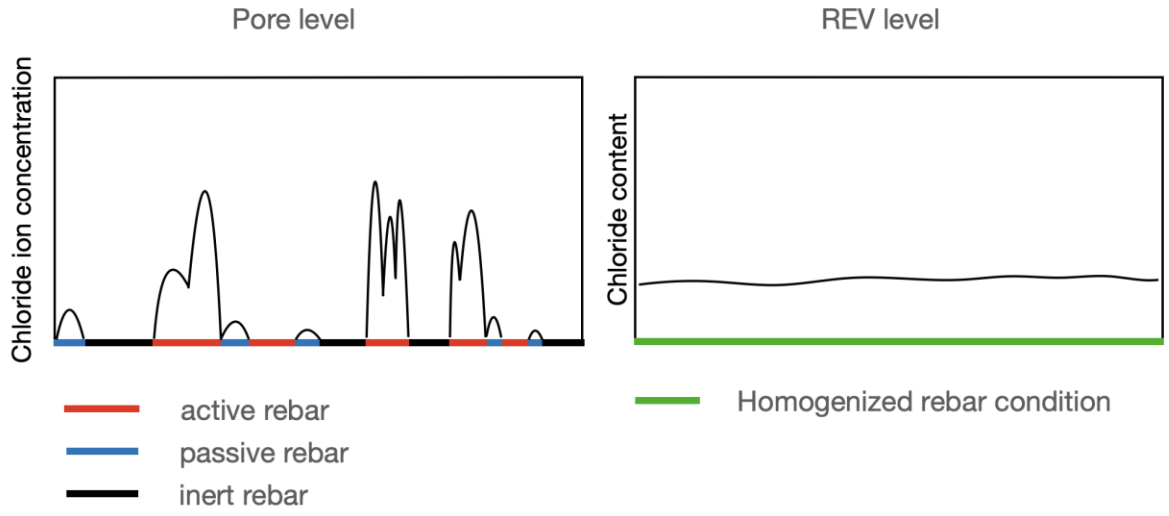


Figure 6.1 Schematic of rebar condition and chloride concentration distribution: localized pore level non-uniformity vs. REV-level homogenization.

We usually do not have access to the microscopic chloride concentration profile on the rebar-pore solution interface. However, the averaged total chloride content from the regular drilling test (ASTM C1152) provides the chloride content level. This is a volume-averaged value and is related to the localized non-uniform chloride concentration. Similarly, the averaged corrosion characteristic is also associated with the rebar-solution interface in the pores in certain respects.

However, rebar corrosion kinetics cannot be treated the same way as other properties such as chloride content. Simply arithmetically averaging the values such as potential and current density, assuming localized uniform corrosion with averaged anodic and cathodic reactions over the whole surface does not reflect the interactive coupling of localized corrosion kinetics. It is not well

understood how this “averaged kinetics” is related to localized corrosion kinetics. It is questionable if it is even valid to treat rebar corrosion kinetics in a homogenized uniform fashion with the same Butler-Volmer kinetics used in the metal-solution system. In the previous circuit model analysis (Li, Evitts, & Boulfiza, 2018), it has been demonstrated that the homogenized kinetics may significantly deviate from the local kinetics (Butler-Volmer equation, activation control) at given conditions when the resistivity is high.

The overall polarization performance or corrosion kinetics can be directly measured in experiments, as shown by previous studies (Alonso et al., 1998b; Chang et al., 2008b; Elsener, 2005; Garcés et al., 2005; Gulikers, 2005; Locke & Siman, 1980; Sohail, 2013). The homogenized kinetics then become empirical functions of the material properties and chemical species conditions (see Appendix for Chapter 4). It will be tedious and prohibitively time-consuming to capture even a few factors (Chapter 3 and Chapter 4). If we can manage to model the underlying phenomenon that mechanistically controls the overall corrosion kinetics, then we can simulate all the scenarios, which provides a link between the rebar corrosion in pore solution and the rebar corrosion in concrete.

To link the activation-controlled corrosion kinetics in solution to that in concrete, some key differences between these environments need to be considered due to the distribution of water and chloride in the pore system.

- the reactive rebar-solution surface area
- distribution of the active and passive rebar-solution sites
- ionic resistance between the active and passive corrosion sites

It is worth mentioning that the restrained oxygen diffusivity can be captured with a limiting current at a macroscopic level. Therefore, it is not taken into account at the pore level.

6.3.2 Corrosion modelling from microscopic (pore-scale) to mesoscopic scale (REV)

In the proposed model (Figure 6.2), three types of rebar-steel interfaces are considered: “active”, “passive”, and “inert”. The rebar-solution interface is assumed to be “reactive,” with either an active or passive state. Both anodic and cathodic activation-controlled kinetics occur on the same

area on each interface in a fashion of uniform corrosion in the bulk solutions. The rebar surface exposed to the drained pores or solid matrix, i.e., rebar-gas or rebar-solid interface, is assumed to be inert. The homogenized steel-concrete interface of a representative elementary volume (REV) is a combination of these interface types: active rebar-solution, passive rebar-solution, and inert. The distribution of the three types of interfaces would be affected by pH, Cl, water, and pore size distribution at the rebar-concrete interface.

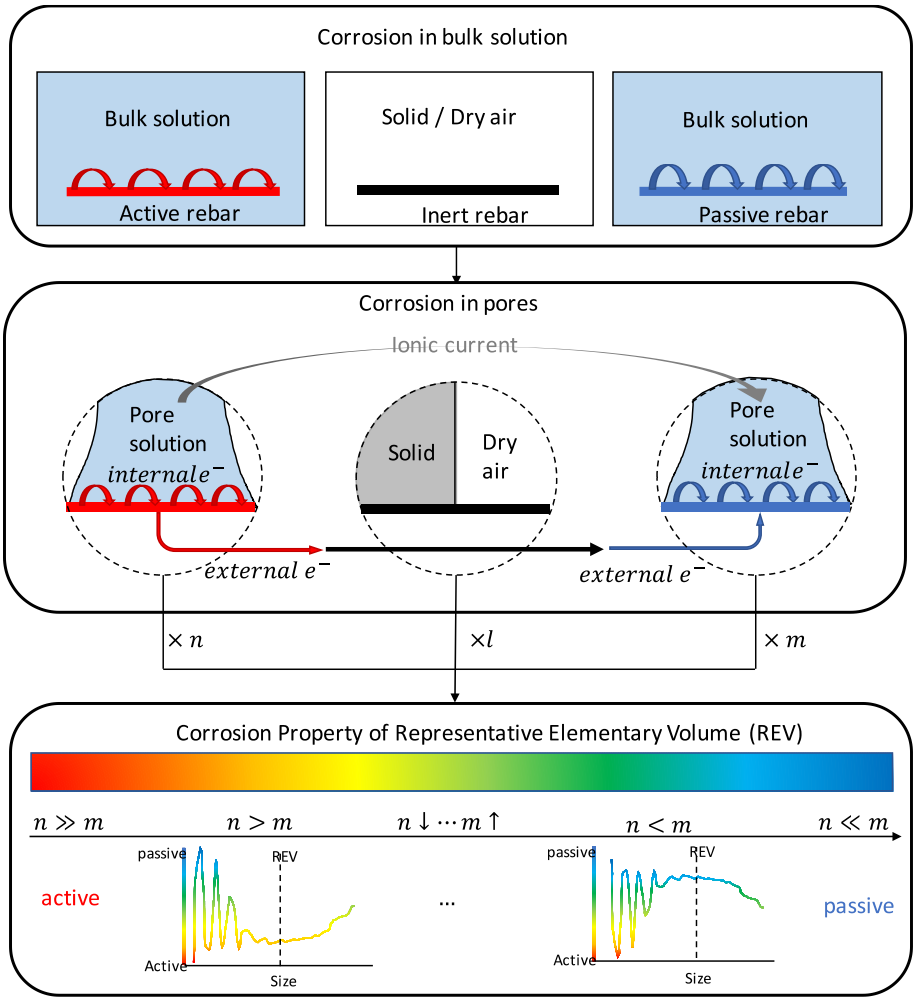


Figure 6.2 A schematic depicting the corrosion of the rebar surface, from pore level to REV. The rebar condition within the REV encompasses a range of conditions, whereas at the pore level, it is binary.

The proposed model, as illustrated in Figure 6.3, starts with quantifying the pore size distribution, which forms the basis for deriving the moisture and chloride distribution based on the known total moisture and chloride content. The moisture and chloride conditions are then used to define the

corrosion cell at the REV scale. Finally, the corrosion cell is used to generate a potentiodynamic scan that characterizes the corrosion kinetics.

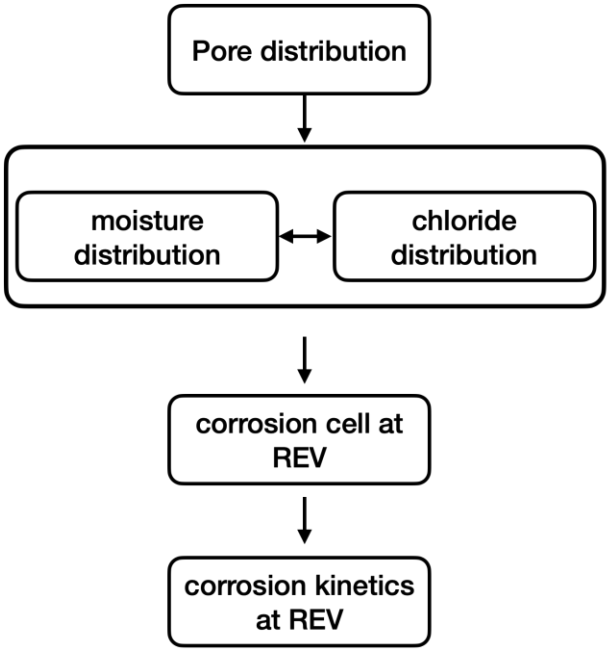


Figure 6.3 Proposed model for characterizing corrosion kinetics in reinforced concrete: schematic representation of the process flow from pore size distribution to corrosion kinetics.

6.4 Pore model development

6.4.1 Pore size distribution and water distribution

The quantification of the pore-water-chloride system starts with determining the pore size distribution. Water is distributed in the pores as adsorbed water, capillary-condensed water, and vapour. The pore distribution can be determined using the volumetric water content retention curve and the classic adsorption-and-condensation model, based on the bundled capillary pore model (Daian, 1988). In this section, we implement this method with improvements in mass conservation, which are solved through an iterative algorithm.

As illustrated in Figure 6.4, all pores in the hardened cement paste can be conceptually connected into one single pore, with the pore diameter varying across the whole range of pore sizes. The pore size distribution can be quantified by the area function $S(r)$ representing the surface area of pores with radii greater than r per unit volume of the porous medium. Equivalently, it can also be

quantified by the volume function $\eta(r)$ which represents the pore volume of the pores with radii smaller than r per unit volume of the porous medium.

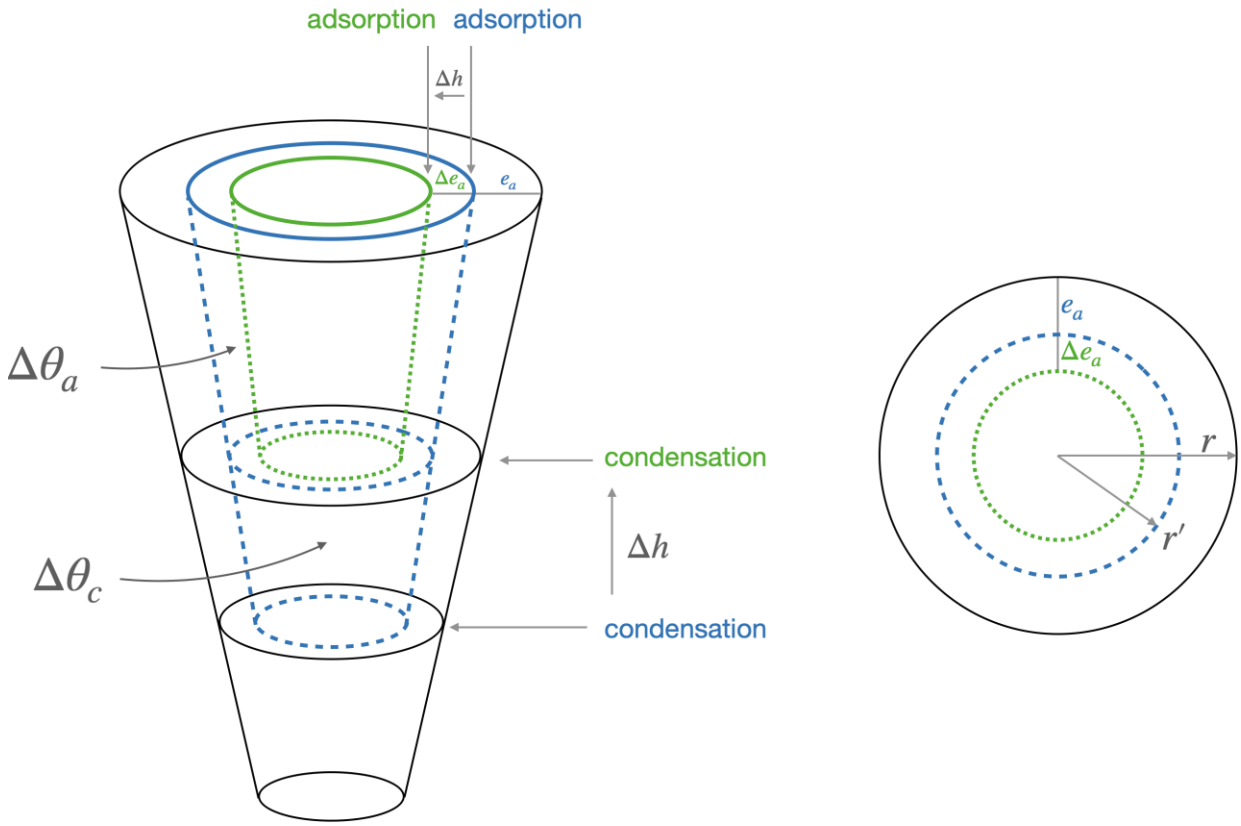


Figure 6.4 Adsorption-condensation model of a capillary pore: Illustration of the conceptualization of pores as a single pore with varying diameter. The change in adsorbed water ($\Delta\theta_a$) and condensed water content ($\Delta\theta_c$) is due to an increase in relative humidity (Δh).

When the water vapour is neglected, the total volumetric water content contains two parts: adsorbed water and condensed capillary water. In the relative humidity (RH) range from zero to near one, there are two regimes: "pure adsorption" and "adsorption + condensation" (Figure 6.5).

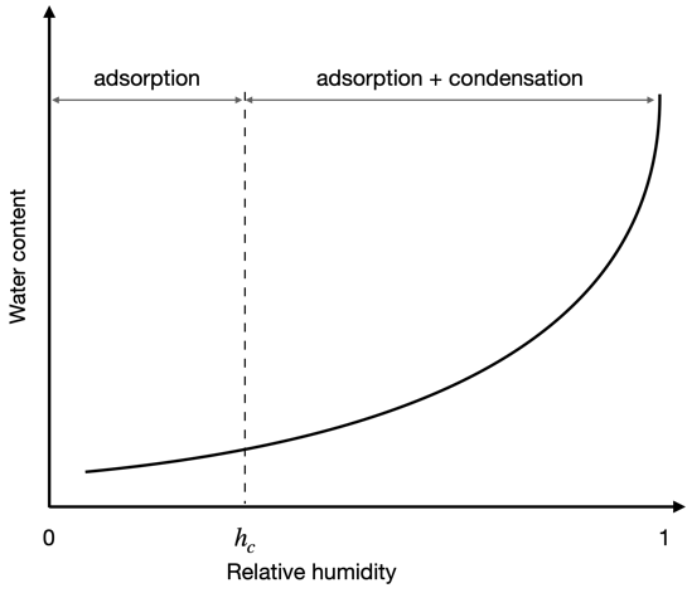


Figure 6.5 Schematic depicting the transition from adsorption to adsorption + condensation regime as relative humidity increases in pores.

6.4.1.1 Pure adsorption regime

In this regime, the adsorbed water is the only contribution to the total water content, given by the equation:

$$\theta = e_a(h) \cdot S_0 \tag{6.1}$$

where, $e_a(h)$ is the water adsorption thickness as a function of the relative humidity,h, S_0 is total pore surface area per unit volume, which equals the maximum value of $S(r)$ when r is taken as the minimum value. In this paper, the value $1.5 \times 10^7\ m^2/m^3$ (Daian, 1988) will be used as an initial guess for S_0 in the iterative calibration process with experimental data.

The volumetric water content θ attributed by pure adsorption is linearly related to the logarithm of the metric potential of capillary water (Daian, 1988):

$$\log|\Phi| = q_{fit} - s_{fit} \cdot \theta \tag{6.2}$$

where, q_{fit} and s_{fit} are fitted parameters and the metric potential of capillary water Φ is expressed by Kelvin-Laplace Equation:

$$\Phi = p_c = \frac{\rho_l RT}{M} \ln(h) \tag{6.3}$$

where, ρ_l is density of water, M is molar mass of water. This linear relationship Equation (6.2) was used to fit the isotherm in the low RH region (adsorption region), and the results were used to construct the water adsorption thickness function $e_a(h)$. It is assumed this relationship is also valid at higher RH where capillary condensation occurs.

6.4.1.2 The transition from pure adsorption to adsorption + condensation

The capillary condensation is caused by the pore water's concave meniscus where, at equilibrium state, the water vapour partial pressure above this meniscus is less than that above a flat water surface. The relative humidity, h is related to the meniscus curvature by Kelvin's Equation:

$$\ln(h) = \ln\left(\frac{p_v}{p_{sat}}\right) = -\frac{2\gamma V_m}{r_{curve} RT} \tag{6.4}$$

where $r_{curve}[m]$ is the radius of the meniscus curvature (negative value for concave meniscus). $p_v[Pa]$ is vapour pressure, $p_{sat}[Pa]$ is saturated vapour pressure above a flat surface at temperature $T[K]$, $\gamma[N/m]$ is surface tension, $V_m[m^3/mol]$ is molar volume, R $[J/\mathrm{K}/\mathrm{mol}]$ is gas constant.

$$r_{curve} = -\frac{2\gamma V_m}{\ln(h) RT} \tag{6.5}$$

r_{curve} is related to the pores size, r by $r_{curve} = r/cos\alpha$. The parameter α is the contact angle determined by the type of liquid and the solid surface. r_{curve} equals the capillary pore radius when the contact angle is 0, which is often used but not experimentally determined. It is more realistic

to assume a contact angle of 60 degrees as derived from (Daian, 1988). The critical pore radius, $R(h)$ for condensation to occur at the relative humidity of h is

$$R(h) = r_{curve} \cdot cos60^0 = -\frac{\gamma V_m}{\ln(h) RT} \tag{6.6}$$

Considering the thickness of adsorbed water, the pores with a radius

$$r < R(h) + e_a(h) \tag{6.7}$$

will be filled with condensed water. The critical RH, h_c, at which the condensation occurs in the smallest pores.

$$r_{min} = R(h_c) + e_a(h_c) \tag{6.8}$$

The transition from pure adsorption to adsorption-condensation happens at a critical RH which depends on the smallest pore size and the adsorption thickness function $e_a(h)$. the critical condensation RH, h_c, and the adsorption function $e_a(h)$ can be solved using an iterative algorithm, given the smallest pore size, r_{min}, and the experimental $\theta - h$ curve.

6.4.1.3 Adsorption + condensation regime

In this regime, as shown in Figure 6.4, for a small increase in RH, Δh, there would be additional condensed water, $\Delta\theta_c$, in empty pores with a corresponding pore radius r':

$$\Delta\theta_c = -\Delta S' \cdot \frac{r'}{2} \tag{6.9}$$

where the $\Delta S'$ is the inner surface change of the adsorbed water layer in the unsaturated pores corresponding to the condensed volume change and is a bit smaller than the unsaturated pore surface change ΔS.

$$\Delta S' = \Delta S(r - e_a)/r \tag{6.10}$$

$$r' = r - e_a \tag{6.11}$$

where e_a is the thickness of the adsorbed layer (e_a is a function of RH, h)

In the unsaturated pores, there is an increase of adsorbed water $\Delta\theta_a$ due to the increased adsorbed water layer thickness Δe_a. The $\Delta\theta_a$ can be derived based on two types of geometry approximations.

1) Thin water layer treated as sheet-like

$$\Delta\theta_a = S(r) \cdot \Delta e_a \tag{6.12}$$

r is the pore radius, ΔS is the change of the empty pore surface area per unit volume.

$$\Delta\theta = \Delta\theta_c + \Delta\theta_a \tag{6.13}$$

$$\Delta S = -\frac{2r}{(r - e_a)^2}[\Delta\theta - S(r) \cdot \Delta e_a] \tag{6.14}$$

2) Alternatively, if the adsorbed thin water layer is treated as pipe-like, which is more accurate when the adsorbed water layer thickness is comparable to the pore size, the increased adsorbed water content becomes dependent on the radius.

$$\Delta\theta_a = \sum_{r}^{r_{max}} \frac{-\Delta S'(r)}{r'}[r'^2 - (r' - \Delta e_a)^2] \tag{6.15}$$

where $\Delta S'$ is the area of the interior surface of adsorbed water in pores with radii greater than r per unit volume of the porous medium and r' is the inner radius of pipe-like adsorbed water layer.

$$\Delta S'(r) = \Delta S(r - e_a)/r \tag{6.16}$$

$$r' = r - e_a \tag{6.17}$$

again recall

$$\Delta\theta = \Delta\theta_c + \Delta\theta_a \tag{6.18}$$

$$\Delta S = -\frac{2r}{(r-e_a)^2}\left[\Delta\theta - \sum_{r}^{r_{max}} \frac{-\Delta S(r-e_a)}{r(r-e_a)}[(r-e_a)^2 - (r-e_a-\Delta e_a)^2]\right] \tag{6.19}$$

The change of saturated pore volume per unit volume of the porous medium can be expressed as

$$\Delta\eta = \Delta S \cdot r/2 \tag{6.20}$$

$\eta(r)$ is the pore volume per unit volume of porous medium with a radius less than r. The volumetric water content can be expressed with

$$\theta = \eta[R(h) + e_a(h)] + e_a(h) \cdot S[R(h) + e_a(h)] \tag{6.21}$$

Boundary conditions:

Smallest pore, r_{min} is assumed as 1 nm according to a MIT test (Daian, 1988).

$$\eta(r_{min}) = 0 \tag{6.22}$$

$$S(r_{min}) = S_0 \tag{6.23}$$

S_0 is calibrated so that the resultant $S(r_{max})$ is positive and approaching zero.

The $S(r)$ is solved iteratively by reverse modelling Equation (6.21) to the experimental water retention data, i.e., $\theta - h$ curve. The numerical solving algorithmic procedure is presented in Figure 6.6.

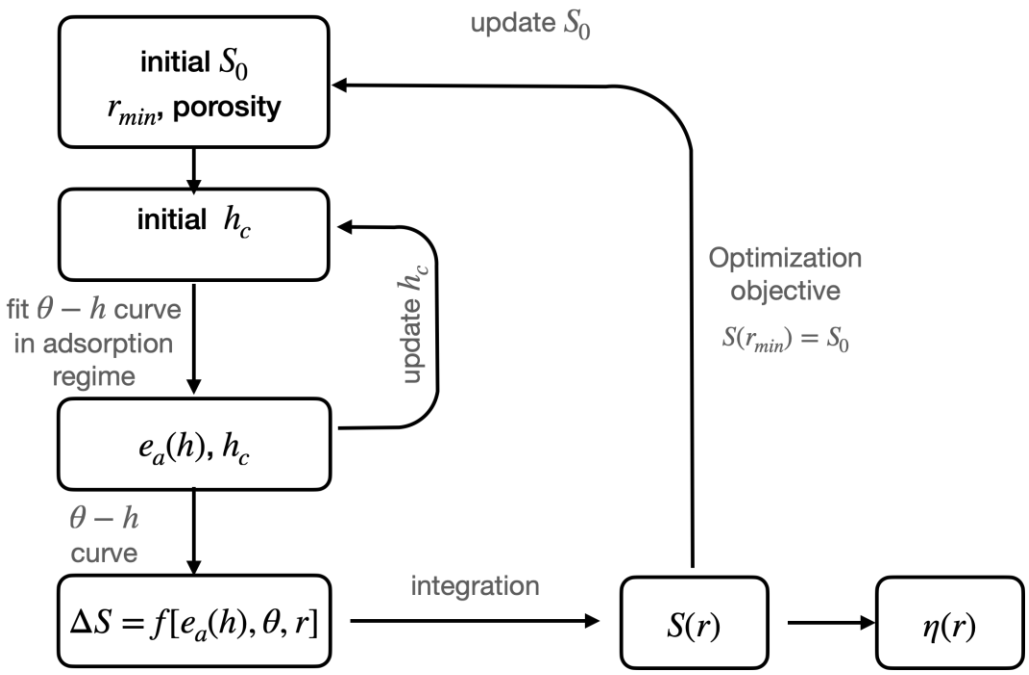


Figure 6.6 Flowchart of the iterative solving procedure to determine the pore distribution from the water retention curve.

In summary, the following improvements has been made compared to (Daian, 1988):

1) In the adsorption regime, the critical RH is calculated iteratively with the fitted adsorption function accounting for the reduction in pore size caused by adsorbed water, which results in a lower critical RH.
2) In the condensation regime, the altered geometry of the adsorbed water volume is treated as pipe-like, rather than sheet-like. The effective pore size changes dynamically with the

adsorbed water layer, leading to a more precise calculation of the condensed water volume change within the pores.

6.4.1.4 Pore size distribution results

The solution of pore size distribution in the form of $\boldsymbol{S(r)}$ or $\boldsymbol{\eta(r)}$ are plotted in **Figure 6.8.**

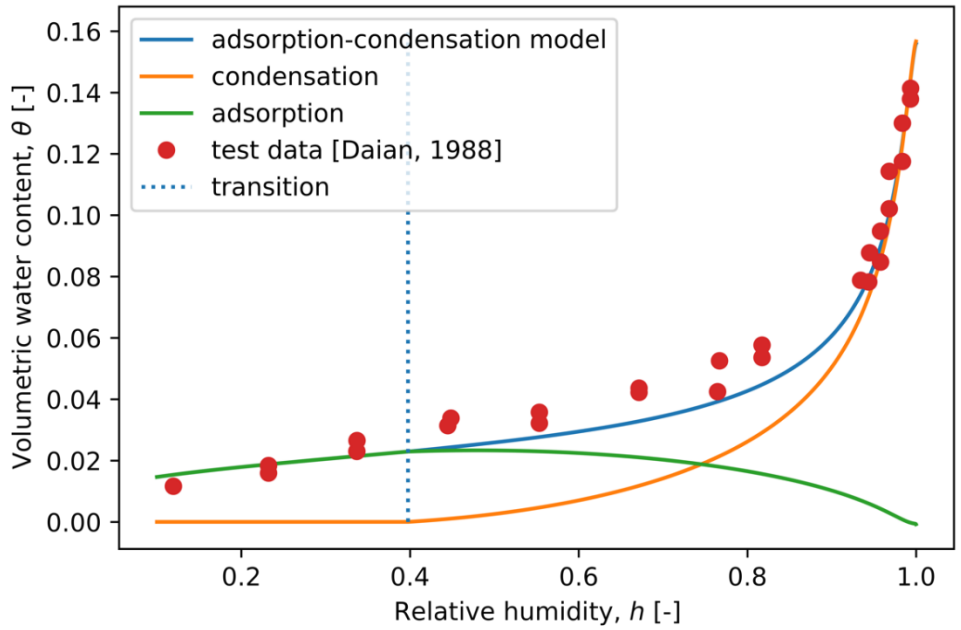
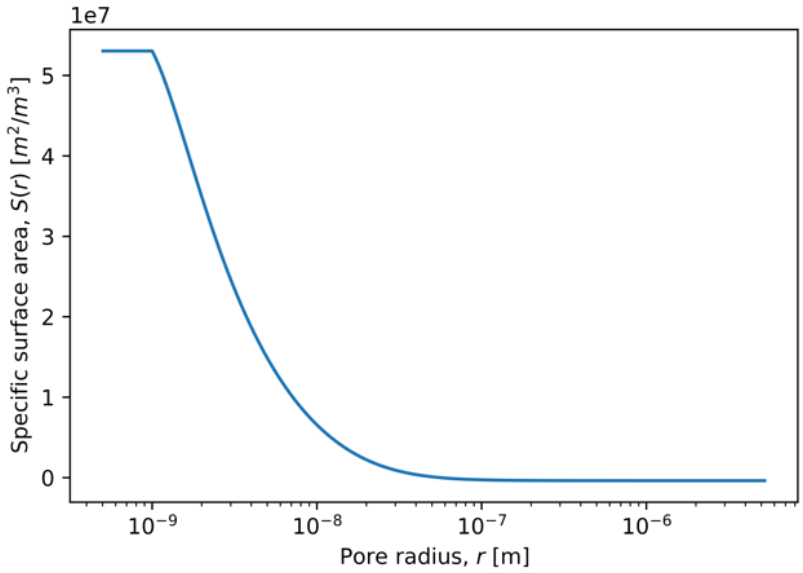
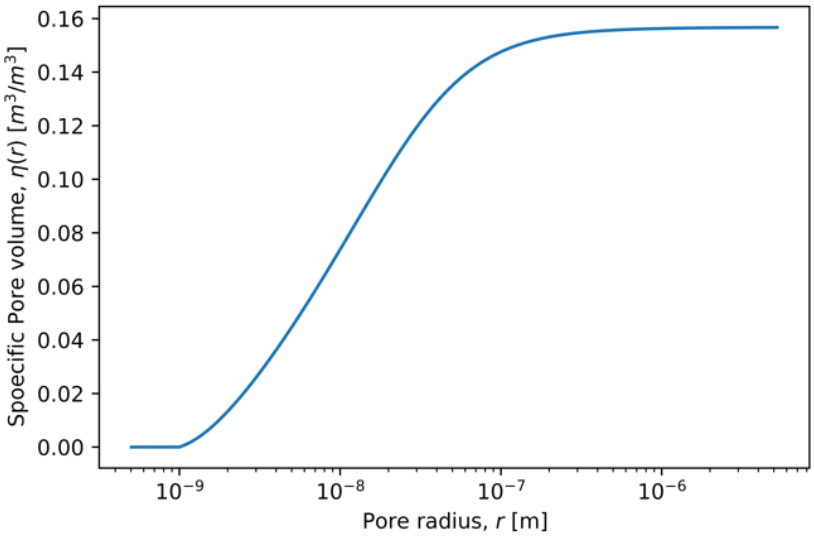


Figure 6.7 Determination of the pore size distribution with the adsorption-condensation model and water retention data. The condensation regime begins at a relative humidity of 0.4.



(a)



(b)

Figure 6.8 Solved pore size distribution (a) surface area per unit volume of mortar for pore size greater than r. (b) pore volume per unit volume of mortar for pore size less than r.

The obtained function of specific surface area or volume effectively serves as the cumulative probability function. It is possible to numerically derive the probability density function from it (see Figure 6.9).

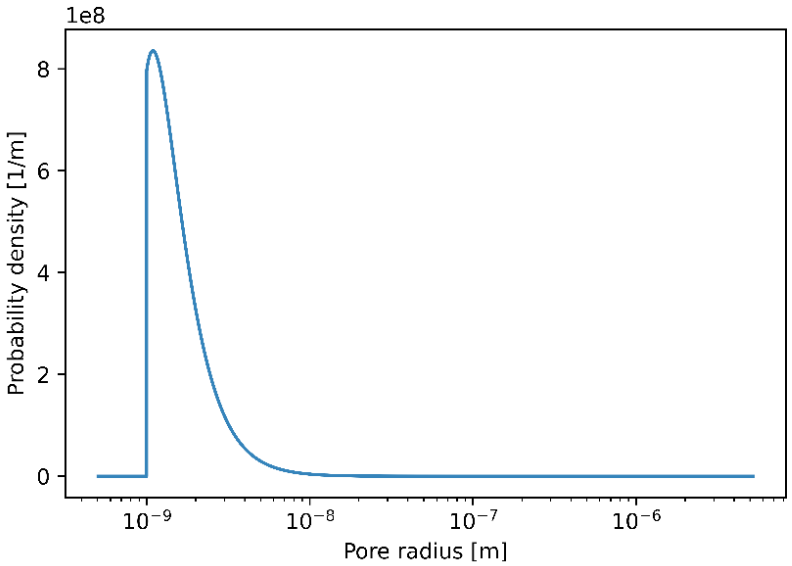


Figure 6.9 The probability density function of the pore size distribution derived from the water retention curve of mortar.

6.4.1.5 Water distribution at the rebar-concrete interface

Given the pore size distribution and the adsorption-condensation model, it is then feasible to calculate the area ratio of the rebar-pore solution to the total rebar-concrete interface.

For saturated pores, the rebar-pore solution area fraction in saturated pores per unit volume of the porous medium, A_{sat} can be calculated as

$$A_{sat} = \theta_{sat} \tag{6.24}$$

$$\theta_{sat} = \eta(r_c) \tag{6.25}$$

Where θ_{sat} is the volumetric water content of water in saturated pores per unit volume of the porous medium, r_c is the pore radius where condensation starts to occur at the relative humidity, h, besides the adsorption.

$$r_c = R(h) + e_a(h) \tag{6.26}$$

For unsaturated pores, the pore solution exists in the adsorbed layer, which extends to the rebar surface in the unsaturated pores, the rebar- pore solution (adsorbed) interface area fraction per unit volume of porous medium, A_{unsat} can be expressed as

$$A_{unsat} = \theta_{unsat} \tag{6.27}$$

$$\theta_{unsat} = S(r_c) \cdot e_a(h) \tag{6.28}$$

where θ_{unsat} is the volumetric water content of adsorbed water in unsaturated pores per unit volume of the porous medium.

Now we have the total rebar-pore solution area per unit volume of concrete.

$$A_{react} = A_{sat} + A_{unsat} \tag{6.29}$$

Therefore, the relationship can be found between the local volume averaged corrosion current density at rebar-pore solution interface, $i_{corr,pore}$, and the total volume averaged corrosion current density at the rebar-mortar interface, $i_{corr,mortar}$.

$$i_{corr,mortar} = \frac{i_{corr,pore}}{A_{react}} \tag{6.30}$$

The corrosion state of the rebar-pore solution interface depends on the competing factors: the pH value and the chloride concentration. The allocation of active and passive rebar-pore solution

interfaces will be determined by the non-uniform chloride concentration distribution, which is discussed in Section 6.4.2.

6.4.1.6 Determine the proper size of the REV

At the pore scale, the pore size is sampled from the pore size distribution. The chloride concentration is also affected by the water saturation condition of the selected pores (adsorbed water or condensed water). These randomnesses in sampling will lead to the uncertainty in arranging the active and passive rebar-pore solution interface and the resistance between them. Therefore, efforts should be made to determine the minimum size of the REV where the volume-averaged porosity stabilizes and hence stable water content, chloride content. The typical length scale of mortar is listed in Table 6.1. The REV size is selected as 1 mm which satisfies the length scales criterion $K^{1/2} \ll d < l \ll L$ for the validity of local volume averaging treatment (Kaviany, 2012).

Table 6.1 Length scales in a mortar

Brinkman screening distance $K^{1/2}$	Pore size d	Representative elementary volume l	Linear dimension of the system L
10^{-9}– 10^{-10} m (Mehta & Monteiro, 2006)	10^{-9}– 10^{-3} m (Mehta & Monteiro, 2006)	$O(10^{-3})$ m	0.1 m

To verify the selected REV size, the sampling was performed for various volumes. A number of pores were randomly sampled from the pore size distribution curve for a given volume, and the volume-averaged porosity, $\langle \epsilon_p \rangle$ is then calculated as

$$\langle \epsilon_p \rangle = \frac{1}{A_{X-REV}} \sum_{1}^{n} \pi r_i^2 \tag{6.31}$$

where A_{X-REV} is the cross-sectional area of the REV perpendicular to the pore direction in the bundled capillary pore model, r_i is the i_{th} of the total n sampled pore radius from the pore size

distribution. The sampling process was repeated 30 times and performed for various REV sizes. The total number of samples, n, was proportional to the selected REV volume size. The maximum n value was determined such that the mean value of $\langle \epsilon_p \rangle$ for the largest REV approaches the macroscopic total porosity.

It can be seen from Figure 6.10, 1 mm^2 size on the rebar-mortar interface significantly reduces the coefficient of variation to around 0.01 (Table 6.2), which justifies the selection of the REV size as representative for the macro-scale property. In this paper, the REV size is selected as 1 mm^3 and the rebar-mortar interface area in the REV is 1 mm^2.

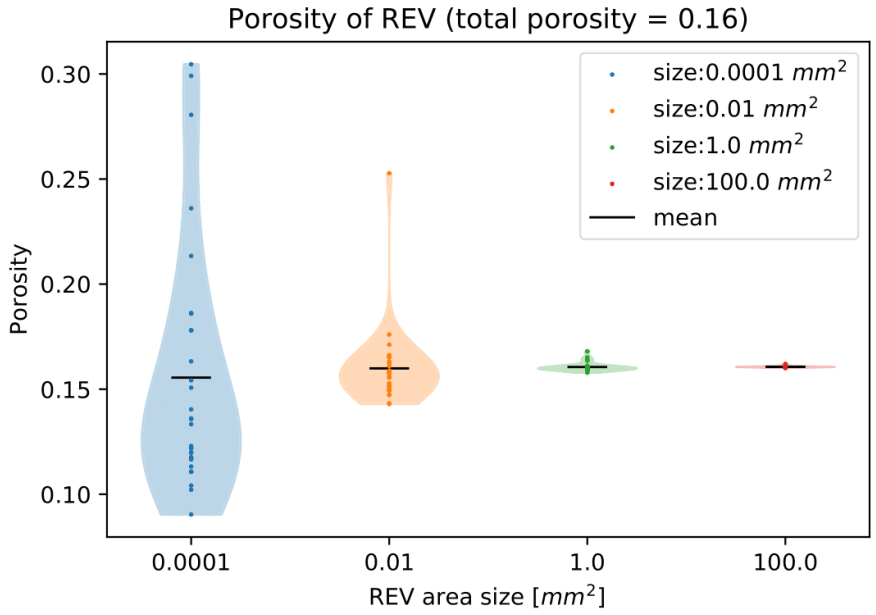


Figure 6.10 Effect of the selected REV size on the scatteredness of volume-averaged porosity at the rebar-mortar interface. Each violin plot shows the probability density of the porosity of the sampled volume.

Table 6.2 Coefficient of variation of the 30 repeatedly sampled total porosities for selected REV sizes

REV size mm^2	Coefficient of variation (CV)
0.0001	0.368
0.01	0.119
1.0	0.0126
100.0	0.00227

6.4.2 Chloride ion in the pore solution

While the rebar-pore solution is reactive in corrosion, the corrosion state could be active or passive depending on the local pH-Cl ratio. There have been studies in this regard in bulk simulated pore solutions. However, there are additional mechanisms involved for the dissolved chloride ions in pore solution in concrete. First, the total chloride ion amount in the pore solution is less than the total chloride content in concrete due to the chloride bonding effect (Glass et al., 1996; Pargar et al., 2017). Second, the chloride ions would concentrate in finer pores due to the double layer effect (Adamson & Gast, 1997), and it has been demonstrated that the chloride ion concentration is significantly higher in pores with a diameter of a few nanometers (Friedmann et al., 2008).

Chloride exists in various forms in the concrete. The free chloride (including partially loosely bound chloride) is affected by the pore wall distance with respect to the electrical double layer formed by the charged solid and adsorbed chloride. The free chloride concentration is higher near the pore wall, which is considered the main contributor to the "chloride concentrate" phenomenon at a macroscopic scale. The raised concentration is more pronounced near the pore wall, leading to a much higher chloride concentration in gel pores (a few nanometres) and smaller capillary pores.

6.4.2.1 Total free chloride: chloride isotherm

The chloride ions in concrete can be classified into four categories: free, loosely bound, bound and strongly bound (Glass et al., 1996). The free chloride is dissolved in the solution and can equalize

with the external bulk solution; the loosely bound chloride is electrostatically adsorbed on the hydration products and can be extracted under high pressure. The rest of the chloride is chemically bonded with the chloroaluminate hydrates (AFt and AFm) and forming Fridel's salt, and they can be extracted by acid solution (bound) or strong acid solution (strongly bound). The controlling mechanism for each category is summarized in Table 6.3.

Table 6.3 Classification of chloride, summarized from (Glass et al., 1996; Pargar et al., 2017)

Cl category	Bound type	Mechanism
Free	Dissolved in solution	Dissolved, high activity
Loosely bound	Physically bound	Electrostatically with C-S-H, Friedel's salt, Portlandite, adsorbed, low activity
Bound and strongly bound	Chemically bound	Friedel's salt: chloroaluminate hydrates, with AFt, AFm

The physical chloride adsorption is dominated by C-S-H as the C-S-H gel is the most abundant hydration product with a very high specific area. This dominance was shown by comparing a surface complexation model and experimental results (Elakneswaran et al., 2009).

Therefore, the main sources of chloride bonding effects are the C-S-H adsorption (physical) and Friedel's salt formation (chemical). These effects were evaluated in an analytical model based on the cement composition (Method 2.3 in Baroghel-Bouny et al., 2012).

$$s_{Cl} = s_{Cl}^{P} + s_{Cl}^{c} = N_{C-S-H} \cdot \frac{\alpha c_{Cl^-}}{c_{OH^-} + \beta c_{Cl^-}} + \delta N_{(C_3A)_{eq}} \tag{6.32}$$

$$N_{(C_3A)_{eq}} = N_{C_3A} + 0.5 \cdot N_{C_4AF} \tag{6.33}$$

where $s_{Cl}^{P}\ [mol/kg]$ is the physically bound chloride by C-S-H adsorption per unit weight of cement, which is a function of the C-S-H quantity, $N_{C-S-H}[mol/kg\ ]$ per unit weight of cement, chloride concentration in solution $c_{Cl^-}[mol/m^3]$, and hydroxide ion concentration $c_{OH^-}[mol/$

$m^3]$. the chemically bound chloride by Friedel's salt formation per unit weight of cement $s_{Cl}^{c}[mol/kg]$ is calculated by the residual quantity of equivalent aluminates per unite weight of cement, $N_{(C_3A)_{eq}}[mol/kg]$, which is a function of the residual quantity of the C_3A, N_{C_3A} and N_{C_4AF}. The coefficient 0.5 is arbitrarily chosen to account for weaker contribution from C_4AF. The coefficients $\alpha = 0.0603$, $\beta = 0.0698$ and $\delta = 1.916$ are calibrated parameters based on the experiment (Baroghel-Bouny et al., 2012).

It is noted that Friedel's salt formation is only considered from the residual calcium aluminates. The ion exchange from hydrated AFt and AFm is not taken into account. However, this part of the bound chloride is mathematically included on the physical adsorption term. The chloride binding isotherm is shown in Figure 6.11 for cement with N_{C-S-H} = 1.3893 mol/kg $N_{(C_3A)_{eq}}$=0.0458 mol/kg.

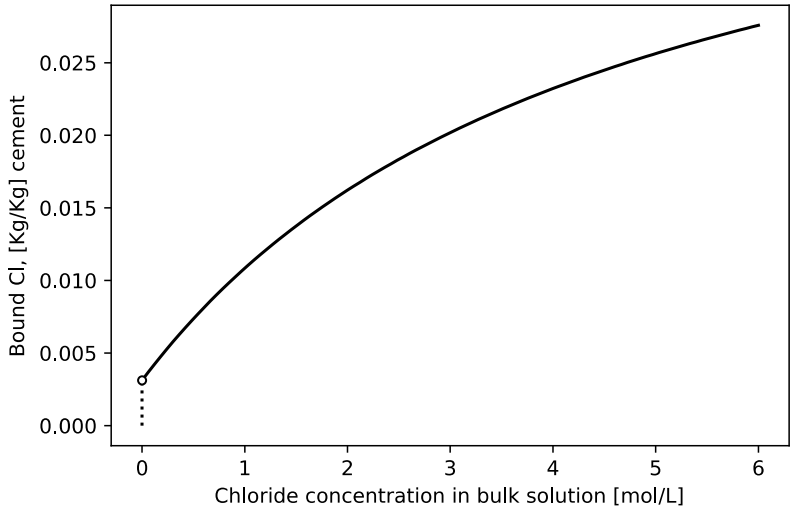


Figure 6.11 Chloride binding isotherm representing the relationship between chloride concentration in solution and chloride content in hardened cement paste. The point marked with a circle is excluded in the curve.

6.4.2.2 Chloride concentrate effect in fine pores

With the chloride binding isotherm, the total free chloride content can be calculated from the total chloride content. As mentioned in the introduction of this section, the adsorbed ions are separated from the free ions in the diffusion layer by the outer Helmholtz plane (OHP) due to the electrical double layer (EDL) phenomena. the concentration of the counter-ion increases as it approaches the OHP, and the concentration of the co-ion decreases (see Figure 6.12).

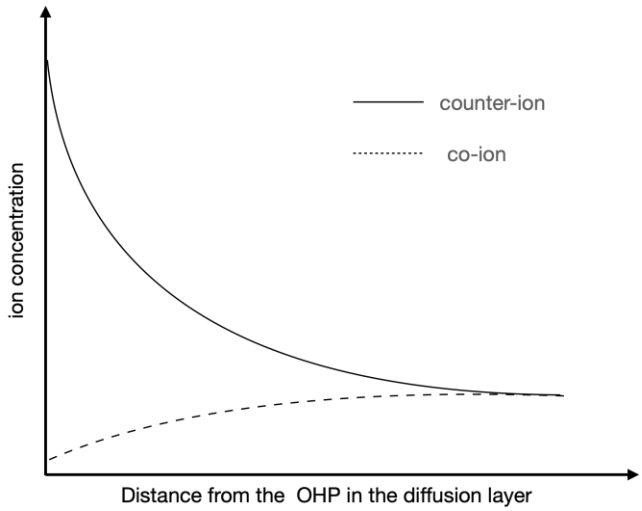


Figure 6.12 Schematic of the EDL effect on the 1-D infinite concentration profile in the diffusion layer from OHP.

The chloride concentrate phenomenon is caused by the positively charged surface of the hydrated cement. In small pores, the concentrate effect is more pronounced. The mean value of the chloride ion concentration, as co-ion, in the overlapped diffusion layer of a pore can be approximated by an analytical expression, Equation (6.34) (Dufrêche et al., 2001; Friedmann et al., 2008)

$$\frac{\bar{c}_{c0}}{c_b} = \frac{Ad_+^2}{1 + ABd_+ + Ad_+^2} \tag{6.34}$$

with

$$A = \frac{1}{8\pi^2} a \tag{6.35}$$

$$B = 4\left[exp\left(\frac{\delta\varphi_{+0}}{2z}\right) - 1\right] \tag{6.36}$$

$$F_0 = \frac{qFd}{4zRT\varepsilon} \gg 1 \tag{6.37}$$

where, $\bar{c}_{c0}[mol/m^3]$ is the mean concentration, $c_b[mol/m^3]$ is the concentration in bulk solution, $F[C/mol]$ is Faraday constant, z is the number of charges of the ion, $R[J/K/mol]$ is gas constant, $T[K]$ is temperature, $d_+[-]$ is the dimensionless length of the pore diameter, and it is related to the pore diameter $d[m]$:

$$d_+ = \kappa d \tag{6.38}$$

$\kappa = \sqrt{\frac{2F^2 c_b}{RT\varepsilon}}$ is Debye constant, $1/\kappa$ represents the extension length of the diffuse layer (Adamson & Gast, 1997). $\delta\varphi_{+0}[V]$ is dimensionless potential at OHP to the reference of the electrical potential in the bulk solution, related to the potential at OHP, $\delta\varphi_0$:

$$\delta\varphi_{+0} = \frac{F}{RT}\delta\varphi_0 \tag{6.39}$$

$q[C/m^2]$ is the surface charge density of EDL and can be expressed as (Galus, 1994)

$$q = -\sqrt{8RT\varepsilon c_b}\, sinh\left(\frac{\delta\varphi_{+0}}{2}\right) \tag{6.40}$$

where, $\varepsilon[F/m]$ is the dielectric permittivity of the solution, which depends on the ionic concentration of the solution. The relationship is interpolated from the relative dielectric permittivity of NaCl solution data from (Hasted et al., 1948) and calculated with the known dielectric permittivity of vacuum $\varepsilon_0 = 8.8541878128 \times 10^{-12} F/m$.

In summary, with the bound chloride isotherm, the total chloride in the pore solution can be determined from the total chloride. The solving procedure is summarized in a flow chart shown in Figure 6.13. The concentration of free chloride ions in the pore solution is a function of pore size. It can be expressed in terms of chloride concentration in bulk solution (or largest pore), c_b, and the pore size distribution and pore water saturation level (covered in previous sections). The value of c_b is unknown and can be solved iteratively such that the sum of the total free chloride in the solution and the corresponding bound chloride content equals the total chloride content. When the c_b is solved, the averaged chloride concentration in pores, $\bar{c}_{c0}$, are simultaneously determined.

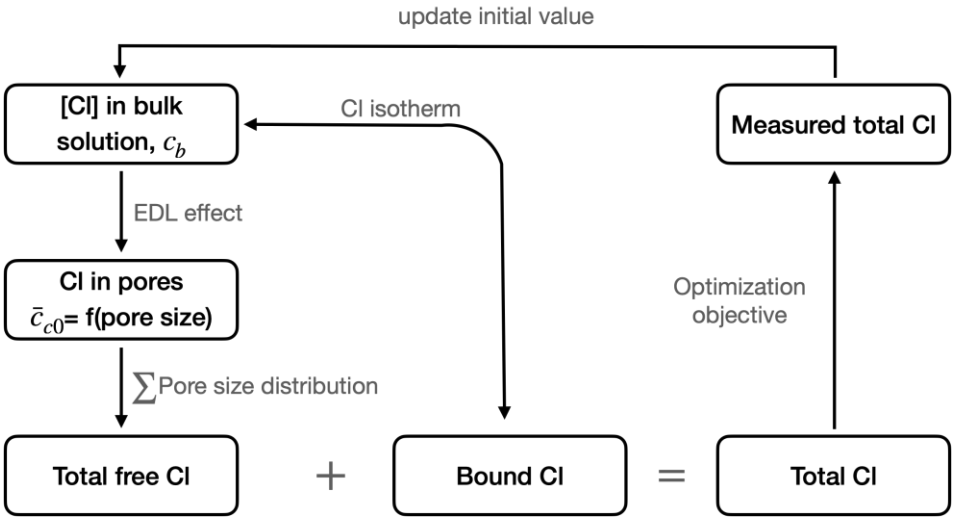


Figure 6.13 Flowchart of the iterative solving procedure to determine the chloride concentration distribution in pore solution.

The chloride concentration distribution in pores is presented for concrete with a total chloride content of 0.6% by weight of cement in Figure 6.14. The chloride ion has a much higher concentration in fine pores, which leads to uneven distribution of chloride content at pore level even in connected saturated pores.

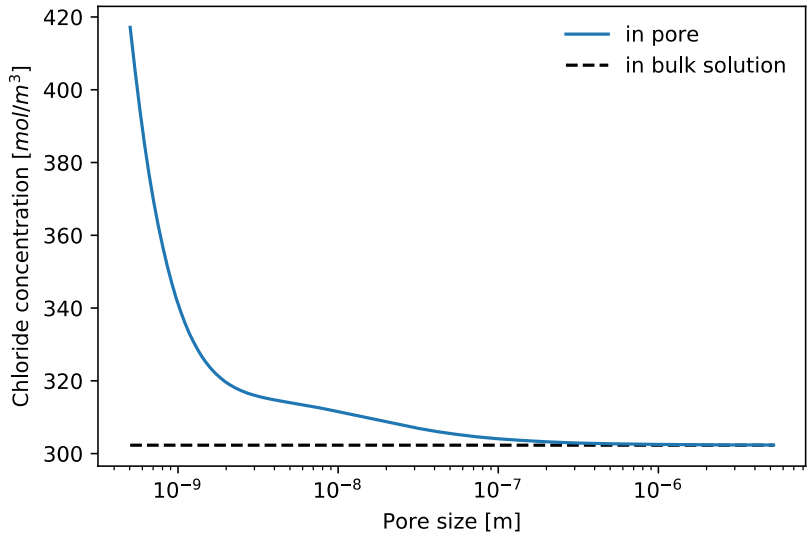


Figure 6.14 Chloride concentration distribution in saturated pores for concrete with a total chloride content of 0.6% by weight of cement.

6.4.2.3 Chloride in unsaturated pores

The movement of water in concrete has two forms: vapour diffusion and liquid conduction. When the concrete is contaminated with chloride and undergoes a drying process, the concentration of chloride ions is increased due to the drying of water in vapour form. The water loss due to liquid conduction is considered not changing the chloride concentration in the pore solution since the chloride ions are moved by advection. Therefore, with these two mechanisms, the total chloride in the adsorbed water layer is reduced but with a higher concentration.

While there is little research on the quantification of these two mechanisms on the contribution of the chloride movement, we may consider advection ratio ranges from 0 to 1, with 0 being pure vapour transport with no chloride moving out the drying pore and 1 being all chloride ions moved out of the drying pore through advection. Although moisture transport through vapour phase was reported as negligible in a sand column experiment when moisture is greater than 4% by weight (Samson et al., 2005), it is suspected that the contribution ratio of liquid movement or advection

may significantly decrease in drier concrete where the liquid phase continuity is significantly reduced.

In addition, when the chloride concentration reaches a saturated level during the drying process, and it may also transport through salt creeping (Van Enckevort & Los, 2013). In this paper, an arbitrary factor of 0.5 is used to account for the free chloride amount reduction in drained pores due to advection and salt creep. The remained chloride ions remain in the saturated pores with finer radius and cause the increase in the chloride concentration.

The chloride concentrations in the adsorbed water layers are considered to remain the same as the saturated condition and are not changed by the concentration gradient-driven diffusion because the mobility of ions in an adsorbed layer is reduced at least by five orders of magnitude(Tokunaga et al., 2017).

6.4.3 Allocation of the active and passive rebar-pore solution interface

With the previous solution for the pore size distribution and the adsorption-condensation model (Section 6.4.1), plus the chloride concentration distribution as a function of the pore size (Section 6.4.2), it is, therefore, possible to quantify the local saturation condition and the chloride concentration in all pores. The rebar-pore solution interfaces are categorized into four types based on the pore saturation condition and $[Cl^-]/[OH^-]$value (Table 6.4).

Table 6.4 Rebar-solution interface types

	$[Cl^-]/[OH^-] \geq$ threshold	$[Cl^-]/[OH^-] <$ threshold
Saturated pore (condensed water)	Active	Passive
Unsaturated pore (adsorbed water)	Passive	Passive

The $[Cl^-]/[OH^-]$ ratio in the solution is a criterion determining if the protective film once formed in the high pH solution is stable. When the $[Cl^-]/[OH^-]$ exceeds the threshold value, the protective film is broken, and the rebar-solution interface becomes actively corroding. However, in unsaturated pores with only the adsorbed water, even with the possible high $[Cl^-]/[OH^-]$ ratio, the rebar is considered passive due to the rust effect. This dense rust formed in the transient from

wet to dry in the unsaturated pores will effectively passivate the steel surface, and the high electrical conductivity of the Fe^{3+}-rich rust makes it a poor cathode for oxygen reduction (Stratmann, 1990). The dried rusted rebar surface also features a more noble corrosion potential, i.e., $-100\ mV_{SCE}$ in chloride-contaminated concrete (Alonso et al., 1998b). Such a passivated interface will turn active only if the pore is re-saturated with the maintained high $[Cl^-]/[OH^-]$. Based on the previous rebar corrosion test in the simulated pore solution (Chapter 3), the threshold $[Cl^-]/[OH^-]$ is set as 0.54.

The combined effect of water and chloride distribution on the corrosion state at the pore level can then be quantified with the pore distribution, water distribution and chloride distribution results. The area of the active and passive ratio can be calculated following the classification shown in Table 6.4.

6.5 Implementation of the corrosion model at the REV level

The area of those two rebar-pore solution corrosion conditions: active and passive, is calculated based on the water and chloride distribution. The corrosion kinetics of the rebar in the simulated pore solution (active and passive rebar) in the previous studies (Chapter 3) will be used to model the micro-corrosion cells of the two rebar states and the coupling within the REV.

6.5.1 Corrosion circuit model

The information of the active and passive rebar-solution area ratio has been calculated. We can incorporate this information into the existing circuit model, which includes active and passive units with equivalent resistors (see Chapter 5). It is worth mentioning that although the same model scheme is used, the meaning of the "unit" is slightly different. In the previous applications, the unit contains the inert interfaces (rebar-solid, and rebar-air), and the corrosion kinetics for each unit are the ones measured from the rebar-concrete interface with a linear dependence on the water saturation level. In the application for the corrosion model within REV, the unit represents the rebar-pore solution interface with the corrosion kinetics which is independent of the water saturation level (Figure 6.15). The same solving scheme was used to solve this model (Li, Evitts, & Boulfiza, 2018).

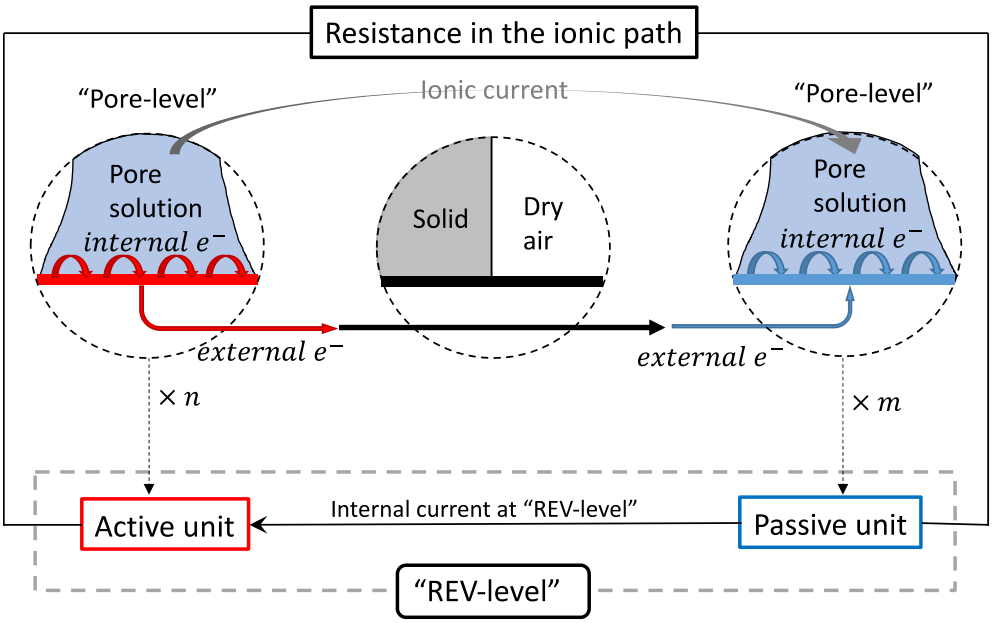


Figure 6.15 The illustration of the updated corrosion circuit model that homogenize the corrosion kinetics in REV from pore level.

6.5.2 On the limitation of the corrosion circuit model

Although using equivalent resistors can describe the resistance between the coupled non-uniform rebar segments, there is no analytical solution for the equivalent resistance due to the complicated potential field in 3-D volume, i.e., concrete. The saturation significantly affects the resistivity of concrete and, at the same time, affects the rebar surface exposed to the solution-filled pores. Also, the non-uniform active units and passivated units could distribute randomly on the rebar while retaining the same area ratio. This geometrical uncertainty may lead to other effects on the homogenized corrosion kinetics of the rebar.

The rebar's homogenized polarization behaviour may also be affected by the shape, orientation, and distance of the coupling electrode by altering the potential field and current distribution in the concrete. The coupling electrode could be the counter-electrode attached to the concrete cover surface, similar to the mesh anode used in the impressed-current cathodic protection application. It could be an electrically connected rebar with a different electrode rest potential forming a macro-corrosion cell.

However, in the application for the rebar-concrete interface in a chosen REV size, i.e. 1 mm^3, the equivalent resistor and the homogenized corrosion kinetics can be insensitive to the active and passive rebar-pore solution interface area sizes and their allocations randomness and counter electrode size and location and deviate from the Butler-Volmer equation. It is assumed that is true and we proceed to apply the circuit model for the corrosion cell simulation in the REV, and then this assumption will be examined by checking the difference between the corrosion current density generated from the model and the corrosion current density fitted from the simulated potentiodynamic scan curve. The assumption would be valid if the relative error between the two is small.

6.6 Corrosion kinetics in solution

The corrosion parameters used for the active rebar-solution interface and passive rebar-solution interface are from the previous corrosion test for rebar in simulated pore solutions (Chapter 3). The values are listed in Table 6.5.

Table 6.5 Corrosion parameters of an active/passive unit

Corrosion unit	E_{corr} $[V_{SCE}]$	$i_{corr}[A/m^2]$	β_a[V/dec]	β_c[V/dec]
Active	-0.616	0.91×10^{-2}	0.3	-0.30
Passive	-0.200	0.05×10^{-2}	0.595	-0.24

6.6.1 Mortar REV resistivity

The resistivity of concrete depends on the resistivity of the pore solution and is dramatically affected by the water saturation level. Although the bundled capillary pore model can address the pore size distribution problem it does not describe the interconnectivity of these pores. Therefore, the complexity of the serial or parallel arrangement of the pores is unknown. There are empirical models such as (Weiss et al., 2013) that uses a single exponential component to account for pore solution's volume, concentration, and the interconnectivity. In this paper, the dependence of the electrical conductivity of concrete is divided into two parts: the conductivity of the pore solution and the volume and connectivity of the pore solution. This can be described by

$$\sigma_{rev} = a \cdot \langle \sigma_{sol} \rangle \cdot Se^{b} \tag{6.41}$$

where σ_{rev} is the electrical conductivity of the porous medium, $\langle \sigma_{sol} \rangle$ is the local volume-averaged electrical conductivity of the pore solution which depends on the water saturation level, a, b are fitted parameter with a accounting for the baseline pore connectivity and pore solution volume fraction, and b accounting for the change in interconnectivity due to the change in water saturation level. For the resistivity test results of non-carbonated mortar sample with variable chloride w/c=0.7, $a = 0.0125917$, $b = 4.12841666$.

The volume-averaged electrical conductivity, $\langle \sigma_{sol} \rangle$ can be calculated as

$$\langle \sigma_{sol} \rangle = \frac{1}{V_{sol}} \int_{\Omega_{sol}} \sigma_{sol} \tag{6.42}$$

where, σ_{sol} is electrical conductivity of pore solution, Ω_{sol} is the domain of pore solution, V_{sol} is the total volume of pore solution. Since the water distribution in the bundled capillary pore model has been determined in Section 6.4.1, the electrical conductance be determined based on the cross-sectional area of the pore solution and their electrical conductance for all the pores.

$$\langle \sigma_{sol} \rangle = \frac{1}{Se} \left\{ \sum_{r_{min}}^{r_{sat}} \sigma_{sol} \cdot \pi \cdot r^2 \cdot \Delta L(r) + \sum_{r_{sat}}^{r_{max}} \sigma_{sol} \cdot \pi \cdot [r^2 - (r - e_a)^2] \cdot \Delta L(r) \right\} \tag{6.43}$$

where, r_{sat} is the largest saturated pore radius, $\Delta L(r)$ is the pore length per unit volume with diameter ranges from $[r, r + \Delta r\ ]$, σ_{sol} is the electrical conductivity of the pore solution in a single pore. σ_{sol} can be calculated by a semi-empirical model that calculates the electrical conductivity of pore solution with ionic concentrations (Snyder et al., 2003).

Ionic strength:

$$I_m = \frac{1}{2}\sum z_i^2 \cdot c_i \tag{6.44}$$

where z_i is the charge number of i_{th} ion, c_i is concentration of i_{th} ion. To account for ions with non-infinite dilution and dependence, the σ_{sol} is written as

$$\sigma_{sol} = \sum z_i \cdot c_i \cdot \lambda_i \tag{6.45}$$

$$\lambda_i = \frac{\lambda_i^0}{1 + G_i \cdot \sqrt{I_m}} \tag{6.46}$$

where G_i is the conductivity coefficient, λ_i^0 is the equivalent conductivity of i_{th} ion at infinite dilution, the values of these parameters can be found in (Snyder et al., 2003) for common ion species in the pore solution: OH^-, K^+, Na^+, Cl^-, Ca^{2+}, SO_4^{2-}.

The concentration of the chemical species such as OH^- K^+, Na^+, Ca^{2+}, SO_4^{2-}, are based on the simulated pore solutions used in the previous studies (Chapter 3) and Cl^- concentration is calculated from the total chloride content (Section 6.4.2).

6.6.2 Equivalent resistors in the circuit model

In the equivalent circuit model (Figure 6.16), the internal resistance between the active and passive units near the rebar, R_0, is approximated with R_{rev}, and the external resistance through the cover $R_e = R_1R_2/(R_1 + R_2)$ is approximated by the infinite grid of resistors with a resistance of R_{rev}. The total resistance of the infinite grid between two neighbouring points is $1/2 \cdot R_{rev}$. Therefore, the external resistance R_e is also R_{rev} and $R_1 = R_2 = 2R_{rev}$. The total resistance between two units is 1.5 R_{rev}. This estimation is thought to be reasonable because in a 3-D concrete domain with a volume of 10 cm × 4 cm × 4 cm, the numerical solution of the resistance between the rebar surfaces in two neighbouring REVs is $1.74 \cdot R_{rev}$.

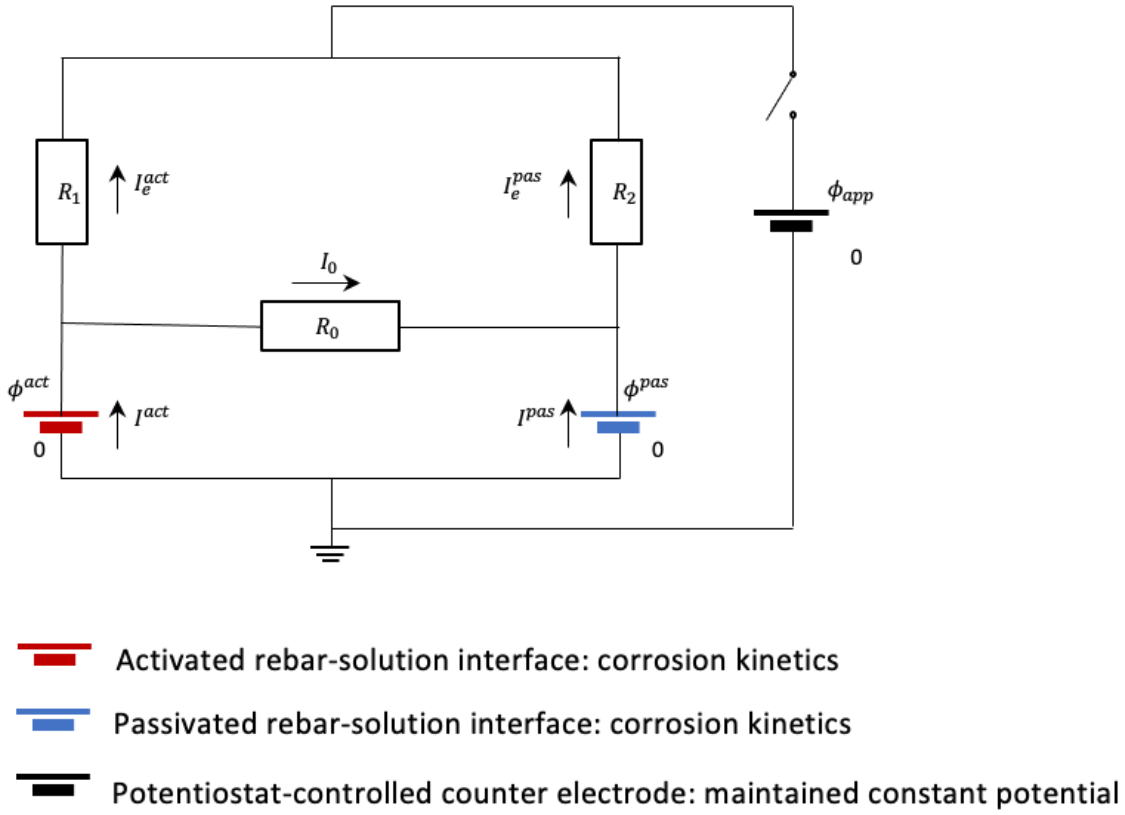


Figure 6.16 Equivalent circuit of the corrosion model at the REV level.

6.7 Simulation results

6.7.1 The effects of water saturation and chloride content on active rebar-solution interface area ratio in concrete

The area ratio of the active rebar-solution interface as a function of the water saturation level and total chloride content is shown in Figure 6.17. The values of water saturation level and chloride content were evenly distributed within the ranges of 0.2–0.95 and 0.1%–2.5% by weight of cement, respectively.

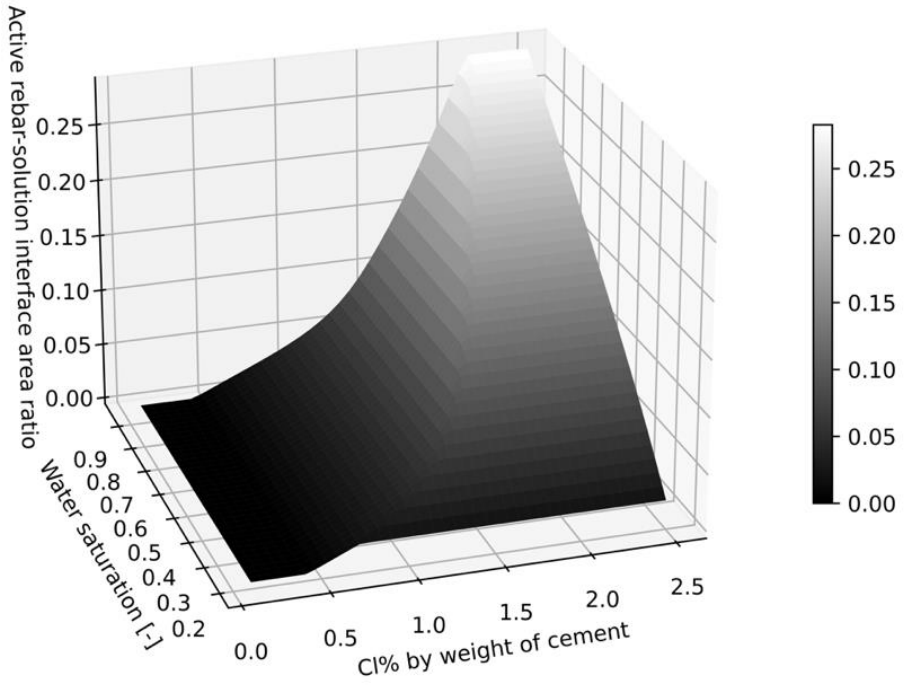


Figure 6.17 Active rebar-solution surface area ratio as a function of water saturation and total chloride content.

As shown in Figure 6.17, at high chloride content levels (Cl% >2%), rebar in the saturated pores with high chloride ion concentration is active and remains passive in the unsaturated pores. The increase of water saturation level leads to a nearly linear increase in active rebar area. At low chloride content levels (Cl%<0.4%), the rebar-solution surface is mainly passivated, and the active rebar area remains near zero. At moderate chloride content levels, the active rebar-solution area ratio can be affected by both chloride content and the water saturation. At these chloride content levels, the active rebar-solution increase with the increase in water saturation level but starts to become independent to the water saturation level when the saturation reaches a threshold value. This saturation threshold value increase with the chloride content.

6.7.2 Simulated homogenized corrosion parameters for rebar-concrete interface

The combined effects of water saturation and chloride on the active and passive area and their ratio will significantly affect the overall corrosion kinetics at the REV level. With the same solving

scheme introduced in Chapter 5, the corrosion circuit model returns corrosion parameters such as corrosion potential, E_{corr} , corrosion current density, i_{corr} , and potentiodynamic scan curves.

E_{corr} and i_{corr} are affected by total chloride content and water saturation. They are plotted in Figure 6.18 together with the experimental test results from Chapter 4. The simulated E_{corr} and i_{corr} as a function of chloride content captured the pattern in the experiment data.

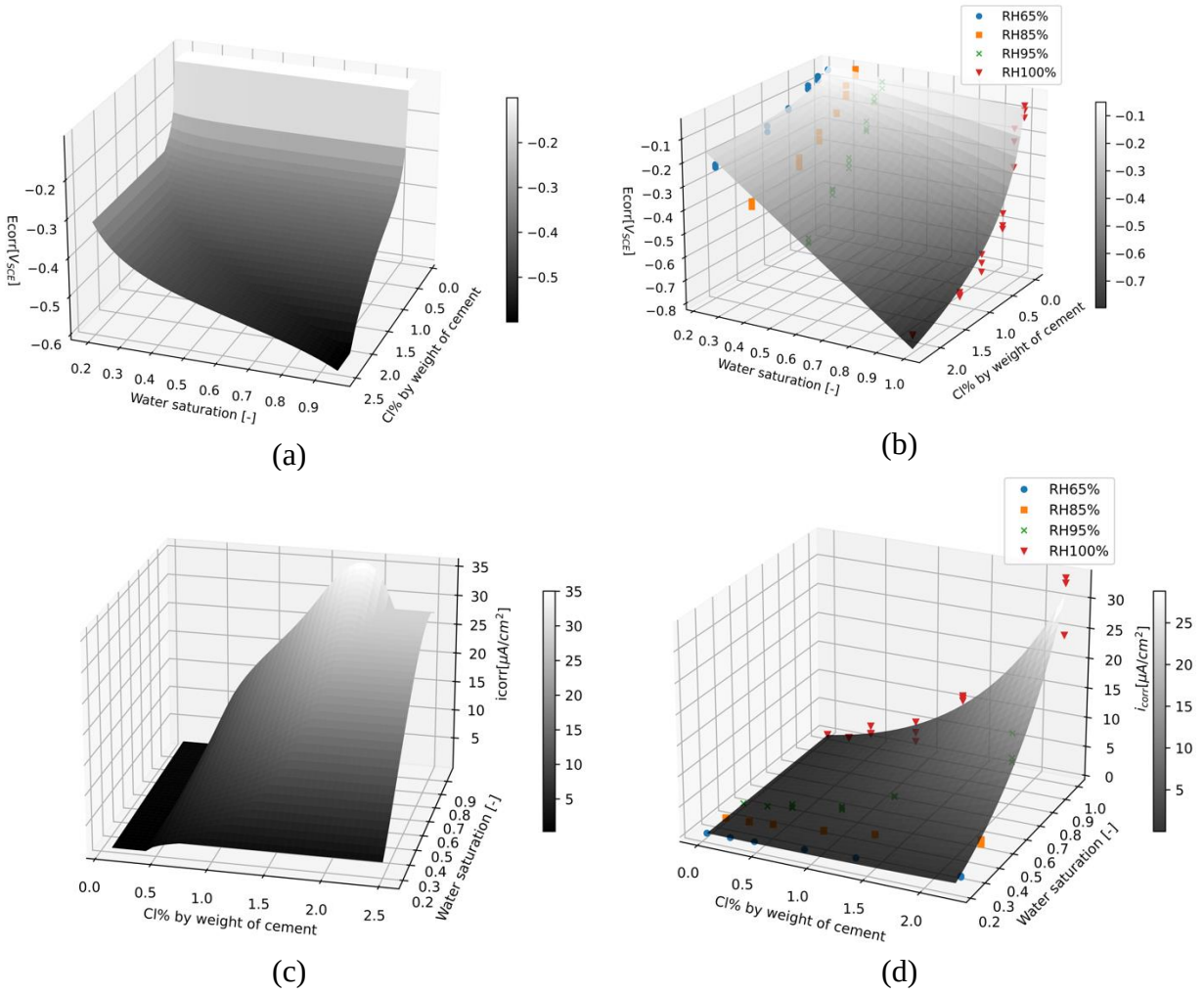
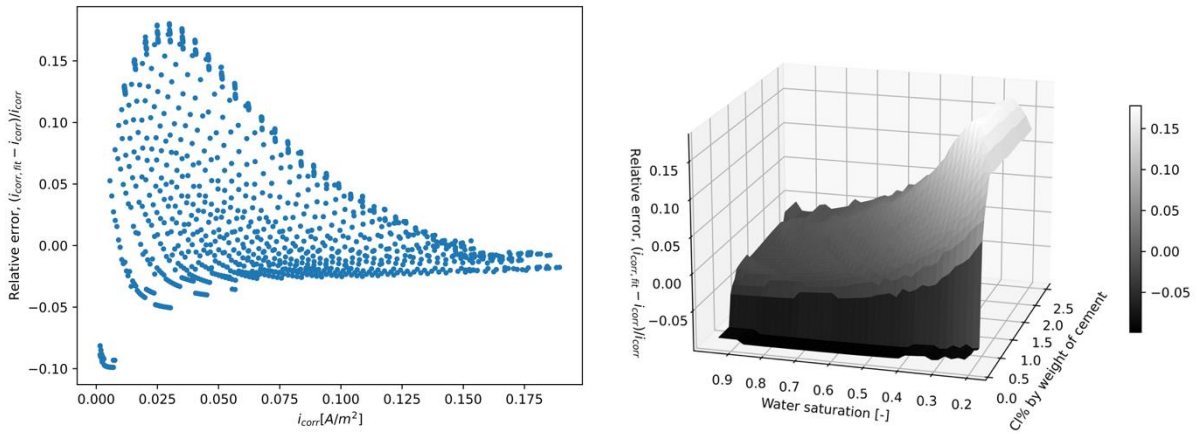


Figure 6.18 Corrosion parameters of rebar-concrete interface affected by the total chloride content and water saturation. (a) simulated corrosion potential E_{corr}, (b) experimental corrosion potential E_{corr}, (c) simulated corrosion current density i_{corr}, (d) experimental corrosion current density i_{corr}. Marks represent experimental data points.

It was previously assumed that the homogenized corrosion kinetics of the rebar-concrete interface simulated in the REV does not deviate from the Butler-Volmer equation. This assumption is now examined by checking the difference between the corrosion current density generated from the model and the corrosion current density fitted from the simulated potentiodynamic scan curve using the Butler-Volmer equation. The relative error between the two is plotted in Figure 6.19. The relative error of the corrosion current density is mostly within $\pm$10%. Higher relative error values (10%–17.9%) are observed among the low i_{corr} values under the condition of high chloride content and low water saturation levels, which results in a decreased absolute error. This error is considered insignificant comparing to the experimental random error. Therefore, the homogenized corrosion kinetics in REV can be described with the Butler-Volmer equation. This allows the application of the homogenized corrosion kinetics in the form of Butler-Volmer equation on the rebar-concrete interface similar to the application of volume-averaged properties such as porosity, permeability, and diffusivity in cracked concrete structures, where the spatial variation of these factors is common.



(a) (b)

Figure 6.19 The relative error of corrosion current density generated from the model and the corrosion current density fitted from the simulated potentiodynamic scan curve using the Butler-Volmer equation. (a) relative error – i_{corr} relationship (b) relative error- water saturation and chloride content relationship.

6.8 Conclusion

This paper aims to model the averaged rebar corrosion behaviour in the representative elementary volume (REV) in concrete from total water and chloride content and the rebar corrosion characteristics in solution.

A novel and comprehensive model framework was developed to link the corrosion-affecting factors, such as water content and chloride content, at a macroscopic level to their non-uniform distributions at the pore level in a quantitative manner.

The pore size distribution determined by the water retention curve was used to quantify the water distribution at a given total moisture content. The water distribution at the pore level was determined through an adsorption-condensation model, and the rebar-solution area for mortar in different water saturation levels was quantified. The chloride distribution at the pore level captured the bonding effect, chloride concentrate effect due to the electrical double layer.

The rebar corrosion states at the pore level were determined by the combined effects of the non-uniform distribution of water and chloride concentration at the pore level, which were derived from the macroscopic water content and chloride content. The water distribution controlled the rebar-solution area, while the water saturation and chloride concentration at the pore level controlled the active and passive areas.

The corrosion model implemented the simulated allocation of the active and passive rebar-solution interfaces at pore level within the REV (l mm^3) and modelled the homogenized corrosion parameters as a function of the water saturation level, and chloride content. These corrosion parameters can be used to model rebar corrosion in cracked concrete structures, where the spatial variation of these factors is often present.

6.9 Epilogue

Chapter 6 provides a crucial bridge between the homogenized corrosion parameters discussed in Chapter 5 and the localized corrosion parameters at the pore scale. By leveraging experimental insights and the corrosion circuit model, this chapter provides a modelling approach to derive the environment-dependent corrosion kinetics that can be used in the comprehensive corrosion model. The next chapter will discuss the simulation of the microenvironment evolution at the rebar depth of concrete structure in its service conditions: the “environment model”.

7 Effect of Accurate Microclimate Representation on the Corrosion Performance of a Bridge Subjected to Carbonation and Chloride Attack

7.1 Preface

In line with the previous chapter's focus on environment-dependent corrosion kinetics, Chapter 7 continues on the simulation of the microenvironment evolution at the rebar depth of concrete structures under service conditions. The developed model is named "environment model" in the project objectives. In this model, a series of transport models for porous media were employed to simulate the transport of water, chloride ions, and carbon dioxide gas in concrete. The model uses a realistic representation of the microclimate immediately surrounding the bridge in terms of temperature, relative humidity, and precipitation. Sensors were also installed for in-situ monitoring of oxygen content, moisture, chloride, and carbonation in the concrete cover.

This model was applied to an uncracked RC structure — an old concrete arch bridge subjected to service conditions with the source of chloride, carbonation, varying air relative humidity, and exposure to unevenly sheltered run-off water. The modelled microenvironment at the rebar depth determines the rebar corrosion rate by employing empirical uniform corrosion kinetics specifically for the century-old steel rebar. It also explores the impact on rebar section loss from different climate scenarios, such as elevated CO_2 levels and changes in precipitation.

Simulation results for the past 100 years of service agree well with the current condition of the bridge while results for 50 years into the future suggest that for this bridge, as well as for other existing bridges, for which design did not consider the effects of climate change may deteriorate more prematurely than originally planned. The implications of climate change on the corrosion performance of concrete structures and their resilience over the next 50 years are discussed.

The manuscript of this chapter titled “Effect of Accurate Microclimate Representation on the Corrosion Performance of a Bridge Subjected to Carbonation and Chloride Attack” was submitted to Case Studies in Construction Materials. Gang Li is the first author who designed and performed most of the experimental study, numerical modelling, and data analysis. Dr. Moh Boulfiza contributed to providing feedback on the experiment design and simulation methods. Both Dr. Moh Boulfiza and Dr. Richard Evitts contributed to the review of the manuscript and provided feedback.

7.2 Introduction

Infrastructure is critical to modern society, and bridges are essential components of the transportation network. The long-term durability of these structures is crucial to ensuring public safety and avoiding costly repairs or replacements. However, the durability of bridges is heavily influenced by the service environment, including factors such as temperature, humidity, precipitation, and exposure to de-icing salts or seawater. With the observed trends of climate change, it is crucial to understand how these environmental factors impact the corrosion performance of bridges.

The service environment can create varying microenvironments around the reinforcing rebars. Exposed weather conditions, chloride from different sources, carbonation of concrete, and the geometry of the structural member all affect the concrete condition and hence the corrosion rate of embedded steel rebars. To model the corrosion performance of bridges accurately, existing approaches have used empirical and numerical models.

Empirical models (Alonso et al., 1988; DURACRETE, 1999; Liu & Weyers, 1998; Vu et al., 2005) relate corrosion rate with various factors but may not capture exposure conditions and concrete conditions in practical environments. Numerical models (Ghods et al., 2007; Isgor & Razaqpur, 2006; Kim & Kim, 2008; Kranc et al., 1992; Muthulingam & Rao, 2015; Sohail, 2013; Yu, Liu, & Li, 2017; Zacaruk, 2013) use electrochemical corrosion kinetics to model the corrosion rate, but rely on accurate corrosion parameters that have a larger scatter, especially in concrete, and face challenges when micro-cell corrosion is dominant (Alonso et al., 1998b; Chang et al., 2008b; Elsener, 2005; Garcés et al., 2005; Gulikers, 2005; Locke & Siman, 1980; Sohail, 2013). Moisture and ion transport in concrete is commonly modelled using the diffusion-advection-dispersion model, but sudden precipitation events require small time steps that makes it computationally expensive to solve these models (Isgor & Razaqpur, 2004; Samson et al., 2005; Scheffler & Plagge, 2010).

In this context, modelling the corrosion of old reinforced concrete bridges, which often experience an overall impact from carbonation and chloride contamination with seasonal precipitation, presents additional challenges. The overall corrosion rate is closer to uniform corrosion, except for the very localized corrosion due to localized damage (cracking). To address these challenges, a

new approach has been proposed, which handles the rapid changing and on-and-off precipitation using a smooth equivalent boundary condition to capture the moisture fluctuation at the rebar depth in a large-scale 3-D model. Given that the bridge under study was built more than 100 years ago (commissioned in 1916), it was imperative to make sure that the material parameters used in the corrosion and transport models provided a realistic representation of the materials used in those days. Cores of actual concrete and rebars were extracted from the arch and used to determine the transport properties of concrete (porosity, permeability, etc.) and the pore solution composition which was used later as the electrolyte in electrochemical potentiodynamic tests to determine the corrosion behaviour of those old rebars. The outcomes of the tests revealed that the concrete and steel reinforcements used were significantly different from modern concrete and steel to warrant site-specific calibration for the material properties that need to be used for a reliable prediction of the future performance of old bridges.

Accurate microclimate modelling can provide valuable insights into the local environment in the vicinity of a structure, and how it affects the structure's durability. In this paper, the effect of accurate microclimate modelling on the corrosion performance of a bridge was investigated. A realistic representation of the microclimate immediately surrounding the bridge, incorporating temperature, relative humidity, and precipitation, and the use of in-situ sensors to monitor critical parameters such as oxygen content, moisture, chlorides, and carbonation in the concrete cover are presented. The study highlights the importance of reliable tools and degradation models to assess the resilience of structures operating in harsh corrosive environments, and the implications of climate change on their long-term durability.

7.3 Research significance

With the observed trends of climate change, it is crucial to understand how environmental factors impact the corrosion rate of reinforced concrete, and this study presents a new approach to accurately model and predict corrosion performance. The proposed methodology overcomes the limitations of existing models, which can be critical for assessing the resilience of structures operating in harsh corrosive environments.

The proposed model used meteorological data to simulate the condition of an old concrete arch bridge. The smoothed equivalent pressure head boundaries were used to reduce the computational cost while retaining adequate accuracy.

The optical fibre oxygen sensor was first employed to measure the actual oxygen content distribution on the rebar of the reinforced arch structure. The test results set oxygen as a controlled variable for the corrosion model.

The rebar corrosion kinetics were tested based on the extracted concrete pore solutions and the site-extracted one-century-old rebars with the oxygen content measured from the arch structure. The synergistic effect of pH and chloride on the corrosion kinetics was quantified in the model.

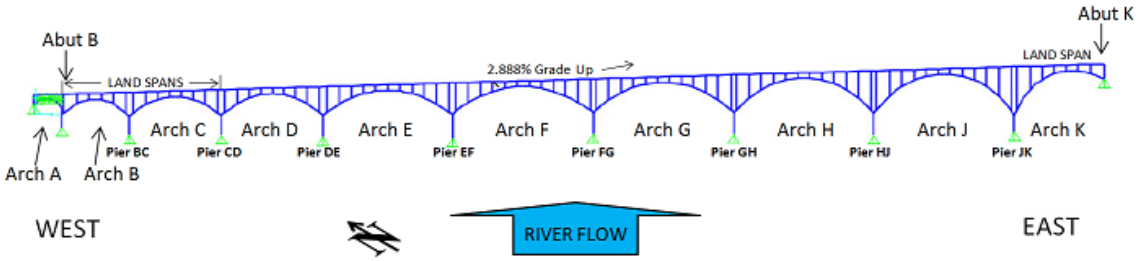
The approach combines field tests and numerical modelling and provides a more reliable prediction of the service life.

Overall, this study highlights the importance of reliable tools and degradation models for ensuring the longevity and safety of critical infrastructure in the face of changing environmental conditions.

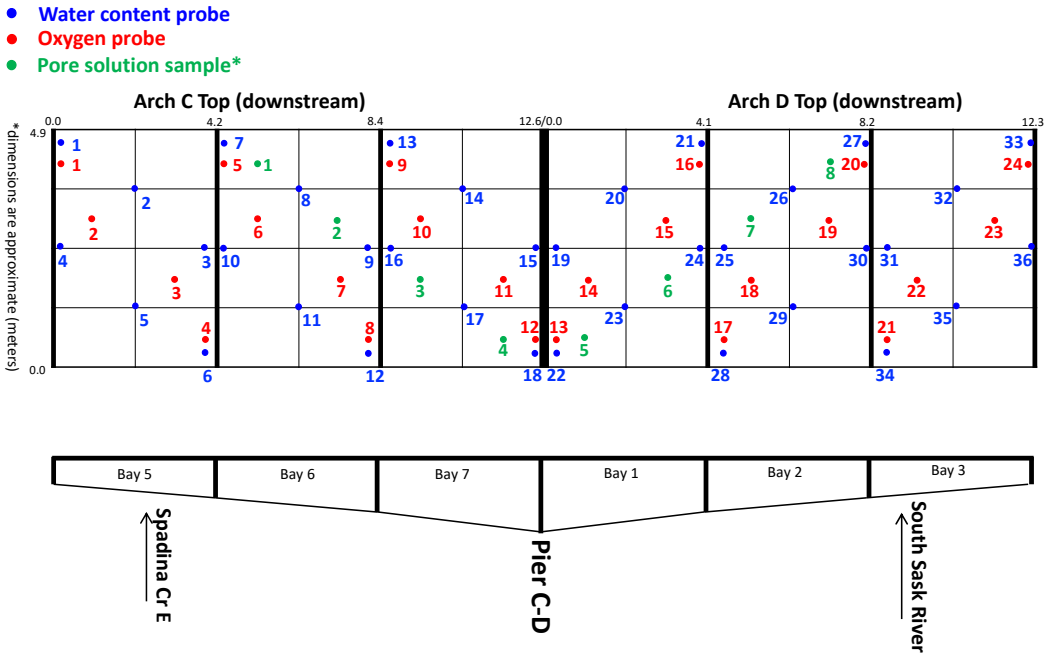
7.4 Field and experimental investigation

The University Bridge in Saskatoon is a multi-span concrete arch bridge that entered service in 1916. It spans the South Saskatchewan River between 25th Street on the West bank and College Drive on the East bank (Figure 7.1a). Over the years, the bridge has undergone repairs and modifications focusing on the deck, sidewalks, railings, and spandrel walls, while the arch component remains a century old.

The moisture content and oxygen content were measured at multiple locations in a diagonal pattern on the arch. The locations of the probes are shown in Figure 7.1b. The average moisture content through the concrete cover was measured with time-domain reflectometry (TDR) probes that were calibrated for the old concrete. The oxygen level was measured by the optical oxygen sensors at the rebar depths. The measured moisture and oxygen levels in August 2014 are shown in Figure 7.2a and Figure 7.2b.



(a)



*Four additional sampling cores on Arch B with the same pattern as Arch C
Core size: 2 inch (5 cm)

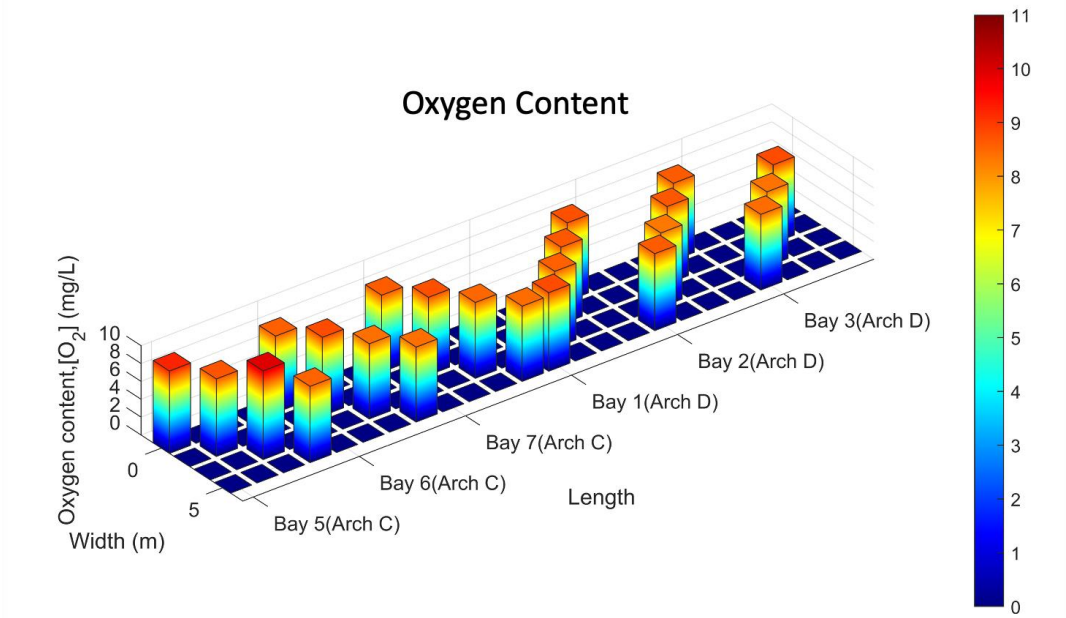
(b)

Figure 7.1 Field program (a) location of Arch C and D of the University Bridge (b) location of water content and oxygen content probes and pore solution extraction cores on Arches C and D.

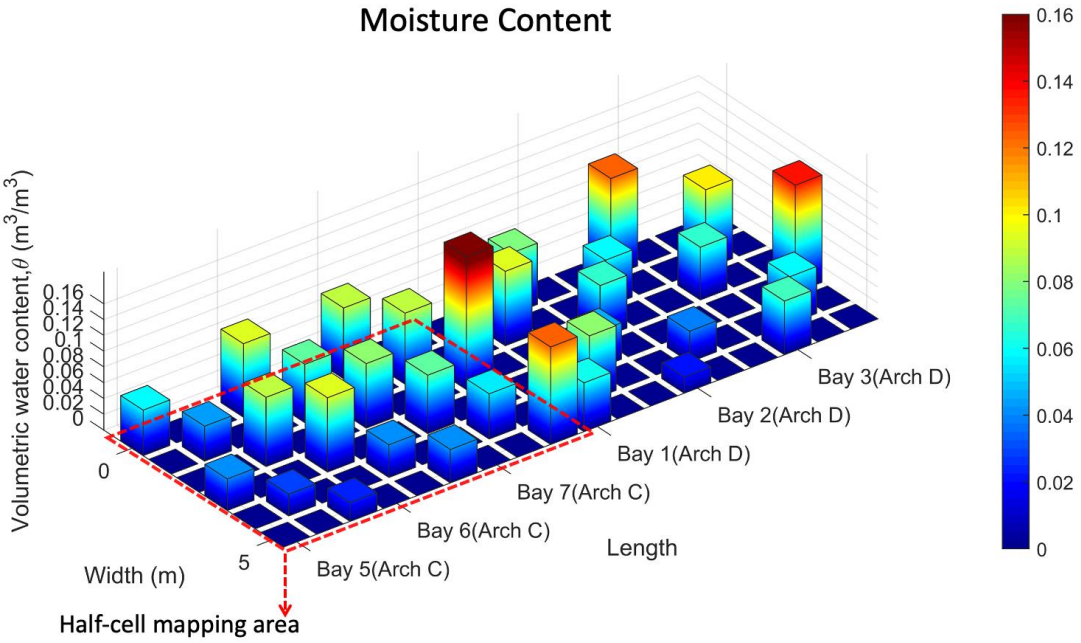
These moisture results show a trend that was consistent throughout the duration of the measurements of elevated water contents on portions of the arch closer to the exterior edge of the

arch and closer to the pier with the largest opening under the spandrel wall. There was one outlier that had high moisture content near the top of the arch and was located towards the interior of the arch. This outlier was later identified due to a leakage in the drainage pipe near that location. These results demonstrate that the deck, curtain walls, and spandrel walls provide varying sheltering for the arch from the rain or run-off water from the deck. The uneven moisture distribution may cause varying corrosion conditions in the arch. At the location of the highest moisture content near the pier, the mapped half-cell potential (Figure 7.2c) shows very negative readings (-350 – -416 mV CSE), which corresponds to a corrosion probability of "more than 90%" according to the ASTM standard (ASTM International, 2015).

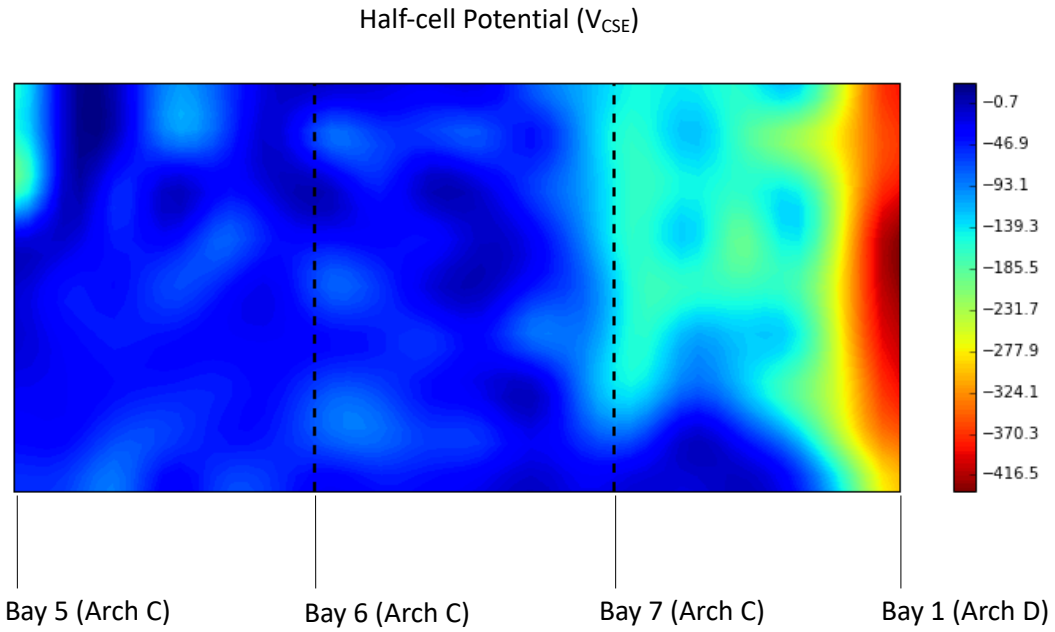
The oxygen content results show a uniform saturated oxygen content (8.2mg/L at 25 °C) at all locations regardless of the varying moisture content and the possible varying corrosion rate. This shows that for the current condition of the arch, the concrete cover does not limit oxygen transport in the corrosion process. It is even this case for the location where the possible high corrosion rate (high consumption rate of oxygen) and high moisture content (low gas permeability) coexist. This observation ruled out the possibility that the air-dried concrete may limit oxygen transport when the corroding rebar has a larger area (Hussain, 2011). Therefore, the oxygen content was set to 100% saturated, a controlled variable, when constructing the corrosion model.



(a)



(b)



(c)

Figure 7.2 The field program shows the distribution of (a) oxygen content, (b) moisture content, and (c) half-cell potential over the arches.

The experimental investigation then provides the material properties for the numerical model. Twelve concrete cores were removed from the arch. The locations of these cores are shown in Figure 7.1b. The pore solutions were extracted using the centrifuge technique (see Appendix 7.9.1). The chemical composition was tested (see Appendix 7.9.2), based on which the simulated pore solution was produced. The simulated pore solution was then used for the electrochemical test of the rebar to provide input to the corrosion model. These concrete cores were also used to test the transport characteristics such as permeability, chloride diffusivity (Ramadani, 2013), and water retention curve (Appendix 7.9.3) for the numerical model.

7.5 Numerical modelling

The model consists of two components: the transport model and the corrosion model. The 3-D transport model defines the geometry of the arch component, where the governing equations that describe the transport of water, chloride ion, and carbon dioxide gas were applied. Different boundary conditions were also selected to properly reproduce the exposure environment of the arch. The corrosion model predicts the corrosion rate based on the microenvironment at the rebar depth simulated by the transport model. The models were developed with a commercial FEM software, COMSOL Multiphysics and solve on the Plato high-performance computing cluster at the University of Saskatchewan.

7.5.1 Transport model

7.5.1.1 Water flow in concrete

In many practical conditions, a porous medium like concrete is not fully saturated. The degree of saturation depends on the surrounding environmental moisture conditions. When concrete is not fully saturated, a negative capillary pressure develops and facilitates the absorption of water, and this pressure is the driving force of water flow within concrete. Under such conditions, the permeability becomes dependent on the water content. Darcy's law has been extended to describe water flow for this kind of problem in unsaturated porous media. One of the earliest and most widely used models for water flow in unsaturated porous media is known as Richard's equation. When coupled with the mass balance equation for an unsaturated porous medium, the governing equations become

$$C\frac{\partial H_p}{\partial t} = \nabla \cdot \left[K_s k_r \nabla\left(H_p + Z\right)\right] \tag{7.1}$$

where, $H_p\,[m]$ is pressure head, $Z\,[m]$ is elevation, $t\,[s]$ is time, $K_s\,[m/s]$ is hydraulic conductivity of concrete at full saturation and $k_r\,[-]$ is relative permeability, defined as the ratio between the hydraulic conductivity at volumetric water content θ, $K(\theta \leq \theta_s)$ to $K(\theta = \theta_s)$, where $\theta_s\,[-]$ is saturated volumetric water content or the total connected porosity, $C\,[1/m]$ is defined as a change in storage due to change in pressure head.

$$C = \frac{\partial \theta}{\partial H_p} \tag{7.2}$$

A constitutive relationship or model that has been shown to work with concrete is the van Genuchten-Mualem model (Kumar, 2010; Li, 2014), given by Equations (7.3)–(7.6) (Mualem, 1976; Van Genuchten, 1980).

$$C = S_e \cdot S_s + \frac{\alpha m}{1-m}(\theta_s - \theta_r) S_e^{\frac{1}{m}} S_e \left(1 - S_e^{\frac{1}{m}}\right)^m \tag{7.3}$$

$$k_r = S_e \left(1 - \left(1 - S_e^{\frac{1}{m}}\right)^m\right)^2 \tag{7.4}$$

$$S_e = \begin{cases} \left[1 + \left|\alpha H_p\right|^n\right]^{-m}, & H_p < 0 \\ 1, & H_p \geq 0 \end{cases} \tag{7.5}$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \tag{7.6}$$

where, $k_r\,[-]$ is relative permeability, $S_e\,[-]$ is effective saturation, $\theta_s\,[-]$ is saturated water content, $\theta_r\,[-]$ is residual water content, $S_s\,[1/m]$ is the storage coefficient. $l\,[-]$, $m\,[-]$, $n\,[-]$,

and $\alpha\,[1/m]$ are empirical parameters ($m = 1 - 1/n$). These empirical parameters were estimated from a water retention test specifically for the extracted concrete from the arch.

These empirical parameters for the site-extracted concrete were determined by a water retention test conducted in the environmental lab of the University of Saskatchewan. This test examined how the degree of saturation depends on the surrounding environmental moisture conditions. The extracted concrete from the University bridge arch was sliced and put in humidity chambers with relative humidity (RH) ranging from 14.5%-100% to reach the equilibrium with the surrounding water vapour. The relationship between pressure head and RH can be derived from the Kelvin and Young-Laplace equation:

$$\ln(RH) = \left(\frac{H_p V_m \rho g}{RT}\right) \tag{7.7}$$

where $RH\ [-]$ is relative humidity, $H_p[m]$ is pressure head, $V_m[m^3/mol]$ is the molar volume of liquid, $\rho\ [kg/m^3]$ is the density of water, $g[m/s^2]$ is gravitational accretion rate, $R\ [J/mol/K]$ is gas constant, and $T[K]$ is temperature.

The RH was then converted into the pressure head (H_p) in these sliced concrete samples in each humidity chamber. The effective saturation (S_e) of each sample was determined by measuring the sample weight change at the equilibrium state in each humidity chamber with respect to its vacuumed-water-saturated state. The empirical parameters α=0.0018 [1/m], and n=1.7 were then determined by curve fitting the H_p-S_e data with Equation (7.5), as shown in Figure 7.3.

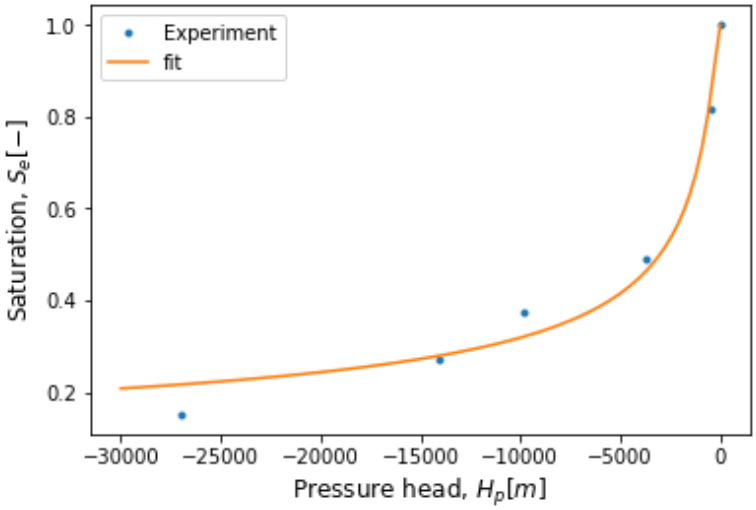


Figure 7.3 Water saturation as a function of pressure head. The experimental data were fitted with van Genuchten-Mualem model.

7.5.1.2 Chloride ion transport

The transport of dissolved chloride in concrete is governed by the diffusion-advection equation, whereby:

$$\frac{\partial(\theta c)}{\partial t} = -\nabla \cdot \boldsymbol{J} \tag{7.8}$$

$$\boldsymbol{J} = -D_{eff}\frac{\partial c}{\partial z} + \boldsymbol{u}c \tag{7.9}$$

$$D_{eff} = D_s \cdot \Phi(S_e) \tag{7.10}$$

where $\boldsymbol{J}$ $[kgm^2/\mathrm{s}]$ is the total volume-averaged mass flux of solute in concrete, $\boldsymbol{u}$ $[m/\mathrm{s}]$ is Darcy's velocity, c $[kg/m^3]$ is the liquid phase-averaged solute concentration, D_{eff} $[m^2/\mathrm{s}]$ effective diffusivity, $D_s[m^2/\mathrm{s}]$ is effective diffusivity under fully saturated conditions, $\Phi(S_e)$ is a

reduction function that accounts for discontinuity in the liquid phase, determined by correlating to the reduction in resistivity due to the increase in water saturation (Ramadani, 2013).

7.5.1.3 Carbonation

Carbonation is modelled as gas diffusion with the sink term representing the consumption of CO_2 due to the carbonation reaction (Equation 7.11 and 7.12). The calculated concentration of carbon dioxide is calibrated to pH in a carbonated synthetic pore solution.

$$\frac{\partial[\epsilon_g\langle\rho_{CO_2}\rangle^g]}{\partial t} = \nabla\cdot\left[D_{eff}^g\nabla\langle\rho_{CO_2}\rangle^g\right] + \text{sink} \tag{7.11}$$

$$\text{sink} = -\Phi'(S_e)[1-\Phi(\epsilon_s)]\epsilon_g\langle\dot{m}_{CO_2}\rangle^g \tag{7.12}$$

where, ϵ_g is the gas volume fraction, ϵ_s is the solid volume fraction, $\langle\rho_{CO_2}\rangle^g$ is the CO_2 density in the gas phase, D_{eff}^g is effective diffusivity of CO_2 in the gas phase, $\langle\dot{m}_{CO_2}\rangle^g$ is the rate of the mass density change of the CO_2 in the gas phase due to carbonation reaction, $\Phi(\epsilon_s)$ and $\Phi'(S_e)$ are factors considering the effect of carbonation completeness and moisture content. It is assumed that CO_2 will dissolve in the pore solution and instantaneously react with $Ca(OH)_2$ dissolved in the solution. The resultant $CaCO_3$ precipitates onto the solid surface. The total reaction is described as

$$Ca(OH)_2(s) + CO_2(g) \xrightarrow{H_2O} CaCO_3(s) + H_2O(l) \tag{7.13}$$

The Deposition of $CaCO_3$will increase the solid volume. The carbonation reaction keeps going until certain layers of $CaCO_3$ form on the surface of concrete pores. It is assumed that the partially covered solid surface will not sufficiently supply $Ca(OH)_2$ to the pore solution hence a reduced CO_2 consumption rate. This phenomenon is accounted for by a reduction function of carbonation degree $\Phi(\epsilon_s)$, which is the ratio of the increased solid volume fraction to its maximum, expressed in Equation (7.15). Eventually, a maximum solid volume fraction will be reached, and the

carbonation reaction stops. The effect of moisture content is accounted for by the factor Φ‘(Se) (Saetta et al., 1993).

The consumption rate of $CO_2$with its reaction with dissolved $Ca(OH)_2$ can be expressed as

$$\langle \dot{m}_{CO_2} \rangle^{\mathrm{g}} = \Phi'(S_e)\, k_{CO_2}(T) \cdot \langle \rho_{CO_2} \rangle^{g} \tag{7.14}$$

$$\Phi(\epsilon_s) = \frac{\epsilon_s - \epsilon_0}{\epsilon_{smax} - \epsilon_0} \tag{7.15}$$

where, $k_{CO_2}(T)[1/s]$ is reaction coefficient, a function of temperature, $\epsilon_0[-]$ is the initial solid volume fraction, $\epsilon_{smax}[-]$ is the maximum solid volume fraction. The solid volume change can be expressed as

$$\frac{\partial \epsilon_s}{\partial t} = \frac{\langle \dot{m}_{\mathrm{CO_2}} \rangle}{M_{\mathrm{CO_2}}} \cdot \left[\mathrm{V_{m.CaCO_3}} - \mathrm{V_{m.Ca(OH)_2}} \right] \tag{7.16}$$

where, $\mathrm{V_{m.CaCO_3}}[m^3/mol]$ and $\mathrm{V_{m.Ca(OH)_2}}[m^3/mol]$ are molar volumes, $M_{\mathrm{CO_2}}[kg/mol]$ is molar mass.

7.5.1.4 The implementation of boundary conditions to different surfaces of the arch

The field investigation showed that the structure was sheltered with varying effectiveness. It is necessary to mark the arch component into three categories (Figure 7.4):

- **Sheltered area**: all the columns, curtain walls, arch soffit, and selected arch top area that are well protected by spandrel walls and deck.
- **Unsheltered area**: sides of the arch exposed to the water.
- **Partially sheltered area**: the selected arch top that is not sufficiently protected by the spandrel walls and deck.

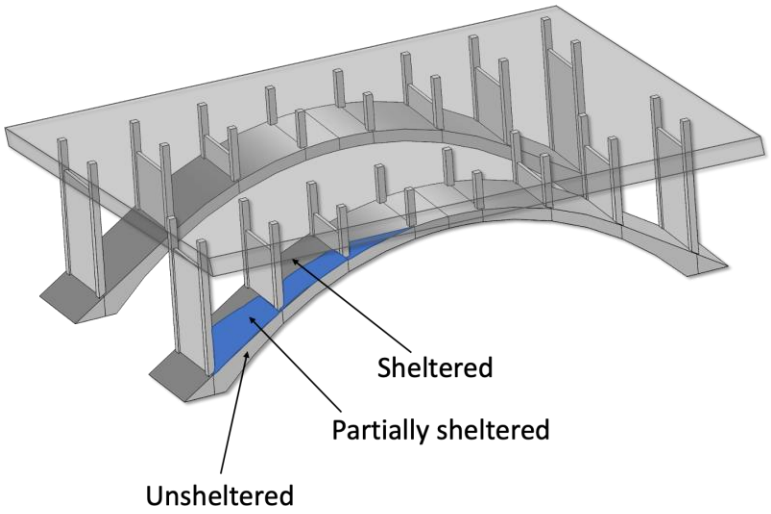


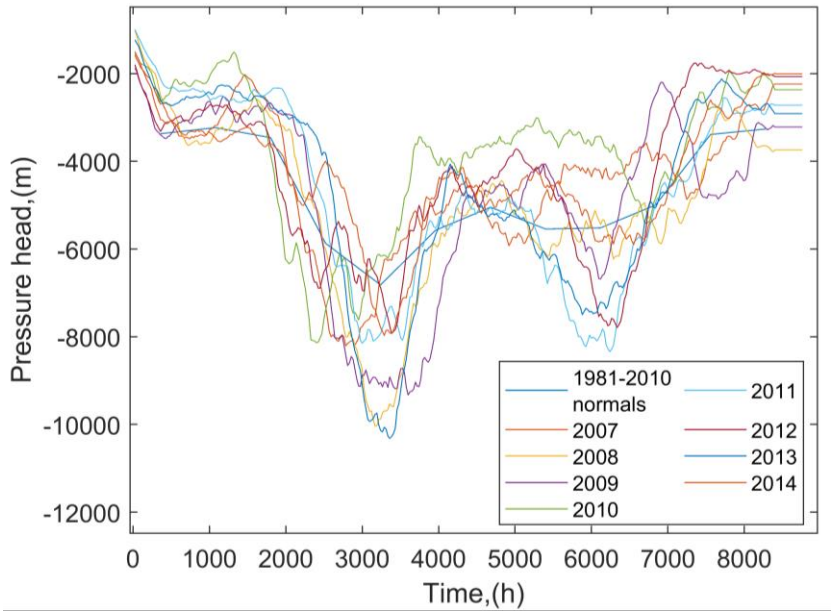
Figure 7.4 Sheltering categories of the arch: Sheltered, Unsheltered, and Partially sheltered areas based on field investigation findings.

7.5.1.5 Boundary conditions for water flow in concrete

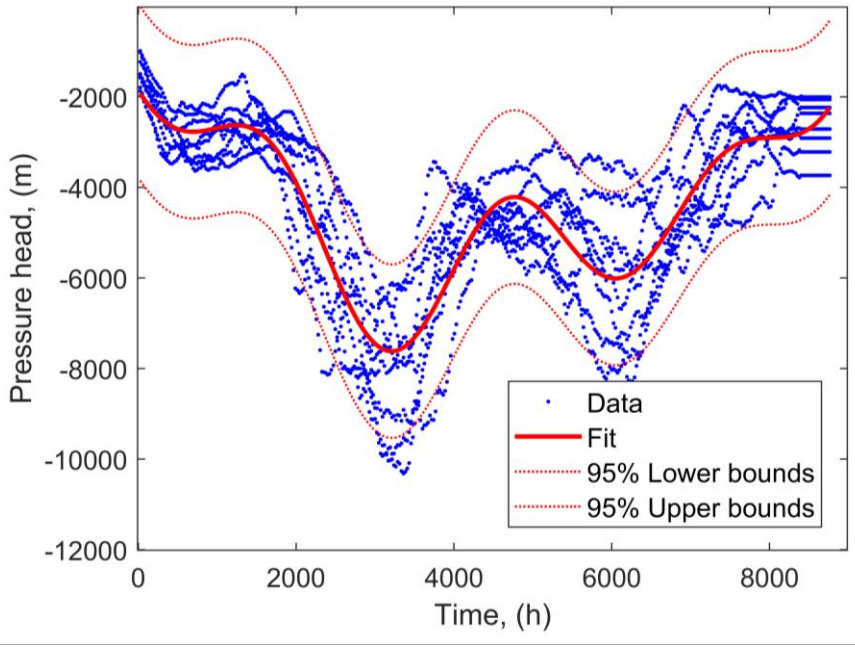
Pressure head is a function of temperature and relative humidity as expressed in Equation (7.7). It was chosen as the boundary conditions, which represent the moisture environments the concrete arches were exposed to, and fundamentally affect the transport properties of arch concrete. The raw weather data (temperature, RH, precipitation) were collected from two weather stations: SASKATOON DIEFENBAKER INT'L A, and SASKATOON RCS. All raw hourly data were smoothed through a moving average filter.

- **Sheltered area**

Hourly and daily data from the years 2007–2014 and normalized monthly data from year 1981 to year 2010 were used to determine the averaged boundary conditions for the sheltered portion of the arch. The averaged value of the pressure head was determined through non-linear regression analysis (Figure 7.5). The averaged boundary conditions were applied to the sheltered area (Figure 7.4).



(a)



(b)

Figure 7.5 Moisture boundary conditions for the sheltered portion of an arch: (a) Raw data of pressure head obtained from weather stations; (b) averaged data fitted from the raw data.

- Unsheltered areas

The unsheltered areas are exposed to both air moisture and rain. Figure 7.6 shows the unsheltered boundary condition for the year 2013, where the vertical lines switch pressure head instantly to 0 representing soaking wet on the concrete surface when the rain amount is greater than 1 mm. These

on-and-off rain incidents, however, are not favourable in terms of computation cost: smaller time step needs to be used to prevent skipping the rain incidents, and during the rain incidents because of the drastic change of the boundary condition, extra smaller time steps were needed for the solver to find converged solutions. The timing of rain incidents is usually handled directly by imposing a water flux (Künzel, 1995) or setting the pressure head as zero. As mentioned in the introduction, this rapid change in the boundary condition is not suitable for the large-scale 3-D model. There are practices to allocate the monthly total rain hours evenly throughout the month, hence repeated wetting-drying cycles in a given month (Shimomura, 2015). Although their sensitivity analysis shows that this approximation gives acceptable accuracy, this approximation still retains rapid change of the boundaries.

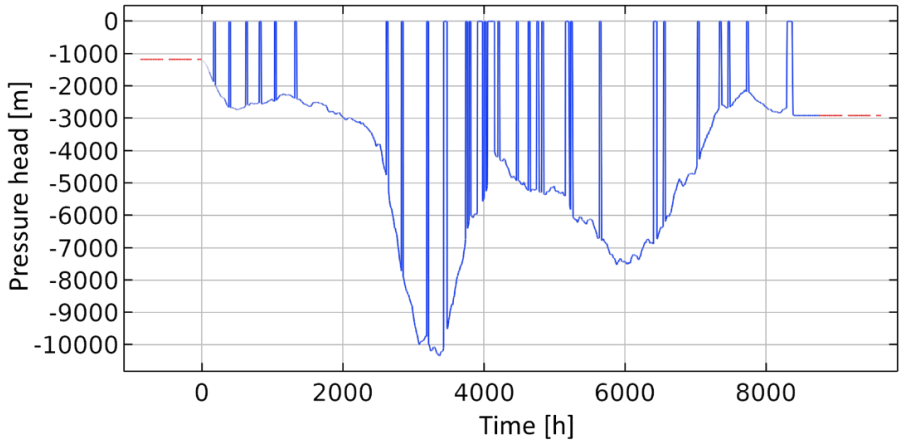


Figure 7.6 Moisture boundary conditions for the unsheltered portion of the arch (eg, Year 2013).

In this paper, an equivalent boundary condition is proposed to speed up the solving in the large-scale 3-D model. The equivalent boundary condition was sought to produce similar moisture content near the rebar depth (10 cm) as the original "spiky" boundary conditions in a 1-D model where the top surface is unsheltered, subject to rain and air, and the bottom surface is sheltered subject to only air. A reverse modelling iterative workflow was developed to find such boundary conditions (BC). As illustrated in Figure 7.7, the saturation results from the 5 cm depth were used as the initial value for the updated top BC, and the 10 cm depth saturation level is set as the target value for the updated model to approach at the same depth. The deviation of the saturation level at the 10 cm depth in the updated model was used to adjust the previous BC assignment with a

relaxation constant ranging from 0 to 1. This iterative workflow ensures that the saturation level acquired by the equivalent BC gives a similar saturation as the original boundary conditions with the occurrence of rain incidents.

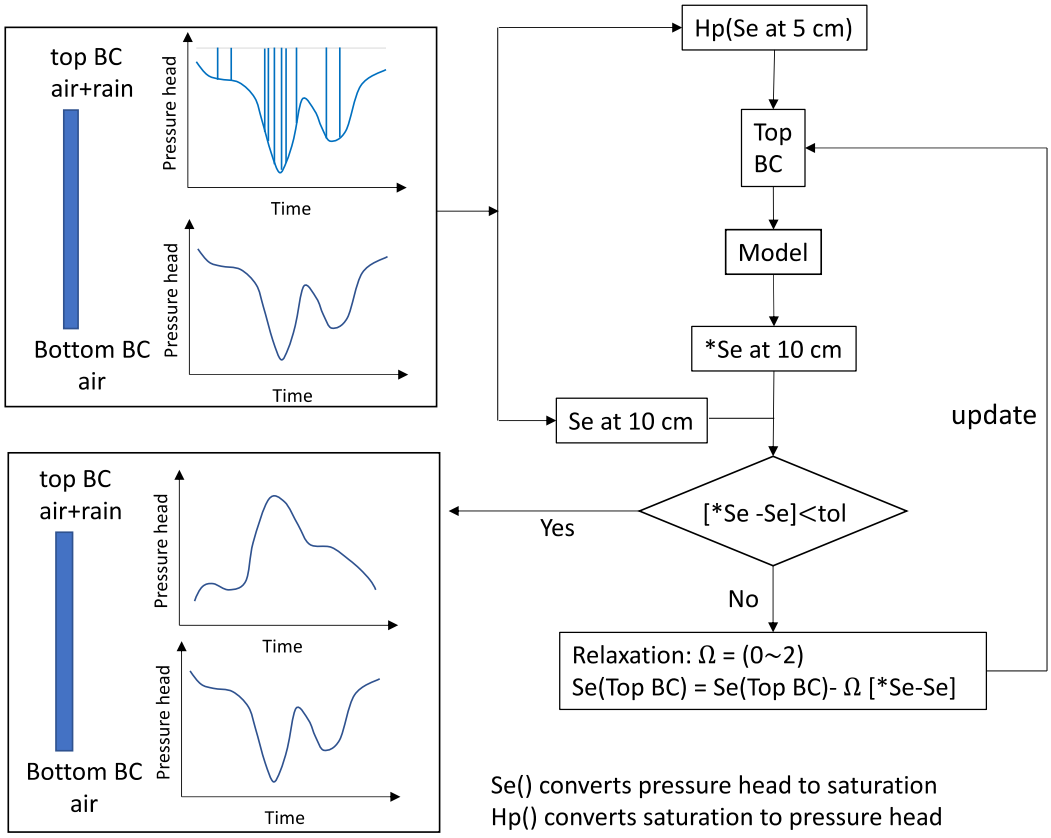


Figure 7.7 Workflow to get the equivalent boundary conditions for the arch top surface exposed to ambient humidity and rain.

Following this workflow, the original boundary condition data from 2009–2014 was used to create the original water saturation level at 5 cm, 10 cm, and 15 cm. The average saturation level at 10 cm was then fitted and used as the target for the equivalent BC to approach. The simulations ran for three repeating years for each year's BC to get a stable moisture distribution. The saturation levels at three depths were compared between the full spiky BC and the equivalent BC, as shown in Figure 7.8. The smooth equivalent boundary captured the seasonal moisture fluctuation at all the selected depths.

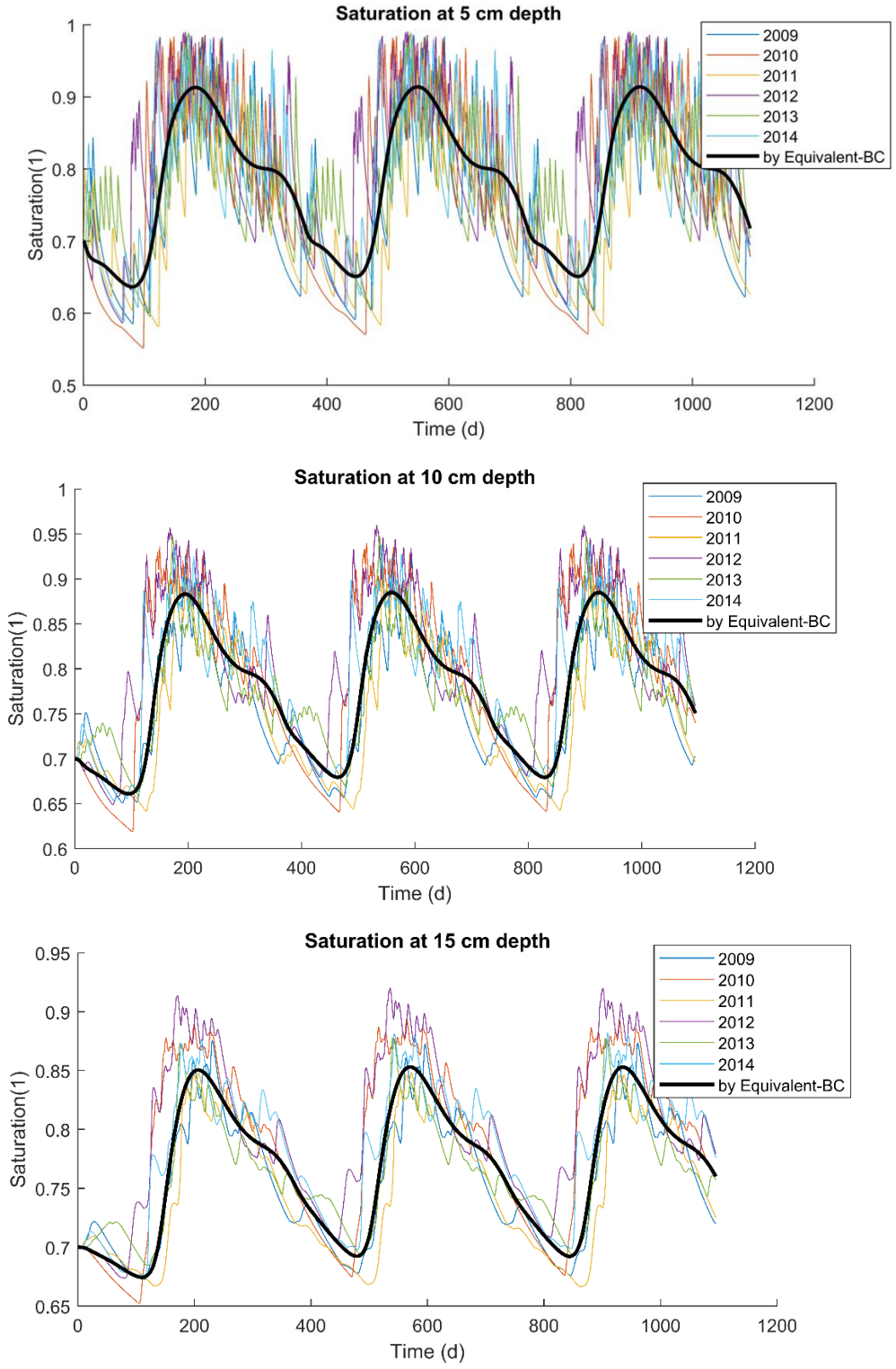


Figure 7.8 Comparison of the saturation results simulated from the equivalent boundary conditions vs. the spiky boundary conditions of year 2009-2014 for the unsheltered portion of the arch.

- Partially sheltered area

For the areas exposed to a varying combination of air moisture and rain, three 2-D linear ramp functions were used to describe the varying combinations, such that the weights of the sheltered conditions and unsheltered conditions were 0 and 1 on the edge of the arch and 1 and 0 on the diagonal line on the arch (Figure 7.4).

7.5.1.6 Boundary conditions for chloride ion transport and carbonation

Boundary conditions for chloride ion transport consisted of two conditions: “concentration” and “no-flux”. Under the “concentration” condition, the concrete surface was subject to 3% wt. NaCl solution, while in “no-flux,” there was zero flux of chloride ions across the surface and existing chloride ions on the surface transport with the concentration gradients and water flow in the concrete domain. The “concentration” boundary condition was implemented for 90 days of simulated annual time from May 15 to July 15 due to a high volume of precipitation and the washed-off accumulated de-icing salts on deck from the winter season. For the rest of the time, “no-flux” was implemented. These conditions were applied to the sheltered and partially sheltered regions as illustrated in Figure 7.4.

For the carbonation process, a constant $0.035\%\ CO_2$ concentration was applied to all exposed arch areas, including curtain walls, columns, arch top and soffit.

7.5.2 Corrosion model

In this large-scale 3-D model, the corrosion rate of rebar in concrete was estimated based on the chloride concentration, pH, and saturation in concrete at rebar depths. The microenvironment difference in concrete resulting from the sheltering effect is expected to have a large length scale over the arch. Therefore, the macro-cell coupling in the rebar mesh is considered negligible, and the microcell form is the major contributor to total corrosion.

The microcell corrosion model was constructed, avoiding the use of modelling of microcell corrosion with the macro-cell fashion, where the single set of corrosion parameters and discrete anodes and cathodes and their area ratios are used. Therefore, the associated limitations such as the need for define the anode and cathode area ratio, discretizing the anode and cathode surface at

a microscopic scale, were avoided. Instead, finding and solving the corrosion current density for rebar exposed to different chemical environments was done by the potentiodynamic scans of rebars in the simulated pore solution with three levels of chloride concentration (0 M, 0.01 M, 0.2 M), three levels of pH (8.5, 12, 13.5), which was described in the previous work (Kennell et al., 2015).

The simulated pore solution used in the test was formulated based on the pore solution from the concrete cores from the arch of the bridge. Pore solution was extracted with the centrifuge technique (Appendix 7.9.1), and the chemical composition of the major ions are tested with spectrophotometry and flame-photometry and are listed in Table 7.1.

Table 7.1 Pore solution composition of the arch concrete

	Cl^- [M]	SO_4^{2-} [M]	Mg^{2+} [M]	Ca^{2+} [M]	Na^+ [M]	K^+ [M]	pH [-]
Mean	0.0246	0.0193	0.0035	0.0121	0.0841	0.0297	12.0
Standard deviation	0.0415	0.0259	0.0059	0.0069	0.0197	0.0087	0.2065

The measured corrosion current density is a function of chloride concentration and pH; the interpolated surface plot is shown in Figure 7.9. The effect of moisture on the rebar corrosion rate in concrete was accounted for by a reduction factor (Figure 7.10), based on the measurement of reinforced concrete corrosion rate at different saturation levels from Tuutti's work (Dangla & Dridi, 2009; Tuutti, 1982).

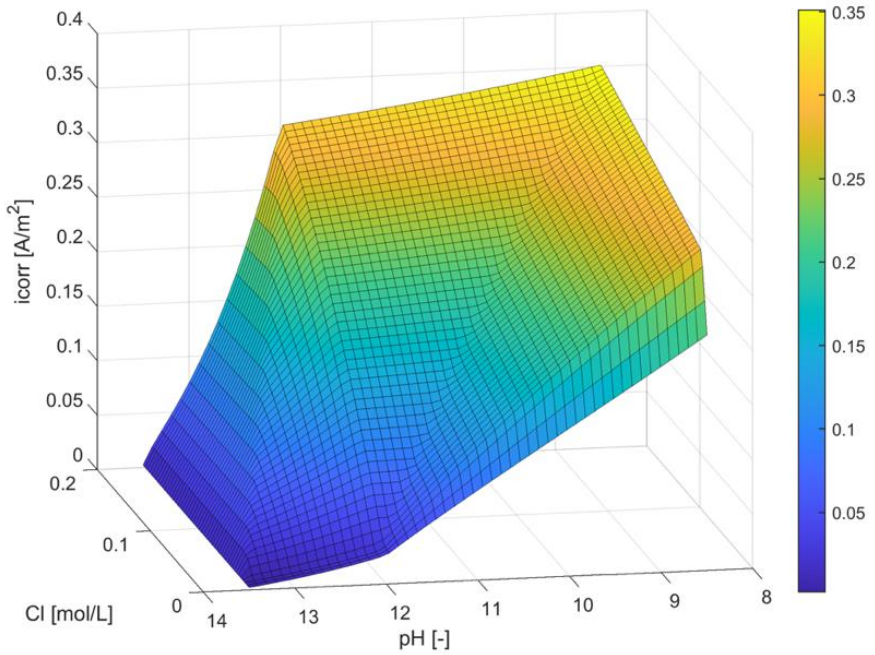


Figure 7.9 Corrosion current density as a function of pH and chloride concentration in the simulated pore solutions.

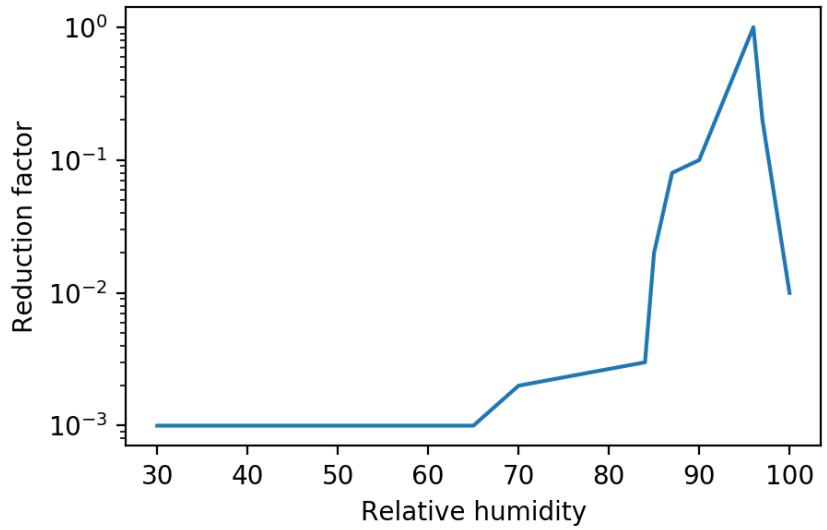


Figure 7.10 Reinforced concrete corrosion rate reduction factor relative to saturated condition, adapted from Tuutti's work (Dangla & Dridi, 2009; Tuutti, 1982).

7.6 Simulation results and discussion

For the simulated case of 100 years of exposure, the predicted distribution of pH and concentration of chlorides are plotted in Figure 7.11 and Figure 7.12, respectively. As the carbonation and chloride ion ingress proceeded through the concrete cover, the once passivated rebar became active and thus had a higher corrosion rate occurred. The corrosion loss in thickness was calculated by the integration of the corrosion rate over time. The results of corrosion loss in thickness on the rebar after 100 years of exposure are plotted in Figure 7.13 and the prediction of additional corrosion between 101 years and 150 years of service is plotted in Figure 7.14.

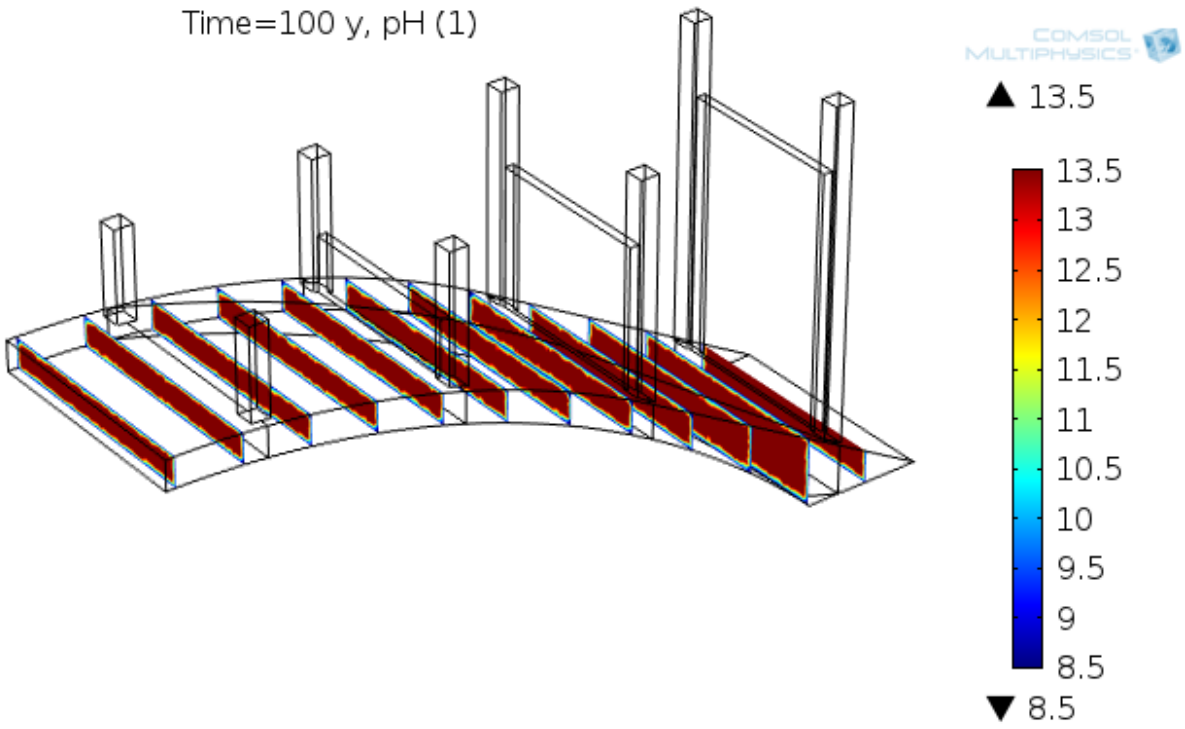


Figure 7.11 pH distribution in Arch C (Bay 5, Bay 6, Bay 7) after 100 years of carbonation.

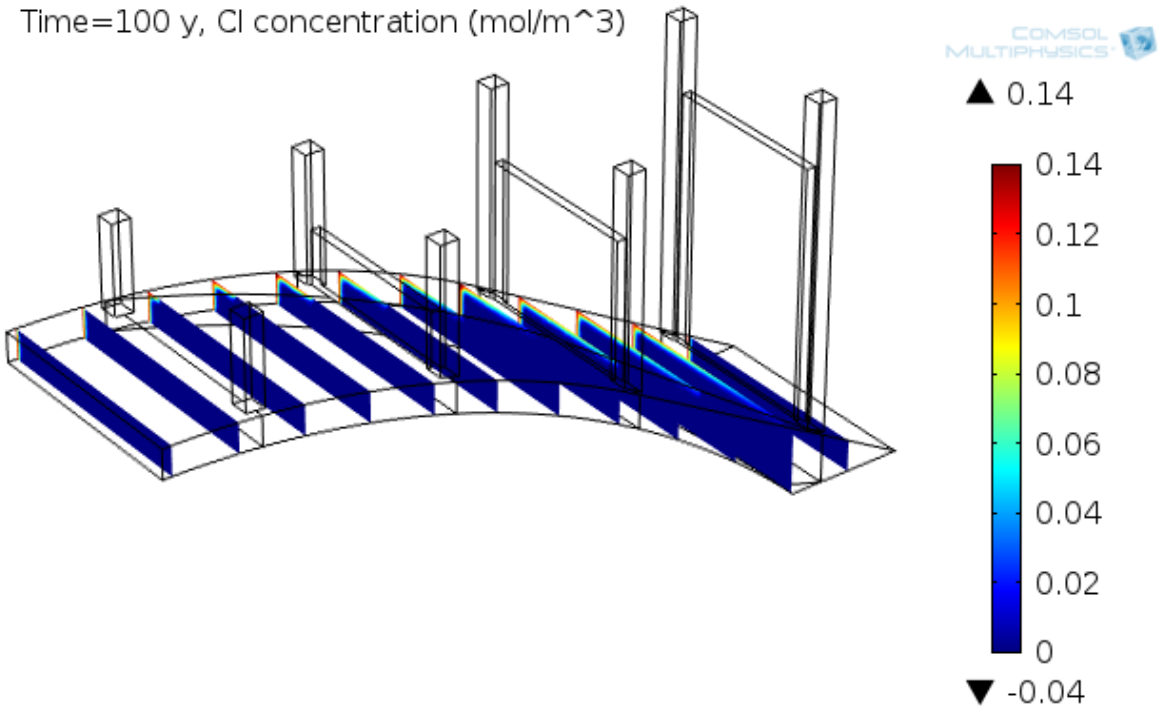
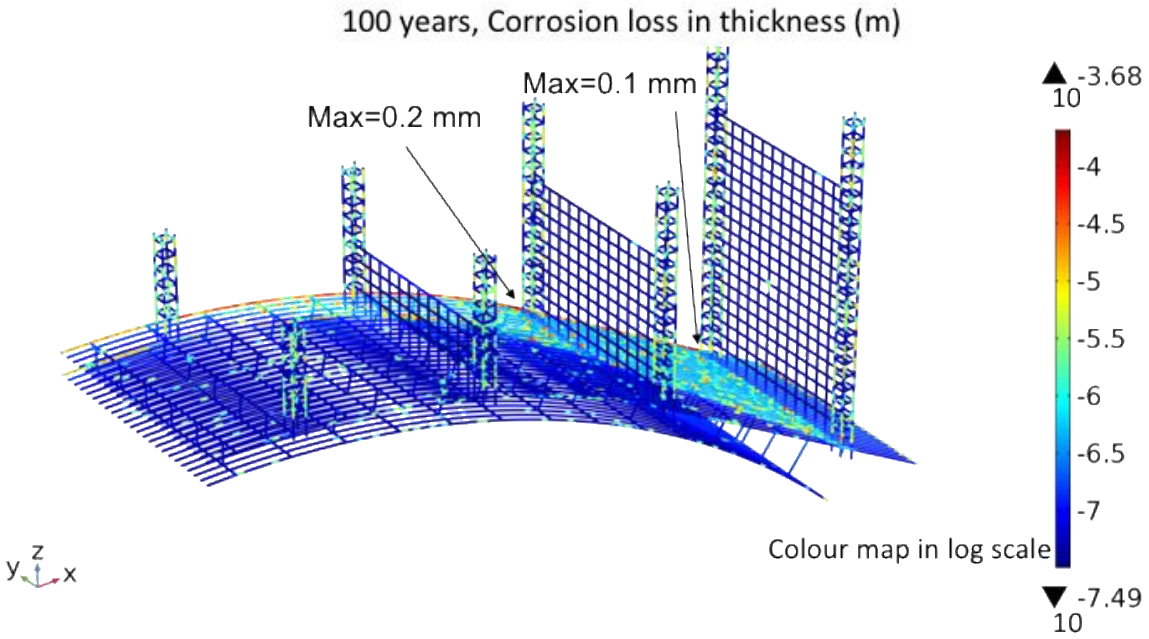
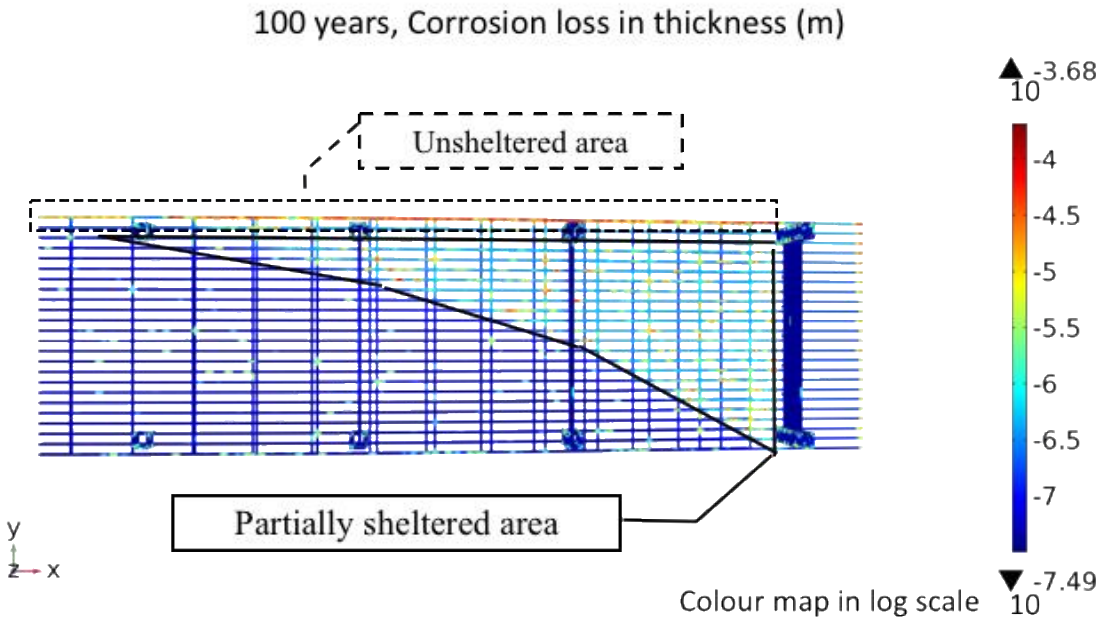


Figure 7.12 Chloride concentration distribution in Arch C (Bay 5, Bay 6, Bay 7) after 100 years of exposure.

As shown in Figure 7.13, where the corrosion thickness loss is plotted on log scale, the majority of corrosion is predicted to occur on the upper layer of longitudinal rebars that are close to the most exposed arch edge where the highest localized corrosion (0.2 mm and 0.1 mm loss in thickness) is predicted on that layer of rebars near the columns. High corrosion losses are mainly found on the two longitudinal rebars near the most exposed edges, and occasionally on the upper layer rebars in the partially sheltered area near the curtain walls. These results agreed well with the current condition of the arch, where recent concrete cover patches were found near the high corrosion loss region.



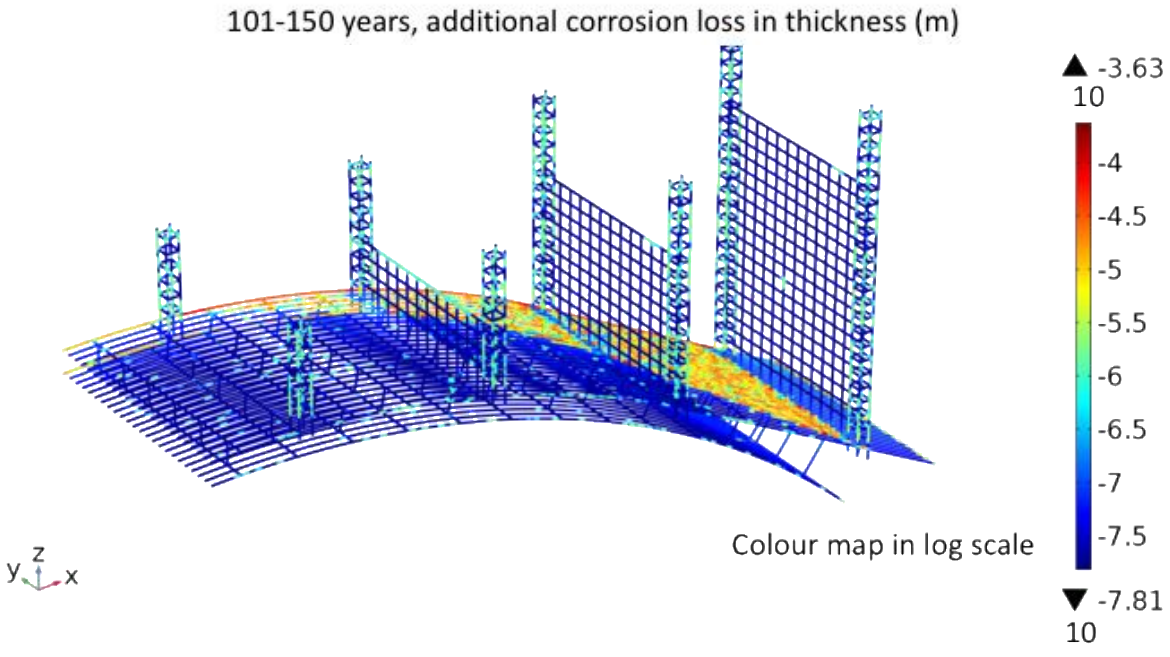
(a)



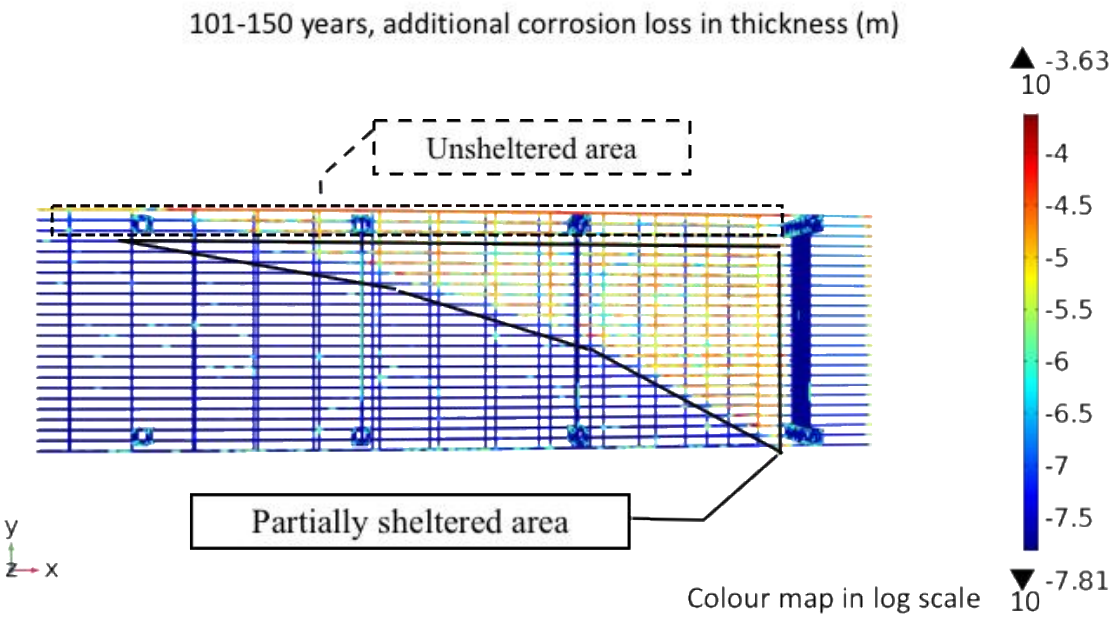
(b)

Figure 7.13 View of the rebar in Arch C (Bay 5, 6, and 7) showing corrosion loss in thickness after 100 years of exposure, plotted in log scale: (a) 3D isometric view, (b) plan view of arch.

The upper layers of rebar in the partially sheltered area have both carbonation and chloride exposure to a lesser extent than the arch edge rebars, and thus have a moderate predicted corrosion loss. In the sheltered area of the arch, a low corrosion rate shows that without exposure to direct rain and runoff water, chloride transport has a limited effect on corrosion. The vertical column rebars have an insignificant corrosion loss due to limited exposure to moisture, even though these rebars have the least concrete cover thickness and the carbonation-induced pH drop would increase the overall corrosion rate. If this part of the bridge remains untreated for another 50 years, corrosion loss is predicted to increase significantly due to deeper carbonation and more accumulation of chloride ions. As shown in Figure 7.14, the maximum additional loss in thickness is slightly larger than that in the previous 100 years altogether. The severity of corrosion in a partially saturated area approaches the condition in the most exposed area. It is suggested to take proactive rehabilitation actions in the next few years to prevent major repairs in the long run. Such actions could include one or more of the following: lowing the moisture content in the concrete by the application of surface sealant, cathodic protection for the most severe hot spots, or increasing the pH, by re-alkalization of the old concrete cover.



(a)



(b)

Figure 7.14 View of the rebar in Arch C (Bay 5, 6, and 7) showing additional corrosion loss during 101 years –150 years of exposure, plotted in a log scale: (a) 3D isometric view, (b) plan view of arch.

7.6.1 Climate change: the effect of elevated CO_2 concentration and change in precipitation on rebar corrosion

The impact of climate change on the built environment may exacerbate the problem of rebar section loss in arches. Climate change will lead to more frequent and intense extreme weather events, changes in temperature and precipitation patterns, which can all affect the corrosion and mechanical properties of rebar. In particular, changes in moisture and CO_2 can alter the microenvironment in concrete that affects rebar corrosion and section loss.

In this study, we explored the impact of various climate scenarios on the probability density distribution of rebar section loss over a 150-year period, encompassing both the present (100 years) and the next 50 years. Simulation results for the past 100 years of service agree well with the field-surveyed condition of the arch in 2014, where concrete patches were made in the most exposed area with high rebar section loss/rust accumulation (Figure 7.15). For the next 50 years into the future, we consider three different climate scenarios, as shown in Table 7.2. Scenario 1: baseline, Scenario 2: with increased CO_2 and higher precipitation, Scenario 3: with increased CO_2 and lower

precipitation due to drought. The increasing CO_2 emission is linearly extrapolated from the historic atmospheric CO_2 concentration (US Department of Commerce). The change of CO_2 concentration is shown in Figure 7.16.



Figure 7.15 Field observation showing concrete patches for corrosion-induced damage on Arch C in the unsheltered area where high corrosion section loss/rust accumulation was predicted.

Table 7.2 Climate scenarios regarding CO_2 emission and precipitation over the next 50 years

Climate scenario	CO_2	Precipitation
Scenario 1 (Baseline)	CO_2 remains at its current level of 400 ppm	Not changed
Scenario 2 (Increased CO_2, wet)	CO_2 emission increases at the current rate, increases up to 525 ppm in 50 years	Precipitation increased by 20%
Scenario 3 (Increased CO_2, drought)	CO_2 emission increases at the current rate, increases up to 525 ppm in 50 years	Precipitation decreased by 20%

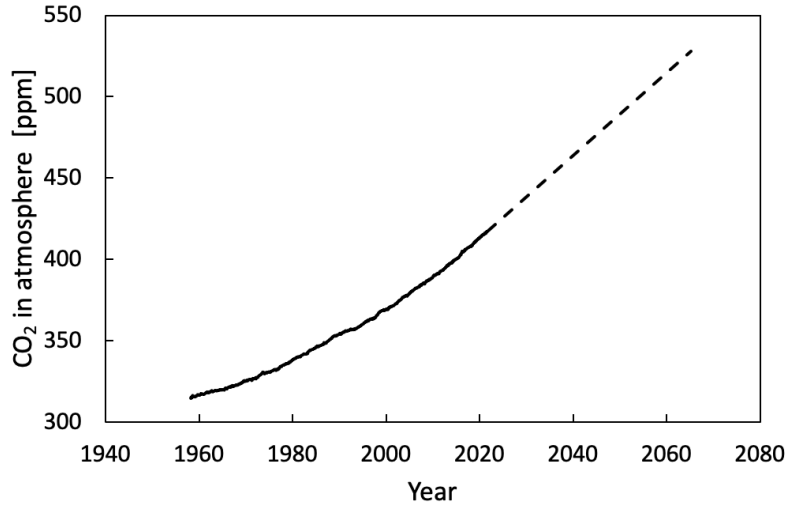


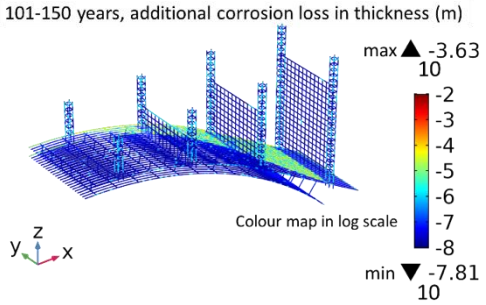
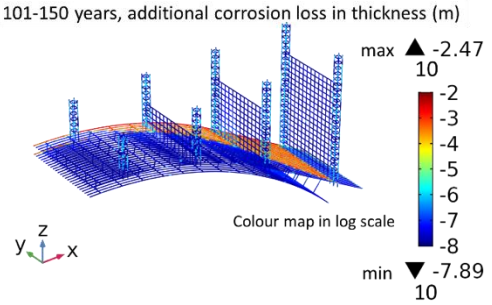
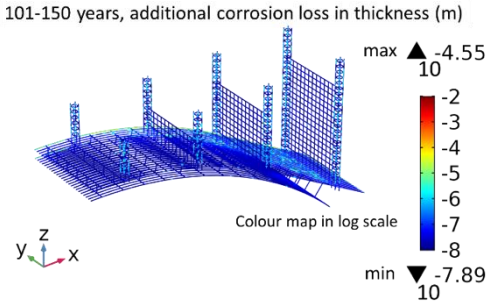
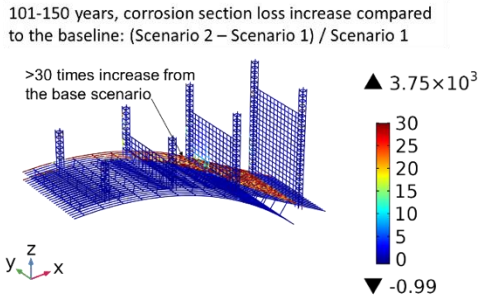
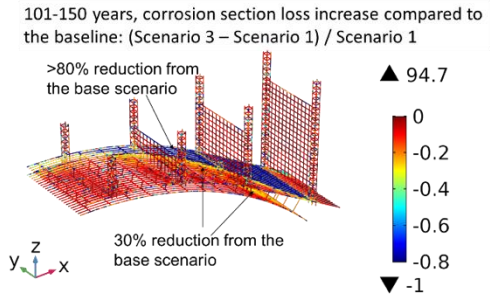
Figure 7.16 Time series plot of monthly mean CO_2 concentration from 1958 to 2021 (US Department of Commerce)**. The dotted line extrapolates with a linear upward trend in CO_2 concentration.**

A comparison of rebar section loss over a service life in the next 50 years in these three different climate scenarios are presented in Table 7.3. The section loss difference in Scenario 2 and Scenario 3 compared to the baseline scenario is calculated as a percentage change.

In Scenario 2, the carbonation and chloride penetration progress faster compared to the baseline Scenario 1. The rebar depassivation occurs earlier than the baseline scenario. The total rebar section loss in 50 years significantly increases in the sheltered and partially sheltered area, by more than 30 times for the maximum section loss. Scenario 2 not only increases the maximum section loss but also exacerbates broader corrosion section loss in the partially sheltered area. The sheltered area showed little difference in rebar section loss.

In Scenario 3, the depassivation of rebar is expedited due to faster carbonation, despite the reduced chloride concentration at rebar depth, when compared to Scenario 1 (the baseline). However, reduced moisture levels would keep the active corrosion rate at a lower rate. Scenario 3 reduced the accumulated section loss. The greatest reduction (greater than 80%) is found in sheltered and partially sheltered areas. It also brought about a 30% reduction in section loss for the rebar under or near the sheltering-influence area, such as the bottom layer rebar and adjacent rebars.

Table 7.3 Comparison of Rebar Section Loss in Three Climate Scenarios over 50 Years

50 years into future	Scenario 1 baseline	Scenario 2 increased CO_2, wet (20% more precipitation)	Scenario 3 increased CO_2, drought (20% less precipitation)
Additional Section Loss in 50 years			
Section loss difference comparing to Scenario 1		 (Scenario 2- Scenario 1)/ Scenario 1	 (Scenario 3- Scenario 1)/ Scenario 1

To visualize the distribution of section loss for the three Scenarios over time, a violin plot (Figure 7.17) was used to show the distribution of section loss on all rebars at a given time, with mean values connected by dashed lines. The findings suggest that the severity and frequency of rebar section loss have increased in the 50 years, particularly with Scenario 2 (increased CO_2 and participation) where the rebar section loss is significantly increased. In Scenario 3, the distribution curve shape of rebar section loss remains largely unchanged when compared to Scenario 1. However, a slight decrease in the distribution of high section loss values has led to a noticeable reduction in the mean value. As such, while Scenario 3 effectively suppresses high rebar section loss, it does not have a significant impact on overall corrosion deterioration.

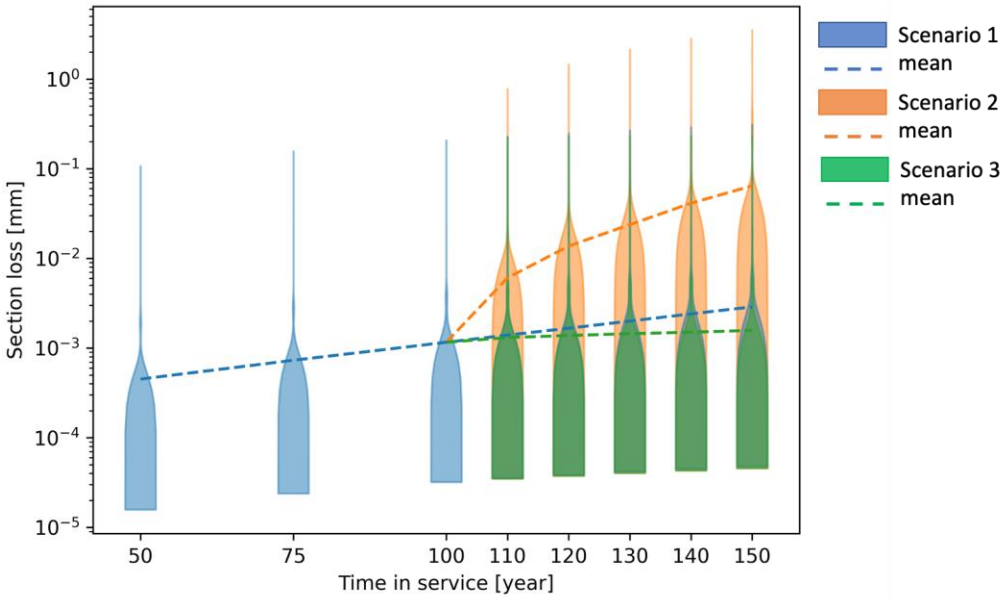


Figure 7.17 Rebar section loss across the arch over 150 years in service. The last 50 years has different climate scenarios. Scenario 1: Baseline, Scenario 2: increased CO_2 and wet (20% more precipitation), Scenario 3: increased CO_2 and drought (20% less precipitation). Each violin represents the distribution density of the section loss over all rebars at a given time. Mean values of section loss are connected by dashed lines.

The findings indicate the importance of considering climate change in the maintenance and management of infrastructure. For example, in Scenario 2, where increased CO_2 and precipitation exacerbate rebar section loss, it is recommended to prioritize maintenance and repair activities in areas prone to corrosion. This includes the implementation of regular inspections, and timely repair

of corroded areas. Additionally, it is essential to improve drainage systems by cleaning off debris and bird droppings in the way. In Scenario 3, where high rebar section loss is suppressed but overall corrosion deterioration remains unaffected, it is recommended to allocate budget to further large-scale repair and replacement.

7.7 Conclusions

A 3-D transport model was developed to simulate the microenvironment changes at the rebar depth in a concrete arch bridge exposed to various environmental factors such as chloride, carbonation, humidity, and runoff water. It incorporates real-world data from in-situ sensors monitoring oxygen content, moisture, chlorides, and carbonation in the concrete cover to create a computationally efficient and realistic microclimate depiction around the bridge. The corrosion model expressed the corrosion current density as a function of pH and chloride concentration, capturing their synergistic effect, was directly from the electrochemical test of the old rebar and formulated pore solution with in-situ concrete pore solution composition. It predicted rebar corrosion rates using century-old steel rebar corrosion kinetics and examined the impact of climate scenarios like elevated CO_2 levels and precipitation changes on rebar deterioration.

The outcomes of this study hold significant implications for the understanding and management of corrosion in reinforced concrete structures. Some of the main findings are as follows:

1) The field optical fibre sensor measurements verified the oxygen saturation level and eliminated the possibility that there was a limited cathodic current caused by the hindered oxygen transport.

2) The microenvironment, including chloride content, pH, and water saturation may not be uniform across the whole reinforced concrete structure due to different exposure conditions.

3) In the simulated University Bridge arch component, a combination of carbonation and chloride ingress accelerates the corrosion rates of rebar. Ineffective sheltering from the rain or more rainwater retention on the arch facilitates this penetration and leads to higher corrosion rates.

4) The study also investigated the effects of different climate scenarios on rebar section loss in arches over a 50-year period. Elevated CO_2 level and increased participation result in a substantial increase in rebar section loss, exacerbating broader corrosion section loss, whereas elevated CO_2 level and decreased participation suppresses high rebar section loss but does not have a significant impact on overall corrosion deterioration. Accordingly, different maintenance and management of infrastructure are recommended.

Overall, this study's findings provide crucial insights into the complex interactions between climate, microenvironment, and corrosion behavior in reinforced concrete structures. By incorporating these findings into practical applications, such as infrastructure maintenance and management, it is possible to enhance the resilience and durability of these structures, ensuring their long-term performance and safety.

7.8 Epilogue

Chapter 7 presents a comprehensive 3-D transport model to simulate the microenvironment evolution at the rebar depth of concrete structure in its service conditions. This model was applied to an uncracked RC structure subjected to service conditions with the source of chloride, carbonation, varying air relative humidity, and exposure to unevenly sheltered run-off water. To formulate realistic boundary conditions for wetting and drying scenarios, historical weather data and field monitoring data were utilized.

Although for this uncracked arch component, the empirical corrosion kinetics specifically for the century-old steel rebar was used, the presented modelling method can be applied to simulate the microenvironment evolution at the rebar depth in cracked concrete under realistic exposure conditions. A comprehensive corrosion model can then be developed in the next chapter by connecting the environment-dependent corrosion kinetics with the environment model.

7.9 Appendix for Chapter 7

7.9.1 Pore solution extraction with centrifuge technique

Twelve two-inch cores were extracted from upstream Arch C and Arch D in a diagonal pattern. The concrete cores were then immediately wrapped with plastic film to prevent any further carbonation. The cores were further cut into shorter cylinders with a height of 5 cm. For each core,

two cylinders at a depth from 5 cm to 10 cm and from 10 cm to 15 cm were selected as the pore solution composition samples, which represent the pore solution conditions at a rebar depth. As shown in Figure 7.18, these samples were saturated with distilled water in a vacuumed dedicator according to ASTM C1202−1.

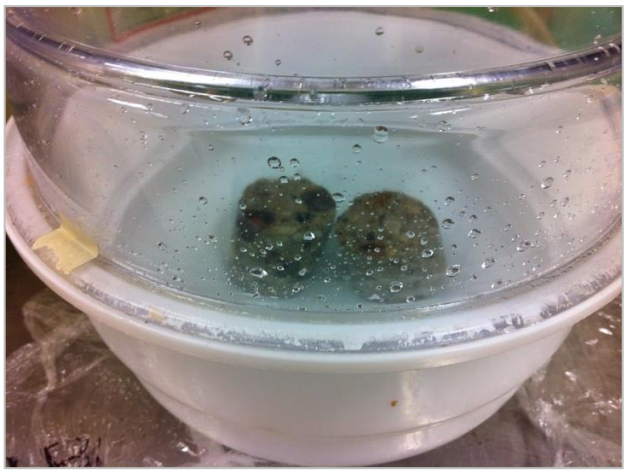
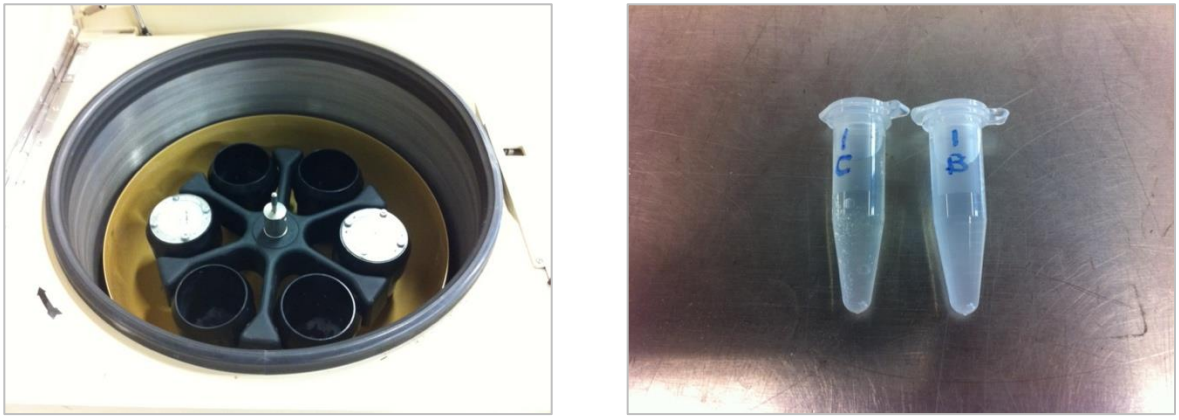


Figure 7.18 Vacuum-saturation of the extracted concrete cores.

The centrifuge technique was employed to extract the pore solution from those saturated concrete cylinders. As illustrated in Figure 7.19, the saturated concrete cores were placed in the centrifuge swing buckets. The centrifuge operated at 3000 rpm for six hours before the pore solution was finally collected from the buckets.



(a) (b)

Figure 7.19 Centrifuge technique (a) Two concrete cores placed in the swing buckets in a centrifuge machine (BECKHAM J6-HC) (b) Pore solution collected from two concrete cores.

7.9.2 Pore solution composition results

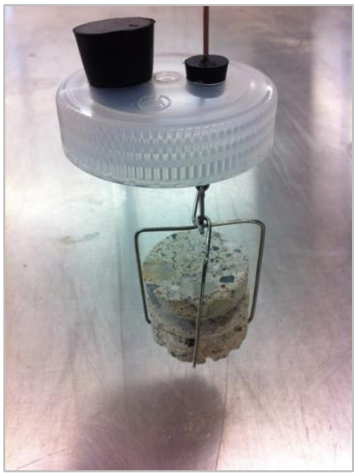
The common species found in concrete such as K^+, Na^+, Ca^{2+}, Mg^{2+}, Cl^-, SO_4^{2-}, OH^- were analyzed. The pore solution composition in as-received partially saturated conditions was derived from the pore solution of the concrete in fully saturated conditions with 0.26 volumetric water content. The test results are summarized in Table 7.4.

Table 7.4 Pore solution composition

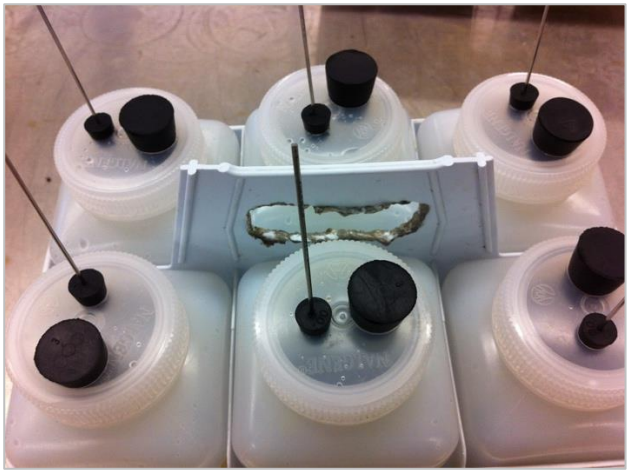
Sample	Cl^-[M]	SO_4^{2-}[M]	Mg^{2+}[M]	Ca^{2+}[M]	Na^+[M]	K^+[M]	pH [-]
1	0.0423	0.0128	0.0029	0.0189	0.0768	0.0220	11.6
2	0.1468	0.0806	0.0186	0.0169	0.1204	0.0330	12.0
3	0.0069	0.0000	0.0004	0.0063	0.0747	0.0293	12.0
4	0.0046	0.0010	0.0001	0.0016	0.1080	0.0440	12.0
5	0.0043	0.0005	0.0001	0.0154	0.0685	0.0293	11.8
6	0.0110	0.0290	0.0033	0.0139	0.0664	0.0232	12.0
7	0.0085	0.0177	0.0025	0.0149	0.0644	0.0232	12.0
8	0.0493	0.0408	0.0125	0.0238	0.0685	0.0183	11.8
9	0.0036	0.0000	0.0001	0.0067	0.0810	0.0305	12.1
10	0.0032	0.0005	0.0000	0.0015	0.1163	0.0476	12.3
11	0.0113	0.0487	0.0021	0.0148	0.0851	0.0244	12.3
12	0.0036	0.0005	0.0001	0.0105	0.0789	0.0317	12.2

7.9.3 Water retention test

The water retention test examined how the degree of saturation of the concrete cores depend on the surrounding environmental moisture conditions. The extracted concrete from the University bridge arch were sliced and put in humidity chambers with relative humidity (RH) ranging from 14.5%–100% to reach the equilibrium with the surrounding water vapour (Figure 7.20). After corresponding the water content of concrete to the humidity of the air in the chamber, the parameters α=0.0018 [1/m] and n=1.7 were then determined as described in Section 7.5.1.1.



(a)



(b)

Figure 7.20 Water retention test setup (a) suspended sliced concrete cores to be placed in a humidity chamber (b) humidity chambers with varying relative humidity.

8 Comprehensive Corrosion Model: Presentation and Validation

8.1 Preface

In Chapter 7, we explored the application of a comprehensive 3D transport model to simulate the microenvironment evolution at the rebar depth in concrete structures under various service conditions. Continuing the investigation from the previous chapter, Chapter 8 focuses on the formulation of a comprehensive corrosion model that incorporates crack configuration into the existing environment model. This model simulates the evolution of moisture and chloride content on embedded rebars in cracked concrete. It employs homogenized corrosion kinetics as functions of the microenvironmental conditions on the embedded rebar to solve the potential and current field in the concrete domain and rebar surface.

To validate the model, a second phase experimental program was designed, specifically targeting the validation of segmented rebars in cracked concrete specimens subject to wetting and drying cycles with chloride salt solutions.

With the validated model, the next chapter will be devoted to a numerical exploration of the possible mechanisms causing the marked different observations in rebar corrosion in cracked concrete.

8.2 Introduction

The most significant effect of concrete cracking on rebar corrosion is the formation of a macro-cell that may significantly increase the corrosion rate. It has been observed that passive reinforcement in the non-cracked part of an RC structure and bare steel can act as effective cathodes if there is an electrical contact to the reinforcement at the cracks when these cathodic areas have sufficient oxygen supply. This is typically the case for cathodic areas located in the splash zone or exposed to air where the corrosion around cracks can be accelerated due to macro-cell corrosion. However, under submerged conditions, the cathodic oxygen reduction reaction is limited by the transport of dissolved oxygen to the rebar surface (Hussain, 2011). It was found that the main effect of macro cracks in submerged concrete is a reduction of the initiation time. Once the active rebar corrosion was initiated, the corrosion rate started high, but rapidly decreased and approached the active corrosion rate typically found in uncracked concrete (Alekseev et al., 1993).

According to a study (Wilkins & Lawrence, 1980), the cross-sectional static cracks, even with up to 1 mm crack width, would not lead to corrosion at a significant level in the case of submerged concrete.

The effect of cracks on the corrosion rate in cracked concrete could also depend on the crack width and concrete type. For high-performance concrete under wetting and drying conditions (splash zone), it has been observed that although the chloride threshold level and the initiation time may be close to zero for crack widths up to 0.8 mm and a concrete cover of 30 mm, the active corrosion rate will become so small that it may be considered as practically harmless (Pettersson et al., 1996). In a study (Lu et al., 2017) under wetting and drying cycles (ratio 1:1), the effect of transverse cracks on chloride-induced steel corrosion was found to depend on both the cover depth and crack parameters such as width and frequency/spacing. The corrosion rate increases more rapidly with increasing crack width and frequency and decreasing cover depth. Under various cover depth and crack frequency scenarios, it has been observed that there exists a certain critical crack width below which the effect of the crack is negligible. However, the reasons for the existence of that critical width and its dependence on exposure conditions and the crack parameters are not clear.

Two different mechanisms of corrosion, the micro-cell and the macro-cell have been identified in cracked concrete specimens (Subramaniam & Bi, 2010). The two mechanisms primarily differ in the length scales and the physical distance between the underlying electrochemical reactions. In the microcell mechanism, the anodic and cathodic reaction sites are microscopic and are located adjacent to each other. In macro-cell corrosion, the anode and cathode are spatially separated by a finite macroscopic distance resulting in a flow of electric charges over a physical distance. In the case of chloride-induced corrosion, the anodic reaction is localized, confined to the area of steel where the chloride threshold level for depassivation is exceeded. The cathodic reaction may significantly occur in all areas that are not permanently water-saturated and cathodic current density decreases as the distance from the crack increases (Raupach, 1996). That explains the experimental observation that crack frequency could be identified as a governing factor in the rate of corrosion (Arya & Ofori-Darko, 1996), as it enables more passive areas near those cracks to be polarized to its cathodic direction and hence larger total cathodic current. Poursaee et al. (2010) studied corrosion of multi-rebar mortar samples subject to various types of de-icing salt solutions, where an active rebar near the surface was subject to the solution and the other three were at a

deeper location. It was found the micro and macro corrosion rates were similar. However, another similar test (Hansson et al., 2006) suggested that macro-cell corrosion is the main mechanism of rebar corrosion in ordinary corrosion. Although these tests were not in cracked concrete, these findings could suggest that the contributions from micro-cell and macro-cell depend on the area of active rebar and passive rebar and their arrangement. The contribution of micro-cell corrosion could be the dominant mechanism when the active rebar area is comparable to the passive rebar and the distance between these two is large.

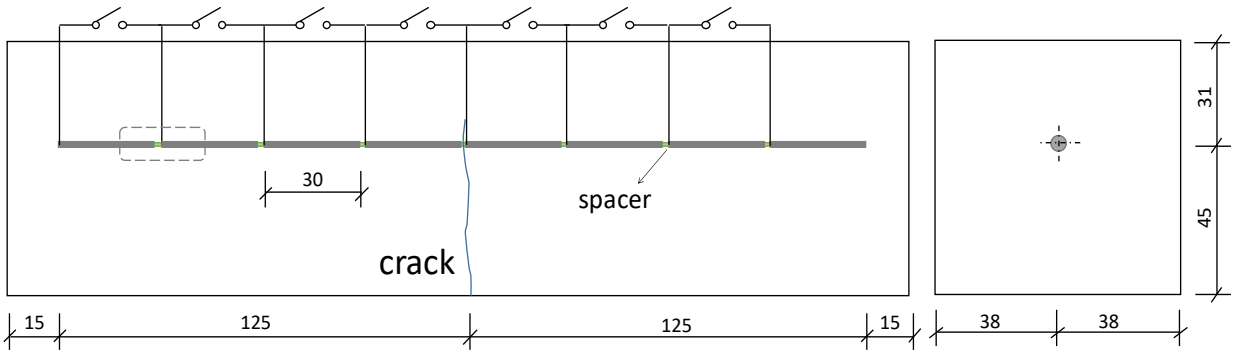
In conclusion, most of the previous research on corrosion in cracked concrete has been limited to measuring either potentials or corrosion rates in either field structures (submerged or splashed zone) or preconditioned samples (often subjected to wetting/drying cycles) in the lab but often without monitoring of the distribution around the crack of any of the environmental parameters that can affect the corrosion process, such as water saturation, oxygen, and chloride. Under different exposure conditions, the properties of cracks such as crack width, crack frequency, the properties of concrete, and cover depths would have very different effects on the corrosion rate. A clear understanding of the relative magnitudes of the microcell and macro-cell mechanisms in a corroding system and what creates the conditions for any of the two to become the dominant mechanism is still currently not available. The role of each mechanism, its contribution to the total corrosion rate and its dependence on the local micro-environment (moisture, oxygen, and chloride) created by the very existence of cracks are not clear yet.

In this chapter, a comprehensive corrosion model is built by coupling the corrosion models developed in previous chapters with environmental models that simulate the evolution of moisture and chloride content on embedded rebars. The corrosion models employ the corrosion kinetics of the embedded rebar which are function of the micro-environment conditions at the rebar location such as moisture content and chloride content. Therefore, the comprehensive model can simulate rebar corrosion rate from the exposure scenario that involves concrete cover cracking, non-uniform water distribution, and chloride contamination. An experimental program was conducted to validate this model.

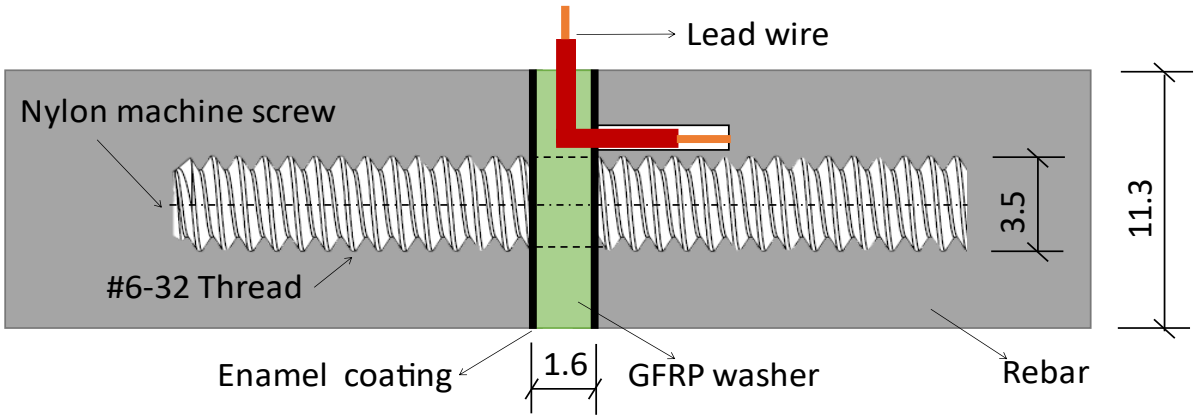
8.3 Experimental program

8.3.1 Material and sample preparation

Concrete beams were made with Type GU cement with a sand/cement ratio of 1.65 and a water-cement ratio of 0.7 and 0.5. The same batch was used to cast the cylinder samples when measuring the steel corrosion parameters (Chapter 4). The configuration of the cracked beam sample is shown in Figure 8.1a. A No.10 rebar was cut into 30 mm-long segments and embedded in mortar. These rebar segments were electrically insulated from each other but were connected through wiring that extended out of the beam so that they could be electrically connected externally. As shown in Figure 8.1b, the rebar segments were machined with internal threads and nylon screws were used to connect them. This connection method allowed the segmented rebar to withstand some tensile strength when the crack was created at the mid-span. The enamel coating and a GFRP washer spacer ensured the rebar bar segments were electrically isolated. Figure 8.1c shows a photo of the rebar.



(a)



(b)



(c)

Figure 8.1 Cracked reinforced concrete beam sample and rebar configuration (a) reinforced concrete beam dimensions (b) segmented rebar connection design (c) Photo of the segmented rebar. Dimension unit: mm.

The beam samples and cylinder samples were cured at 25 ℃ and relative humidity greater than 90% for two months. The beam samples were then pre-cracked to introduce natural cracks at the mid-span of the beams by a three-point bending test (Figure 8.2). The beam samples were then coated with epoxy on all sides except the surface with the open crack. This helps the formation of a 2-D moisture and chloride penetration profile around the crack. The cracks were reopened to 0.3 mm and 1.0 mm after the epoxy coating using aluminum shims. A number of cylinders were cast with the same mixes used for the beams to monitor the penetration of moisture and chloride ion. The cylinders' side surfaces were also coated with epoxy to facilitate approximate 1-D species transport. Figure 8.3 shows the finished beam samples and cylinder samples.

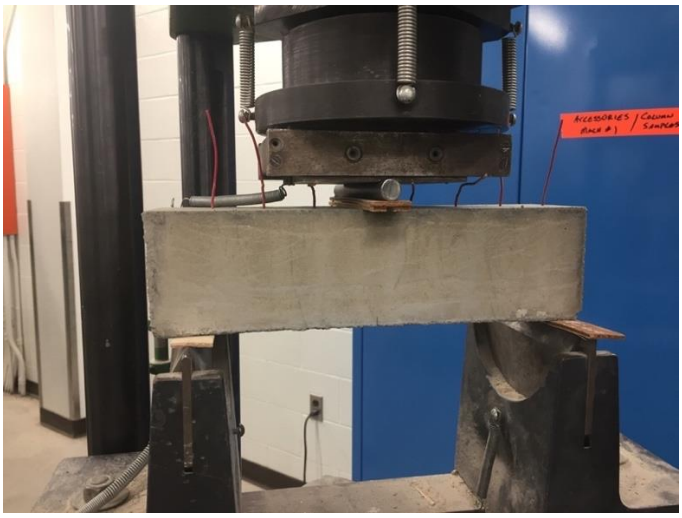


Figure 8.2 Three-point loading to create vertical cracks.



Figure 8.3 Epoxy-coated beam samples and cylinder samples.

8.3.1.1 Sample exposure condition

The beam samples and cylinder samples were subject to wetting and drying conditions. The samples were initially set as fully saturated by submersion in fresh water for two weeks and then dried for six days in the ambient environment of the Structures Laboratory. Starting at the second wetting event, samples were submerged in a 5% wt. NaCl salt solution. The first wetting event started on Day 7 for 24 hours. After a 13-day drying period, the second wetting event started on Day 20 for 48 hours and following that was a 43-day drying period. The weights of the beam samples were measured at the end of each wetting session and periodically during each drying session. The water loss for each sample was recorded as the weight loss of the beam samples from their saturated condition. The water loss during the wetting and drying process is plotted in Figure 8.4.

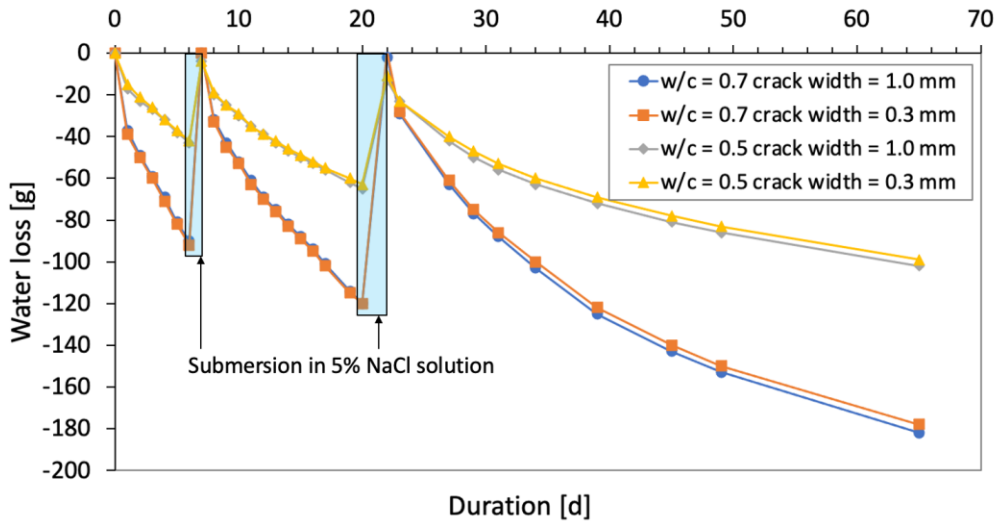


Figure 8.4 Water loss of beam samples during the wetting and drying. Legend: w/c is the water/cement ratio, and crack width is the surface crack opening size.

8.3.1.2 Current and potential measurement

The corrosion condition of the segmented rebar was recorded in two modes: disconnected mode and connected mode. In the disconnected mode, as shown in Figure 8.5 and Figure 8.6, the open circuit potential and the polarization resistance of each disconnected rebar segment were measured. The potential is open circuit corrosion potential for the rebar segments in its sounding environment, and the polarization resistance can give information about the resting corrosion current density.

In the connected mode, the rebar pieces were connected, and the ampere meter was installed across the whole rebar ends. The current flowing between the neighbouring rebar segments was measured by turning off the switch between those two neighbouring rebar segments so the current flows through the ampere meter. Figure 8.7 shows the measurement of the current flowing between the first and second rebar segments and the measured current was denoted as $I_{1,2}$. The current between the second and third rebar segments was denoted as $I_{2,3}$ and so forth.

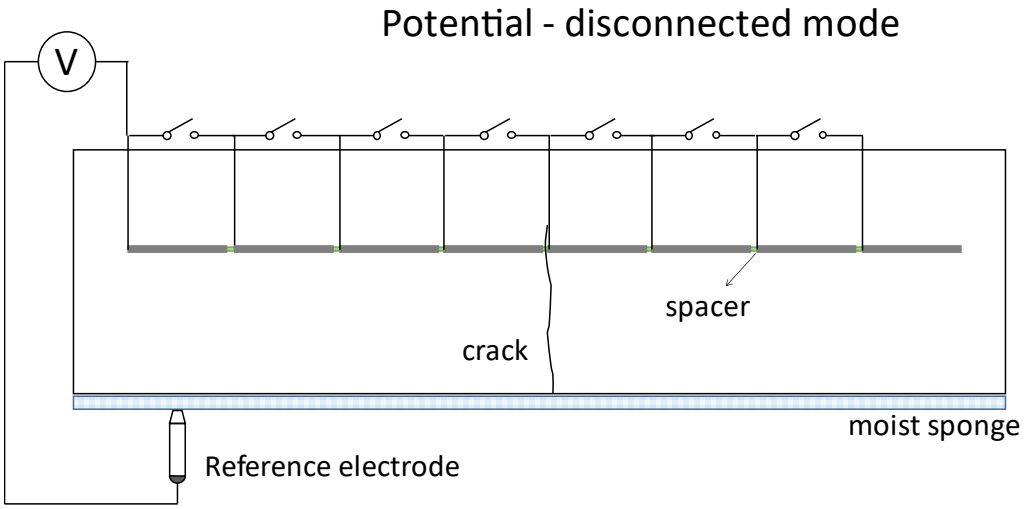


Figure 8.5 Open circuit potential test in the disconnected mode.

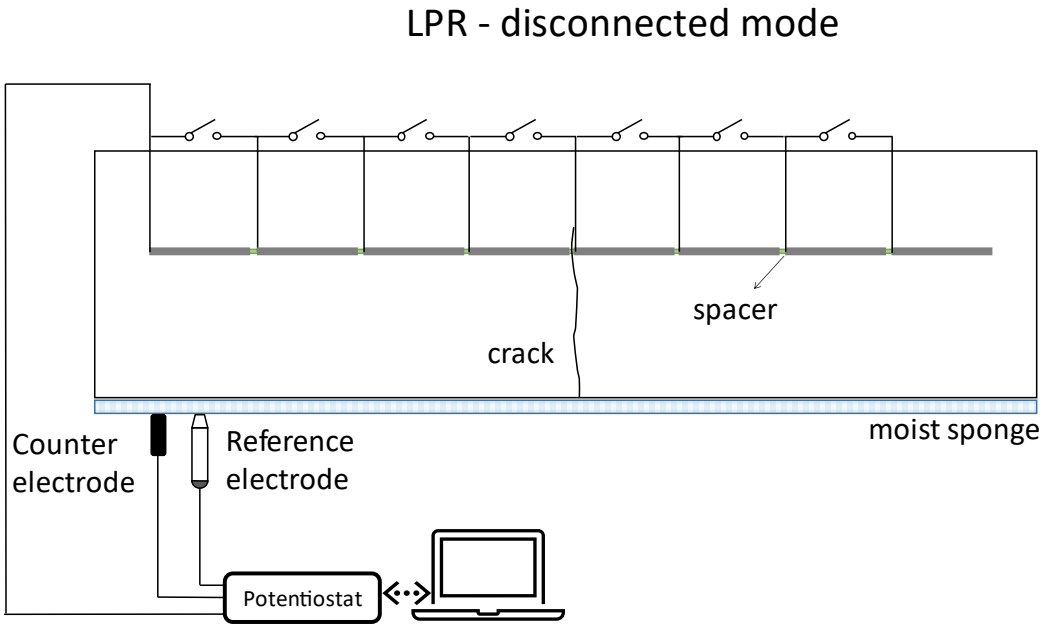


Figure 8.6 Linear polarization test in the disconnected mode.

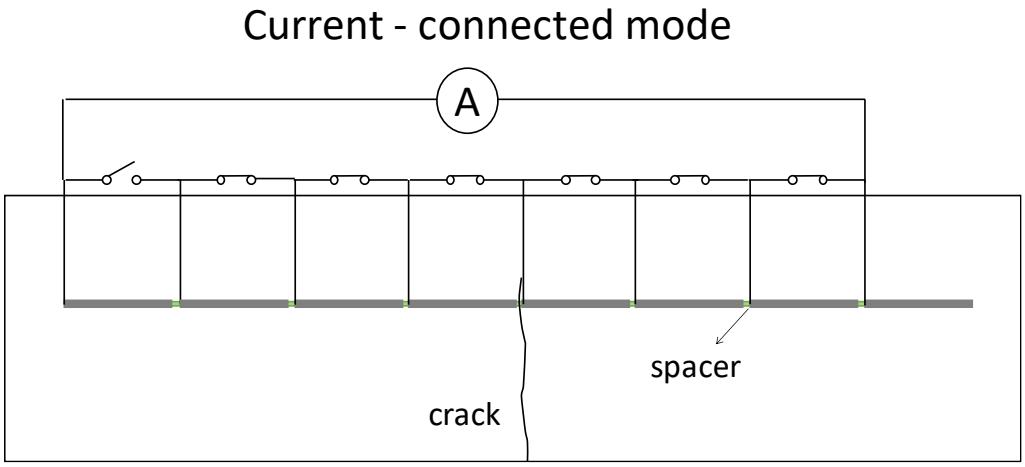


Figure 8.7 Current test between first and second rebar segment.

The current that went in (-) or out (+) denoted as I_1, I_2 … I_8 from each rebar segment can be derived. Figure 8.8 shows the assigned positive direction of the currents.

For the first rebar segment,

$$I_1 = I_{1,2} \tag{8.1}$$

for the last rebar segment,

$$I_n = -I_{(n-1),n} \tag{8.2}$$

for the rebar segment in between,

$$I_n = I_{n,(n+1)} - I_{(n-1),n} \tag{8.3}$$

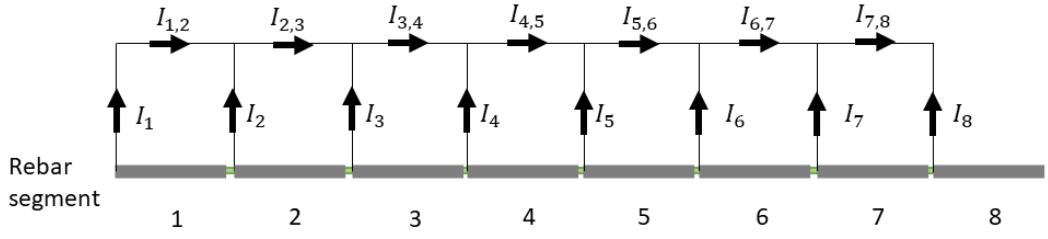


Figure 8.8 Definition of the positive current direction for the connected mode.

8.4 Numerical corrosion model

8.4.1 Geometry setup

An 3-D numerical model were implemented in a commercial FEM software, COMSOL Multiphysics. The model replicates the sample geometric configuration used in the experimental program described earlier. The crack line was implemented as a flat surface, as previous simulation studies have indicated that the roughness of the crack line has minimal influence on the water transport profile around the crack (Li, 2014). Half of the beam (Figure 8.9) was used for the simulation to take advantage of the symmetry and save computation time.

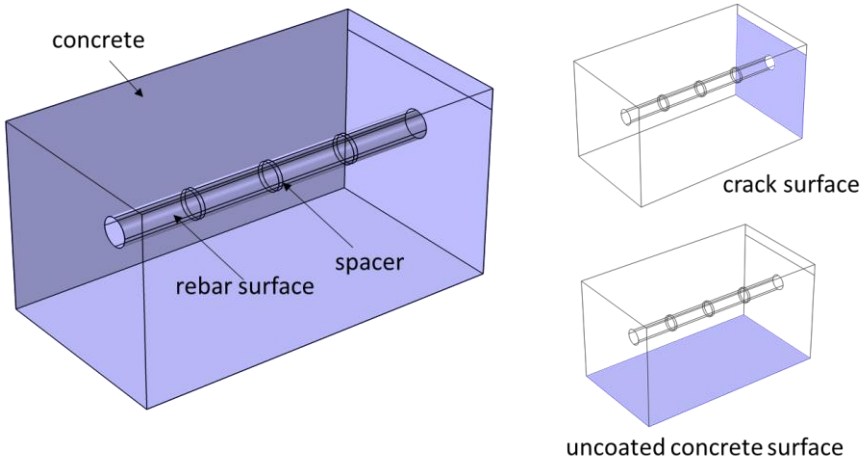


Figure 8.9 3-D model geometry implementing half of the beam sample with the segmented rebar.

8.4.2 Water flow and transport of species

The non-uniform microenvironment on the rebar was simulated with the transport of moisture and chloride ion in the cracked concrete. The boundary conditions for the exposed uncoated beam concrete surface and crack surface are aligned with the wetting and drying cycles and salt exposure used in the corresponding experiment.

8.4.2.1 Governing equations

The governing equations used for the moisture and chloride transport are described in detail in Chapter 7 and restated here for convenience.

8.4.2.2 Water flow in concrete

In many practical conditions, a porous medium like concrete is not fully saturated. The degree of saturation depends on the surrounding environmental moisture conditions. When concrete is not fully saturated, a negative capillary pressure develops and facilitates the absorption of water, and this pressure is the driving force of water flow within concrete. Under such conditions, the permeability becomes dependent on the water content. Darcy's law has been extended to describe water flow for this kind of problem in unsaturated porous media. One of the earliest and most widely used models for water flow in unsaturated porous media is known as Richard's equation,

$$C\frac{\partial H_p}{\partial t} = \nabla \cdot \left[K_s k_r \nabla\left(H_p + Z\right)\right] \tag{8.4}$$

where H_p [m] is pressure head, Z [m] is elevation, t [s] is time, K_s[m/s] is hydraulic conductivity of concrete at full saturation and k_r is relative permeability, defined as the ratio between the hydraulic conductivity at volumetric water content θ, $K(\theta \leq \theta_s)$ to $K(\theta = \theta_s)$, where θ_s[-] is saturated water content or the total connected porosity, C [1/m] is defined as a change in storage due to change in pressure head,

$$C = \frac{\partial \theta}{\partial H_p} \tag{8.5}$$

A constitutive relationship or model that has been shown to work with concrete is the van Genuchten- Mualem model (Kumar, 2010; Li, 2014), given by Equations (8.6 – 8.9) (Mualem, 1976; van Genuchten, 1980):

$$C = S_e \cdot S_s + \frac{\alpha m}{1-m}(\theta_s - \theta_r) S_e^{\frac{1}{m}} S_e \left(1 - S_e^{\frac{1}{m}}\right)^m \tag{8.6}$$

$$k_r = S_e \left(1 - \left(1 - S_e^{\frac{1}{m}}\right)^m\right)^2 \tag{8.7}$$

$$S_e = \begin{cases} \left[1 + \left|\alpha H_p\right|^n\right]^{-m}, & H_p < 0 \\ 1, & H_p \geq 0 \end{cases} \tag{8.8}$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \tag{8.9}$$

where, k_r $[-]$ is relative permeability, S_e $[-]$ is effective saturation, θ_s $[-]$ is saturated water content, θ_r $[-]$ is residual water content, S_s $[1/m]$ is the storage coefficient. l $[-]$, m $[-]$, n $[-]$, and α $[1/m]$ are empirical parameters ($m = 1 - 1/n$)

The hydraulic conductivity in the crack is expressed with the cubic law (Witherspoon et al., 1980):

$$T_f = \frac{\rho g}{12\mu} a^3 \tag{8.10}$$

$$K_f = \frac{\rho g}{12\mu} a^2 \tag{8.11}$$

where T_f $[m^2/s]$ is transmissivity, K_f $[m/s]$ is the hydraulic conductivity of the crack, ρ is the density of fluid, a $[m]$ is the aperture of the crack, g is the acceleration of gravity.

8.4.2.3 Chloride ion transport in concrete

The transport of dissolved chloride in concrete is governed by the diffusion-advection equation, whereby

$$\frac{\partial(\theta c)}{\partial t} = -\nabla \cdot \boldsymbol{J} \tag{8.12}$$

$$\boldsymbol{J} = -D_{eff}\frac{\partial c}{\partial z} + \boldsymbol{u}c \tag{8.13}$$

$$D_{eff} = D_s \cdot \Phi(S_e) \tag{8.14}$$

where $\boldsymbol{J}$ $[kg \cdot m^2 s^{-1}]$ is the total volume-averaged mass flux of solute in concrete, $\boldsymbol{u}$ $[m \cdot s^{-1}]$ is Darcy's velocity, c $[kg \cdot m^{-3}]$ is the liquid phase-averaged solute concentration, D_{eff} $[m^2 \cdot s^{-1}]$ effective diffusivity, D_s is effective diffusivity under fully saturated conditions, $\Phi(S_e)$ is a reduction function that accounts for discontinuity in the liquid phase, determined by a resistivity-based technique (Ramadani, 2013).

The diffusivity of chloride transport in a crack is

$$D_f = D_{aq} \tag{8.15}$$

where D_{aq} is the diffusivity of chloride ions in an aqueous solution.

8.4.2.4 Gas transport in concrete

Because the bulk movement of gas in concrete is rare in practical scenarios, the advection and dispersion effects were neglected. The diffusion of oxygen or carbon dioxide were modelled as driven by the gas density/partial pressure in the gas phase.

$$\frac{\partial[\epsilon_g \langle \rho_{O_2,CO_2} \rangle^g]}{\partial t} = \nabla \cdot \left[D_{eff}^{O_2,CO_2} \nabla \langle \rho_{O_2,CO_2} \rangle^g\right] + \text{sink}_{O_2,CO_2} \tag{8.16}$$

where, ϵ_g is the gas volume fraction, ϵ_s is the solid volume fraction, $\langle \rho_{O_2,CO_2} \rangle^g$ the density of O_2 or CO_2 in the gas phase, $D_{eff}^{O_2,CO_2}$ is effective diffusivity of O_2 or CO_2 in the gas phase, the sink term, sink_{O_2,CO_2} represents the consumption rate for carbon dioxide or oxygen. The consumption rate of carbon dioxide is determined by the saturation level and pH in the pore solution and the consumption rate of oxygen is determined by the oxygen-reduction cathodic reaction of the rebar corrosion.

8.4.2.5 Boundary conditions

The two drying and wetting cycles for the uncoated concrete surfaces have the same setting as the experiment after the initial saturation. It started with a six-day drying session and a one-day wetting session and followed by a two-day wetting session and a 43-day drying session. The wetting condition corresponds to water submergence, and the drying condition corresponds to a relative humidity of 25% at 25 ℃. The relationship between hydraulic pressure head and RH and temperature can be derived from the Kelvin equation (8.17) and Young-Laplace equation (8.18):

$$\ln(RH) = \ln\left(\frac{p_v}{p_{sat}}\right) = \frac{2\gamma V_m}{rRT} \tag{8.17}$$

where $p_v[Pa]$ is vapour pressure, $p_{sat}[Pa]$ is saturated vapour pressure above a flat surface at temperature $T[K]$, $\gamma[N/m]$ is surface tension, $V_m[m^3/mol]$ is molar volume, R $[J \cdot mol^{-1} \cdot K^{-1}]$ is gas constant, $r[m]$ is radius of a droplet (negative value for concave meniscus), which is related to the largest capillary pores size, r_p by $r = r_p/cos\alpha$. The parameter α is the contact angle determined by the type of liquid and the solid surface.

$$p_c = \frac{2\gamma \cos(\alpha)}{r} \tag{8.18}$$

where $p_c[Pa]$ is the capillary pressure and it is related to pressure head $H_p[m]$ by

$$H_p = \frac{p_c}{\rho g} \tag{8.19}$$

where $\rho[kg/m^3]$ is density of water, g is gravitational acceleration rate.

$$H_p = \ln(RH) \cdot \frac{RTcos(\alpha)}{V_m \rho g} \tag{8.20}$$

The wetting and drying conditions were therefore implemented as hydraulic pressure head of -19465.43 m for drying corresponding to a relative humidity of 25% at 25℃, and 0 for wetting corresponding to submersion condition. The chloride concentration on the exposed surface is set as 5 wt% NaCl in wetting cycles and no-flux in drying cycles. A summary of boundary condition settings is shown in Figure 8.10.

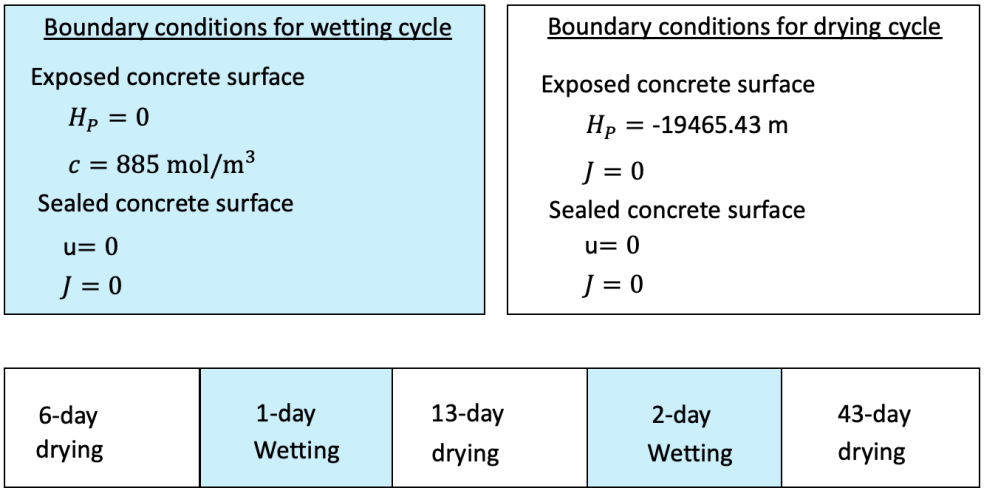


Figure 8.10 Water and chloride transport boundary conditions for exposed and sealed concrete surface in wetting and drying cycles.

8.4.2.6 Calibration of transport parameters

As described in Section 8.3.1.1, the mortar cylinder samples were prepared and underwent the same wetting and drying cycles as the beam samples. The cylinder samples were coated with epoxy except for the two ends, which approximates a 1-D transport problem. The cylinder weights were measured to record water content change over the wetting and drying cycles. A 1-D model was used to simulate water and chloride ions transport of these cylinders. The measured water content changes and chloride content profiles were used to calibrate the transport parameters in the model.

Figure 8.11 shows the averaged volumetric water content for cylinder samples and simulated results from the calibrated models. The calibrated moisture transport parameters are shown in Table 8.1.

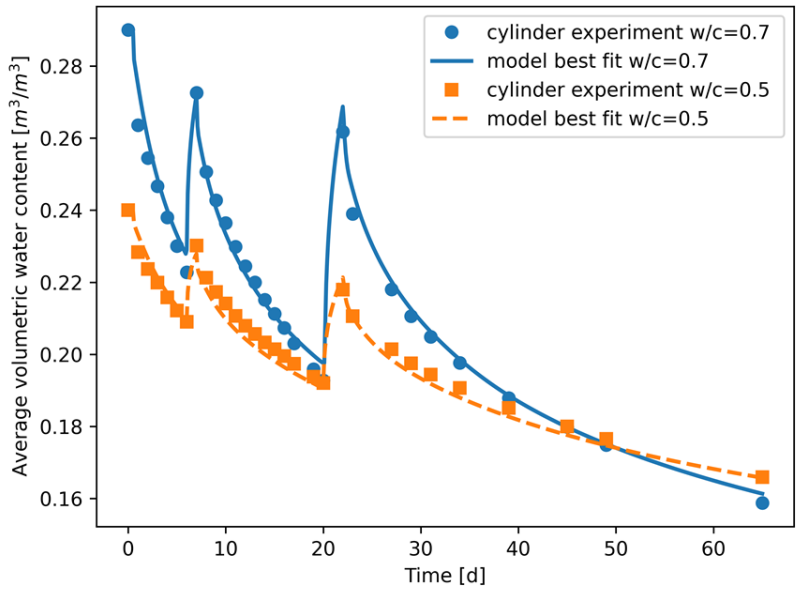


Figure 8.11 Calibration of water transport parameters using a 1-D model implementation with the water content change in cylinder samples.

Table 8.1 Moisture transport parameters

symbol	Value (w/c=0.7)	Value (w/c=0.5)	Unit	Description
θ_s	0.29	0.24	-	Saturated volumetric water content
n	1.5	1.35	-	van Genuchten parameter n
α	0.0018	0.0018	m^{-1}	van Genuchten parameter α
K_s	8×10^{-12}	3×10^{-12}	m/s	Saturated hydraulic conductivity

The chloride effective diffusivities under fully saturated conditions, D_s, for the cylinder samples were calibrated with chloride profiles in the cylinder samples at 9, 24, and 65 days, which correspond to two days after each wetting cycle and at the end of the second drying cycle (Figure 8.12). The water-soluble chloride contents were measured according to ASTM C1218 with the digested aqueous chloride concentration determined using Mercuric Thiocyanate-iron spectrophotometry. The reduction function $\Phi(S_e)$ that accounts for discontinuity in the liquid phase representing dependency of chloride diffusivity on the water content, is determined by a

resistivity test (Ramadani, 2013). The calibrated chloride diffusivities of the saturated concrete are listed in Table 8.2.

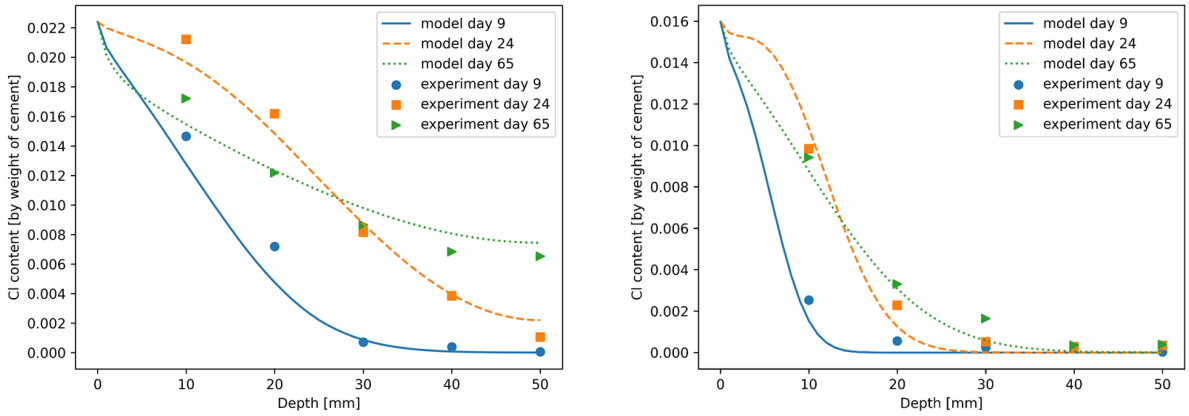


Figure 8.12 Calibration of chloride penetration model using a 1-D model implementation with the experimental chloride profile in mortar cylinder samples. (a) w/c = 0.7, (b) w/c = 0.5.

Table 8.2 Chloride transport parameters

symbol	Value (w/c=0.7)	Value (w/c=0.5)	Unit	Description
D_s	7.86×10^{-11}	7.28×10^{-12}	m^2/s	Effective diffusivity under water saturated condition
$\Phi(S_e)$	$Se^{4.12}$	$Se^{4.35}$	-	Relative diffusivity, Se is water saturation

8.4.3 Electrochemical process

8.4.3.1 Activation polarization

The electrochemical process was simulated with the secondary current distribution where the electrochemical reaction at the electrode-electrolyte interface is activation-controlled. The current density is a function of the overpotential using the Butler-Volmer equation.

For iron dissolution and deposition

$$i_{net}^{Fe} = i_a^{Fe} + i_c^{Fe} = i_0^{Fe} \cdot 10^{(E-E_{eq}^{Fe})/\beta_a^{Fe}} + i_0^{Fe} \cdot \left[-10^{(E-E_{eq}^{Fe})/\beta_c^{Fe}}\right] \quad (8.21)$$

where i_{net}^{Fe} is net current density, i_a^{Fe} is anodic current density of iron dissolution, i_c^{Fe}(negative) is cathodic current density of iron deposition, i_0^{Fe} is exchange current density, E_{eq}^{Fe} is equilibrium potential, E is electrode potential, β_a^{Fe} is anodic Tafel slope and β_c^{Fe}(negative) is cathodic Tafel slope.

For oxygen evolution and reduction

$$i_{net}^{O_2} = i_a^{O_2} + i_c^{O_2} = i_0^{O_2} \cdot 10^{(E-E_{eq}^{O_2})/\beta_a^{O_2}} + i_0^{O_2} \cdot \left[-10^{(E-E_{eq}^{O_2})/\beta_c^{O_2}}\right] \quad (8.22)$$

where $i_{net}^{O_2}$ is net current density, $i_a^{O_2}$ is anodic current density of oxygen evolution, $i_c^{O_2}$(negative) is cathodic current density of oxygen reduction, $i_0^{O_2}$ is exchange current density, $E_{eq}^{O_2}$ is equilibrium potential, E is electrode potential, $\beta_a^{O_2}$ is anodic Tafel slope and $\beta_c^{O_2}$(negative) is cathodic Tafel slope.

For the mixed electrode reaction of iron dissolution and oxygen reduction, neglecting iron dissolution reaction and oxygen evolution reaction when the electrode potential E satisfies $E_{eq}^{Fe} \ll E \ll E_{eq}^{O_2}$.

$$i_{net} = i_a^{Fe} + i_c^{O_2} = i_0^{Fe} \cdot 10^{(E-E_{eq}^{Fe})/\beta_a^{Fe}} + i_0^{O_2} \cdot \left[-10^{(E-E_{eq}^{O_2})/\beta_c^{O_2}}\right] \quad (8.23)$$

When the coupled reactions are balanced with no net accumulation of electrons, $i_{net} = 0$, let the solution be $E = E_{corr}$ (corrosion potential), $i_a^{Fe} = -i_c^{O_2} = i_{corr}$ (corrosion current density). Equation (8.23) can be re-written in terms of i_{corr} and E_{corr}

$$i_{net} = i_a^{Fe} + i_c^{O_2} = i_{corr} \cdot 10^{(E-E_{corr})/\beta_a} + i_{corr} \cdot \left[-10^{(E-E_{corr})/\beta_c}\right] \quad (8.24)$$

where the superscripts of β_a^{Fe} and $\beta_c^{O_2}$ are dropped for simplicity.

Equation 8.24 is the Butler-Volmer equation with the corrosion parameters E_{corr}, i_{corr}, β_a, β_c which can be directly measured from a corroding rebar in its corrosive environment. The discussion of these parameters and their dependence on the exposure conditions were elaborated in Chapter 3 and Chapter 4. The corrosion kinetics of the rebar were formulated based on the model functions which describe the corrosion potential and corrosion current density as functions of the volumetric water content and chloride content. The functions can be found in the Appendix of Chapter 4 and their surface plots are shown in Figure 8.13 and Figure 8.14.

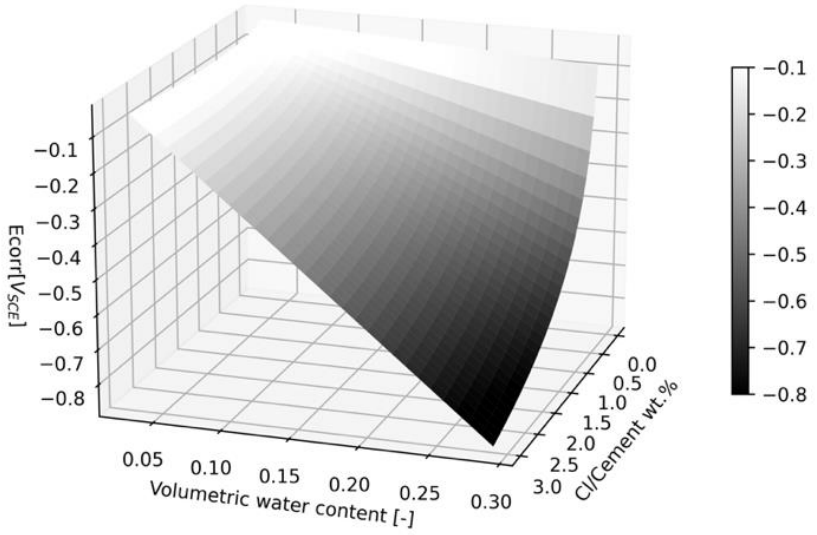


Figure 8.13 Corrosion potential as a function of chloride content and water content.

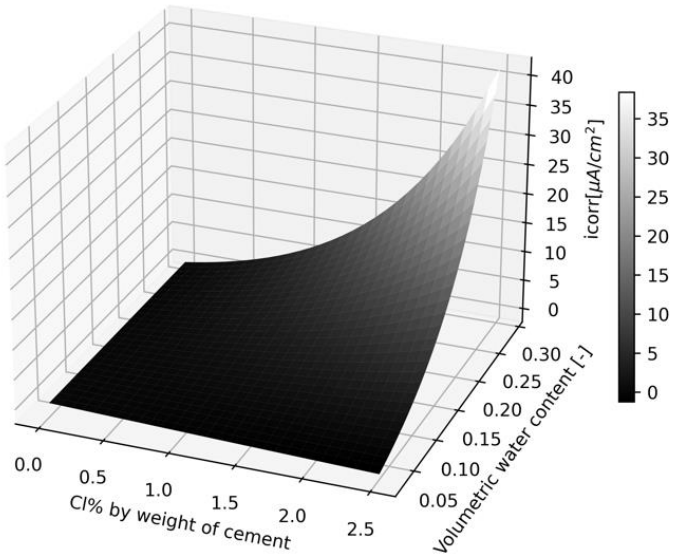


Figure 8.14 Corrosion current density as a function of chloride content and water content.

The anodic Tafel parameter is 0.5 V and the cathodic Tafel parameter is -0.25 V. These represent the averaged Tafel slopes from the experiment (Chapter 3.5.6) since there was no clear trend found as a function of the water saturation and chloride content.

8.4.3.2 Concentration polarization

Concentration polarization is caused when the electrochemical reaction is significantly faster than the mass transport of these species. In reinforced concrete corrosion, the limited oxygen diffusion is often considered for the concentration polarization effect. The oxygen has a lower concentration c_s adjacent to the rebar surface. The oxygen reduction rate becomes slower, and the equilibrium potential becomes lower. Therefore, more overpotential (concentration overpotential) is needed to achieve the same cathodic current normally driven at a bulk concentration c_0. A first-order dependence of concentration on the activation-controlled kinetics was assumed (Frankel, 2016):

$$i_c^{O_2} = i_{corr} \cdot \frac{c_s}{c_0}\left[-10^{(E-E_{corr})/\beta_c}\right] \tag{8.25}$$

It is worth noting that this equation corresponds to a concentration polarization overpotential component $\eta_{conc} = \beta_c \log_{10} \frac{c_0}{c_s}$. In the numerical model, c_0 is solved by coupling the electrode kinetics and the mass transport of oxygen.

Equation (8.24) becomes

$$i_{net} = i_a^{Fe} + i_c^{O_2} = i_{corr} \cdot 10^{(E-E_{corr})/\beta_a} + i_{corr} \cdot \frac{c_s}{c_0}\left[-10^{(E-E_{corr})/\beta_c}\right] \tag{8.26}$$

In the electrolyte domain, the current and potential is governed by Ohm's Law

$$\boldsymbol{i} = -\sigma \nabla \phi_l \tag{8.27}$$

where $\boldsymbol{i}$ is current density vector, σ is the electrical conductivity of the electrolyte, ϕ_l is electrolyte potential.

The sum of all electrode reaction currents is implemented on the interface between the electrode and electrolyte domain.

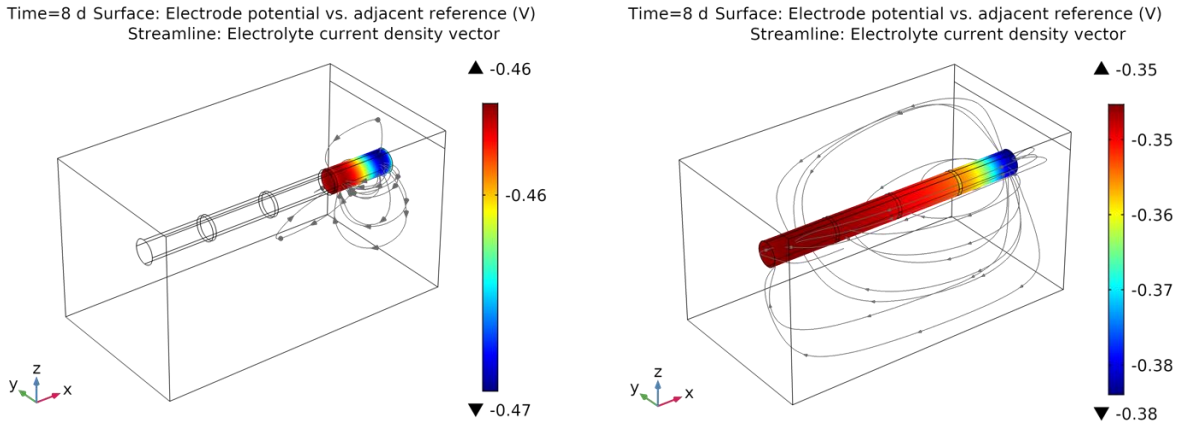
$$i_\perp = i_{net} \tag{8.28}$$

where $i_\perp$ is the $\boldsymbol{i}$ component perpendicular to the electrode-electrolyte interface and $\phi_l = -E$ at the electrode-electrolyte interface.

8.5 Validation of the corrosion model

The simulated results were validated with the experimental results at selected times. These selected times represent the conditions with various moisture and chloride contents in the cracked beams. Figure 8.15 shows examples of the solved model for disconnected mode and connected mode, respectively. Four results were extracted from the model: corrosion potential and corrosion current density of each rebar segment in disconnected mode, rebar potential and total current of each rebar

segment in the connected mode. As shown in Figure 8.16, the potential values were extracted from the concrete cover above each rebar segment. The current density is the area-averaged value on each rebar surface, and the total current is the surface-integrated value on each rebar segment surface.



(a) (b)

Figure 8.15 Example of solved models in (a) disconnected mode for rebar #4 and (b) connected mode for rebar #1 – 4 at Day 8 (one day after the first wetting cycle). Surface plot shows the electrode potential and streamline plot shows the electrolyte current density vector.

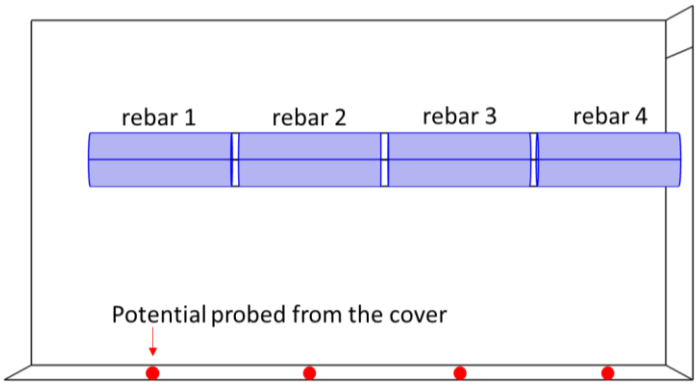


Figure 8.16 The locations of potential data extraction in the 3-D model during the post-processing procedure.

8.5.1 Disconnected mode: open-circuit corrosion potential

The open-circuit potential reflects the corrosion state of the rebar segment in the corrosive environment. The open circuit potential experiment measurements are compared with the simulated results. Four model setups were used to acquire the open circuit potential of each rebar segment from the numerical model. In each setup, the corrosion kinetics were applied to an individual rebar segment while the rest of the rebar surfaces remain insulated. The comparison between the simulated results and the test results are presented in Figure 8.17.

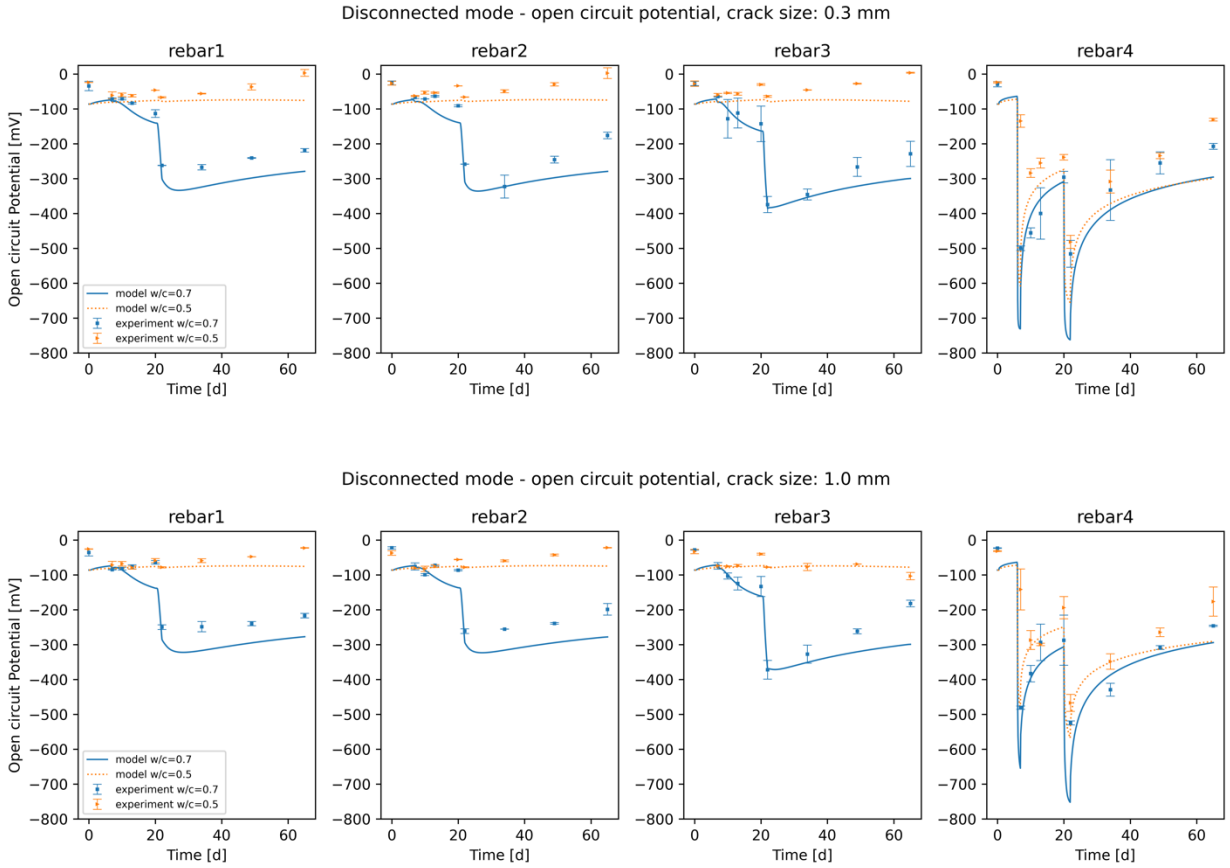


Figure 8.17 Validation of open circuit potential in the disconnected mode.

Rebar 4 segment near the crack responds sensitively to the drying and wetting cycle due to the rapid change of chloride and moisture near the crack. For beams with w/c of 0.7, rebar1-3 are not immediately affected by the first 1-day wetting session but are quickly affected by the second 2-day wetting session when the chloride concentration reached the critical value at the rebar level. The crack size does not significantly affect the potential evolution around the crack. However,

denser concrete slows down the penetration of chloride so that the rebar 1-3 remains not affected for w/c of 0.5 concrete samples.

8.5.2 Disconnected mode: corrosion current density

In the disconnected mode, the rebar segments are electrically insulated. The corrosion current density on each rebar segment is an area-averaged value over the rebar segment surface. The corrosion current density of the rebar is plotted in Figure 8.18.

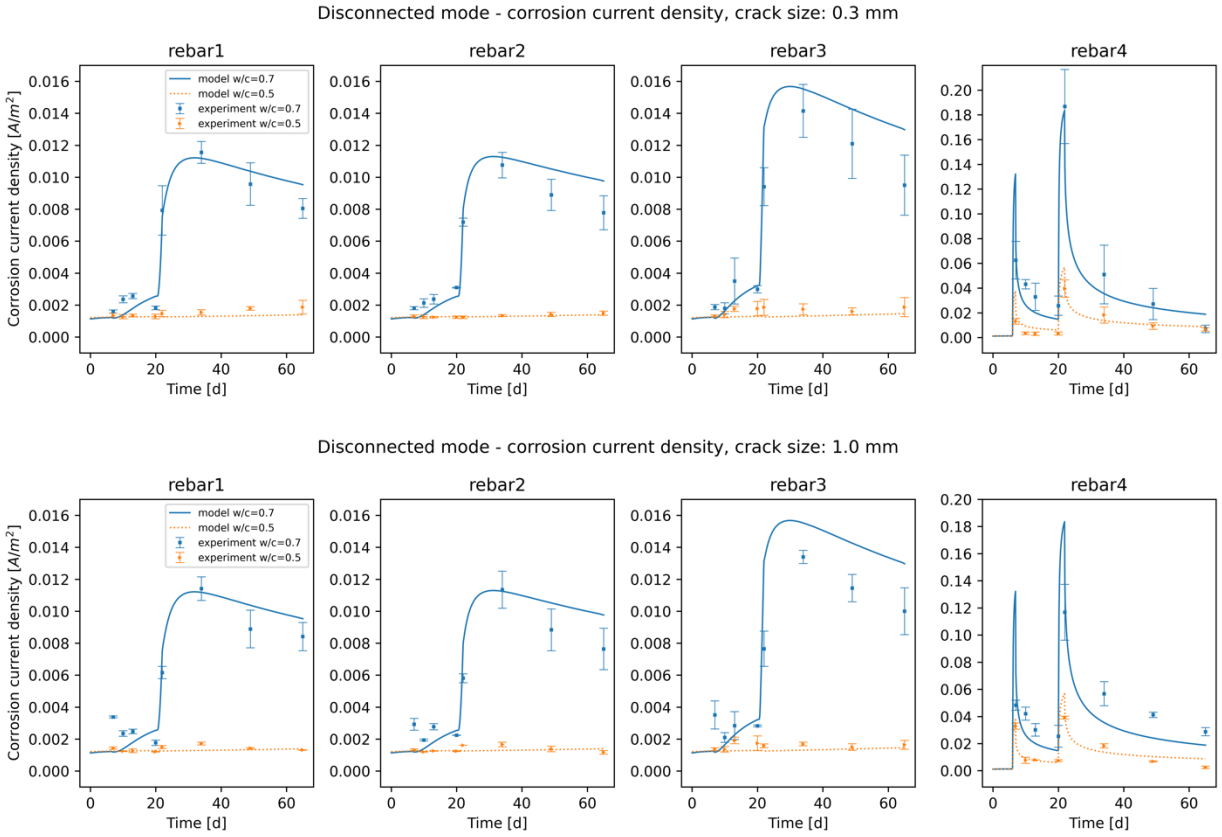


Figure 8.18 Validation of the current density of the rebar segments in the disconnected mode.

8.5.3 Connected mode: rebar segment potential

In the connected mode, the rebar segment potentials are measured from the concrete cover and the potential data were also extracted from the model in the same manner. The comparison results are presented in Figure 8.19.

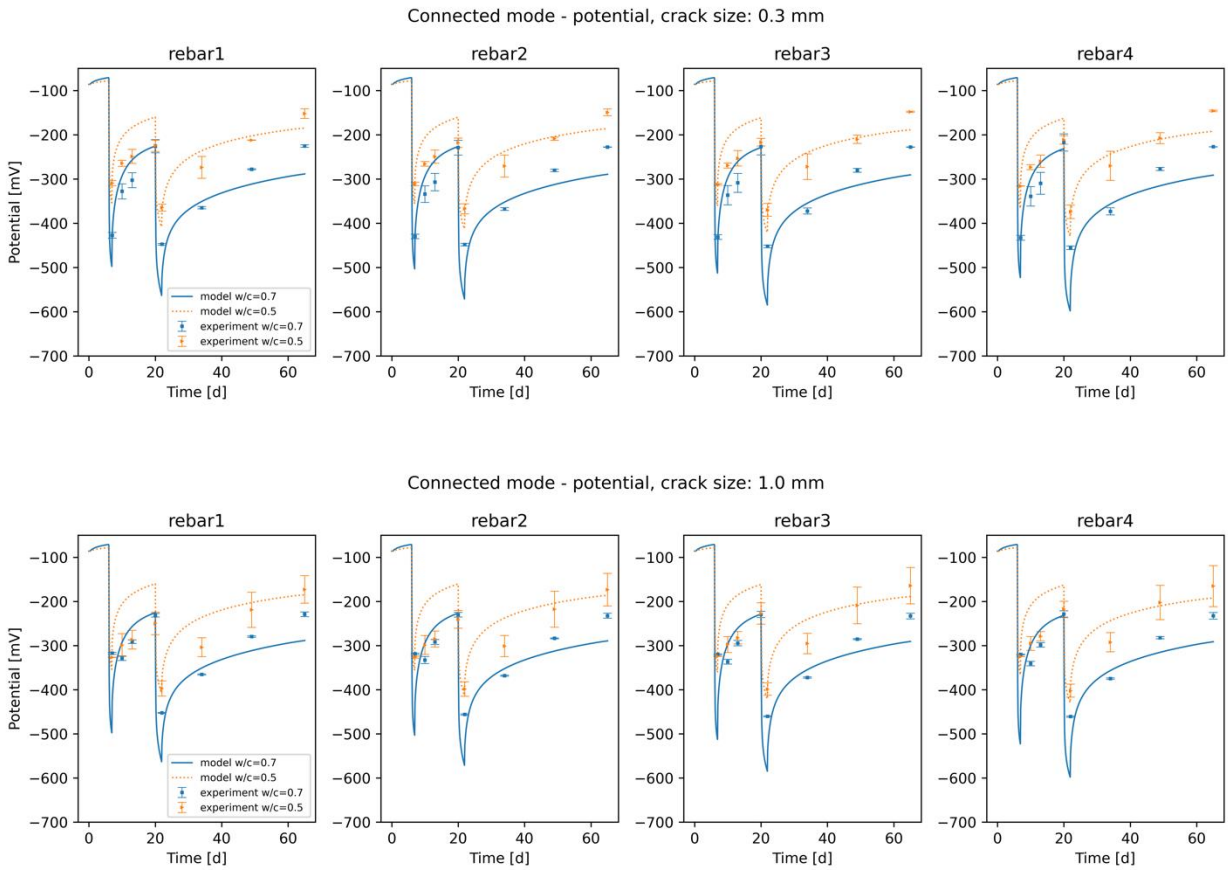


Figure 8.19 Validation of the potential of each rebar segment in the connected mode.

8.5.4 Connected mode: net current of each rebar segment

In the connected mode, all rebar segments are electrically connected. The total current entering (-) or exiting (+) each rebar segment is calculated by integrating the net current density over the rebar surface. The comparison with the experimental measurement is plotted in Figure 8.20.

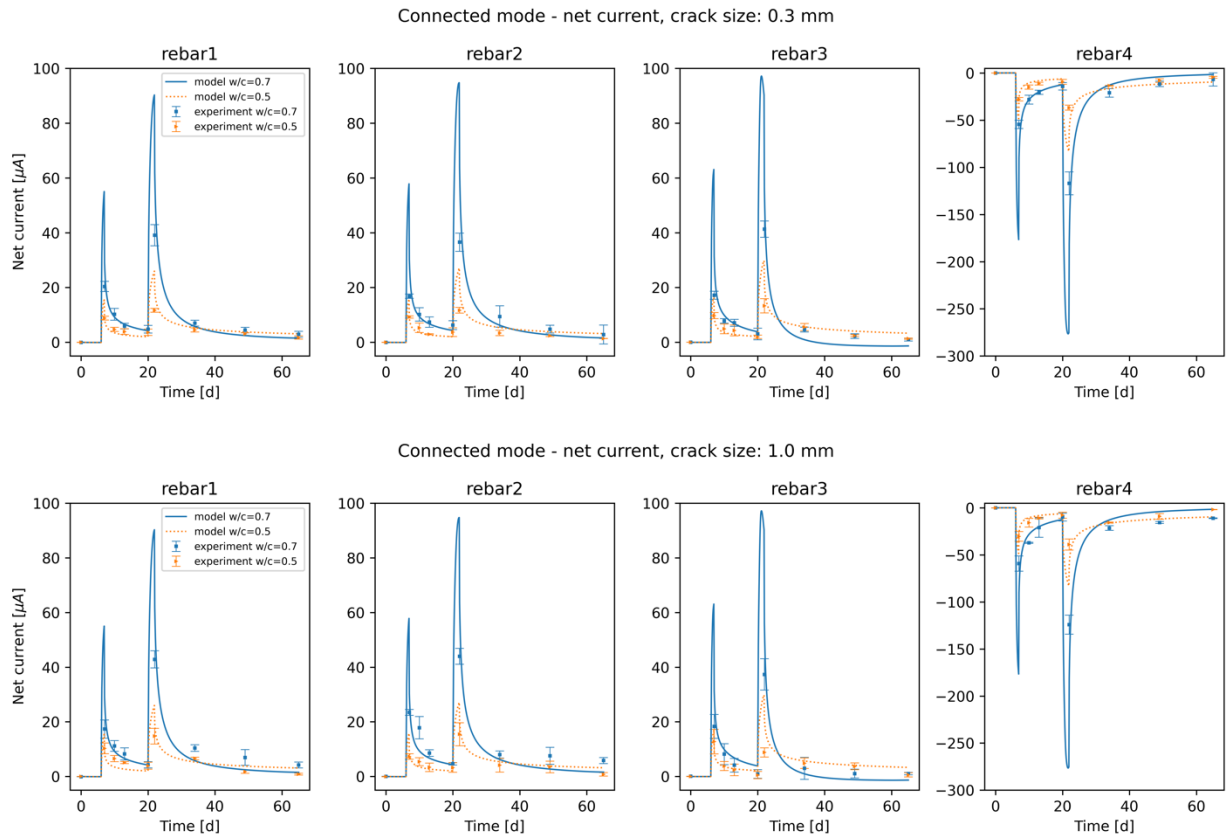


Figure 8.20 Validation of the net current of each rebar segment in the connected mode.

8.6 Conclusion

A comprehensive corrosion model has been developed. It simulates the evolution of moisture and chloride content on embedded rebars in cracked concrete and uses homogenized corrosion kinetics as functions of the microenvironmental conditions on the embedded rebar to solve the potential and current field in the concrete domain and rebar surface.

The corrosion model was validated by comparing simulated results with experimental results at various moisture and chloride contents for segmented rebar in cracked beams. The segmented beam sample was tested in disconnected mode and connected mode in wetting and drying cycles. The measured results, corrosion potential, corrosion current density in disconnected mode, and rebar potential, total current in connected mode, were compared to a 3-D model representing the same experimental setup.

The water-cement ratio had the greatest influence on the corrosion condition of the rebar in the cracked beams. The tested crack sizes (0.3 mm and 1.0 mm) had similar effects on increasing the non-uniform corrosion rate near the cracks. The simulated corrosion results were validated by experimental studies. The comprehensive corrosion model can predict the rebar corrosion in the cracked beams subject to wetting and drying cycles with the contamination of chloride.

8.7 Epilogue

Chapter 8 focused on the formulation of a comprehensive corrosion model that incorporates crack configuration. This model simulates moisture and chloride content evolution on embedded rebars in cracked concrete, using homogenized corrosion kinetics. The model was validated with the corrosion test results from segmented rebars in cracked concrete subjected to wetting and drying cycles with chloride salt solutions. This comprehensive corrosion model will contribute to our understanding and practical applications in assessing and designing reinforced concrete structures in a corrosive environment such as marine environment, potash mine, bridge deck with the application of de-icing salts. In the next chapter, the validated model will be utilized to explore the mechanisms behind the observed variations of rebar corrosion in cracked concrete.

9 Numerical Investigation of Effects of Crack on Rebar Corrosion in Concrete

9.1 Preface

In this chapter, the validated model with a 2-D implementation is utilized to explore the intricate relationship between cracks, microenvironmental factors, and rebar corrosion. The model allows for a comprehensive examination of key parameters such as moisture, chloride accumulation, oxygen availability, electrode potential, and corrosion current density of the rebar. The numerical exploration provides a mechanistic explanation of the observed rebar corrosion behaviour in cracked concrete, with particular emphasis on corrosion reduction over time and the effect of crack size. This chapter provides a deeper understanding of the relationship between cracks, microenvironmental factors, and rebar corrosion, which can provide a more accurate assessment of corrosion and select optimized corrosion mitigation strategies in practical applications.

9.2 Summary of typical and peculiar observed behaviours (Lab and/or field)

The variation in the effects of crack on the rebar in cracked concrete was reviewed in Chapter 8 (Section 8.2) and is summarized as follows. Typically, macro-cell corrosion will accelerate the active corrosion rate at the crack where the protective film is damaged due to chloride ions. The active corrosion may start with a high rate, but it will decrease over time, resulting in a small and harmless rate for submerged concrete structures. However, in the partially saturated case, the corrosion rate also decreases with time and there is a peculiar behaviour where there exists a critical width below which corrosion is considered harmless. In the subsequent sections, we will examine how the cracks affect the microenvironment and corrosion of rebar in cracked concrete in submerged conditions and wetting and drying conditions so that it provides a mechanistic explanation of the observed rebar corrosion behaviour in cracked concrete.

9.3 Numerical investigation of effects of crack on corrosion

For computational efficiency, a 2-D model implementation was used to explore the effect of cracking on the microenvironment of concrete in the cracked concrete. As shown in Figure 9.1, the geometric configuration was similar to the 3-D model used in the validation tests, but with a longer length of the concrete beam. The rebar was represented as a line in the concrete domain. The governing equations for transport of water, chloride ion, and oxygen in the concrete domain

can be found in Section 8.4.2. The governing equations for the electrochemical process can be found in Section 8.4.3.

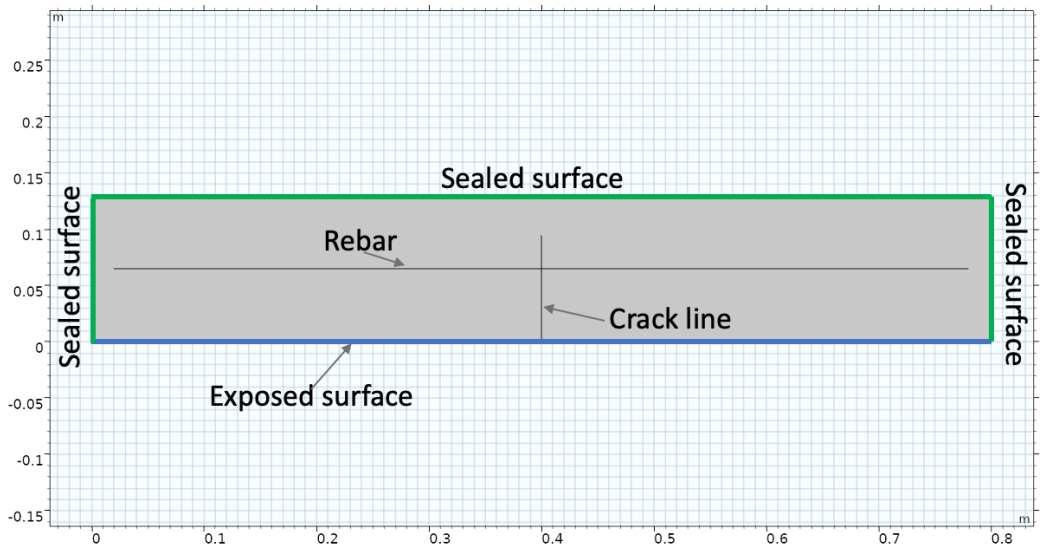


Figure 9.1 Geometric configuration of the 2-D corrosion model of the cracked beam.

The numerical investigation will explore, under two types of exposure conditions, how the evolution of the microenvironment affects the corrosion of steel reinforcement in cracked concrete. The focus will be placed on the key factors such as moisture, chloride accumulation, oxygen availability and the resultant electrode potential, and corrosion current density of the rebar. The investigation will focus on two types of cracks: a crack with a crack width of 1 mm on the surface, and a very fine crack size of 0.1 mm on the surface. The crack widths decrease linearly along the depth of the crack with the crack width of zero at the crack tip. The hydraulic conductivity along the crack line is depended on the crack width, defined by the cubic law, Equation (8.11). The chloride diffusivity along the crack line equals the chloride diffusivity in the aqueous solution during the wetting session and matches that in concrete during the drying session. The oxygen diffusivity along the crack equals the oxygen diffusivity in air during the drying session and is equivalent to the oxygen diffusivity in concrete during the wetting session. The boundary conditions for wetting/submerged condition and drying condition are summarized in Figure 9.2.

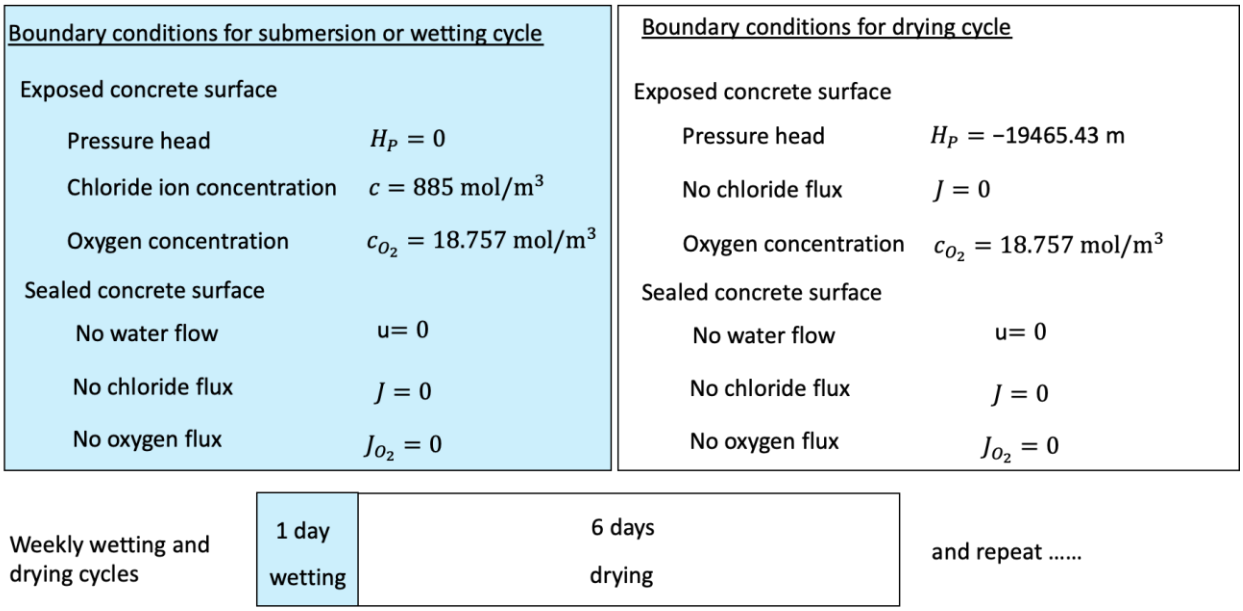


Figure 9.2 Model boundary condition settings for the investigated exposure conditions: submerged condition and wetting and drying cycles.

9.3.1 Submerged condition

9.3.1.1 Submerged condition: 1 mm crack

As shown in Figure 9.3, under the submerged condition, concrete remains saturated. At the early stage (Day 30), chloride ions diffuse to the rebar surface through the crack, which causes localized depassivation of the rebar in the cracked zone. The corrosion kinetics on the rebar are mainly influenced by the chloride content. As shown in Figure 9.4, the equilibrium potential on the rebar is affected by the chloride content and water content (Figure 8.13), and hence, has very different negative value at the crack due to high chloride content. The actual electrode potential on the rebar is determined by the electrochemical reaction (Section 8.4.3). The rebar has an electrode potential (-0.42 – -0.56 V) with the most negative potential at the crack. This potential variation is small due to the low resistivity of the water-saturated concrete. The rebar near the crack has a positive overpotential (electrode potential > equilibrium potential) and the net electrolyte current flows from the cracked area to the uncracked area of the rebar, which confirms that most anodic reaction (iron dissolution) occurs at the crack. The most cathodic reaction occurs on the rebar not at the

crack but in the vicinity of the crack with lower chloride content, which is indicated by the reduced oxygen saturation level in that area.

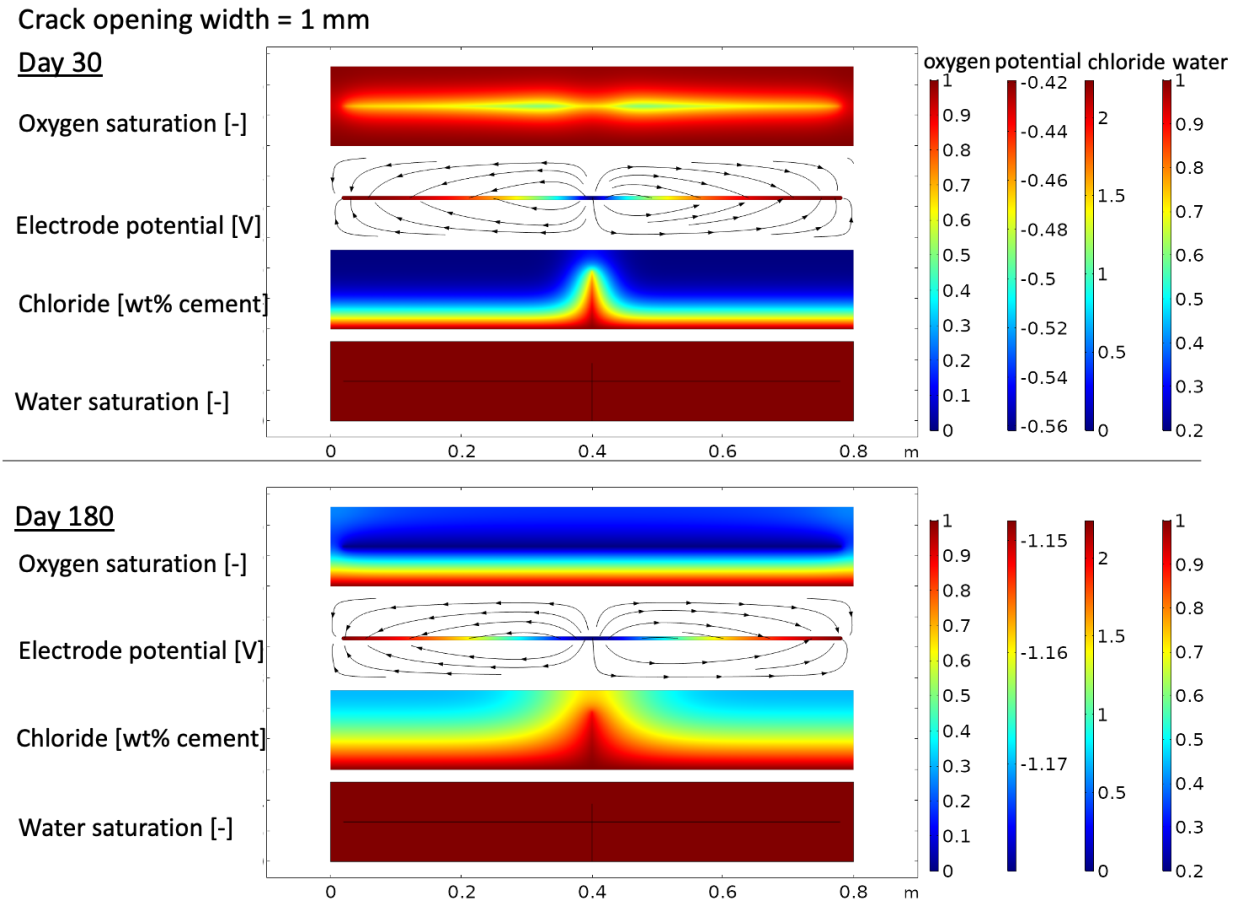


Figure 9.3 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the submerged condition at the early stage (day 30) and developed stage (day 180).

With the continuing chloride ingress and corrosion, chloride content will increase along the whole rebar, as shown in Figure 9.3 at Day 180 (called here the developed stage). The difference in the degree of depassivation and equilibrium potential at the crack and away from the rebar becomes less significant, but still, the electrolyte current flows from the cracked area to the uncracked area of the bar meaning the rebar at the crack is more anodic. The oxygen is almost depleted on the rebar surface, which limits the cathodic reaction (oxygen reduction) and leads to concentration polarization with a very negative potential (-1.15 – -1.17 V). The electrode potential on the rebar becomes more uniform.

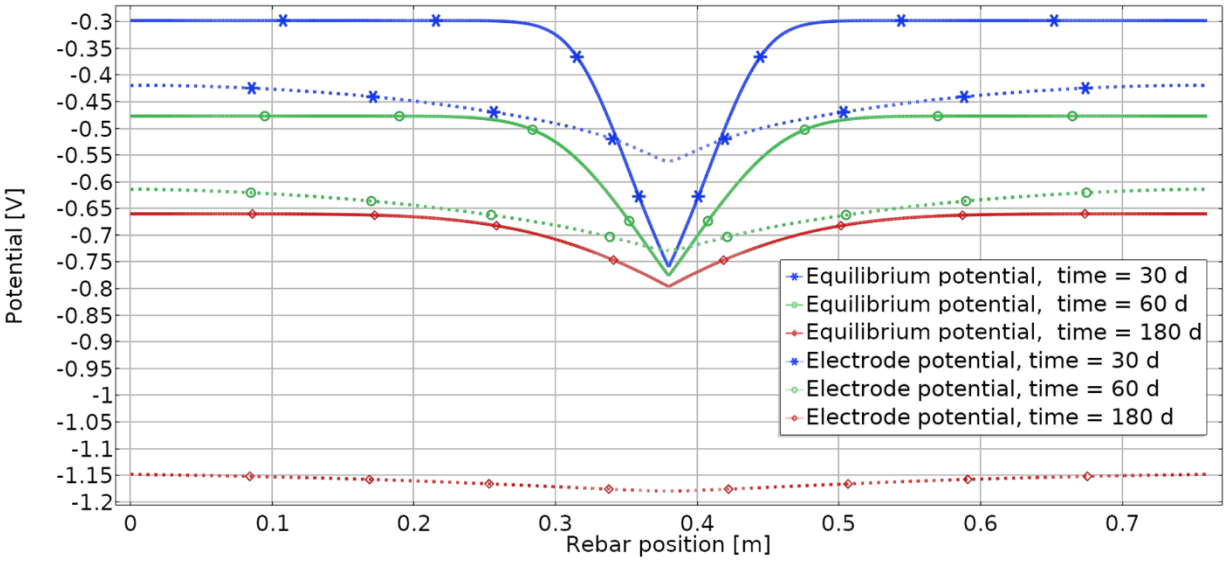


Figure 9.4 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential.

9.3.1.1.1 Corrosion current

From the early stage to the developed stage, the intensity, and the distribution of corrosion current density (anodic current density) on the rebar change over time. As shown in Figure 9.5, the corrosion current density at the crack decreased significantly over time. Corrosion becomes less localized, and the current density spread out to a larger corrosion area with a value comparable to the corrosion of a passivated rebar. As shown in Figure 9.6, the line-averaged rebar corrosion current density in the cracked zone increases to a peak value and then decreases over time. The cracked zone is defined as 10 cm across the crack, the typical sampling size of the rebar by the experimental measurement (linear polarization resistance or LPR). The total anodic current density on the rebar in the cracked zone has two contributions: the micro-cell component which has the corresponding cathodic current from the rebar in the cracked zone and the macro-cell component which has the corresponding cathodic current from the rebar outside of the cracked zone. As the chloride penetration and oxygen consumption proceed, the macro-cell component decreases to a similar weight as the micro-cell component, which is in agreement with the spread-out corrosion

current distribution along the whole rebar. The averaged corrosion current in the cracking zone decreases with time due to limited oxygen supply despite an increase in chloride content.

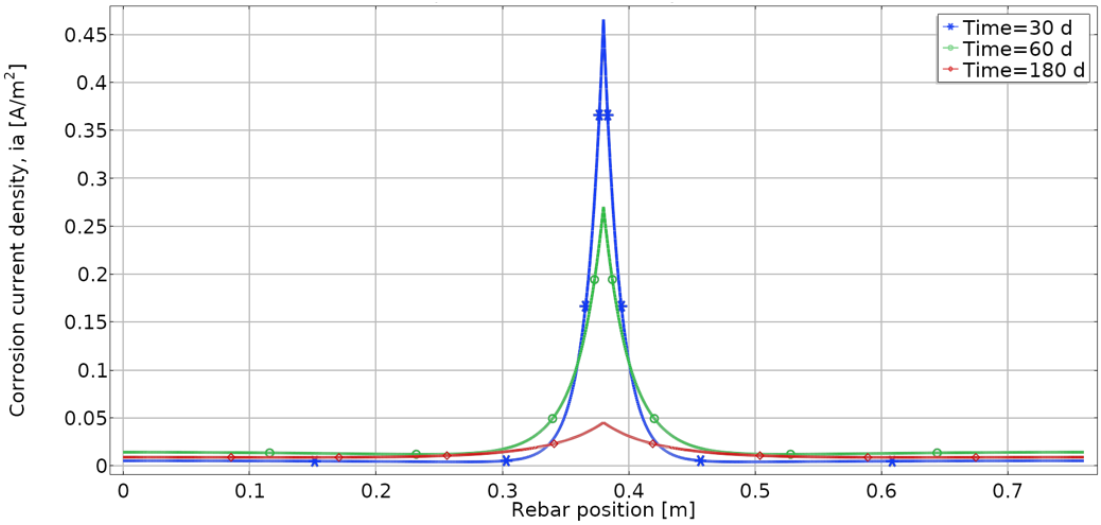


Figure 9.5 Corrosion current density distribution along the rebar at different times.

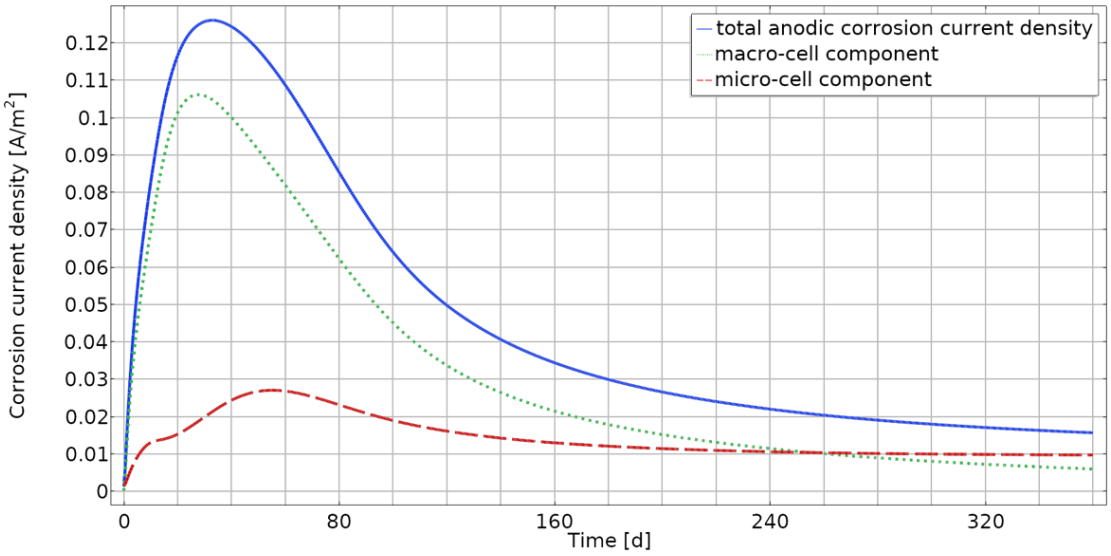


Figure 9.6 Averaged corrosion current in the cracking zone at different times for the submerged condition with 1 mm crack.

9.3.1.2 Submerged condition: 0.1 mm crack

As shown in Figure 9.7, for a much finer crack (0.1 mm), the diffusion of chloride ions through the crack is reduced. The penetration rate of the chloride ions through the crack relative to their penetration through uncracked concrete is significantly decreased in this case.

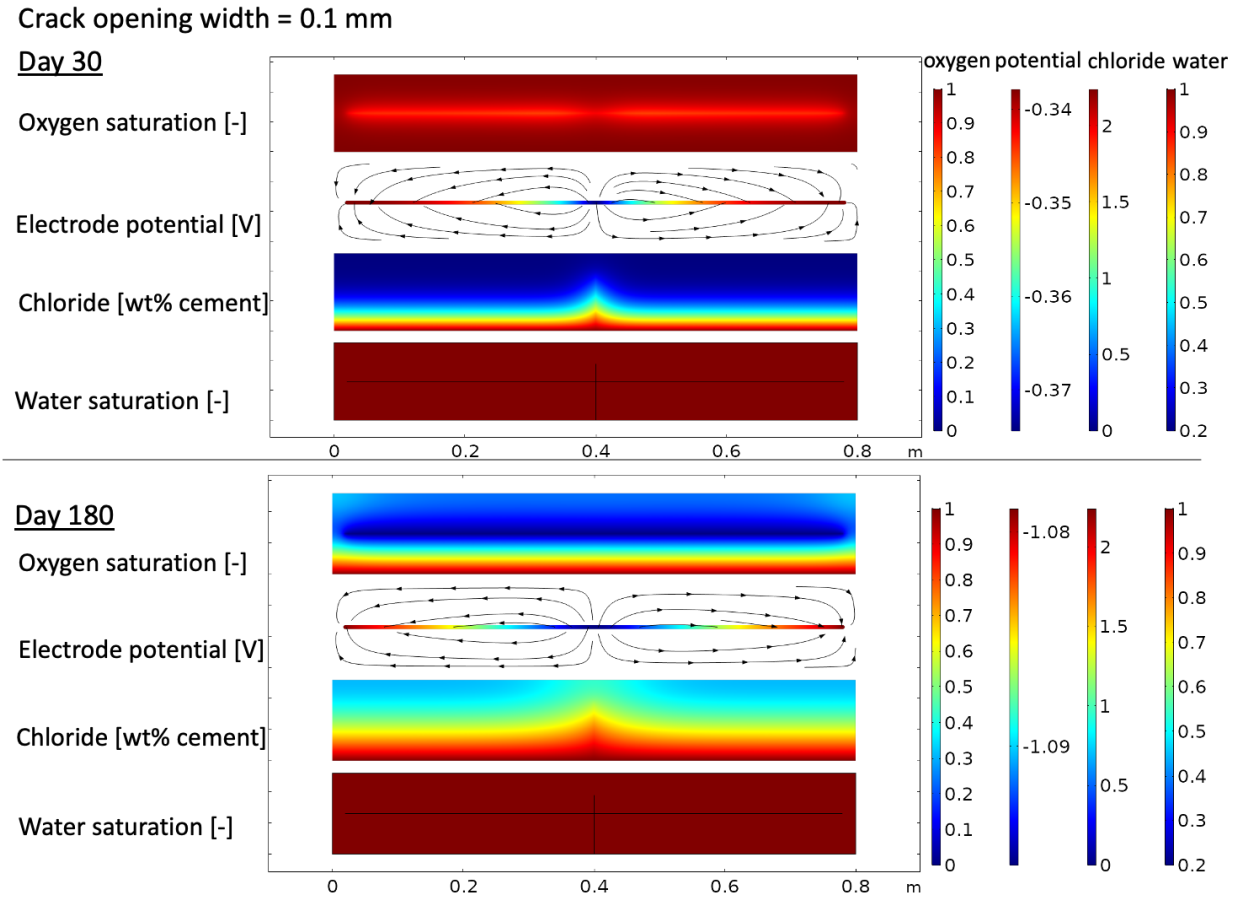


Figure 9.7 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the submerged condition at the early stage (Day 30) and developed stage (Day 180). Case: 0.1 mm crack.

As a result, compared to the 1-mm crack case, at the early stage (Day 30), the localized high corrosion current density of the 0.1-mm case is reduced because the chloride content at the crack is only slightly higher than that on the rest of the rebar and the rebar at the crack is less anodically polarized (Figure 9.8).

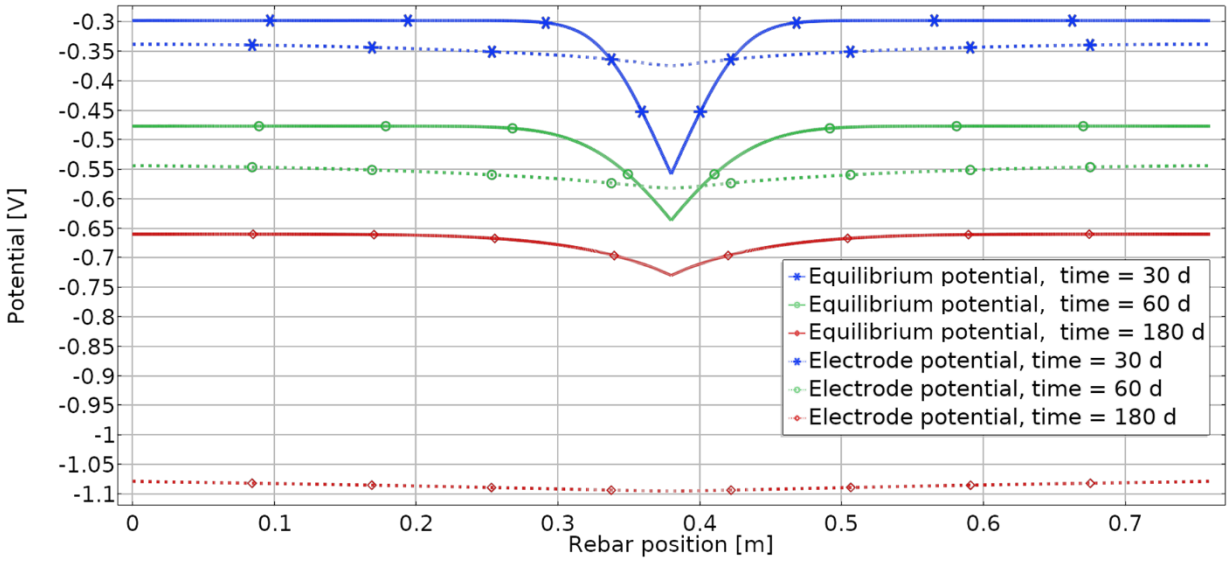


Figure 9.8 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack.

For the same reason, the peak corrosion current density (0.5 A/m^2) is also reduced by approximately 50%. The overall reduced corrosion also means a reduced cathodic reaction. Therefore, the depletion of oxygen is also delayed, which leads to a slower reduction of corrosion current density over time at the crack, as shown in Figure 9.9 but the corrosion current density finally levels off and is reduced to a similar value as the one seen in the 1 mm crack case. In addition, Figure 9.10 shows that for this finer crack case, the micro-cell component has a much higher weight of contribution to the total corrosion current density in the cracked zone. The micro-cell corrosion starts becoming dominant after the corrosion current density peaks at Day 60.

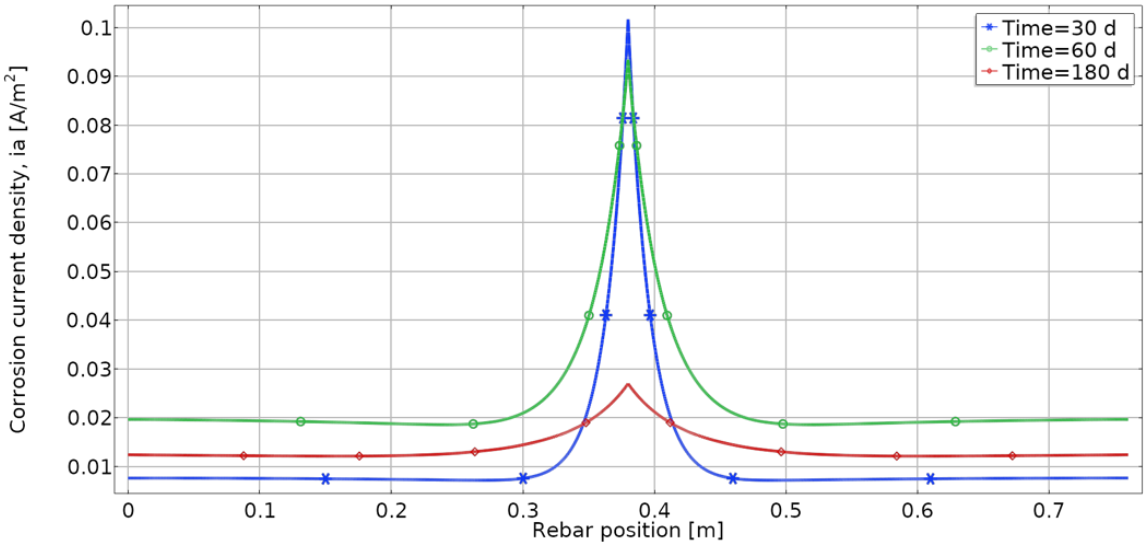


Figure 9.9 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack.

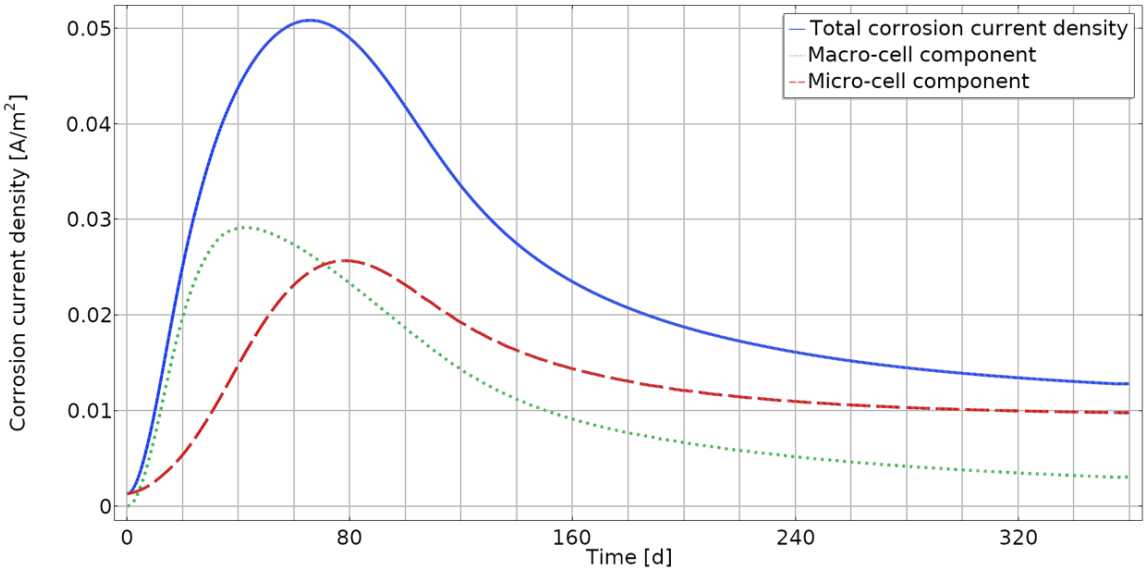


Figure 9.10 Averaged corrosion current in the cracking zone decreases with time due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack.

In summary, the corrosion model captures the field observation of the dependence of crack size and corrosion rate reduction over time. For the submerged condition, cracks affect the corrosion by affecting the chloride diffusion rate through the crack. Cracked concrete with a smaller crack size has reduced peak corrosion current and slower reduction of corrosion rate over time, an overall more uniform corrosion with more micro-cell corrosion. Eventually, the corrosion will reduce to

a similar value when the whole rebar is contaminated with chloride and the corrosion rate is controlled by the limiting diffusion of oxygen.

9.3.2 Wetting and drying condition

When cracked reinforced concrete is exposed to the wetting and drying scenario with a chloride source during the wetting cycles, the corrosion kinetics of the rebar is influenced by both water content and chloride content. The change in water content will significantly affect chloride transport and oxygen diffusion. The chloride can penetrate concrete from the crack via advection in the wetting cycle after a drying cycle. The oxygen depletion could be less of a concern for mostly dry conditions (caused by higher oxygen diffusivity in drier concrete). We will look at scenarios with two crack sizes while tracking the availability of oxygen.

9.3.2.1 Wetting cycles: 1 mm crack

The weekly wetting and drying cycles were set up as one day of wetting and six days of drying. The solution for Day 28.5 and Day 175.5 (half-day into the wetting cycle) are selected to represent the early stage and developed stage of corrosion.

As shown in Figure 9.11, in the early stage, rapid chloride penetration occurs at the crack via advection in the wetting cycle. The rebar at the crack is anodically polarized, exerting net anodic current. The electrolyte current flows out from the rebar at the crack. Similar to the submersion case, the most cathodic reaction occurs on the rebar not at but near the crack. The oxygen has a higher saturation level on the rebar than the submersion case with a 1 mm crack. This is because oxygen is replenished partially during the drying cycle where the dried crack and dried concrete near the crack have a much higher oxygen diffusivity. In the developed stage, the development of the chloride profile shows slower continuing chloride penetration due to no flux in the drying period and reduced diffusivity of chloride in drier concrete. In the drying cycle, the chloride continues to diffuse into deeper concrete while there is no supply on the concrete surface.

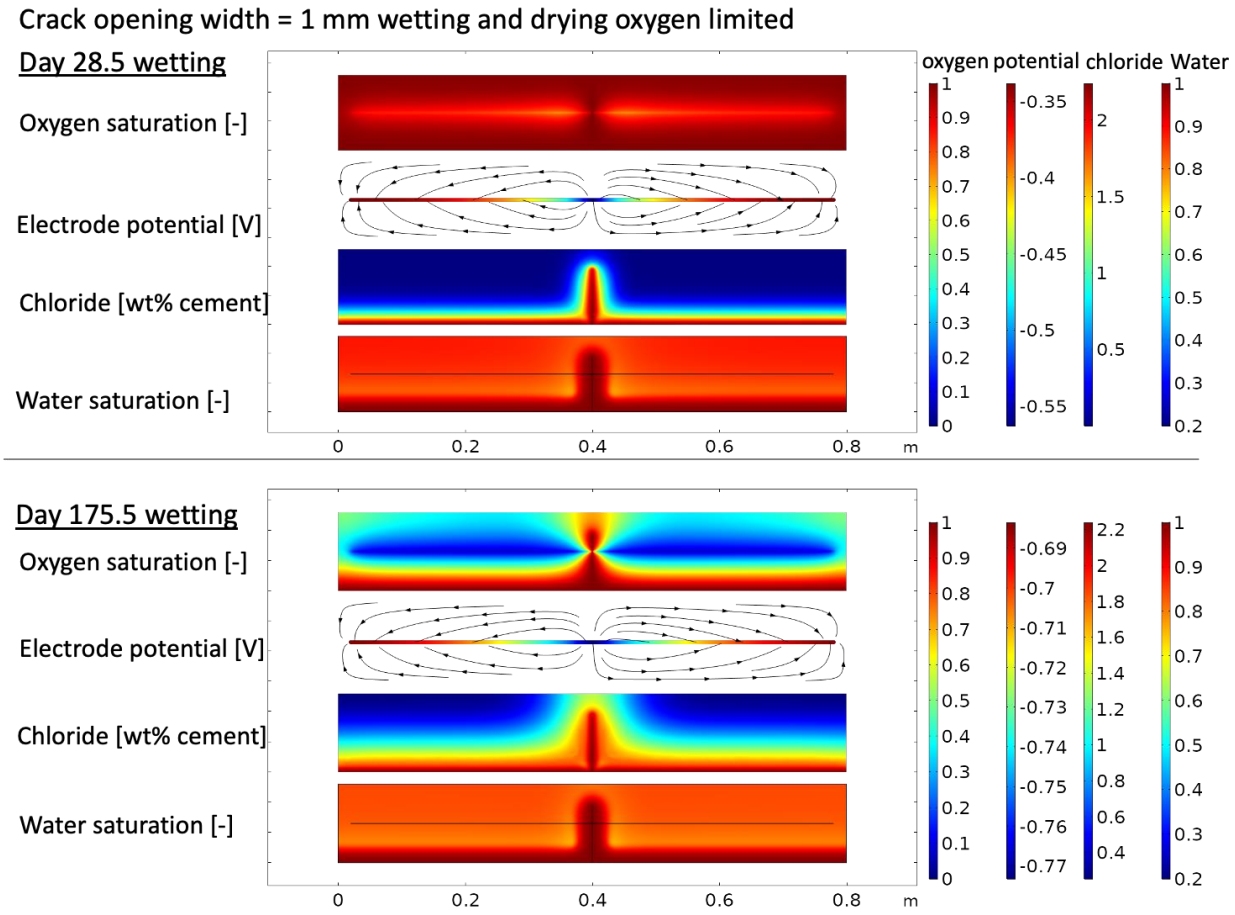


Figure 9.11 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the wetting cycles at the early stage (day 28.5) and developed stage (day 175.5).

As shown in Figure 9.12, the equilibrium potential difference between the rebar at the crack and rebar away from the crack is reduced, which results in a reduced anodic polarization of the rebar at the crack, which led to a reduced anodic corrosion current density at the rebar. The accumulated oxygen consumption on the rebar has caused oxygen to deplete on the rebar away from the crack. The oxygen-limited cathodic reaction also contributes to the reduction of the anodic reaction at the crack. Therefore, the corrosion current at the crack decreases over time (Figure 9.13). It is worth mentioning that the oxygen at the crack still has a higher saturation level because the rebar at the crack is anodically polarized, and the oxygen reduction reaction rate is suppressed to a low level. The additional oxygen supply through the crack is not sufficient to replenish the oxygen consumed away from the crack.

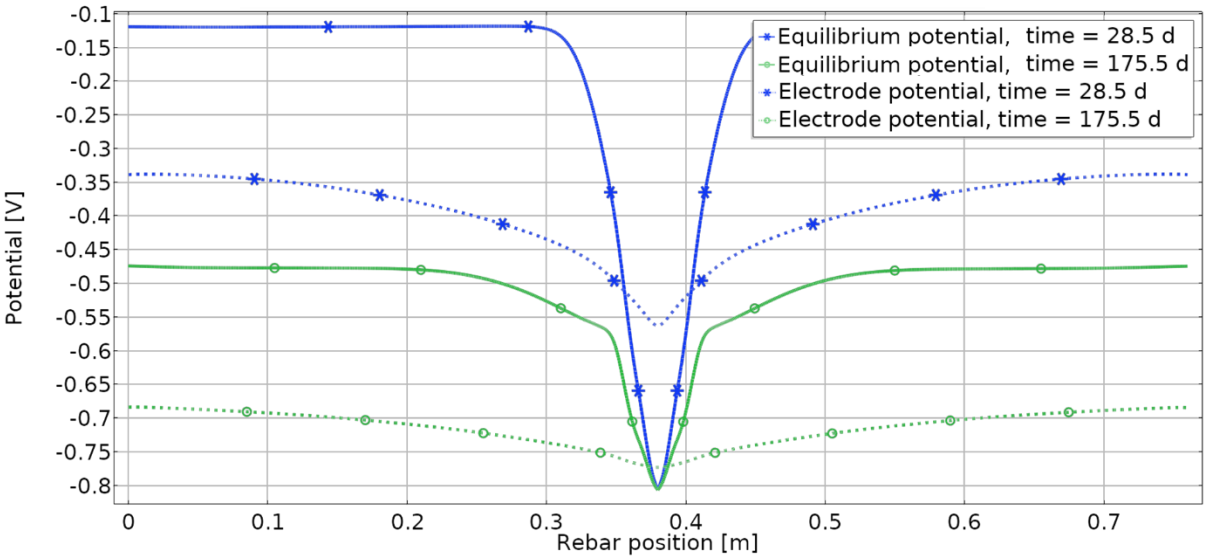


Figure 9.12 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 1 mm crack, wetting cycles.

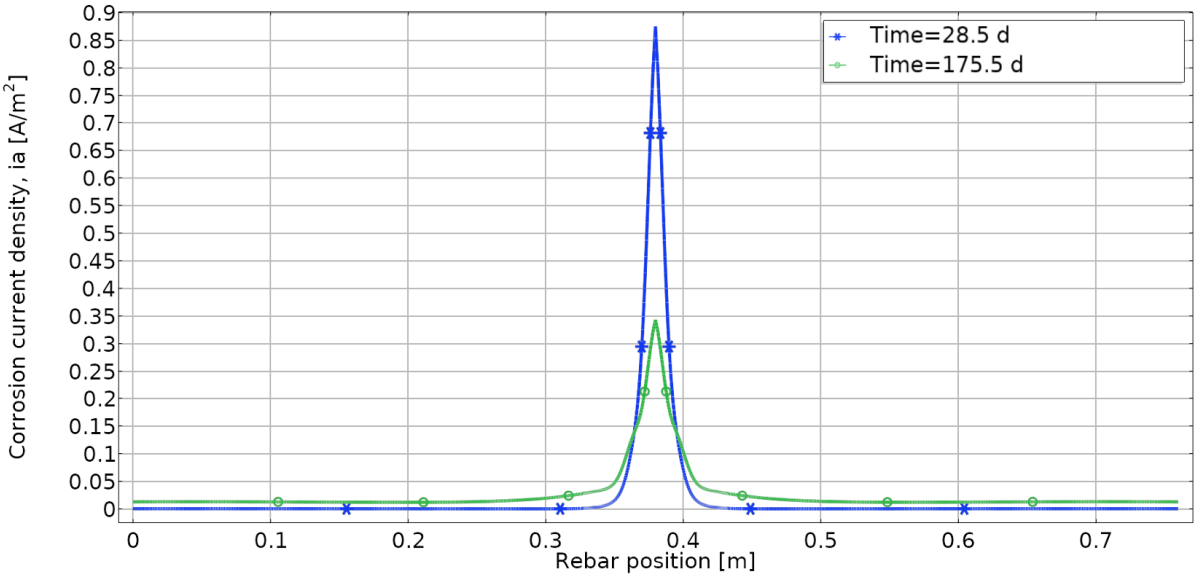


Figure 9.13 Corrosion current density distribution along the rebar at different times. Case: 1 mm crack, wetting cycles.

9.3.2.2 Drying cycles: 1 mm crack

The solution for Day 35 and Day 182 (the end of the drying cycle) are selected to represent the early stage and developed stage of corrosion. As shown in Figure 9.14, at the end of each drying cycles the water content near the crack is low, and the chloride content profile near the crack is close to 2D diffusion. The chloride content on the concrete surface and crack surface is slightly less than the case in the wetting cycles due to no flux boundary condition on these surfaces during the drying cycles. Rebar at the crack becomes less anodic when exposed to low water content and high chloride content. The rebar in the cracked zone could have more positive equilibrium potential than the rest of the rebar so that the net effect of rebar at the crack is providing cathodic current.

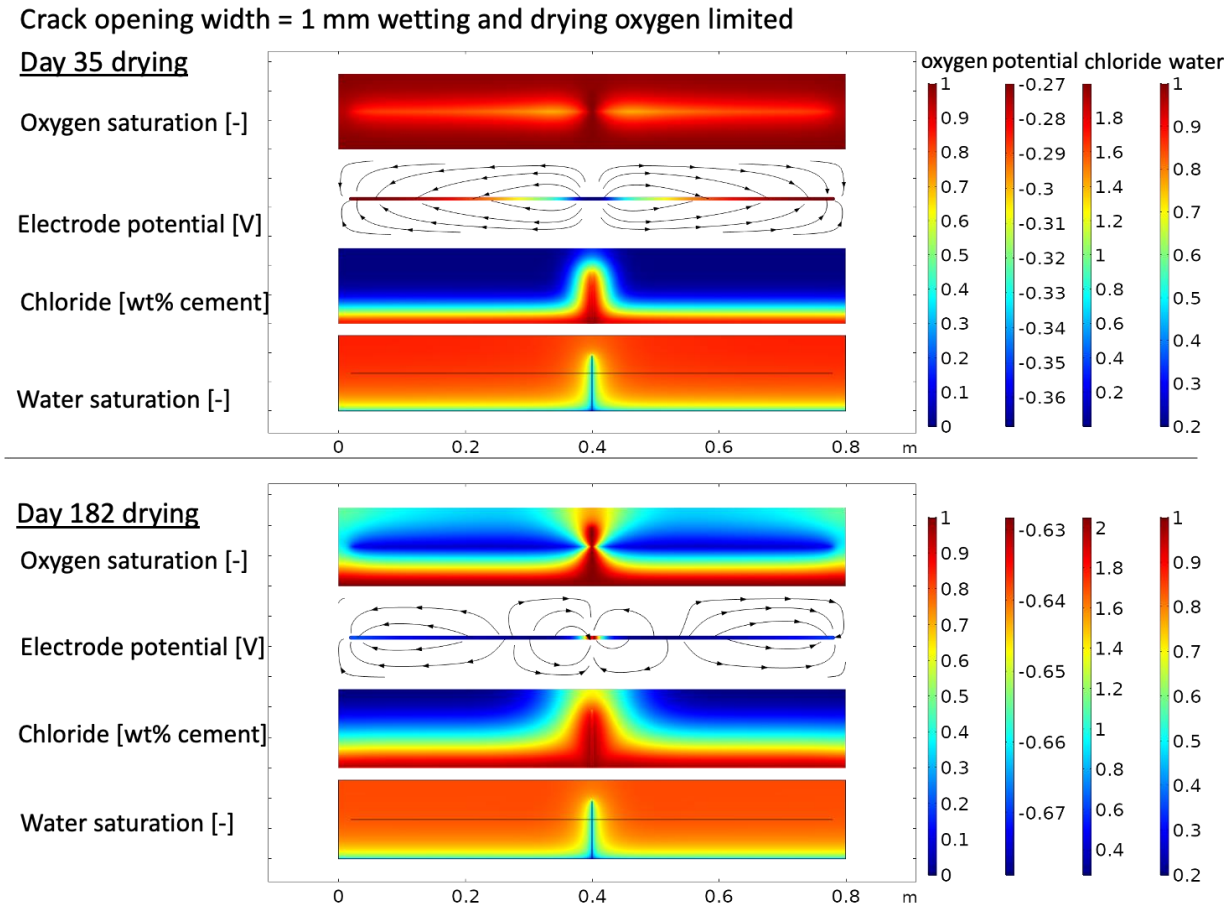


Figure 9.14 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the drying cycles at the early stage (day 32) and developed stage (day 182).

This is the case in the developed stage at Day 182 (Figure 9.15), where chloride content on the rebar away from the crack increases to an extent, the rebar away from the crack can be more anodic than the rebar at the crack. The most anodic area is found around 15 cm away from the rebar, and the electrolyte current flows from that spot to the rebar at the crack and the rebar far away from the crack.

Because of the low water content near the crack in drying cycles, the corrosion current density is low at the crack and the maximum corrosion current density is found immediately near the crack within 5 cm (Figure 9.16). The corrosion current density distribution spread out to a larger area away from the crack with a significant reduction in the intensity of the corrosion current density.

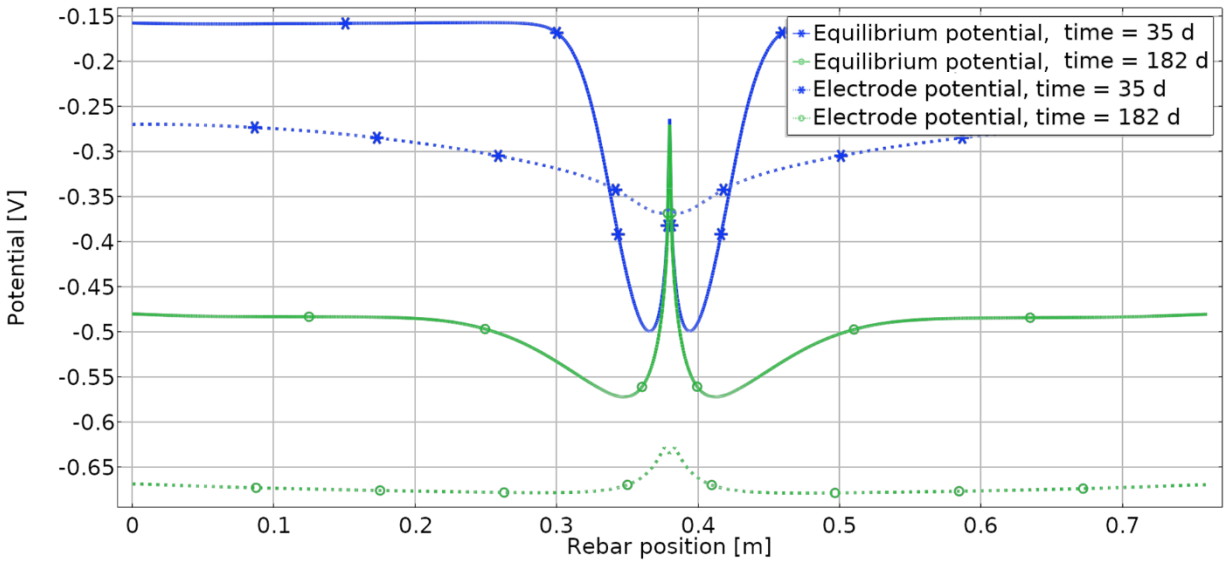


Figure 9.15 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 1 mm crack, drying cycles.

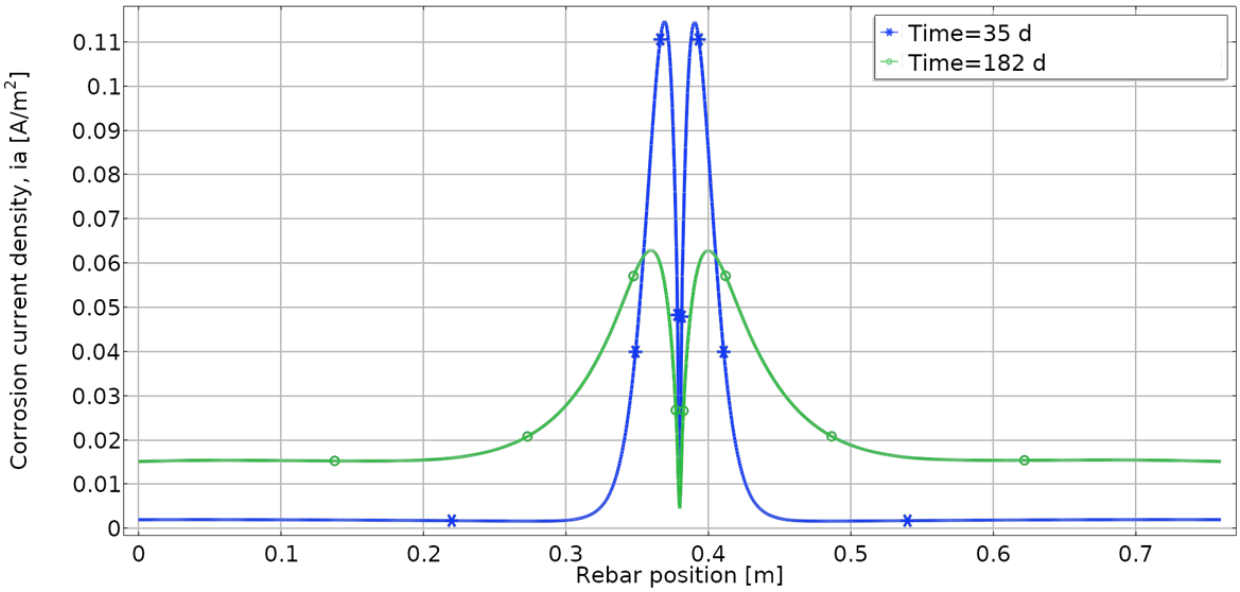


Figure 9.16 Corrosion current density distribution along the rebar at different times. Case: 1 mm crack, drying cycles.

9.3.2.3 Corrosion current change over the wetting and drying cycles: 1 mm crack

As shown in Figure 9.17, the line-averaged corrosion current density in the cracked zone fluctuates intensively with the wetting and drying events. For consecutive wetting and drying cycles, it peaks during the wetting cycle and decreases in the drying cycle. Over a number of wetting and drying cycles, the corrosion current density in the cracked zone peaks at the early stage and then decreases over time to a smaller steady rate. The anodic current density is dominated by the macro-cell mechanism in the early stage and in the developed stage, the micro-cell anodic current surpasses the macro-cell counterpart and becomes the controlling mechanism. It is worth noticing that in the drying cycles in the developed stage, the micro-cell cell is dominant because the rebar exposed to dried chloride-contaminated concrete at the crack effectively acts as a cathode.

The maximum corrosion current density of the whole rebar is shown in Figure 9.18. The maximum corrosion rate is located at the crack during the wetting cycles, in the cracking zone in the early stages, and in the uncracked zone in the developed stage. The peak values in the wetting cycles at the crack keep decreasing from the beginning.

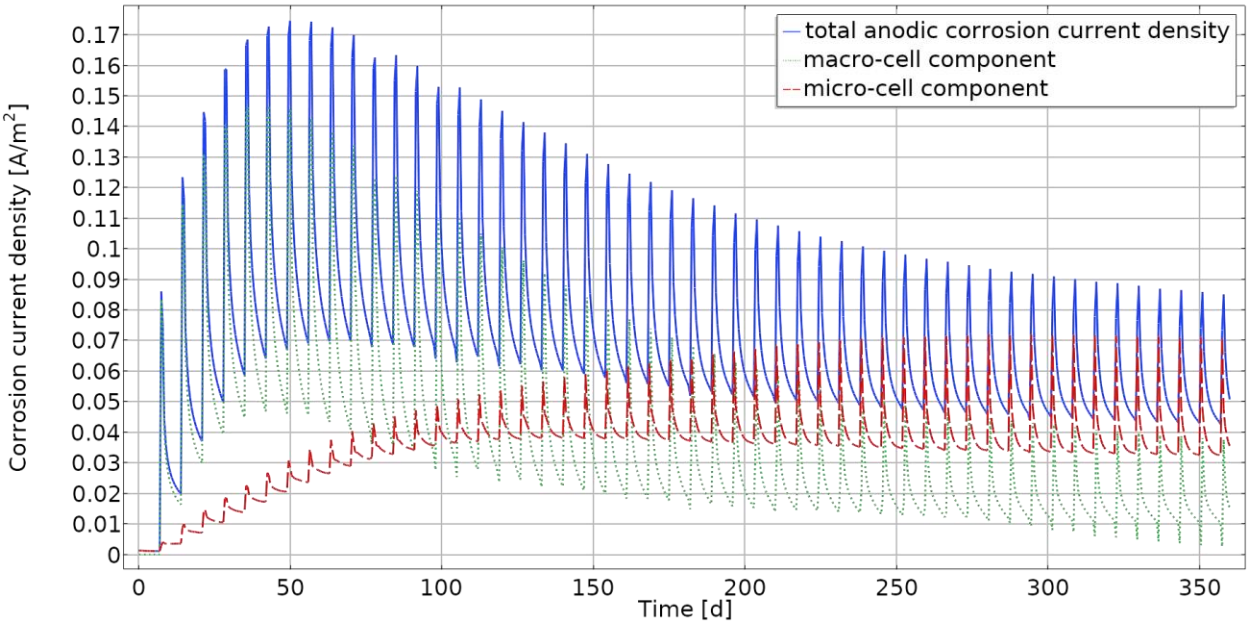


Figure 9.17 Averaged corrosion current in the cracking zone during wetting and drying cycles. Case: 1 mm crack, wetting and drying cycles.

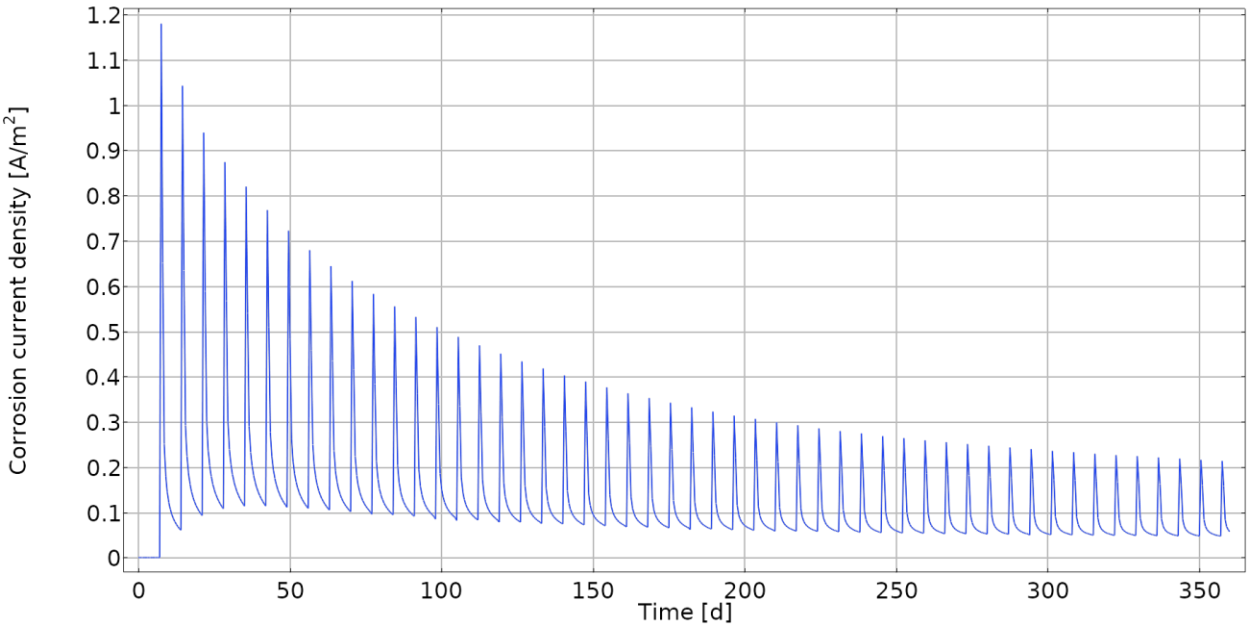


Figure 9.18 Maximum corrosion current during wetting and drying cycles. Case: 1 mm crack, wetting and drying cycles.

9.3.2.4 Wetting cycles: 0.1 mm crack

As in the 1-mm crack case, the weekly wetting and drying cycles were set up as one day of wetting and six days of drying. The solution for Day 28.5 and Day 175.5 (half-day into the wetting cycle) are selected to represent the early stage and developed stage of corrosion, as shown in Figure 9.19. The equilibrium potential and the corrosion current density distribution along the rebar at different times are shown in Figure 9.20 and Figure 9.21.

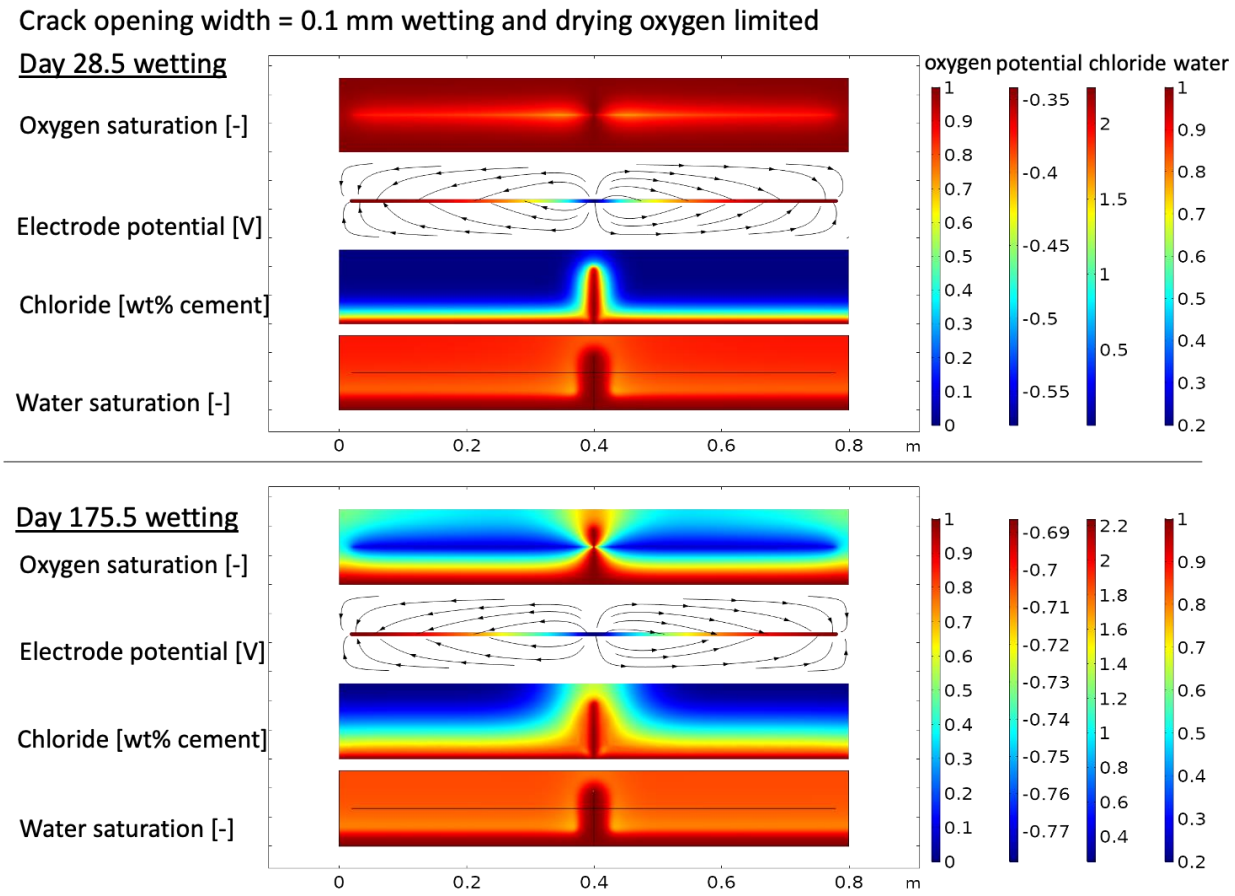


Figure 9.19 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the wetting cycles at the early stage (Day 28.5) and developed stage (Day 175.5).

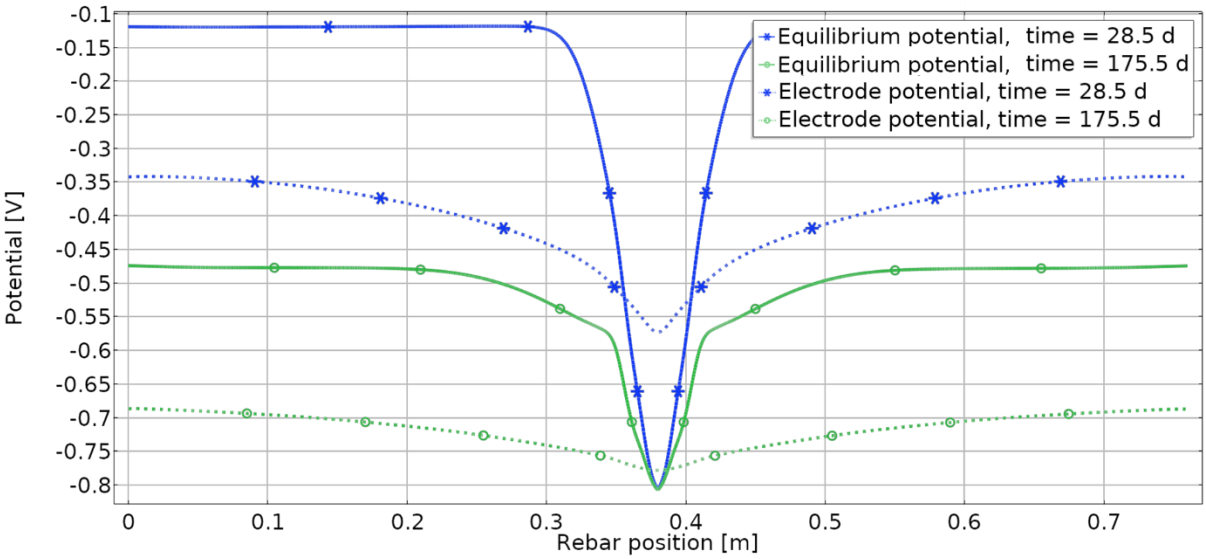


Figure 9.20 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack, wetting cycles.

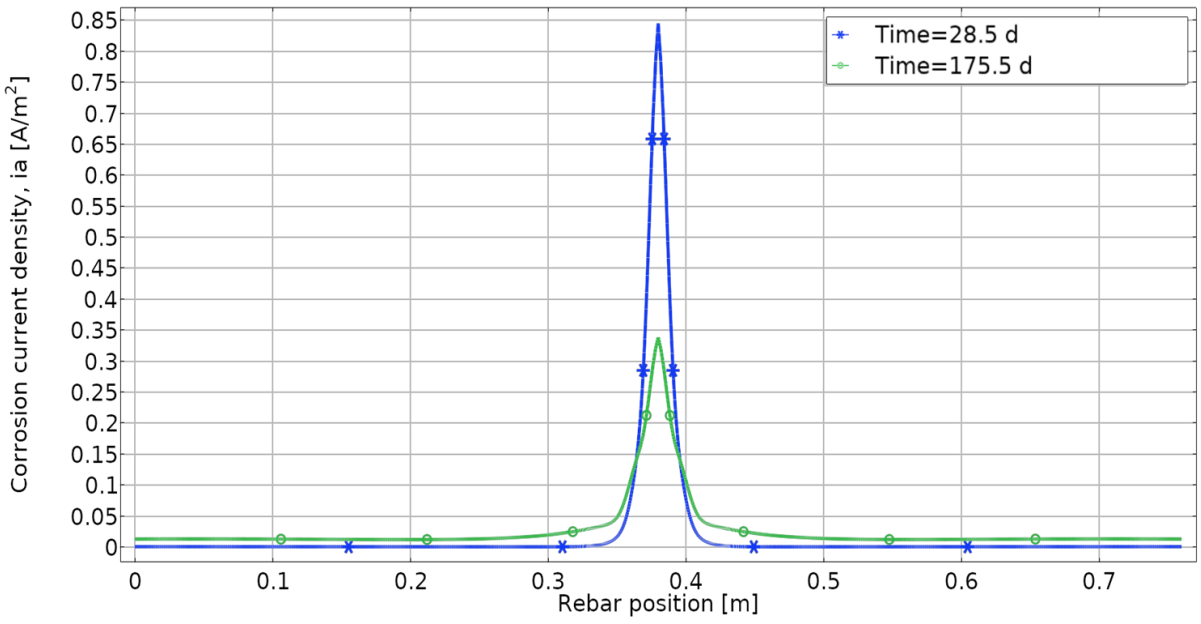


Figure 9.21 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack, wetting cycles.

9.3.2.5 Drying cycles: 0.1 mm crack

As in the case of the 1-mm crack, the solution for Day 35 and Day 182 (the end of the drying cycle) are selected to represent the early stage and developed stage of corrosion, as shown in Figure 9.22. The equilibrium potential and the corrosion current density distribution along the rebar at different times are shown in Figure 9.23 and Figure 9.24.

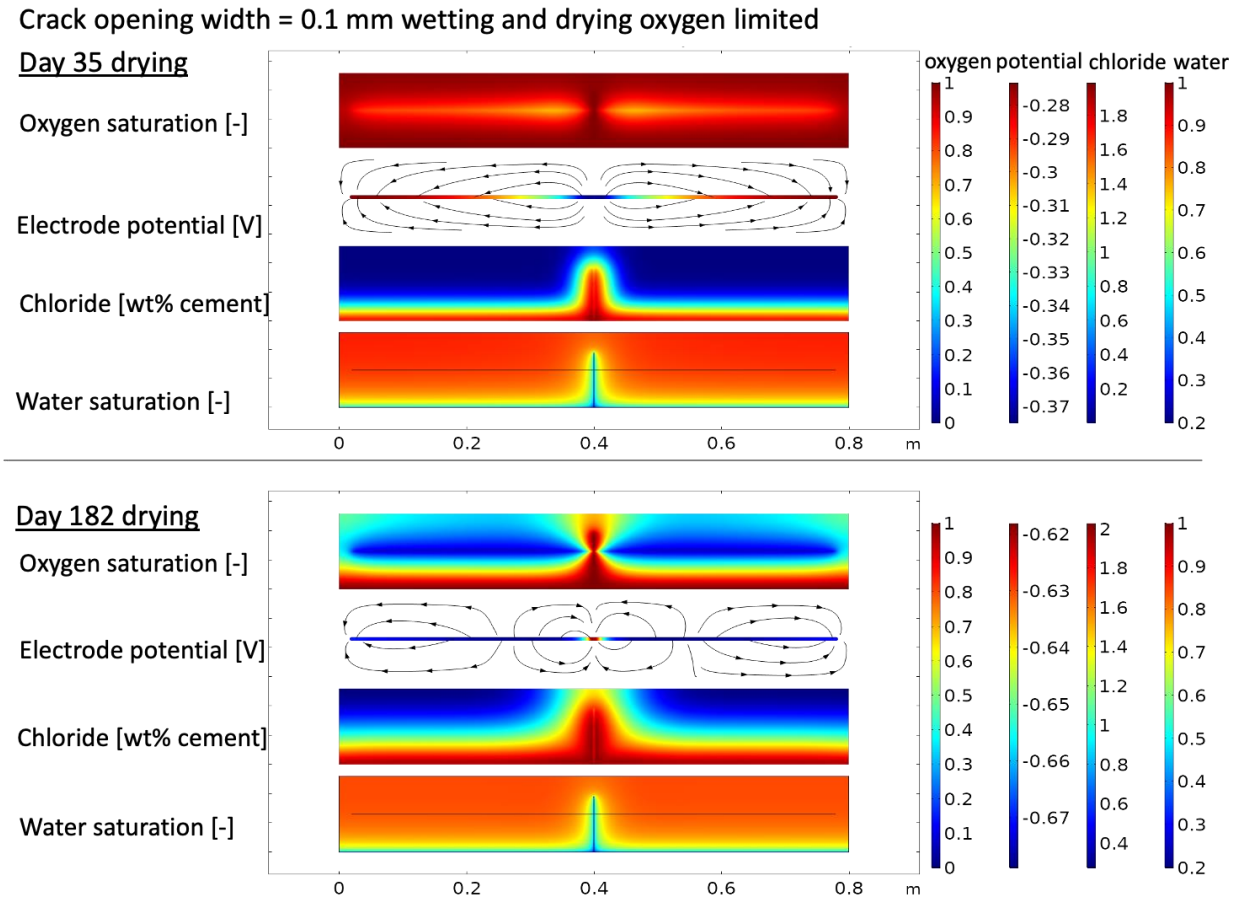


Figure 9.22 Water saturation, chloride content, electrode potential and oxygen saturation in the cracked concrete under the drying cycles at the early stage (day 32) and developed stage (day 182).

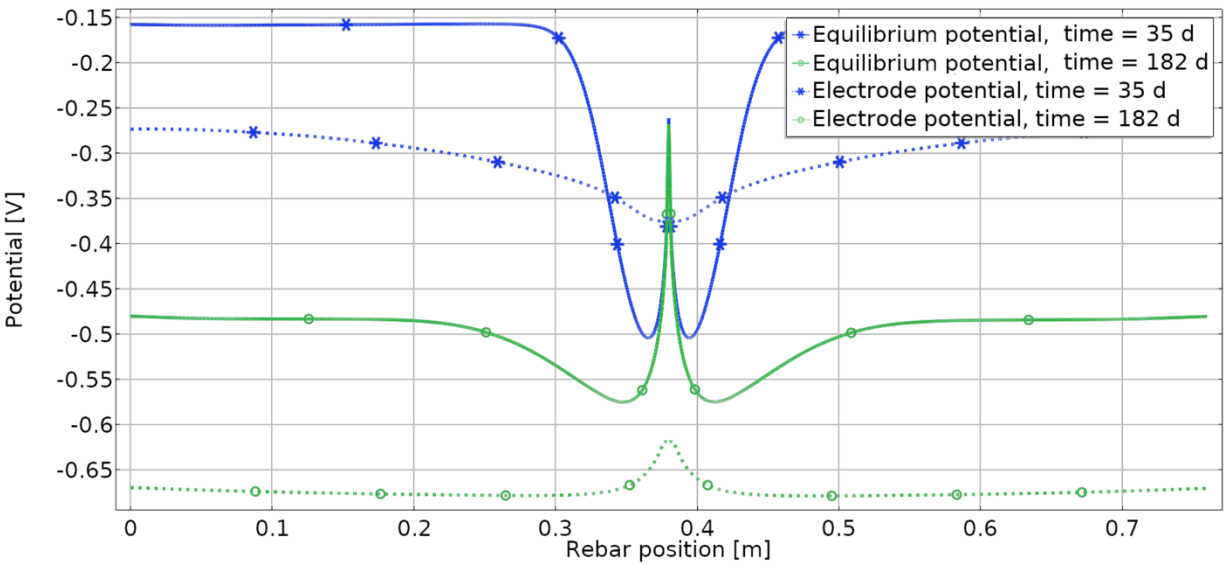


Figure 9.23 Equilibrium potential of the rebar determined by microenvironment and electrode potential of the rebar determined by the electrochemical reaction; the overpotential = electrode potential – equilibrium potential. Case: 0.1 mm crack, drying cycles.

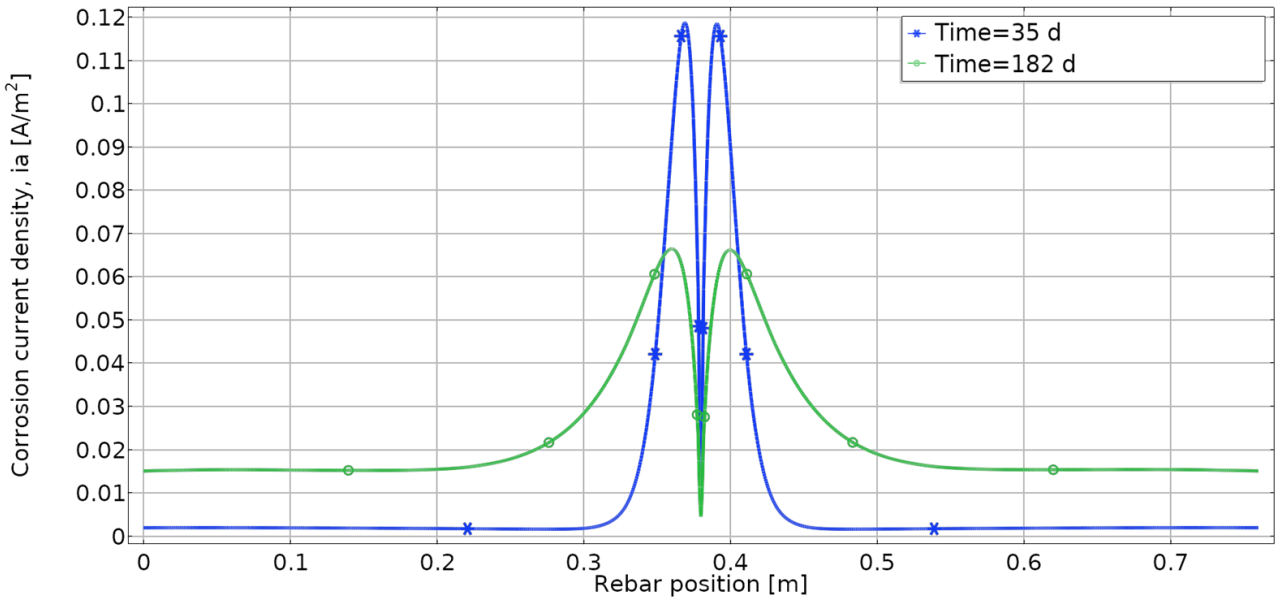


Figure 9.24 Corrosion current density distribution along the rebar at different times. Case: 0.1 mm crack, drying cycles.

9.3.2.6 Corrosion current change over the wetting and drying cycles: 0.1 mm crack

The line-averaged corrosion current density in the cracked zone through the wetting and drying events is plotted in Figure 9.25. The maximum corrosion current density of the whole rebar is shown in Figure 9.26.

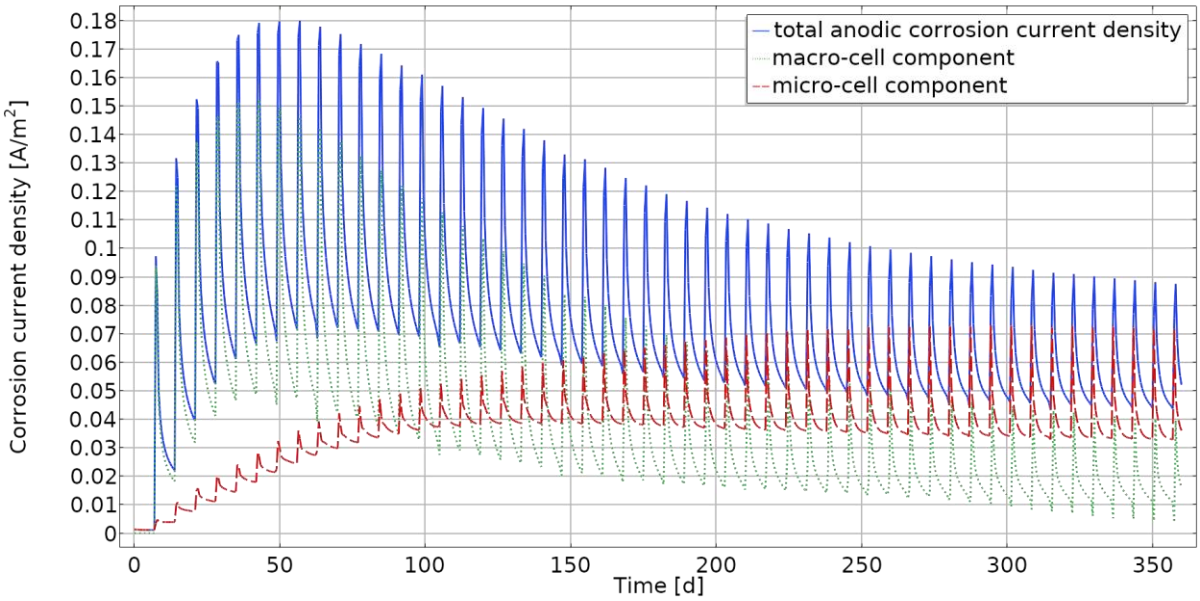


Figure 9.25 Averaged corrosion current in the cracking zone decreases in the drying cycle and decreases overtime due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack, wetting and drying cycles.

The case of a 0.1 mm crack exhibits an almost identical solution to the 1 mm crack case. The reason behind this is that unlike in submerged conditions where a smaller crack size reduces the progress of chloride diffusion through the crack, in wetting and drying cycles, the crack opening size does not have a significant effect on the chloride profile around the crack, even with the same crack length. The rapid penetration of chloride is primarily attributed to the wicking effect of the dried concrete surrounding the crack.

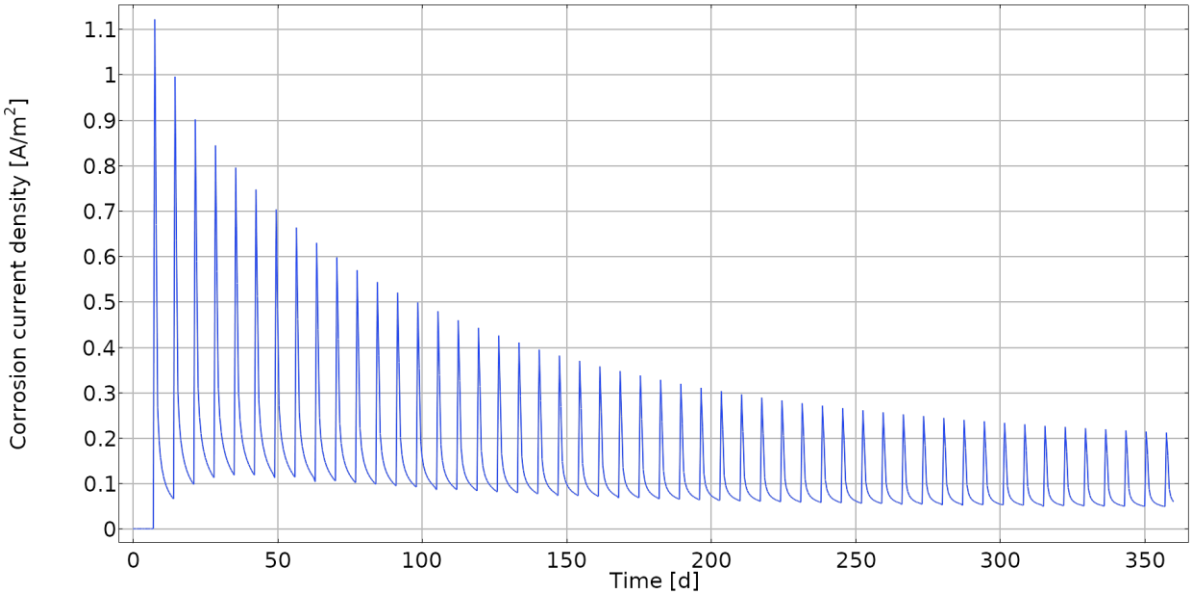


Figure 9.26 Maximum corrosion current on the rebar fluctuates in the wetting and drying cycles and decreases with time due to limited oxygen and more exposure to chloride. Case: 0.1 mm crack, wetting and drying cycles.

In summary, the model suggests that the corrosion kinetics of the rebar in the cracking zone are affected by chloride and moisture, with the possibility of becoming a cathode in drying cycles. The average corrosion rate of the rebar in the cracking zone peaks during the wetting cycle and decreases in the following drying cycle, and peak values decrease over the repeated wetting and drying cycles. The final reduced corrosion rate is controlled by the chloride-contaminated area on the rebar and the reduced oxygen concentration on the rebar. The micro-cell corrosion eventually becomes the dominant contributor to the rebar corrosion in the cracking zone.

For the wetting and drying scenario, the chloride profile is not significantly affected by the crack opening size in the wetting and drying cycles which is different from the submerged case. As a result, the model at this stage, does not capture the field observation where smaller cracks are less harmful.

9.3.3 Crack healing and critical crack width

A substantial amount of laboratory and field experiments have shown that the width of surface cracks in reinforced concrete plays a crucial role in determining the corrosion rate of the steel reinforcement. Standard codes in many countries have adopted a threshold crack width to reflect

experimental observations that cracks below a certain width do not significantly affect the corrosion rate of steel reinforcement in concrete. This threshold crack width typically ranges from 0.1 mm to 0.2 mm. Although a number of possible mechanisms (effect of crack width on the ingress of aggressive agents, self-healing, etc.) have been suggested, mostly by speculation, as possible reasons for the difference in corrosion behaviour between thin cracks and thick cracks, there is no clear understanding about which parameters are actually affecting the difference in corrosion behaviour between thin cracks and wide cracks. Hence, there is a need to shed more light on the intricate relationship between crack characteristics and corrosion behaviour, a compelling yet complex aspect of reinforced concrete durability that remains poorly understood.

The current model assumes that the cracks remain open indefinitely without any healing mechanism, which may not be a realistic representation of the long-term behaviour of concrete structures. In reality, concrete has the ability to heal thin cracks over time through a process known as autogenous healing. This process occurs when the moisture in the environment reacts with the unhydrated cement particles in the concrete, leading to the formation of calcium-silicate hydrate (C-S-H) that can fill in small cracks and pores. In the next section, the crack healing mechanism will be added to the current model to investigate its effect on corrosion in cracked concrete.

Therefore, to more accurately model the long-term effects of cracks in concrete structures, the model should incorporate the concept of crack healing. The crack can be assumed to heal only up to a maximum crack opening. Since the crack opening changes with depth, fractures with wide surface crack openings would seal the crack only up the height corresponding to this max “healable” opening. This concept allows a unified treatment of wide cracks and thin ones. After healing the cracks would remain at their new reduced depth and have, essentially the effect of a reduced concrete cover at their locations.

The maximum healable crack size in concrete could be up to 0.2 mm in width in freshwater environments under atmospheric conditions, where calcite precipitation is identified as the main cause of autogenous self-healing. However, the efficiency of healing may vary depending on several factors such as crack width, roughness, tortuosity, the presence of liquid water, water pressure, water chemistry, temperature, aggregate size, gradation curve, water-cement ratio, cement content, and exposure conditions (Lahmann et al., 2022).

To investigate the crack healing effect on corrosion in the existing corrosion model, the healable crack width was set as 0.08 mm and it heals linearly with time up till Day 60. The healing effect was expressed as the reduction of the transport properties of water, chloride ions and oxygen gas. When the crack is completely healed, its transport parameters on the crack line become negligible. Figure 9.27 shows a surface plot of the reduction function with the reduction to 10^{-15} .

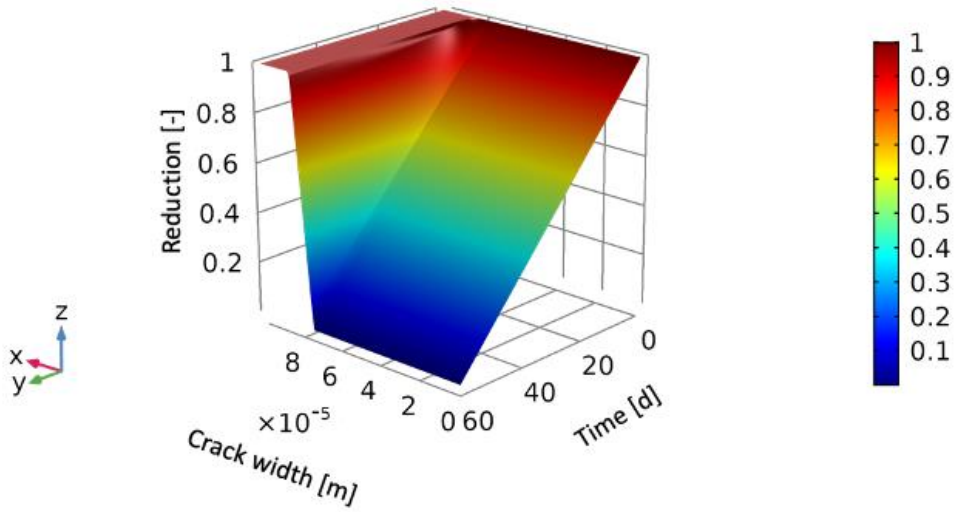


Figure 9.27 Linear reduction of transport properties such as hydraulic conductivity and chloride and oxygen diffusivity as a function of time when crack width with a healable crack width of 0.08 mm.

For the model with a 0.1 mm crack surface opening, the crack width decreases linearly to zero to the tip of the crack. As shown in Figure 9.28 and Figure 9.29, the hydraulic conductivity and chloride and oxygen diffusivities reduce over time for the crack size less than 0.08 mm, so effectively the rebar regained some concrete cover at the crack (the crack width at the rebar depth is less than 0.08 mm, the healable crack width).

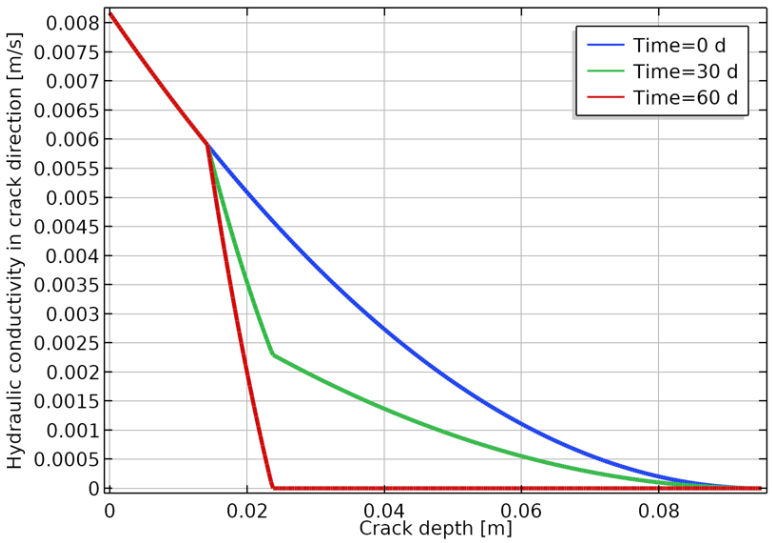


Figure 9.28 The hydraulic conductivity along the crack line changes due to crack healing.

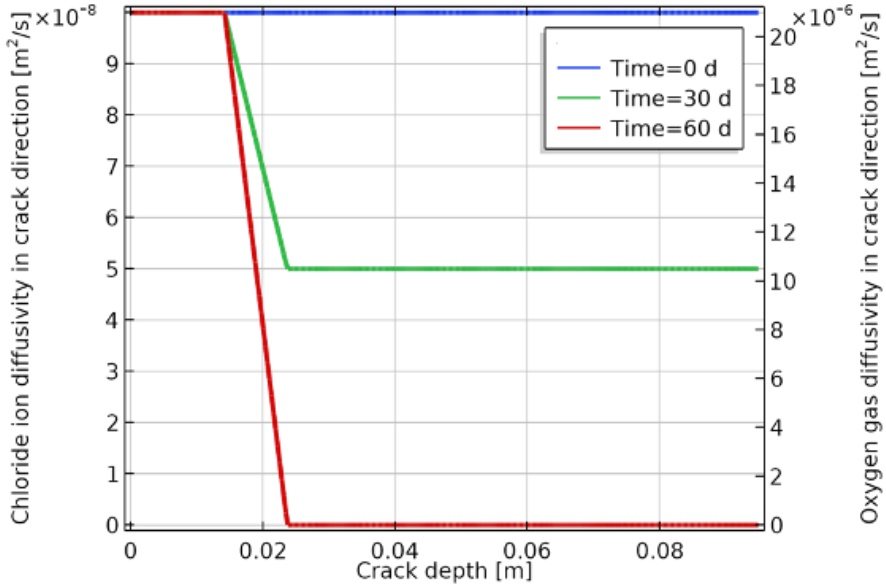


Figure 9.29 The diffusivity of chloride ions and oxygen gas along the crack line changes due to crack healing.

This crack-healing implementation accelerates the corrosion rate reduction over the wetting and drying cycles (Figure 9.31). The effect is brought by the reduced chloride content gradient, and the reduced oxygen saturation level in the cracking zone. Because the healing is controlled by the crack width, finer cracks will have more healing than bigger cracks resulting in more recovered concrete cover for the rebar at the crack. For the investigated simulation, the crack has a surface

opening of 0.1 mm. The crack completely heals at Day 60 for a crack width less than the maximum healable width (0.8 mm). In addition, a less permeable high-performance concrete (HPC) was implemented. It has reduced hydraulic conductivity, K_s and chloride diffusivity, D_s of the control concrete by a factor of 0.2. Low oxygen diffusivity, $D_{eff}^{O_2}$ refers to the diffusivity that caused oxygen to be depleted eventually on the rebar when there is no crack healing effect and high $D_{eff}^{O_2}$ is scaled up by ten.

For less permeable concrete, the effect of the crack healing on reducing corrosion would be more pronounced so the finer cracks become less harmful. As shown in Figure 9.30, the initial high concentration of chloride at the crack dissipates over time with a diluted chloride content at the crack and this reduction is more effective for the less permeable HPC with a smaller accumulation of chloride content at the crack. The rebar at the crack will become less anodic or even enter the passive zone with the restoration of the passive film. This prediction aligns with the field observation that for less permeable high-performance concrete, the allowable crack width is larger concerning corrosion damage.

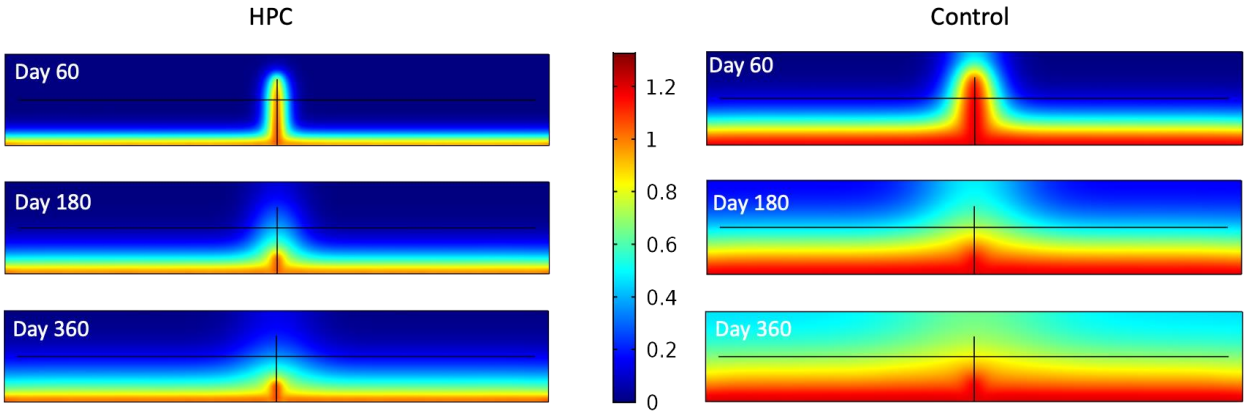


Figure 9.30 The evolution of chloride content profile (wt.% Cl by weight of cement) around the crack (0.1 mm surface opening) after complete crack healing (Day 60) with a width up to 0.08 mm and crack width greater than 0.08 mm remains unhealed. Left: high performance concrete (HPC), Right: control concrete.

The averaged corrosion current density in the cracking zone is plotted in Figure 9.31. After the complete crack healing at Day 60, there is a major corrosion reduction in the cracking zone for

both types of concrete. The reduced corrosion current density is similar to that in the drying cycle due to healed crack shielding the sudden water suction and the companion chloride ion advection in the wetting cycles. The corrosion current density continues to steadily decrease and levels off to a stable low corrosion current density.

For the control concrete, chloride content increased significantly at the crack and overall on the entire rebar. The rebar at the crack has a similar exposure compared to the rest of the rebar. The rebar corrosion in the cracking zone has a major contribution in the form of micro-cell corrosion. The stabilized low corrosion rate depends on the availability of oxygen. As shown in Figure 9.31, low oxygen diffusivity has led to a much lower corrosion current density corresponding to oxygen depletion on the rebar surface.

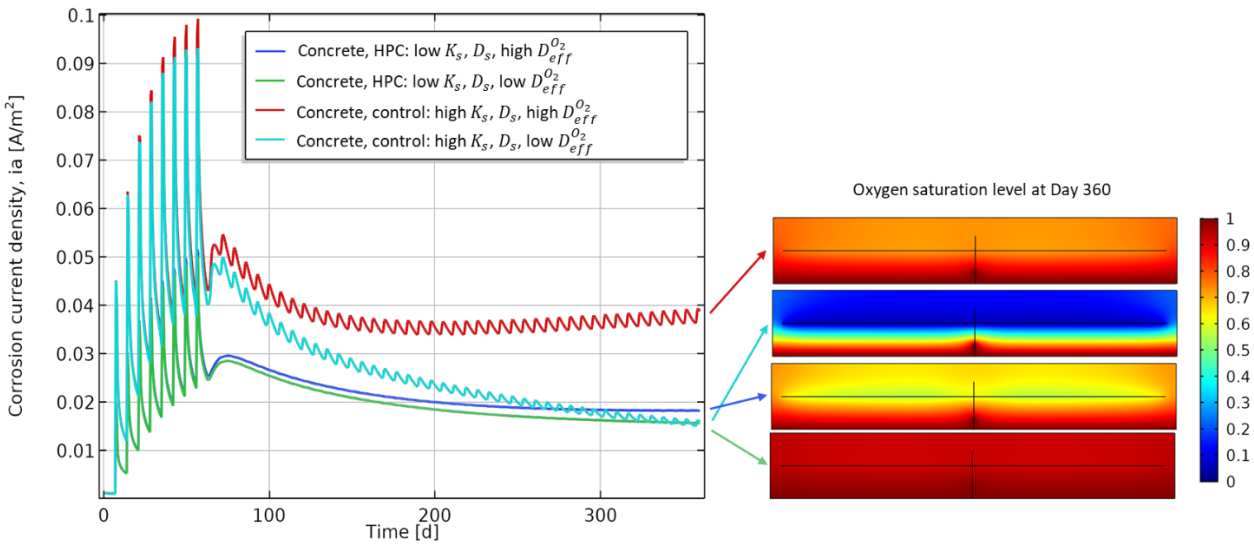


Figure 9.31 The effects of crack healing on the reduction of corrosion rate in the cracking zone. The surface crack width is 0.1 mm; the salt solution in the wetting condition is 3.0 wt.% NaCl. HPC (high performance concrete) has lower hydraulic conductivity and chloride diffusivity by a reduction factor of 0.2 to the control concrete. Low oxygen diffusivity $D_{eff}^{O_2}$ refers to the diffusivity that could have caused oxygen to be depleted eventually on the rebar if there is no crack healing effect.

For the HPC, this stable corrosion rate is not significantly affected by oxygen diffusivity. The corrosion rate is so small that the corresponding oxygen reduction reaction will not cause a significant decrease in oxygen content on the rebar. The corrosion current is low due to the dissipated high chloride from the cracking zone.

9.3.4 Critical crack width

The “critical crack width” or “threshold crack width” is described as a maximum allowable surface crack opening, below which the crack is considered harmless. The numerical investigation has provided a deeper mechanistic understanding on the intricate relationship between crack healing, exposure conditions, and corrosion behaviour. Building upon the insights gained from this numerical study, this critical crack width for reinforced concrete in a corrosive environment can be sought in the following way:

First, the corresponding crack width at the rebar depth should be less than the maximum healable crack width so that the crack healing process can recover some of the concrete cover at the crack. This value can be determined by the crack geometry and the thickness of the concrete cover, and it is not conservative.

Second, with the recovered concrete cover over the crack, it should bring a significant reduction in the corrosion current density, potentially to a negligible value. However, this depends on the existing corrosive microenvironment around the crack at the time of significant crack healing, continuous exposure conditions in the service condition, concrete properties, and corrosion kinetics of the embedded metal. The evaluation of this complex corrosion activity on the rebar is necessary to determine this value, which can be done using the proposed comprehensive corrosion model.

9.4 Conclusion

The comprehensive corrosion model that was presented and validated in chapter 8 was used in this chapter to assess the impact of cracks on corrosion. A 2-D model was used to examine how the cracks affect the microenvironment and corrosion of rebar in cracked concrete in submerged conditions and wetting and drying conditions. It examines the key microenvironment factors such as moisture, chloride accumulation, oxygen availability and the resultant electrode potential, and corrosion current density of the rebar and provided a mechanistic explanation of the observed rebar corrosion behaviour in cracked concrete: corrosion reduction over time and the crack size effect.

9.4.1 Submerged condition

In the submerged condition, cracks affect corrosion by affecting the chloride diffusion rate through the crack. A thinner crack results in a reduced diffusion influx of chloride ions through the crack. Concrete that is cracked with a smaller crack size has a reduced peak corrosion current and slower reduction of the corrosion rate over time, resulting in overall more uniform corrosion with more micro-cell corrosion. Eventually, the corrosion will reduce to a similar value when the entire rebar is contaminated with chloride, and the corrosion rate is controlled by the limiting diffusion of oxygen.

9.4.2 Wetting and drying condition

9.4.2.1 Corrosion reduction over a drying cycle and wetting and drying cycles

For wetting and drying conditions, the model suggests that the corrosion kinetics of the rebar in the cracking zone are affected by chloride and moisture, with the possibility of becoming a cathode in drying cycles. The average corrosion rate of the rebar in the cracking zone peaks during the wetting cycle and decreases in the following drying cycle, and peak values decrease over the repeated wetting and drying cycles. The final reduced corrosion rate is determined by the chloride-contaminated area on the rebar and the reduced oxygen concentration on the rebar. Micro-cell corrosion eventually becomes the dominant contributor to the rebar corrosion in the cracking zone.

9.4.2.2 Crack size and crack healing

For the wetting and drying scenario, unlike the submerged case, the model shows that the crack opening size does not significantly affect the chloride influx through the crack. To capture the effect of crack size on corrosion, the model needed to incorporate the crack healing mechanism, which is a more realistic representation of the long-term behaviour of concrete structures.

This crack healing implementation allowed the crack to heal when the crack opening was less than the critical “healable size”, and the healable portion of the crack had decreasd transport properties over time. The crack healing can slow down future chloride transport and potentially allow high chloride content to dilute near the crack and limit oxygen diffusion through the crack. The less permeable concrete has a more significant chloride content reduction due to crack healing. The

crack healing implementation accelerated the corrosion rate reduction over the wetting and drying cycles. For a given critical healable crack size, a finer crack has a more healable crack length than a wider crack resulting in more recovered concrete cover for the rebar at the crack.

For the same crack geometry, the effect of crack healing on the reduction of corrosion also depends on other factors such as the permeability of concrete. For the control concrete, the stabilized low corrosion rate is dependent on the availability of oxygen. However, for HPC, the stable corrosion rate is not significantly affected by oxygen diffusivity, but mainly due to the dissipation of high chloride concentration from the cracking zone.

9.4.2.3 Implication on the "critical crack width" – the maximum allowable surface crack opening, below which the crack is considered harmless in a corrosive environment.

The critical crack width should be small enough such that the corresponding crack width at the rebar depth is greater than the maximum healable crack width. Therefore, the crack healing process can recover some of the concrete cover at the crack. This value is determined by the crack geometry and the thickness of the concrete cover.

The critical crack width should be even smaller so that with the recovered concrete cover over the crack, it should bring a significant reduction in the corrosion current density, potentially to a negligible value. The resultant corrosion rate depends on many factors such as the existing corrosive microenvironment around the crack at the time of significant crack healing, continuous exposure conditions in the service condition, concrete properties, and corrosion kinetics of the embedded metal. This is the reason the "critical crack width" value has a large variation for different cover thicknesses, different concrete types, and different exposure conditions. The evaluation of this complex corrosion activity on the rebar is necessary to determine this value, which can be done using the proposed comprehensive corrosion model.

9.5 Epilogue

The comprehensive corrosion model provides a mechanistic understanding of the complex behaviour of rebar corrosion in cracked reinforced concrete structures. By considering factors such as crack width, healing, and microenvironment-dependent corrosion kinetics, the model enables the assessment of corrosion rate in cracked and uncracked RC structures in realistic service

environments. In addition, the determination of the critical crack width for specific reinforced concrete configurations and their service environments becomes possible.

10 Conclusions and Recommendations

10.1 Summary of work

This research work aimed to develop a comprehensive mechanistic corrosion model for rebar corrosion in non-uniform microenvironments caused by cracking and exposure conditions in reinforced concrete structures. The developed model is a versatile tool to perform in-depth corrosion assessment for RC structures and service life prediction. It allows preventive action to take place and avoids unnecessary major expensive repairs, and hence, a reduced life cycle cost. It would also help policymakers to regulate the evaluation criteria and choose optimum rehabilitation strategies for RC structures exceeding their design service life that do not necessarily need to be replaced.

Both experimental work and numerical modelling work have been conducted to overcome the challenges that current numerical models face in describing the corrosion cell of rebars in concrete in the presence of spatial variations of the microenvironment due to cracking and service exposure conditions. This research work was organized into nine chapters that established a logical progression in addressing specific objectives and connecting the research topics.

A thorough literature review was conducted on three topics of corrosion of cracked reinforced concrete: 1) the common and contradictive experimental and field observations of non-uniform corrosion in cracked concrete, 2) fundamentals on corrosion modelling and various numerical treatments of the corrosion cell, and 3) fundamentals on mass transfer models describing the evolution of corrosive microenvironment in concrete affected by various service exposure conditions.

The phase one experimental study focused on characterizing the corrosion parameters of rebar in simulated pore solutions and mortar using an improved test method and investigating the synergistic effects of chloride, carbonation, and water on rebar corrosion. The phase two experimental program performed a corrosion test of the segmented rebars in cracked concrete under wetting and drying cycles with chloride salt solutions. The measured potential and current distributions on these rebar segments were used to validate the developed corrosion model.

In the modelling part, a circuit model was developed to simulate the polarization behaviour of rebar with non-uniform conditions. This model was used to explore the relationship between overall corrosion parameters and non-uniform corrosion parameters at multilevel scales. At the pore level, a model was developed to simulate the non-uniform allocation of active and passive rebar-solution interface in a representative elementary volume based on non-uniform moisture distribution and free chloride ion distribution at a pore level.

The environment model was developed to simulate the microenvironment evolution in concrete by simulating the transport of water, chloride ions, oxygen, and carbon dioxide gases in concrete. This model was applied to an old concrete arch bridge under various service conditions, including chloride sources, carbonation, moisture fluctuation, and exposure to different environmental factors. The predictions of the model aligned well with the current condition of the bridge, and simulations for future climate scenarios provided insights into the implications of climate change on rebar section loss.

A comprehensive corrosion model was then developed by integrating crack configuration into the previously developed environment model to simulate the moisture and chloride content evolution on embedded rebars in cracked concrete. It employed homogenized corrosion kinetics dependent on the microenvironmental conditions to solve the potential and current fields on both the concrete domain and rebar surface. With the validated model, a numerical investigation was performed to explore the mechanisms of how the cracks affect the microenvironment and the corrosion behaviour in the cracked concrete, which provided a mechanistic explanation of the observed rebar corrosion behaviour in cracked concrete, including corrosion reduction over time and the influence of crack size and crack healing.

10.2 Conclusions

The project addressed the synergistic effects of chloride, moisture, and carbonation on corrosion parameters, explored the relationship between overall and non-uniform corrosion parameters, and developed a mechanistic corrosion model that considers the impact of cracking and exposure conditions. The findings contribute to a better understanding of rebar corrosion in complex

microenvironments, offering insights into the design and maintenance of reinforced concrete structures. The detailed findings regarding the specific sub-objectives are as follows:

10.2.1 The synergistic effect of chloride, moisture, and carbonation on the corrosion parameters of rebar in concrete.

10.2.1.1 Test methods

Accurate quantification of corrosion parameters is a vital prerequisite to corrosion modelling. An improved three-electrode potentiodynamic test procedure was developed to measure the corrosion parameters of as-received rebar in simulated pore solutions and mortar. It was found that the use of a Faraday shield and floating ground significantly reduced the large electrical potential oscillations caused by real-time IR compensation. The air-dried rebar-embedded mortar samples should be tested in a non-submerged setting to avoid the open circuit potential drifting. It was necessary to perform separate anodic and cathodic scans to minimize the alteration of the rebar surface.

10.2.1.2 Corrosion parameters in simulated pore solution (SPS) and carbonated simulated pore solution (CSPS) with various chloride concentrations, the significance of depassivation time

The continuous processes of depassivation and rust build-up were found to have an impact on corrosion parameters during the observed periods (short term: 36 hours, 72 hours, and long term: 20 days). Over the 20 days of conditioning of rebar in SPS and CSPS with various chloride concentrations, depassivation slightly decreased the corrosion potential (E_{corr}) for both SPS and CSPS, while significantly increasing the corrosion current density (i_{corr}) for rebar in CSPS. A chloride concentration was identified (2% wt.% for SPS and 1% wt.% for CSPS), beyond which higher chloride concentrations had no significant effect on E_{corr} and i_{corr} over the long term (20 days). The results of the repeated tests showed decreased random uncertainties for long-term conditioning compared to short-term ones.

The anodic Tafel slope in SPS decreased with the chloride concentration at 72 hours and reached a similar value (0.2–0.3 V/dec) for all chloride concentrations at 20 days. In CSPS, the anodic

Tafel slope reached similar values for all chloride concentrations and drifted to a higher value (0.4–0.5 V/dec) at 20 days with the rust build-up.

The cathodic Tafel slope $|\beta_c|$ tends to increase with chloride concentration in both SPS and CSPS but with a large variance for short-term conditioning. More stable and less scattered values were found at 20 days when the disruption of passive oxide film stabilized over time. In CSPS, $|\beta_c|$ increased with the chloride concentration (0.13–0.55 V/dec). In SPS, it only increased with chloride concentration up to 2% (wt.%) from 0.23 V/dec to 0.38 V/dec and then levelled off at higher chloride concentrations.

10.2.1.3 Corrosion parameters for rebar in uncarbonated and carbonation mortars with various chloride contents

For both non-carbonated and carbonated mortars, the corrosion potential E_{corr} decreasd to more negative values with the increase of chloride content. However, unlike SPS, in non-carbonated mortar samples, the corrosion potential (E_{corr}) decreased as the chloride concentration increased, without exhibiting a specific threshold chloride content in partially saturated conditions with chloride content up to 2.24% by weight of cement. However, under saturated conditions, a "threshold" chloride content (0.21% – 0.43% by weight of cement) was observed, similar to the behaviour observed in SPS.

In carbonated mortar samples, on the contrary, this chloride content threshold was observed under partially saturated conditions, but E_{corr} remained similar (around -550 mV – -600 mV) across all chloride levels under saturated conditions. Additionally, E_{corr} values in carbonated mortars were more noble compared to non-carbonated mortars, resembling the behaviour observed in SPS and CSPS.

In drier conditions with lower chloride contents, the corrosion current densities i_{corr} in carbonated samples were considerably higher compared to non-carbonated samples. However, this difference was reduced with the increase in chloride content and relative humidity (RH).

In non-carbonated mortar, i_{corr} increased with higher chloride content across all humidity conditions, which aligned with the inverse trend of E_{corr}. However, in carbonated mortar under

drier conditions (RH 65-85%), the two-stage leveling off trend observed in E_{corr} did not manifest in i_{corr}. Instead, i_{corr} continued to rise with larger chloride content (0.43%-2.24% by weight of cement), while E_{corr} plateaued. This suggests that factors beyond chloride depassivation, such as moisture content and resistivity, contributed to the corrosion current density with a lesser impact on E_{corr}.

In both non-carbonated and carbonated mortar, the anodic Tafel slopes (β_a) did not exhibit a clear dependency on chloride content and conditioning relative humidity (RH), ranging from 0.3–1.0 V/dec. However, exceptionally high β_a values were observed in non-carbonated mortar under submerged conditions with higher chloride content.

In partially saturated conditions, the cathodic Tafel slope ($|\beta_c|$) in non-carbonated mortar increased with chloride content and stabilized at higher chloride levels. Carbonated mortars had significantly larger $|\beta_c|$ values (0.30–0.55 V/dec) compared to non-carbonated mortars (0.12–0.30 V/dec), although no definitive trend was observed in carbonated mortar. In saturated conditions for both types of mortars, $|\beta_c|$ reached a peak value when the chloride content reached 0.43% by weight of cement and then decreased with higher chloride content.

10.2.1.4 The interactive effect of moisture, chloride, and carbonation on rebar corrosion

Carbonation and chloride content affect water content and resistivity in mortars. In carbonated mortars, higher chloride content led to increased water retention at a given RH and decreased resistivity at a given saturation level. However, the effect of chloride content on water retention and resistivity was insignificant in non-carbonated mortars.

Water facilitated the depassivation of rebar and therefore affected E_{corr} in chloride-contaminated mortar in different ways for non-carbonated and carbonated mortar. In non-carbonated mortar, E_{corr} decreased with water saturation when chloride content exceeded a critical value but was less affected at lower chloride levels. In carbonated mortars, the effect of saturation on E_{corr} was strongest without chloride and diminished with increasing chloride content.

Water saturation level affects corrosion rates due to three causes: 1) an area effect by the influenced rebar-pore solution area, 2) an ohmic control effect by the influenced bulk resistivity, and 3) an anode-to-cathode ratio effect.

In non-carbonated mortar, when there is no chloride, the main contributor to rebar corrosion rate is the area effect. When there is chloride present, in addition to the area effect, there also exists an ohmic control effect and anode-to-cathode ratio. The chloride-induced rebar depassivation leads to non-uniform corrosion with a varying anode-to-cathode ratio dependent on water saturation. This non-uniform corrosion cell is at a scale that is under ohmic controlled by a varying bulk resistivity which is also dependent on water saturation.

In carbonated mortar, the rebar corrosion behaviour is affected by both the anode-to-cathode ratio and the surface area. The rust scale on the rebar affects the anode-to-cathode ratio and is influenced by the moisture level. When the rust is dry, it is in a passivation state, and when it is wet, it is in an active state, which results in the dependence of the corrosion potential on the water saturation level in carbonated mortar. This dependence is more pronounced when the chloride content is low, as the conductivity of the rust is highly dependent on water content, and less significant when the chloride content is high. The non-uniform corrosion cell occurs on a scale that is not sensitive to bulk resistivity, hence less ohmic controlled.

10.2.2 Homogenized rebar corrosion parameters in concrete as a function of the variable microenvironment

A circuit model was created to simulate the potentiodynamic scan of non-uniform reinforced concrete, representing the overall polarization behaviour.

It was found that the area-averaged localized potential (homogenized potential) of the rebar was equivalent to the IR-compensated potential measured from the concrete cover. However, the non-uniform local potential on the active and passivated rebar units (rebar-concrete interface at mesoscopic-macroscopic level) and electrical resistance between them may lead to a distorted scan curve. When fitted to the Butler-Volmer equation, it exhibited larger Tafel constants and an overestimation of corrosion current density. This error was more pronounced when the active and passivated rebar unit areas were comparable and the resistance between them was large. Therefore,

corrosion parameters should be measured on a rebar-concrete interface with a uniform condition at mesoscopic–macroscopic scale.

The parametric study performed using this model showed that the homogenized corrosion parameters were functions of concrete resistance and the area ratio of the active or passive rebar-concrete surface area. Therefore, the homogenized corrosion parameters should be a function of the microenvironments around the rebar.

The findings suggested that the homogenized corrosion parameters should be measured in the mortar samples used in the experimental studies that are uniform at a mesoscopic-macroscopic level, and therefore samples were treated with various levels of chloride contents, carbonation states, and water saturation. The homogenized corrosion parameters as a function of the microenvironment should be empirically derived from the test data from experiments similar to those carried out in this project.

A numerical homogenized model was also developed to establish a mechanistic link between the corrosion parameters observed in bulk solution tests and those in mortar. The model quantitatively determined the non-uniform distributions of corrosion-affecting factors, such as water content and chloride content, at the pore level.

The pore size distribution was quantified by an improved adsorption-condensation model with the water retention curve. The distribution of water at the pore level was determined from the pore size distribution and total saturation level. The chloride concentration distribution at the pore level for a given total chloride content was then determined accounting for the bonding effect and chloride concentrate effect due to the electrical double layer. The saturation state and chloride concentration in pores determined the allocation of the area of active and passive rebar-solution interface in a representative elementary volume (REV) of 1 mm^3. A circuit model was then employed to simulate the homogenized corrosion parameters of the rebar-concrete interface in REV from corrosion parameters of rebar-solution interface at the pore level. The simulated corrosion parameters captured the observed pattern from the empirical functions from the experimental data.

10.2.3 Simulation of the evolution microenvironment at the rebar depth of a concrete structure in service conditions.

A comprehensive 3-D transport model was developed to simulate the microenvironment evolution at the rebar depth of a concrete structure in its service conditions. A series of diffusion-advection-reaction transport models were used to simulate the mass transport of water, chloride ions, and carbon dioxide gas.

An optimization routine was developed to form a smooth equivalent pressure head boundary condition to replace the spiky boundary conditions with discrete rain incidence with acceptable accuracy.

This model was applied to an uncracked RC structure — a one-century-old concrete arch bridge subjected to service conditions, including sources of chloride, carbonation, varying air relative humidity, and exposure to unevenly sheltered run-off water. Historical weather data and field monitoring data were used to formulate realistic boundary conditions during wetting and drying scenarios. Laboratory test results from site-extracted concrete and rebar provided corrosion kinetics parameters for the old rebar that was different from modern rebar.

The study also examined the impact of various climate scenarios on rebar section loss in arches over a 50-year projected timeframe. It was observed that elevated CO_2 levels and increased precipitation led to a significant increase in rebar section loss, exacerbating overall corrosion deterioration. Conversely, elevated CO_2 levels with decreased precipitation suppressed high rebar section loss but did not have a significant effect on overall corrosion deterioration. Based on these findings, different maintenance and management approaches for infrastructure were recommended.

10.2.4 Validation of comprehensive corrosion model

A comprehensive corrosion model was developed to solve for the potential and current field in the concrete domain and on the rebar surface for cracked reinforced concrete. It used the homogenized corrosion kinetics as functions of the microenvironmental conditions on the rebar-concrete interface which changed with the evolution of moisture and chloride content on embedded rebars in cracked concrete.

The segmented beam sample was tested in disconnected mode and connected mode in wetting and drying cycles with chloride salt solution.

In disconnected mode, the corrosion potential and corrosion current density for each segment were measured. In connected mode, the rebar electrode potential and total net current leaving or entering each rebar were measured. The 3-D numerical model representing the same experimental setup was validated by comparing the experimental results and the simulated results extracted from the model.

10.2.5 Numerical investigation of effects of crack on rebar corrosion in concrete

A 2-D model was utilized to investigate the impact of cracks on the microenvironment and corrosion of rebar in cracked concrete under submerged condition and wetting and drying cycles. It was used to explain the complex effect of cracks on corrosion, which includes corrosion reduction over time and the effect of crack size and crack healing.

10.2.5.1 Submerged condition

Cracks affected corrosion by affecting the diffusion rate of chloride ions through the crack. Thinner cracks resulted in reduced chloride influx, leading to lower peak corrosion current and slower reduction of corrosion rate over time. This led to more uniform micro-cell corrosion. Eventually, corrosion reached a similar level when the entire rebar was contaminated with chloride, and the corrosion rate was limited by oxygen diffusion.

10.2.5.2 Wetting and drying condition

The model suggested that the corrosion kinetics of rebar in the cracking zone were affected by chloride and moisture, with the possibility of rebar acting as a cathode during drying cycles. The average corrosion rate in the cracking zone peaked during wetting cycles and decreased during subsequent drying cycles. The peak values decreased with repeated wetting and drying cycles. The final corrosion rate was determined by the chloride-contaminated area on the rebar and reduced oxygen concentration. Micro-cell corrosion eventually became the primary contributor to rebar corrosion in the cracking zone.

10.2.5.3 Crack size

In wetting and drying scenarios, the simulation results demonstrated that the crack opening size had a diminished impact on chloride influx through the crack, unlike the submerged case. Based on the extensive numerical investigation, among all parameters that are known to control the corrosion rate, self-healing, which takes place in thin cracks, was the single parameter than can explain the stark difference in corrosion behaviour between thin cracks and thick cracks in the splash zone (wetting and drying cycles). To capture the effect of crack size on corrosion, the model was improved to incorporate crack healing to realistically represent the long-term behaviour of concrete structures in the presence of thin cracks. Crack healing allowed the crack to heal when the opening was smaller than a critical "healable size," resulting in decreased transport properties over time. Crack healing reduced future chloride transport, diluted high chloride content near the crack, and limited oxygen diffusion. Less permeable concrete such as HPC experienced greater chloride content reduction due to crack healing. Crack healing accelerated the reduction of corrosion rate over wetting and drying cycles. Finer cracks had a greater healable crack length, resulting in more recovered concrete cover for the rebar at the crack.

10.2.5.4 Implication on critical crack width

The critical crack width should be small enough so that the crack width at the rebar depth is smaller than the maximum healable crack width. This allows the crack healing process to recover some of the concrete cover at the crack. The critical crack width should be further minimized to a standard that the corrosion current density in the cracking area is significantly reduced, potentially to a negligible value. The resultant corrosion rate depends on factors such as the existing corrosive microenvironment, continuous exposure conditions, concrete properties, and corrosion kinetics. The developed comprehensive corrosion model can be used to determine the uncertain critical crack width for given reinforced concrete configuration and its service environment.

10.3 Recommendations for future research

10.3.1 Effect of temperature on corrosion parameters

More work should be done to investigate the effect of temperature on corrosion parameters. The temperature dependence of the chemical reaction rate is usually described by the well-known Arrhenius equation. For rebar corrosion in concrete, however, there is no direct pure Arrhenius trend because the corrosion rate is affected by multiple factors, pH, chloride, moisture content, resistivity, temperature, etc. and the change in temperature will also affect other important factors such as resistivity. However, there are numerical models that take into account the effect of temperature. For example, in the numerical model feathering discrete separated anodic and cathodic sites (Pour-Ghaz et al., 2009a), the Arrhenius type of dependency on temperature has been reflected on the exchange current density of the two half-cell reactions (Fe^{2+}/Fe and H_2O/OH^-), and another major influencing factor is the resistance change due to temperature between the separated anode and cathode. Although the effect of resistivity is debatable when the distribution of anodic and cathodic sites is unclear, the effect from the resistance can be tuned by the anodic/cathodic area ratio, a user-input in the macro-cell corrosion model, which is effectively a calibration factor. It is worth mentioning that since resistivity is related to the water content, mathematically, one can correlate either one of them to the experimental data regardless of the physical mechanism. A similar model was also developed by Yu et al. (2017), and corrosion rate predictions from both models are reasonably comparable to the experimental result with varying temperature and ambient relative humidity (Pour-Ghaz et al., 2009b; Yu et al., 2017).

Nevertheless, with the developed numerical homogenized model that links the rebar-solution interface corrosion behaviour from the pore level to the rebar-concrete in a REV, it is promising to simulate the effect of temperature on the rebar corrosion in concrete without the anode and cathode area as a user's input. In the homogenized model, besides the dependency of exchanging current density on temperature, temperature would also affect water distribution (liquid volume fracture and continuity), resistivity, and chloride binding isotherm.

10.3.2 Enhancing interpretation of half-cell potential mapping results

A regular typical assessment of an RC structural member involves tests of cover thickness, delamination, carbonation depth, chloride content at various depths, compressive strength of concrete cores, half-cell potential mapping, and occasionally corrosion current density etc. Among those acquired data, the most accessible result is the half-cell potential. The interpretation of the half-cell mapping results can be improved. For example, in a bridge deck inspection, the half-cell test is a routine task, and a single criterion is applied to interpret the data (ASTM C876 - 15. G01.14). Since the homogenized corrosion potential depends on the moisture content, the potential field formed by the coupling of rebar mesh in the deck also depends on the moisture and geometry of rebar and concrete cover. The measured half-cell (potential on the deck surface) is affected by moisture and geometry. Therefore, moisture-and-geometry adjusted criteria should be used.

10.3.3 Dynamic link between corrosion and concrete fracture models

The traditional limit-state for the durability design for RC structure is the onset of active corrosion. It is not uncommon to have existing aged structures in service that already well pass this corrosion-initiation period. These structures are operating at their active corrosion stage and often have visible corrosion and corrosion-induced cracking and delamination. To improve the service life prediction for reinforced concrete structures, it is essential to establish a dynamic link between the corrosion model and a concrete fracture model. This linkage ensures that corrosion-induced cracking and propagation are considered and the change in the crack property can affect the corrosion.

REFERENCES

Adamson, A. W., & Gast, A. P. (1997). *Physical Chemistry of Surfaces, 6th Edition*. Wiley.

Alekseev, S. N., Ivanov, F. M., Modry, S., & Schiessl, P. (1993). *Durability of reinforced concrete in aggressive media. (ed. M. Majithia & S.K. Mallick), Russian Translation Series 96*. A.A. Balkema Publishers.

Alonso, C., Andrade, C., & González, J. A. (1988). Relation between resistivity and corrosion rate of reinforcements in carbonated mortar made with several cement types. *Cement and Concrete Research*, *18*(5), 687–698. https://doi.org/10.1016/0008-8846(88)90091-9

Alonso, C., Andrade, C., Izquierdo, M., Nóvoa, X. R., & Pérez, M. C. (1998a). Effect of protective oxide scales in the macrogalvanic behaviour of concrete reinforcements. *Corrosion Science*, *40*(8), 1379–1389. https://doi.org/10.1016/S0010-938X(98)00040-7

Alonso, C., Andrade, C., Izquierdo, M., Nóvoa, X. R., & Pérez, M. C. (1998b). Effect of protective oxide scales in the macrogalvanic behaviour of concrete reinforcements. *Corrosion Science*, *40*(8), 1379–1389. https://doi.org/10.1016/S0010-938X(98)00040-7

Andrade, C., Gulikers, J., Polder, R., & Raupach, M. (2003). RILEM TC 154-EMC: "Electrochemical Techniques for Measuring Metallic Corrosion" Recommendations Half-cell potential measurements-Potential mapping on reinforced concrete structures Prepared by B. Elsener with contributions from. In *Materials and Structures / Matrriaux et Constructions* (Vol. 36).

Andrade, C., Sarría, J., & Alonso, C. (1999). Relative humidity in the interior of concrete exposed to natural and artificial weathering. *Cement and Concrete Research*, *29*(8), 1249–1259. https://doi.org/10.1016/S0008-8846(99)00123-4

Angst, U., Elsener, B., Larsen, C. K., & Vennesland, Ø. (2009). Critical chloride content in reinforced concrete — A review. *Cement and Concrete Research*, *39*(12), 1122–1138. https://doi.org/10.1016/j.cemconres.2009.08.006

Angst, U., Elsener, B., Larsen, C. K., & Vennesland, Ø. (2011). Chloride induced reinforcement corrosion: Rate limiting step of early pitting corrosion. *Electrochimica Acta*, *56*(17), 5877–5889. https://doi.org/10.1016/j.electacta.2011.04.124

Angst, U. M., Geiker, M. R., Alonso, M. C., Polder, R., Isgor, O. B., Elsener, B., Wong, H., Michel, A., Hornbostel, K., Gehlen, C., François, R., Sanchez, M., Criado, M., Sørensen, H., Hansson, C., Pillai, R., Mundra, S., Gulikers, J., Raupach, M., … Sagüés, A. (2019). The effect of the steel–concrete interface on chloride-induced corrosion initiation in concrete: a critical review by RILEM TC 262-SCI. *Materials and Structures/Materiaux et Constructions*, *52*(4). https://doi.org/10.1617/s11527-019-1387-0

Arya, C., & Ofori-Darko, F. K. (1996). Influence of crack frequency on reinforcement corrosion in concrete. *Cement and Concrete Research*, *26*(3), 345–353. https://doi.org/10.1016/S0008-8846(96)85022-8

ASTM International. (2015). Standard test method for corrosion potentials of uncoated reinforcing steel in concrete. ASTM C876 - 15. G01.14,. *ASTM International*, 1–8. https://doi.org/10.1520/C0876-15

Baroghel-Bouny, V., Wang, X., Thiery, M., Saillio, M., & Barberon, F. (2012). Prediction of chloride binding isotherms of cementitious materials by “analytical” model or “numerical” inverse analysis. *Cement and Concrete Research*, *42*(9), 1207–1224. https://doi.org/10.1016/j.cemconres.2012.05.008

Bažant, Z. P., & Najjar, L. J. (1971). Drying of concrete as a nonlinear diffusion problem. *Cement and Concrete Research*, *1*(5), 461–473. https://doi.org/10.1016/0008-8846(71)90054-8

Bažant, Z. P., & Najjar, L. J. (1972). Nonlinear water diffusion in nonsaturated concrete. *Mat??Riaux et Constructions*, *5*(1), 3–20. https://doi.org/10.1007/BF02479073

Borges, P. H. R., Costa, J. O., Milestone, N. B., Lynsdale, C. J., & Streatfield, R. E. (2010). Carbonation of CH and C–S–H in composite cement pastes containing high amounts of BFS.

Cement and Concrete Research, *40*(2), 284–292. https://doi.org/10.1016/J.CEMCONRES.2009.10.020

Boulfiza, M., Sakai, K., Banthia, N., & Yoshida, H. (2003). Prediction of Chloride Ions Ingress in Uncracked and Cracked Concrete. *ACI Materials Journal*, *100*(100), 38–48.

Cao, C. (2014). 3D simulation of localized steel corrosion in chloride contaminated reinforced concrete. *Construction and Building Materials*, *72*, 434–443. https://doi.org/10.1016/j.conbuildmat.2014.09.030

Cao, C., Cheung, M. M. S., & Chan, B. Y. B. (2013). Modelling of interaction between corrosion-induced concrete cover crack and steel corrosion rate. *Corrosion Science*, *69*(24), 97–109. https://doi.org/10.1016/j.corsci.2012.11.028

Chang, Z.-T., Cherry, B., & Marosszeky, M. (2008a). Polarisation behaviour of steel bar samples in concrete in seawater. Part 1: Experimental measurement of polarisation curves of steel in concrete. *Corrosion Science*, *50*(2), 357–364. https://doi.org/10.1016/j.corsci.2007.08.009

Chang, Z.-T., Cherry, B., & Marosszeky, M. (2008b). Polarisation behaviour of steel bar samples in concrete in seawater. Part 2: A polarisation model for corrosion evaluation of steel in concrete. *Corrosion Science*, *50*(11), 3078–3086. https://doi.org/10.1016/j.corsci.2008.08.021

Daian, J. F. (1988). Condensation and isothermal water transfer in cement mortar Part I - Pore size distribution, equilibrium water condensation and imbibition. *Transport in Porous Media*, *3*(6), 563–589. https://doi.org/10.1007/BF00959103

DaÏan, J. F., & Saliba, J. (1993). Transient moisture transport in a cracked porous medium. *Transport in Porous Media*, *13*(3), 239–260. https://doi.org/10.1007/BF00622445

Dangla, P., & Dridi, W. (2009). Rebar corrosion in carbonated concrete exposed to variable humidity conditions. Interpretation of Tuutti's curve. *Corrosion Science*, *51*(8), 1747–1756. https://doi.org/10.1016/j.corsci.2009.04.029

Davis, S. L., Goldberg, D., DeGood, K., Donohue, N., & Corless, J. (2013). *The Fix We're In For: The State of Our Nation's Bridges 2013*.

Dufrêche, J.-F., Marry, V., Bernard, O., & Turq, P. (2001). Models for electrokinetic phenomena in montmorillonite. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, *195*(1), 171–180. https://doi.org/https://doi.org/10.1016/S0927-7757(01)00840-8

DURACRETE. (1999). *Probabilistic Performance Based Durability Design of Concrete Structures* (p. 86p). http://fib.bme.hu/proceedings/lindvall.pdf

Elakneswaran, Y., Nawa, T., & Kurumisawa, K. (2009). Electrokinetic potential of hydrated cement in relation to adsorption of chlorides. *Cement and Concrete Research*, *39*(4), 340–344. https://doi.org/10.1016/j.cemconres.2009.01.006

Elsener, B. (2005). Corrosion rate of steel in concrete-Measurements beyond the Tafel law. *Corrosion Science*, *47*(12), 3019–3033. https://doi.org/10.1016/j.corsci.2005.06.021

Fabjan, C., Kazemi, M. R., & Neckel, A. (1980). Untersuchungen über den Mechanismus und die Katalyse der kathodischen Sauerstoffreduktion an verschiedenen Metallen. *Berichte Der Bunsengesellschaft Für Physikalische Chemie*, *84*(10), 1026–1031.

Federal Highway Administration. (2002). *Corrosion Cost and Preventive Strategies in the United States (Report FHWA-RD-01-156).*

Figueira, R. B., Sadovski, A., Melo, A. P., & Pereira, E. V. (2017). Chloride threshold value to initiate reinforcement corrosion in simulated concrete pore solutions: The influence of surface finishing and pH. *Construction and Building Materials*, *141*, 183–200. https://doi.org/10.1016/j.conbuildmat.2017.03.004

Frankel, G. S. (2016). Fundamentals of Corrosion Kinetics. In A. E. Hughes, J. M. C. Mol, M. L. Zheludkevich, & R. G. Buchheit (Eds.), *Active Protective Coatings: New-Generation Coatings for Metals* (pp. 17–32). Springer Netherlands. https://doi.org/10.1007/978-94-017-7540-3_2

Friedmann, H., Amiri, O., & Aït-Mokhtar, A. (2008). Physical modeling of the electrical double layer effects on multispecies ions transport in cement-based materials. *Cement and Concrete Research*, *38*(12), 1394–1400. https://doi.org/10.1016/j.cemconres.2008.06.003

Galus, Z. (1994). *Fundamentals of electrochemical analysis*. Ellis Horwood.

Gamry Instruments. (2012). *Reference 600^{TM} Potentiostat/Galvanostat/ZRA Operator's Manual*. www.gamry.com/service-support/

Garcés, P., Andrade, M. C., Saez, A., & Alonso, M. C. (2005). Corrosion of reinforcing steel in neutral and acid solutions simulating the electrolytic environments in the micropores of concrete in the propagation period. *Corrosion Science*, *47*(2), 289–306. https://doi.org/10.1016/j.corsci.2004.06.004

Ge, J., & Isgor, O. B. (2007). Effects of Tafel slope, exchange current density and electrode potential on the corrosion of steel in concrete. *Materials and Corrosion*, *58*(8), 573–582. https://doi.org/10.1002/maco.200604043

Ghods, P., Isgor, O. B., McRae, G. A., & Gu, G. P. (2010). Electrochemical investigation of chloride-induced depassivation of black steel rebar under simulated service conditions. *Corrosion Science*, *52*(5), 1649–1659. https://doi.org/10.1016/j.corsci.2010.02.016

Ghods, P., Isgor, O. B., & Pour-Ghaz, M. (2007). A practical method for calculating the corrosion rate of uniformly depassivated reinforcing bars in concrete. *Materials and Corrosion*, *58*(4), 265–272. https://doi.org/10.1002/maco.200604010

Glass, G. K., & Chadwick, J. R. (1994). An investigation into the mechanisms of protection afforded by a cathodic current and the implications for advances in the field of cathodic protection. *Corrosion Science*, *36*(12), 2193–2209. https://doi.org/10.1016/0010-938X(94)90017-5

Glass, G. K., Page, C. L., & Short, N. R. (1991). Factors affecting the corrosion rate of steel in carbonated mortars. *Corrosion Science*, *32*(12), 1283–1294. https://doi.org/10.1016/0010-938X(91)90048-T

Glass, G. K., Wang, Y., & Buenfeld, N. R. (1996). An investigation of experimental methods used to determine free and total chloride contents. *Cement and Concrete Research*, *26*(9), 1443–1449. https://doi.org/10.1016/0008-8846(96)00115-9

Gu, P., & Beaudoin, J. J. (1998). Obtaining Effective Half-Cell Potential Measurements in Reinforced Concrete Structures. *Construction Technology Update No.18*, 1–4.

Gulikers, J. (2005). Theoretical considerations on the supposed linear relationship between concrete resistivity and corrosion rate of steel reinforcement. *Materials and Corrosion*, *56*(6), 393–403. https://doi.org/10.1002/maco.200403841

Hansson, C. M., Poursaee, a., & Laurent, a. (2006). Macrocell and microcell corrosion of steel in ordinary Portland cement and high performance concretes. *Cement and Concrete Research*, *36*(11), 2098–2102. https://doi.org/10.1016/j.cemconres.2006.07.005

Hasted, J. B., Ritson, D. M., & Collie, C. H. (1948). Dielectric properties of aqueous ionic solutions. Parts I and II. *The Journal of Chemical Physics*, *16*(1), 1–21. https://doi.org/10.1063/1.1746645

Hornbostel, K., Angst, U. M., Elsener, B., Larsen, C. K., & Geiker, M. R. (2015). On the limitations of predicting the ohmic resistance in a macro-cell in mortar from bulk resistivity measurements. *Cement and Concrete Research*, *76*, 147–158. https://doi.org/10.1016/j.cemconres.2015.05.023

Hussain, R. R. (2011). Effect of moisture variation on oxygen consumption rate of corroding steel in chloride contaminated concrete. *Cement and Concrete Composites*, *33*(1), 154–161. https://doi.org/10.1016/j.cemconcomp.2010.09.014

Isgor, O. B., & Razaqpur, a. G. (2004). Finite element modeling of coupled heat transfer, moisture transport and carbonation processes in concrete structures. *Cement and Concrete Composites*, *26*(1), 57–73. https://doi.org/10.1016/S0958-9465(02)00125-7

Isgor, O. B., & Razaqpur, a G. (2006). Modelling steel corrosion in concrete structures. *Materials and Structures*, *53*˙(12), 291–302. https://doi.org/10.1617/s11527-005-9022-7

Jennings, H. M., Kumar, A., & Sant, G. (2015). Quantitative discrimination of the nano-pore-structure of cement paste during drying: New insights from water sorption isotherms. *Cement and Concrete Research*, *76*, 27–36. https://doi.org/10.1016/j.cemconres.2015.05.006

Jennings, S. G. (1988). The mean free path in air. *Journal of Aerosol Science*, *19*(2), 159–166. https://doi.org/10.1016/0021-8502(88)90219-4

Ji, Y. S., Zhao, W., Zhou, M., Ma, H. R., & Zeng, P. (2013). Corrosion current distribution of macrocell and microcell of steel bar in concrete exposed to chloride environments. *Construction and Building Materials*, *47*, 104–110. https://doi.org/10.1016/j.conbuildmat.2013.05.003

Jones, D. A. (1996). *Principles and prevention of corrosion*. Prentice Hall.

Jones, E., Oliphant, T., Peterson, P., & Others. (2001). *SciPy: Open Source Scientific Tools for Python*.

Kaviany, M. (2012). *Principles of heat transfer in porous media*. Springer Science & Business Media.

Kennell, G., Zacaruk, J., Li, G., & Boulfiza, M. (2015). The University Bridge Arch Assessment - A New Approach. *2015 Transportation Association of Canada (TAC) Conference & Exhibition*, 1–14. http://conf.tac-atc.ca/english/annualconference/tac2015/s18/kennell.pdf%0Ahttps://trid.trb.org/view/1399685

Kim, C., & Kim, J. (2008). Numerical analysis of localized steel corrosion in concrete. *Construction and Building Materials*, *22*(6), 1129–1136. https://doi.org/10.1016/j.conbuildmat.2007.02.007

Kontogeorgis, G. M., & Kiil, S. (2016). Fundamental Equations in Colloid and Surface Science. In *Introduction to Applied Colloid and Surface Chemistry* (pp. 74–95). John Wiley & Sons, Ltd. https://doi.org/10.1002/9781118881194.ch4

Kranc, S. C., & Sagüés, A. A. (1994). Computation of Reinforcing Steel Corrosion Distribution in Concrete Marine Bridge Substructures. *CORROSION*, *50*(1), 50–61. https://doi.org/10.5006/1.3293494

Kranc, S. C., Sagüés, A. A., & Sagues, A. (1992). Computation of corrosion macrocell current distribution and electrochemical impedance of reinforcing steel in concrete. *Computer Modeling in Corrosion*, *48*(8), 95–112. https://doi.org/10.1520/STP24689S

Kumar, A. (2010). *Water Flow and Transport of Chloride in Unsaturated Concrete* [Master's thesis]. University of Saskatchewan.

Künzel, H. M. (1995). Simultaneous heat and moisture transport in building components. In *IRB Verlag Suttgart*. https://doi.org/ISBN v.3-8167-4103-7

Lahmann, D., Edvardsen, C., & Kessler, S. (2022). *Autogenous self-healing of concrete: Experimental design and test methods-A review*. https://doi.org/10.1002/eng2.12565

Li, G. (2014). *Effect of Cracks on the Transport Characteristics of Cracked Concrete* [Master's Thesis]. University of Saskatchewan.

Li, G., Evitts, R., & Boulfiza, M. (2018). Homogenized-model-simplified-circuit: code for the simplified circuit model. *Zenodo*. https://doi.org/10.5281/ZENODO.1406210

Li, G., Evitts, R., & Boulfiza, M. (2023). The impact of non-uniformity and resistivity on the homogenised corrosion parameters of rebars in concrete – a circuit model analysis. *Corrosion Engineering, Science and Technology*, *58*(4), 399–409. https://doi.org/10.1080/1478422X.2023.2190442

Li, G., Evitts, R., Boulfiza, M., & Li, A. D. S. (2018). A customized Python module for interactive curve fitting on potentiodynamic scan data. *Zenodo*. https://doi.org/10.5281/ZENODO.1406195

Liu, T., & Weyers, R. W. (1998). Modeling the Dynamic Corrosion Process in Chloride Contaminated Concrete Structures. *Cement and Concrete Research*, *28*(3), 365–379. https://doi.org/10.1016/S0008-8846(98)00259-2

Locke, C. E., & Siman, A. (1980). Electrochemistry of Reinforcing Steel in Salt-Contaminated Concrete. In *Corrosion of Reinforcing Steel in Concrete* (pp. 3–16). ASTM International. https://doi.org/10.1520/STP27465S

López, W., & González, J. A. (1993). Influence of the degree of pore saturation on the resistivity of concrete and the corrosion rate of steel reinforcement. In *Cement and Concrete Research* (Vol. 23, Issue 2, pp. 368–376). https://doi.org/10.1016/0008-8846(93)90102-F

Lu, C., Yang, J., Li, H., & Liu, R. (2017). Experimental Studies on Chloride Penetration and Steel Corrosion in Cracked Concrete Beams under Drying-Wetting Cycles. *Journal of Materials in Civil Engineering*, *04017114*. https://doi.org/10.1061/(ASCE)MT.1943-5533.0001953. ©

Maekawa, K., Ishida, Tetsuya., & Kishi, T. (2008). *Multi-scale modeling of structural concrete*. CRC Press.

Martın-Pérez, B. (1999). *Service life modelling of RC highway structures exposed to chlorides*.

Mays, Geoffrey. (2004). *Durability of concrete structures : investigation, repair, protection*. Taylor & Francis.

Mehta, P. K., & Monteiro, P. J. M. (2006). *Concrete : microstructure, properties, and materials.* (Third Edit). McGraw-Hill. https://doi.org/10.1036/0071462899

Monfore, G. E. (1968). The Electrical Resistivity of Concrete. *Journal of the PCA Research and Development Laboratories*, 35–48.

Moreno, M., Morris, W., Alvarez, M. G. G., & Duffó, G. S. S. (2004). Corrosion of reinforcing steel in simulated concrete pore solutions effect of carbonation and chloride content. *Corrosion Science*, *46*(11), 2681–2699. https://doi.org/10.1016/j.corsci.2004.03.013

Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, *12*(3), 513–522.

Muthulingam, S., & Rao, B. N. (2014). Non-uniform time-to-corrosion initiation in steel reinforced concrete under chloride environment. *Corrosion Science*, *82*, 304–315. https://doi.org/10.1016/j.corsci.2014.01.023

Muthulingam, S., & Rao, B. N. (2015). Non-uniform corrosion states of rebar in concrete under chloride environment. *Corrosion Science*, *93*, 267–282. https://doi.org/10.1016/j.corsci.2015.01.031

Ngala, V. T., & Page, C. L. (1997). Effects of carbonation on pore structure and diffusional properties of hydrated cement pastes. *Cement and Concrete Research*, *27*(7), 995–1007. https://doi.org/10.1016/S0008-8846(97)00102-6

Nguyen, W., Duncan, J. F., Devine, T. M., & Ostertag, C. P. (2018). Electrochemical polarization and impedance of reinforced concrete and hybrid fiber-reinforced concrete under cracked matrix conditions. *Electrochimica Acta*, *271*, 319–336. https://doi.org/10.1016/j.electacta.2018.03.134

Otieno, M., Beushausen, H., & Alexander, M. (2012). Prediction of corrosion rate in reinforced concrete structures - A critical review and preliminary results. *Materials and Corrosion*, *63*(9), 777–790. https://doi.org/10.1002/maco.201106282

Pargar, F., Koleva, D. A., & Van Breugel, K. (2017). Determination of chloride content in cementitious materials: From fundamental aspects to application of Ag/AgCl chloride sensors. In *Sensors (Switzerland)* (Vol. 17, Issue 11). https://doi.org/10.3390/s17112482

Pech-Canul, M. a., & Castro, P. (2002). Corrosion measurements of steel reinforcement in concrete exposed to a tropical marine atmosphere. *Cement and Concrete Research*, *32*(3), 491–498. https://doi.org/10.1016/S0008-8846(01)00713-X

Pel, L., Kopinga, K., & Brocken, H. (1996). Moisture transport in porous building materials. In *Heron* (Vol. 41, Issue 2). http://www.phys.tue.nl/nfcmr/PhD-Pel-1995.pdf

Perez, N. (2004). *Electrochemistry and corrosion science*. Springer.

Pettersson, K., Jorgenson, O., & FidjestolI, P. (1996). The effect of cracks on reinforcement corrosion in high-performance concrete in a marine environment. *Special Publication*, *163*, 185–200.

Pour-Ghaz, M., Isgor, O. B., & Ghods, P. (2009a). The effect of temperature on the corrosion of steel in concrete. Part 1: Simulated polarization resistance tests and model development. *Corrosion Science*, *51*(2), 415–425. https://doi.org/10.1016/j.corsci.2008.10.034

Pour-Ghaz, M., Isgor, O. B., & Ghods, P. (2009b). The effect of temperature on the corrosion of steel in concrete. Part 2: Model verification and parametric study. *Corrosion Science*, *51*(2), 426–433. https://doi.org/10.1016/j.corsci.2008.10.036

Poursaee, A., & Hansson, C. M. (2007). Reinforcing steel passivation in mortar and pore solution. *Cement and Concrete Research*, *37*(7), 1127–1133. https://doi.org/10.1016/j.cemconres.2007.04.005

Poursaee, A., Laurent, A., & Hansson, C. M. M. (2010). Corrosion of steel bars in OPC mortar exposed to NaCl, MgCl2 and CaCl2: Macro- and micro-cell corrosion perspective. *Cement and Concrete Research*, *40*(3), 426–430. https://doi.org/10.1016/j.cemconres.2009.09.029

Pu, Q., Jiang, L., Xu, J., Chu, H., Xu, Y., & Zhang, Y. (2012). Evolution of pH and chemical composition of pore solution in carbonated concrete. *Construction and Building Materials*, *28*(1), 519–524. https://doi.org/10.1016/j.conbuildmat.2011.09.006

Puyate, Y. T., Lawrence, C. J., Buenfeld, N. R., & McLoughlin, I. M. (1998). Chloride transport models for wick action in concrete at large Peclet number. *Physics of Fluids*, *10*(3), 566–575. https://doi.org/10.1063/1.869584

Ramadani, T. (2013). *Effect of Silica Fume on Moisture Flow and the Advective- Dispersive Transport of Ionic Species in Unsaturated Concrete* [M.Sc. Thesis]. University of Saskatchewan.

Raupach, M. (1996). Chloride-induced macrocell corrosion of steel in concrete - Theoretical background and practical consequences. *Construction and Building Materials*, *10*(5), 329–338. https://doi.org/10.1016/0950-0618(95)00018-6

Ryu, D. W., Ko, J. W., & Noguchi, T. (2011). Effects of simulated environmental conditions on the internal relative humidity and relative moisture content distribution of exposed concrete. *Cement and Concrete Composites*, *33*(1), 142–153. https://doi.org/10.1016/j.cemconcomp.2010.09.009

Saetta, A. V, Schrefler, B. A., & Vitaliani, R. V. (1993). The Carbonation of Concrete and the Mechanism of Moisture, Heat and Carbon Dioxide Flow Through Porous Materials. *Cement and Concrete Research*, *23*, 761–772.

Sagüés, A., Lau, K., & Kranc, S. (2008). Modeling of Corrosion of Steel in Concrete with Potential-Dependent Chloride Threshold. *17th Int. Corros. Conf., Paper*, *4006*, 1–16.

Samson, E., Marchand, J., Snyder, K. A., & Beaudoin, J. J. (2005). Modeling ion and fluid transport in unsaturated cement systems in isothermal conditions. *Cement and Concrete Research*, *35*(1), 141–153. https://doi.org/10.1016/j.cemconres.2004.07.016

Scheffler, G. A., & Plagge, R. (2010). A whole range hygric material model: Modelling liquid and vapour transport properties in porous media. *International Journal of Heat and Mass Transfer*, *53*(1–3), 286–296. https://doi.org/10.1016/j.ijheatmasstransfer.2009.09.030

Schiessl, P., Helland, S., Gehlen, C., Nilsson, L. O., & Rostam, S. (2005). fib bulletin 34 - Model code for service life design (MC-SLD). In *Structural Concrete and Time - Proceedings of the fib Symposium* (Vol. 1). fib. The International Federation for Structural Concrete. https://doi.org/10.35789/fib.BULL.0034

Schießl, P., & Raupach, M. (1997). Laboratory studies and calculations on the influence of crack width on chloride-induced corrosion of steel in concrete. *ACI Materials Journal*, *94*(1), 56–61. https://doi.org/10.14359/285

Schmitz, K. S. (2017). Thermodynamics of the Liquid State. In *Physical Chemistry* (pp. 203–260). Elsevier. https://doi.org/10.1016/b978-0-12-800514-9.00005-5

Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with python. *9th Python in Science Conference*.

Shah, V., Scrivener, K., Bhattacharjee, B., & Bishnoi, S. (2018). Changes in microstructure characteristics of cement paste on carbonation. *Cement and Concrete Research*, *109*, 184–197. https://doi.org/10.1016/j.cemconres.2018.04.016

Shi, J., Sun, W., Jiang, J., & Zhang, Y. (2016). *Influence of chloride concentration and pre-passivation on the pitting corrosion resistance of low-alloy reinforcing steel in simulated concrete pore solution*. https://doi.org/10.1016/j.conbuildmat.2016.02.107

Shimomura, T. (2015). Universal moisture transport model in concrete under natural environment. *Proceedings of the 5th International Conference on Construction Materials.*, 1224–1235.

Silva, N., Luping, T., & Rauch, S. (2012). Application of LA-ICP-MS for meso-scale chloride profiling in concrete. *Materials and Structures*, *46*(8), 1369–1381. https://doi.org/10.1617/s11527-012-9979-y

Snyder, K. A., Feng, X., Keen, B. D., & Mason, T. O. (2003). Estimating the electrical conductivity of cement paste pore solutions from OH-, K+ and Na+ concentrations. *Cement and Concrete Research*, *33*(6), 793–798. https://doi.org/10.1016/S0008-8846(02)01068-2

Sohail, M. G. (2013). *Corrosion of steel in concrete: development of an accelerated test by carbonation and galvanic coupling* [Doctoral dissertation, Université de Toulouse]. http://thesesups.ups-tlse.fr/1938/

Stefanoni, M., Angst, U., & Elsener, B. (2015). Local electrochemistry of reinforcement steel - Distribution of open circuit and pitting potentials on steels with different surface condition. *Corrosion Science*, *98*, 610–618. https://doi.org/10.1016/j.corsci.2015.06.004

Stefanoni, M., Angst, U. M., & Elsener, B. (2019). Kinetics of electrochemical dissolution of metals in porous media. *Nature Materials*, *18*(9), 942–947. https://doi.org/10.1038/s41563-019-0439-8

Stratmann, M. (1990). Atmospheric corrosion of iron. A discussion of the physico-chemical fundamentals of this omnipresent corrosion process invited review. *Berichte Der Bunsengesellschaft/Physical Chemistry Chemical Physics*, *94*(6), 626–639. https://doi.org/10.1002/bbpc.19900940603

Subramaniam, K. V., & Bi, M. (2010). Investigation of steel corrosion in cracked concrete: Evaluation of macrocell and microcell rates using Tafel polarization response. *Corrosion Science*, *52*(8), 2725–2735. https://doi.org/10.1016/j.corsci.2010.04.030

Sun, C., Yuan, L., Zhai, X., Qu, F., Li, Y., & Hou, B. (2018). Numerical and experimental study of moisture and chloride transport in unsaturated concrete. *Construction and Building Materials*, *189*, 1067–1075. https://doi.org/10.1016/J.CONBUILDMAT.2018.08.158

Tan, Y. (2012). Heterogeneous Electrode Processes and Localized Corrosion. In *Heterogeneous Electrode Processes and Localized Corrosion*. John Wiley & Sons, Inc. https://doi.org/10.1002/9781118466360

The, L., Dao, N., The, V., Dao, N., Kim, S., & Ann, K. Y. (2010). *Modeling Steel Corrosion in Concrete Structures - Part 2 : A Unified Adaptive Finite Element Model for Simulation of Steel Corrosion*. *5*, 314–326.

Tokunaga, T. K., Finsterle, S., Kim, Y., Wan, J., Lanzirotti, A., & Newville, M. (2017). Ion Diffusion Within Water Films in Unsaturated Porous Media. *Environmental Science and Technology*, *51*(8), 4338–4346. https://doi.org/10.1021/acs.est.6b05891

Tuutti, K. (1982). Corrosion of steel in concrete. *Report 4-82, Swedish Cement and Concrete Research Institute, Sweden*.

US Department of Commerce, N. G. M. L. (n.d.). *Global Monitoring Laboratory - Carbon Cycle Greenhouse Gases*.

Van Enckevort, W. J. P., & Los, J. H. (2013). On the creeping of saturated salt solutions. *Crystal Growth and Design*, *13*(5), 1838–1848. https://doi.org/10.1021/cg301429g

Van Genuchten, M. T. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils 1. *Soil Science Society of America Journal*, *44*(5), 892–898.

Vu, K., Stewart, M. G., & Mullard, J. (2005). Corrosion-induced cracking: Experimental data and predictive models. *ACI Structural Journal*, *102*(5), 719–726. https://doi.org/10.14359/14667

Warkus, J., Raupach, M., & Gulikers, J. (2006). Numerical modelling of corrosion - Theoretical backgrounds. *Materials and Corrosion*, *57*(8), 614–617. https://doi.org/10.1002/maco.200603992

Weiss, J., Snyder, K., Bullard, J., & Bentz, D. (2013). Using a Saturation Function to Interpret the Electrical Properties of Partially Saturated Concrete. *Journal of Materials in Civil Engineering*, *25*(8), 1097–1106. https://doi.org/10.1061/(asce)mt.1943-5533.0000549

Wilkins, N. J. M., & Lawrence, P. F. (1980). Fundamental Mechanisms of Corrosion of Steel Reinforcements in Concrete Immersed in Sea-Water. *Offshore Energy Technology Board.*

Witherspoon, P. a., Wang, J. S. Y., Iwai, K., & Gale, J. E. (1980). Validity of Cubic Law for fluid flow in a deformable rock fracture. *Water Resources Research*, *16*(6), 1016–1024. https://doi.org/10.1029/WR016i006p01016

Yu, B., Liu, J., & Chen, Z. (2017). Probabilistic evaluation method for corrosion risk of steel reinforcement based on concrete resistivity. *Construction and Building Materials*, *138*, 101–113. https://doi.org/10.1016/j.conbuildmat.2017.01.100

Yu, B., Liu, J., & Li, B. (2017). Improved numerical model for steel reinforcement corrosion in concrete considering influences of temperature and relative humidity. *Construction and Building Materials*, *142*, 175–186. https://doi.org/10.1016/j.conbuildmat.2017.03.045

Yu, B., Yang, L., Wu, M., & Li, B. (2014). Practical model for predicting corrosion rate of steel reinforcement in concrete structures. *Construction and Building Materials*, *54*, 385–401. https://doi.org/10.1016/j.conbuildmat.2013.12.046

Zacaruk, J. A. (2013). *Corrosion Cell Formation on a Bar Embedded in Concrete Exposed To Chlorides* (Issue August) [PhD Thesis]. University of Saskatchewan.